

UNIVERSITÉ PARIS XIII – SORBONNE PARIS NORD

École doctorale Sciences, Technologies, Santé Galilée

Du contrôle de qudits aux pertes à deux corps : manipulation du spin nucléaire et photoassociation du ^{87}Sr ultrafroid

From qudit control to two-body losses: nuclear-spin manipulation and
photoassociation of ultracold ^{87}Sr

THÈSE DE DOCTORAT
présentée par

Pauline GUESDON

Laboratoire de Physique des Lasers

pour l'obtention du grade de
DOCTEURE EN PHYSIQUE

soutenue le 14 décembre 2026 devant le jury d'examen constitué de :

MAURIN Isabelle

Laboratoire de Physique des Lasers

Président-e du jury

COMPARAT Daniel

Laboratoire Aimé-Cotton

Rapporteur

BERNON Simon

LP2N, Institut d'optique

Rapporteur

ROBERT-DE-SAINT-VINCENT Martin

Laboratoire de Physique des Lasers

Co-directeur

LABURTHER TOLRA Bruno

Laboratoire de Physique des Lasers

Co-directeur

PASQUIOU Benjamin

Laboratoire de Physique des Lasers

Membre invité

Abstract

Acknowledgements

Contents

Introduction	1
1 Gas preparation and spin control	4
1.1 Fermi gas preparation	9
1.1.1 Blue optical setup	
1.1.2 Transverse cooling	11
1.1.2.1 Operating principle and radiation pressure	
1.1.2.2 Experimental setup and laser configuration	
1.1.2.3 Detuning and power optimization	12
1.1.3 Zeeman cooling	13
1.1.4 Blue MOT	14
1.1.5 Repumper	15
1.1.5.1 Hyperfine structure of Strontium-87	
1.1.5.2 Optimization of the repumper sweep	16
1.1.6 Laser cooling on the intercombination line : the “red MOT”	17
1.1.6.1 Optical setup	
1.1.6.2 Swap cooling, broadband MOT to narrowband	19
1.1.6.3 Stirring laser	22
1.1.6.4 Narrowband MOT (Single-frequency MOT)	25
1.1.6.5 Optimization of narrowband MOT Parameters	
1.1.7 Optical Dipole Trap (ODT) and evaporative cooling	
1.1.7.1 Loading the crossed ODT	
1.1.7.2 Optimization of the evaporation ramps	27
1.1.7.3 Optical setup	28
1.2 Spin Selection	29
1.2.1 Optical Pumping	
1.3 Lasers for spin Manipulation	30
1.3.1 External Cavity Diode Laser (ECDL)	
1.3.1.1 Laser architecture	
1.3.1.2 Problems encountered on the MOGLabs ECDL	31
1.3.2 Optical bench for coherent Raman manipulations	32
1.3.2.1 Physical Concept	
1.3.2.2 Experimental setup of the Raman and PA process	33
1.3.3 Fabry-Pérot cavity and PDH Locking system	35
1.3.3.1 Cavity design and assembly	
1.3.3.2 Pound-Drever-Hall locking system and PID controller analysis	37

1.4	Spin Measurement Scheme	40
2	Quantum state measurement and characterization in qudit systems	42
2.1	Article ‘Coherent control over the high-dimensional space of the nuclear spin of alkaline-earth atoms’ [1]	
2.1.1	Gas preparation and spin control	
2.1.2	Implementation of $\mathfrak{su}(N)$ generators.	44
2.1.3	Experimental setting	45
2.1.4	Coherent manipulation of a nuclear-spin qubit	47
2.1.5	Beyond pseudo-spin 1/2 schemes: qudit interferometry	51
2.1.5.1	Parallel Ramsey interferometers for simultaneous multiple-field sensing.	52
2.1.5.2	Simultaneous measurements of multiple collective atomic observables.	56
2.1.6	Conclusion	61
2.2	Outlook to positive-operator-valued measures (POVMs)	62
2.2.1	Derivation of the experimental POVM framework	63
2.2.1.1	Four-level extended space dynamics	
2.2.1.2	Reduction to the target qubit subspace	64
3	Photoassociation : concept and spectroscopy of ^{88}Sr and ^{87}Sr	66
3.1	Introduction to photoassociation	67
3.1.1	Scattering theory	
3.1.2	PA and complex scattering length	71
3.1.3	General scaling laws for PA	77
3.1.4	Handling multiple molecular potentials	79
3.1.4.1	Angular momentum coupling and state enumeration	
3.1.4.2	Effective hamiltonian and potential modeling	80
3.1.5	Vibrational states estimation theories	82
3.1.5.1	WKB approximation	
3.1.5.2	Leroy-Bernstein scaling	83
3.1.5.3	Mass scaling and isotopic shifts	84
3.2	Spectroscopy protocol and calibration of the photoassociation laser	86
3.2.1	Gas preparation and isotope specificities	
3.2.2	Photoassociation lasers and their spectral characteristics	87
3.2.2.1	ECDL locking and frequency referencing	
3.2.2.2	Spectral imperfections from the locking scheme	90
3.2.2.3	Spectral imperfections from the filter cavity system	
3.2.3	Atomic and molecular lines calibration	94
3.2.3.1	General experimental framework and baseline parameters	
3.2.3.2	Fitting model for molecular resonances	
3.2.3.3	Decay dynamics (dN/dt)	96
3.2.3.4	Determining the optical scattering length l_{opt}	97
3.2.3.5	Estimation of the uncertainties	100
3.2.4	Experimental results on ^{88}Sr	101
3.2.4.1	PA experiments with an alternative laser source	104
3.2.4.2	Noise of the diode	105
3.2.4.3	Decay measurements on ^{88}Sr	107

3.3	^{87}Sr inelastic collisions	111
3.3.1	A protocol to identify molecular lines, learned from misleading signals	112
3.3.1.1	Interpretation of the decay $N(t)$	113
3.3.1.2	Temperature analysis	113
3.3.2	Molecular lines on ^{87}Sr	115
3.3.3	Interpretation of the large linewidth and weak loss rate	123
3.3.3.1	Possible technical causes	
3.3.3.2	Physical hypotheses for the weak loss rate and large linewidth	125
3.3.3.3	Tests to be done in the future	128
	Conclusion	130

List of Figures

1.1	Energy diagram of ^{87}Sr	6
1.2	Extracted and modified from [2] Experimental full setup for the cooling and trapping of the atoms with figures on the velocity, temperature and typical magnetic gradient on each stage.	7
1.3	Experimental sequence for the magnetic field, laser intensities, and laser frequencies	8
1.4	Optical layout of the 461 nm blue laser system	9
1.5	Diagram of the frequencies of each laser composing the blue chain . .	10
1.6	Illustration of the transverse cooling effect	11
1.7	Discussion on TC detuning efficiency (Felix Faisant thesis)	13
1.8	Energy level diagram of ^{87}Sr , 3P_2 transition	16
1.9	Optical setup for the repump and the beat-lock setup of the MOGlabs laser onto the Radiant Dyes.	17
1.10	Optical setup of the 689 nm laser cooling system.	18
1.11	Frequencies of the red lasers used for the red MOT cooling stages . .	19
1.12	Mechanism of SWAP (sawtooth-wave adiabatic passage) cooling . . .	20
1.13	Broadband capture stage of the 689 nm red MOT.	21
1.14	Illustration of the MOT effect on the m_F states of the hyperfine $^3P_1, F = 11/2$ level	23
1.15	Illustration of the melange concept on the $^1S_0(F = 9/2) \rightarrow ^3P_1(F = 9/2)$ transition	24
1.16	Simulation extracted from [3]. Principle of localized 689 nm MOT loading inside the crossed optical dipole trap	26
1.17	Evaporative cooling mechanism in a crossed optical dipole trap	27
1.18	Optical setup of the 1075 nm crossed dipole trap laser system.	28
1.19	Optical pumping scheme on the $^1S_0(F = 9/2) \rightarrow ^3P_1(F = 11/2)$ transition	30
1.20	MOGlabs External Cavity Diode Laser architecture used for spin manipulation	31
1.21	Spin manipulation scheme extracted from [2]	33
1.22	Complete optical setup including the Fabry-Pérot cavity and PDH stabilization loop.	34
1.23	Fabry-Pérot comparison of cavity construction method	36
1.24	Schematic of the Pound-Drever-Hall locking system	37
1.25	Pound-Drever-Hall error signal of the homemade Fabry-Pérot cavity .	38
1.26	Homemade PID scheme	39
1.27	Closed-loop noise analysis of the Fabry-Pérot cavity	39
1.28	Principle of spin-selective detection via coherent momentum transfer. .	41

2.1	Resume of the article experiments implemented over four m_F states . . .	43
2.2	Rabi oscillations driven by $\delta m_F = +1, +2$ Raman coupling and Ramsey interferogram between two Zeeman states	47
2.3	Analysis on phase and contrast of a Ramsey interferogram between two Zeeman states	52
2.4	Simultaneous sensing of two external fields over four m_F states with phase analysis	55
2.5	Measuring two non-commuting observables in a single realization over four states	59
3.1	Illustration of the amplitude's wavefunction of collisional particles . . .	69
3.2	Scheme of the general concept of photoassociation	72
3.3	Hund's case (c) coupling scheme	80
3.4	Density of molecular potentials for ^{173}Yb [4]	81
3.5	Illustration of the WKB semi-classical approximation in a simplified potential well	83
3.6	Plot of the energy levels depending on the isotopic mass of Strontium from mass-scaling article	85
3.7	Beat-note scheme of the PA laser between the "AOM 11/2" light, tuned to the $^1S_0 \rightarrow ^3P_1, F = 11/2$ resonance, and the MOGLabs laser	87
3.8	Frequency schemes for photoassociation (PA) depending on the targeted isotope	89
3.9	Beat note between the MOGLabs laser and the "AOM 11/2" light . . .	90
3.10	Squared modulus of the 0 th - and 1 st -order Bessel functions J_n , which set the power distribution between the carrier and the PDH sidebands.	91
3.11	Scheme of the PDH sidebands addressing the atomic resonance	92
3.12	Atomic loss due to the FP cavity's PDH sideband	93
3.13	Experimental verification of the known resonances of ^{88}Sr	103
3.14	Comparison of the $\nu = -1$ molecular line profile with two different sources	105
3.15	Noise measurement on ^{87}Sr at $\delta = -6$ MHz from the atomic resonance	106
3.16	Twelve m_F states of the $^3P_1, F = 11/2$ hyperfine excited state, measured at $\delta = +6$ MHz and $\delta = +12$ MHz from the atomic resonance.	107
3.17	Atom number decays on the two ^{88}Sr molecular lines	108
3.18	Zoom on the part of the red laser chain used for PA	111
3.19	Evolution of the atom number as a function of the illumination time at $f = -6$ MHz	113
3.20	Temperature of the gas during the decay at -6 MHz (atomic signal)	114
3.21	Characterization of the molecular line at -259.01 ± 0.12 MHz in ^{87}Sr	117
3.22	Characterization of the molecular line at -276.83 ± 0.03 MHz in ^{87}Sr	119
3.23	Characterization of the molecular line at -377.2 ± 0.5 MHz in ^{87}Sr	121
3.24	Experimental sequence for ODT and PA laser chopping	124
3.25	Scheme of predissociation	127

List of Tables

3.1	Summary of the two vibrational states identified for ^{88}Sr , comparing measured line centers and linewidths to the literature values from Zelevinsky et al. [5]. Γ_0 and Γ_{thermal} are estimated from first principles (not extracted from the data). Uncertainties are estimated as described in Sec. 3.2.3.5. *Intensity during the PA pulses: the $\nu = -1$ spectrum was recorded with the ODT/PA chopping sequence (Fig. 3.24, 50% duty cycle), i.e. a time-averaged intensity of $4 \pm 1 \text{ mW/cm}^2$	102
3.2	Comparison of the photoassociation line of the bound state $\nu = -1$ measured with two different laser sources. The excess width is obtained by subtracting in quadrature the natural and thermal widths from the measured one, $\sqrt{\Gamma_{\text{tot}}^2 - (\Gamma_0/2\pi)^2 - \Gamma_{\text{thermal}}^2}$; it gives an order of magnitude of the technical broadening.	104
3.3	Summary of the results obtained on the two ^{88}Sr lines. Both methods account for the measured linewidth (Eq. 3.84): Method 1 uses the depth of the spectrum, Method 2 the decay of the atom number (Sec. 3.2.3.4). *Computed with the time-averaged intensity of the spectrum, $4 \pm 1 \text{ mW/cm}^2$; this value disagrees with Method 2 by a factor $\simeq 50$ (see text).	110
3.4	Summary of the fit parameters and experimental conditions for the three ^{87}Sr lines.	122

Introduction

In the history of science, multidisciplinary research has repeatedly shown the value of crossing field boundaries.

The Saha-Langmuir equation, linking ionization states to local temperature and pressure, was originally developed for physical chemistry. Yet it transformed astrophysics through the work of Cecilia Payne-Gaposchkin [6], who established that stars are predominantly composed of hydrogen and helium. Similarly, X-ray crystallography—devised to probe crystal lattices in solid-state physics—was applied by Rosalind Franklin and Raymond Gosling to determine the helical structure of DNA [7], contributing directly to molecular biology. In the same way, the discovery of radioactivity, polonium, and radium by Marie and Pierre Curie connected physics, chemistry, and medicine [8].

Following this interdisciplinary approach, ultracold atomic physics emerged as a platform for quantum simulation. As proposed by Richard Feynman [9], quantum simulators use controllable atomic systems to study complex many-body problems in condensed matter that remain difficult to model on classical computers. Over the past two decades, cold atom experiments have built a continuous line of progress at the boundary of these fields. I. Bloch *et al.* first demonstrated the transition from a superfluid to a Mott insulator state in an ultracold Bose gas [10], establishing optical lattices as a genuine platform to simulate strongly correlated materials such as transition metal oxides. Building on this proof of principle, systems governed by the Hubbard model were later implemented with both bosons and fermions, enabling the observation of magnetic ordering such as short-range Néel order [11, 12]. Resolving such subtle spatial correlations directly required single-atom, single-site resolution: the quantum gas microscopy technique developed by M. Greiner and I. Bloch [13], first demonstrated on ^{87}Rb condensates and later extended to fermionic systems, made it possible to directly image the antiferromagnetic correlations mentioned above.

Alkaline-earth atoms stand out as a particularly well-suited species to explore these domains further, owing to properties that make them attractive for quantum simulation, optical metrology, and molecular physics alike.

Interest and properties of ^{87}Sr

Strontium is an alkaline-earth element whose stable isotopes are documented in the literature (e.g., [5, 14, 15, 16]). The bosonic isotopes (^{88}Sr , ^{86}Sr , and ^{84}Sr) have zero nuclear spin ($I = 0$).

This work focuses on the fermionic isotope ^{87}Sr , which has a natural abundance of 7.0%. Unlike its bosonic counterparts, ^{87}Sr possesses a nuclear spin $I = 9/2$, giving

access to multi-level quantum dynamics and $SU(N)$ -symmetric interactions (see a few lines below).

In the 1S_0 ground state, the electronic angular momentum is zero ($J = 0$), making the atom almost insensitive to magnetic field fluctuations. However, because there is no electronic magnetic moment, static magnetic fields cannot easily drive spin transitions directly. Optical methods are therefore required to address the nuclear spin states.

Finally, the decoupling between the nuclear spin and the electronic shell in the ground state leads to an $SU(N)$ collisional symmetry: elastic scattering properties of two Zeeman states m_F and m'_F are independent of the nuclear spin projection m_F and m'_F . As a direct consequence, elastic collisions cannot redistribute population among spin states – an arbitrary spin mixture, once prepared, is protected against collisional spin relaxation. This is what makes the optical spin-manipulation and detection techniques presented below reliable: any spin dynamics observed can be attributed to the applied fields rather than to collisional decoherence of the prepared state.

Metrology

Fermionic strontium first attracted interest for metrology because, unlike its bosonic counterparts, spin-polarized samples are protected against collisional shifts of the clock transition frequency by the Pauli exclusion principle [17]. It is now widely used in optical frequency standards, connecting precision measurement with atomic physics:

- **Ultra-narrow clock transition:** The $^1S_0 \rightarrow ^3P_0$ transition features a narrow natural linewidth (~ 1 mHz) shielded from magnetic perturbations due to $J = 0$ in both states [17]. H. Katori and collaborators identified a *magic wavelength* ($\lambda_{\text{magic}} \approx 813$ nm) where the AC Stark shifts of 1S_0 and 3P_0 cancel, enabling trapping in optical lattices without shifting the transition frequency [18].
- **Beating the Standard Quantum Limit (SQL):** Measurement protocols using quantum logic, heterodyne detection, or spin-squeezing aim to reduce phase uncertainty below the SQL ($\Delta\theta \propto 1/\sqrt{N}$) [19]. For example, J. Ye and co-workers reported clock fractional instabilities at the 10^{-18} level [20]. In their experiment, a coherent spin state is prepared via a $\pi/2$ pulse. Cavity-mediated non-destructive measurement generates a spin-squeezed state that reduces phase noise. A second pulse completes the Ramsey sequence, achieving phase uncertainty below the SQL with a squeezing parameter $\xi^2 = 0.63 < 1$.

Photoassociation and dissipative quantum engineering

Optical control of short-range interactions also connects atomic physics with molecular spectroscopy:

- **Interaction control:** Interatomic interactions in degenerate quantum gases can be modified through molecular spectroscopy via photoassociation [21, 22], using optical Feshbach resonances to adjust the s -wave scattering length. Nevertheless these manipulations come with photon scattering responsible of gas heating.

- **Dissipation-assisted state preparation:** Controlled dissipation, such as two-body losses induced by photoassociation, can be used to stabilize target quantum states. Theoretical works (e.g., [23]) indicate that engineered dissipation can prepare Dicke states for precision measurements below the SQL.

Thesis Outline

This manuscript covers experimental atomic cooling, quantum measurement schemes, and ultracold molecular spectroscopy.

The first chapter details the experimental sequence used to prepare a degenerate ^{87}Sr gas, from initial laser cooling to nuclear spin state preparation. Physical models and operating parameters are presented for Transverse Cooling, the Zeeman Slower, Magneto-Optical Traps (MOT), and Evaporative Cooling. This chapter also describes a custom optical spectrum-filtering setup and diagnostic procedures developed for Extended-Cavity Diode Lasers (ECDL, MOGLabs).

The second chapter presents the quantum measurement platform constructed during this thesis, integrating the direct article we produced [1]. We characterize the coherence time of the nuclear spin qudit using Ramsey interferometry, then implement a dual-Ramsey scheme to extract two physical parameters within a single experimental cycle. Finally, we examine non-commuting observables in an initial 2-dimensional Hilbert space that we extend to 4-dimensions subspace and map the protocol onto Positive-Operator-Valued Measures (POVM).

The third chapter reports on the spectroscopy of molecular lines in $^{87}\text{Sr}_2$. Following an overview of photoassociation and ultracold dimer physics, we describe the inelastic collisions of fermions and analyze the technical challenges encountered during their production and detection.

Chapter 1

Gas preparation and spin control

In this chapter, I describe each stage of the cooling and loading process from a practical “user guide” perspective, concluding with an explanation of the spin manipulation schemes tailored to our experimental goals: coherent control of quantum systems, precision measurement, and photoassociation spectroscopy of ^{87}Sr .

Preparation of the cold gas follows the standard scheme of ultracold gas experiments, combining successive stages of laser cooling with a final phase of evaporative cooling in an optical trap. Starting from a thermal atomic beam, the aim of this sequence is to progressively reduce the atoms’ velocity spread and increase their phase-space density, until quantum degeneracy is reached.

The gas preparation begins with an atomic flux effusing from an oven held at $T = 490^\circ\text{C}$, with a mean forward velocity of $v \simeq 500$ m/s. The first cooling stage, 2D transverse cooling (TC), reduces the transverse velocity spread of this beam to $\simeq 20$ m/s, at an estimated temperature $T \simeq 1$ mK – somewhat above the Doppler limit of the $^1S_0 \rightarrow ^1P_1$ transition, since TC deliberately operates in the low-intensity regime discussed below.

Meanwhile, the Zeeman slower and blue Magneto-Optical Trap (MOT) lasers remain on for $t_{\text{BlueMOT}} = 8$ s, both operating on the broad $^1S_0 \rightarrow ^1P_1$ transition (see Fig. 1.1), which both require a high magnetic gradient.

Because the transition is not closed, atoms accumulate in 3P_2 . The dipolar magnetic moment of the atoms trapped in this metastable state help to capture them with a magnetic trap set to $\Delta B = 51$ G/cm for a coil current of 170 A. This trap is crucial for two reasons: it enables to accumulate the more atoms in the metastable state which demand few seconds but also to avoid losing them during the process when they are not addressed by the blue MOT anymore.

They are then repumped in 3.2 ms by a frequency sweep across multiple hyperfine states of the $^3P_2 \rightarrow ^3D_2$ transition, to recover atoms that decayed in metastable -instead of 3P_1 - and immediately recaptured in a MOT on the narrow line. By the end of this stage, approximately 30 million atoms are captured spread in about a centimeter in space.

The atoms are subsequently cooled on the narrow $^1S_0 \rightarrow ^3P_1$ intercombination transition, in two successive red MOT stages: 182 ms of Sawtooth Wave Adiabatic Passage (SWAP) and broadband cooling followed by 250 ms of narrower-band cool-

ing (see below). Throughout both stages, a second “stirring” beam is applied in parallel with the main MOT beam to counteract a loss channel intrinsic to the red MOT’s σ^+/σ^- configuration (see below). The first stage, operated at a magnetic gradient $\Delta B \simeq 1$ G/cm, brings the cloud to $T \simeq 200$ μ K; the resulting Boltzmann-distributed gas has an in-situ spatial spread of $\simeq 1$ mm. The second stage, at the same gradient, further cools the same 30 million atoms to an average temperature of $T \simeq 30$ μ K, with a size of 300 μ m.

The final cooling step is a single-frequency (monofrequency) red MOT, operated at $\Delta B \simeq 2.5$ G/cm at the crossing point of the two arms of an Optical Dipole Trap (ODT), occupying about 30 μ m. Two configurations are used depending on the results targeted in this thesis: in Chapter 2, the atoms are additionally loaded into an optical lattice formed at the crossing point, reaching a colder temperature of $T \simeq 3$ μ K before the gas is further evaporated reaching deep quantum degeneracy at $T \simeq 100$ nK, giving $T/T_F = 0.25$ for a mean geometric pulsation $\bar{\omega} = 2\pi \cdot 35$ estimated at the beginning of the evaporation, in a round surface of about 10 μ m diameter.

In Chapter 3, the atoms are held in one or two arms of the ODT only, within a round region of $\simeq 100$ μ m, reaching $T \simeq 3$ μ K with an atom number of 400 000.

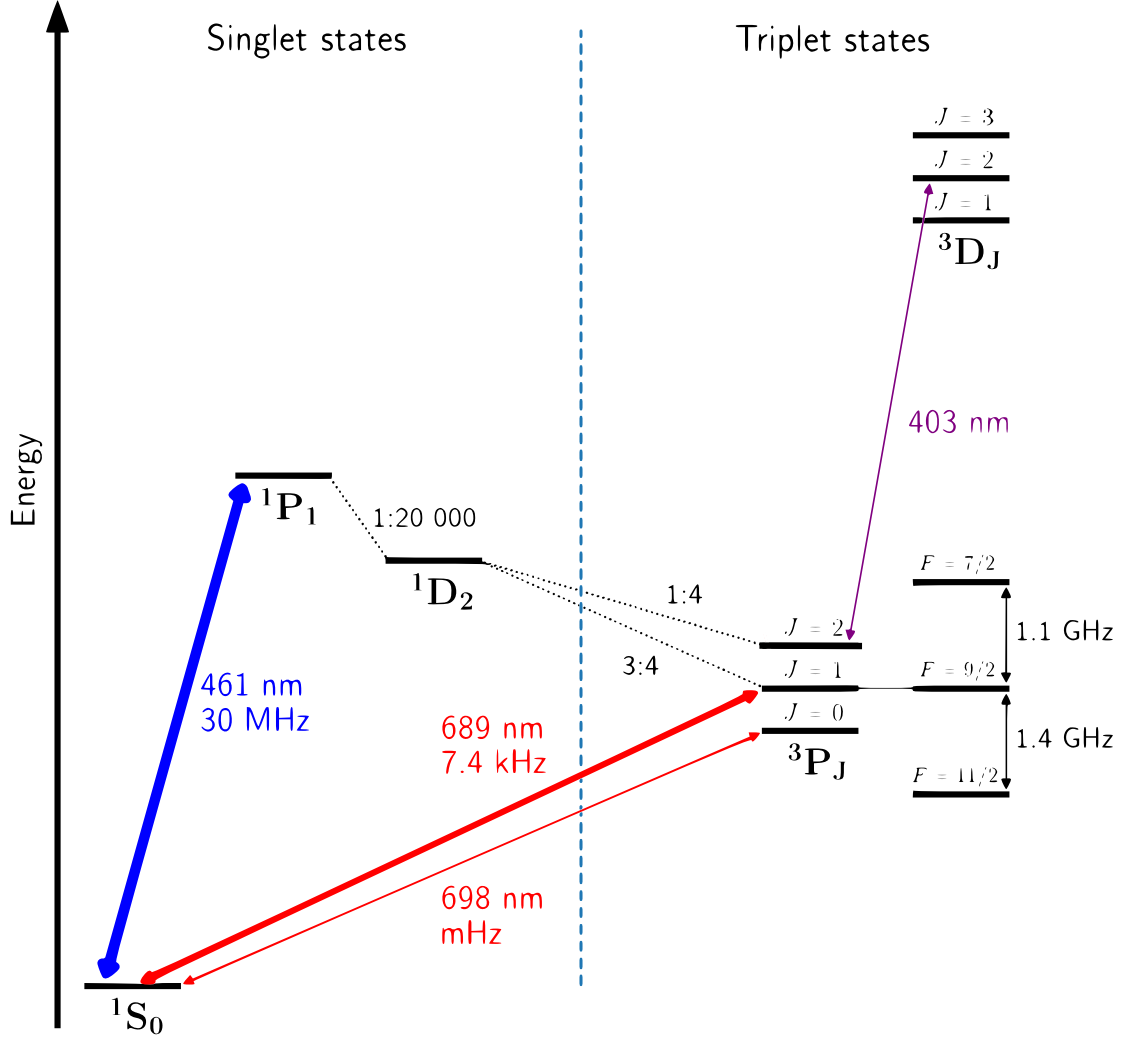


Figure 1.1: Extracted from [3] Energy spectrum of ^{87}Sr including fine and hyperfine interactions. In blue the 461 nm transition used for TC, Zeeman cooling, blue MOT stage and imaging. The purple arrow indicates the repump line at 403 nm from the metastable state 3P_2 used for the atoms leaking from the intermediate 1D_2 state. In red the transition to 3P_1 at 689 nm used for all stages of the red MOT, optical pumping, spin manipulation, coherent manipulation (see Chap.2), photoassociation (see Chap.3, through the different hyperfine states $F = 7/2, 9/2, 11/2$). The 698 nm transition indicates the clock transition which we do not use in this experiment.

Fig 1.2 is a representation of the entire experimental setup that summarizes the speed of the atoms, real magnetic gradient, the temperature and the size of the gas for every cooling stage.

The complete timing sequence – magnetic field ramps, laser intensities, and laser frequencies – for every stage described in this chapter is summarized in Fig. 1.3.

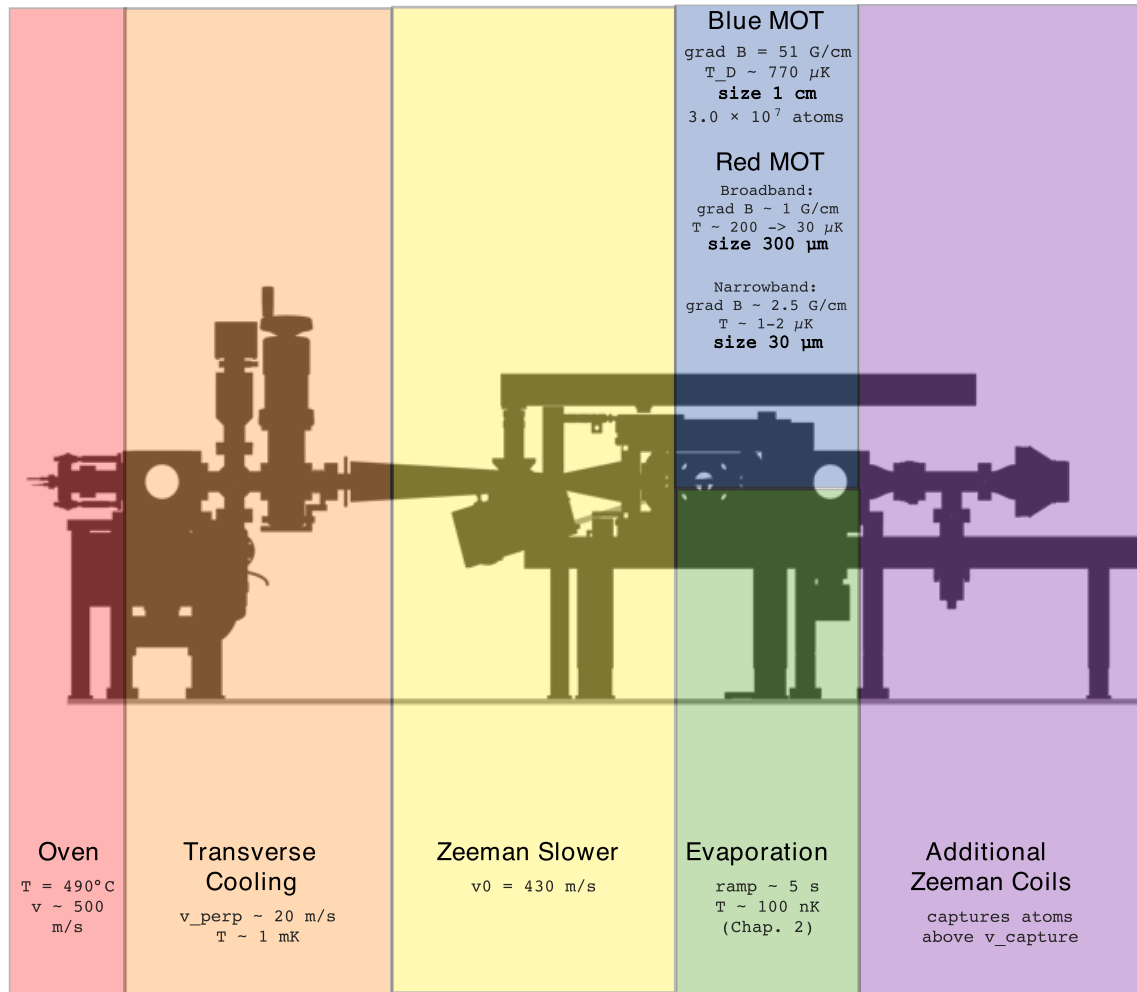
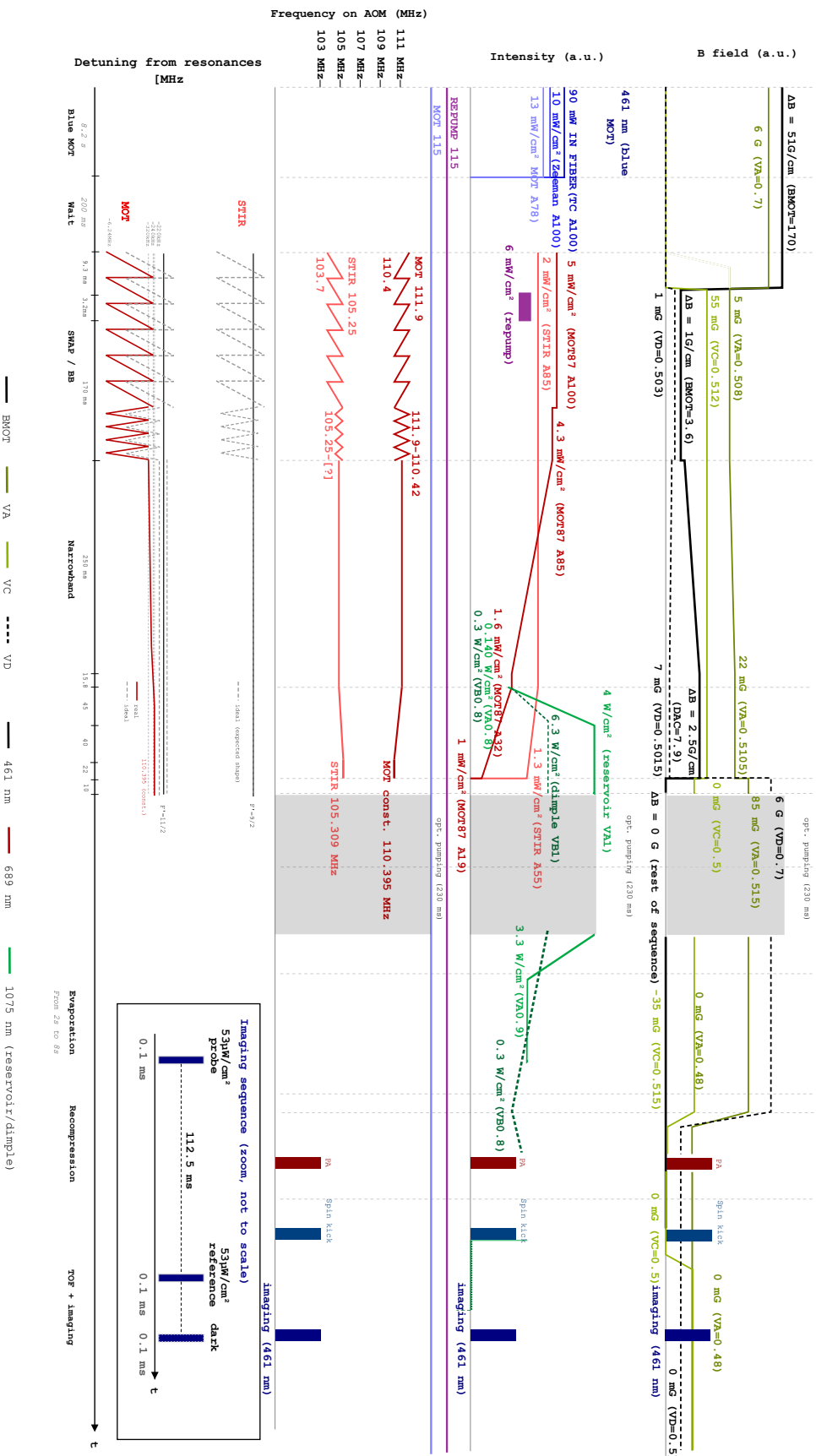


Figure 1.2: Extracted and modified from [2] Experimental full setup for the cooling and trapping of the atoms with figures on the velocity, temperature and typical magnetic gradient on each stage.



1.1 Fermi gas preparation

1.1.1 Blue optical setup

Solid strontium is vaporized in an oven heated to 490°C, outputting an atomic beam through a cluster of fine capillaries that provide initial spatial collimation.

The first cooling stages operate on the main $^1S_0 \rightarrow ^1P_1$ transition using a master-slave laser architecture (see Fig. 1.4). The master laser is an Extended-Cavity Diode Laser (ECDL, New Focus Vortex) frequency-stabilized via saturated absorption spectroscopy on ^{87}Sr [2]. It seeds three high-power slave laser diodes, forcing them to inherit the master's narrow linewidth and central frequency while delivering the high output power needed for the cooling beams.

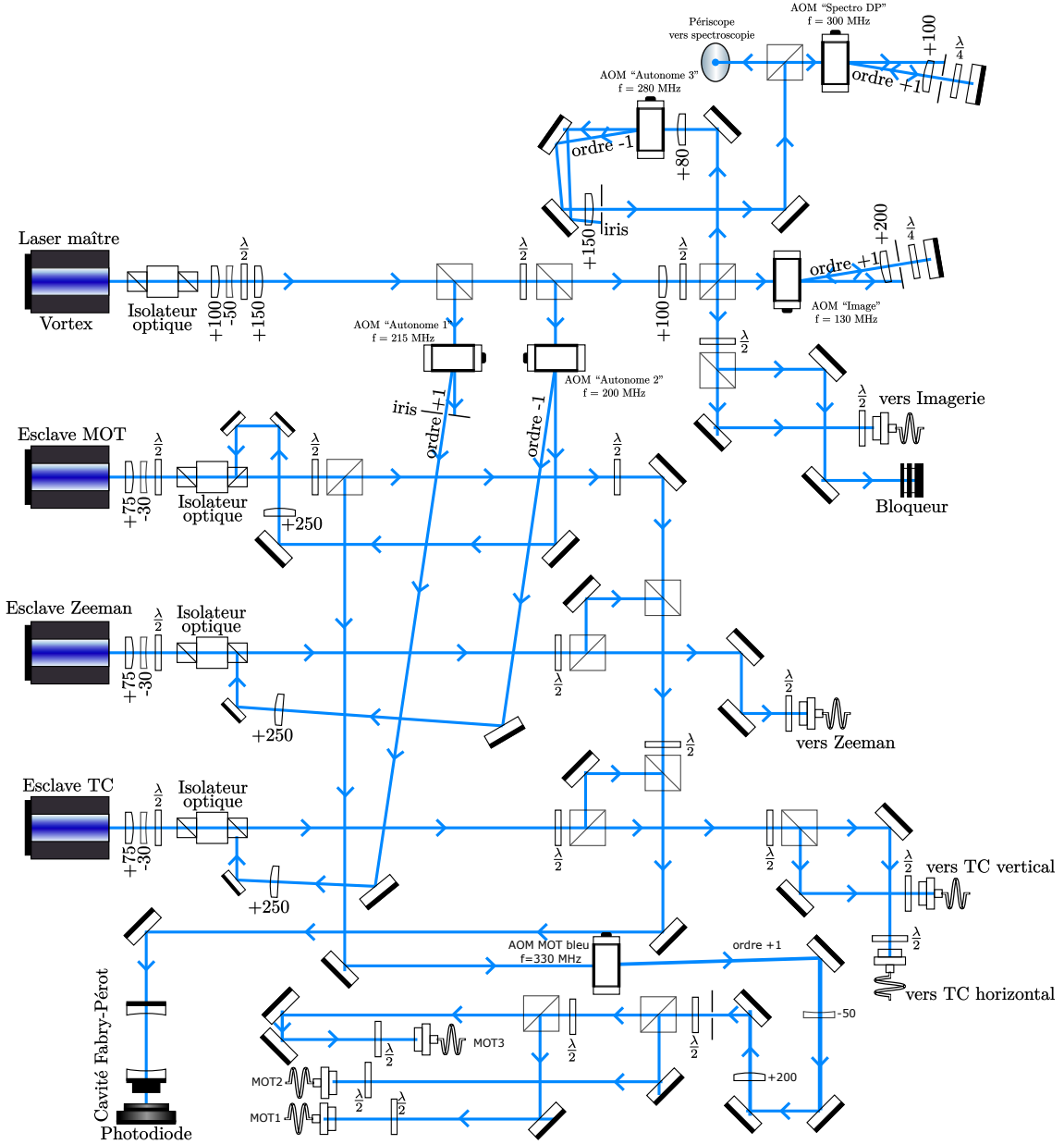


Figure 1.4: Optical layout of the 461 nm blue laser system for transverse cooling, Zeeman slowing, and blue MOT.

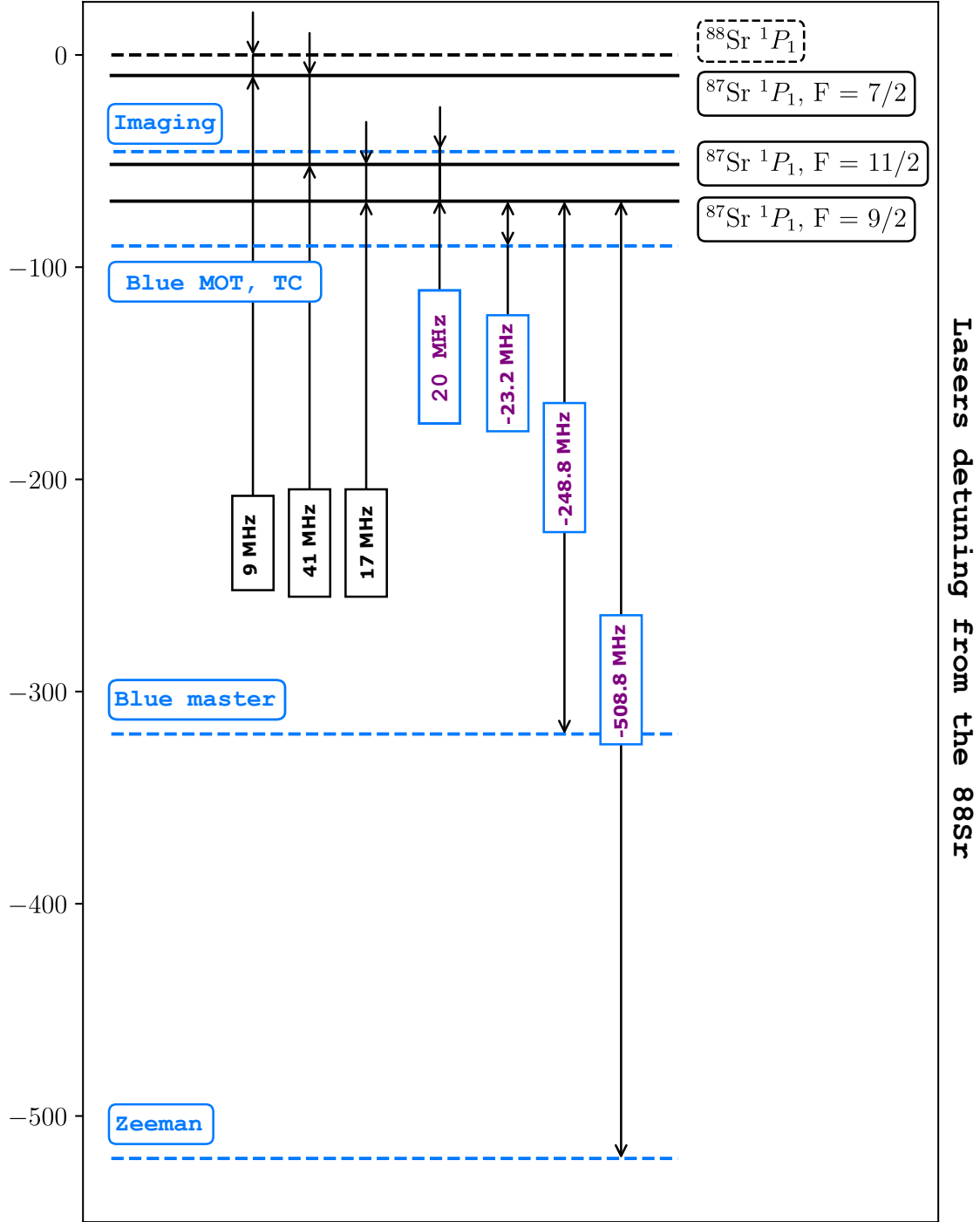


Figure 1.5: Adapted from P.Bataille thesis [24] Diagram of the frequencies of each laser composing the blue chain of the blue MOT broadband cooling. Blue dashed lines indicate the tunable frequencies of the lasers. Plain black lines indicate the transitions of ^{87}Sr and ^{88}Sr isotopes. The frequencies refer mostly to the optical frequency at the output of the laser called ‘Blue master’.

1.1.2 Transverse cooling

Following extraction from the oven capillaries, the atomic beam is further collimated using 2D transverse cooling (TC). This stage employs two orthogonal pairs of counter-propagating laser beams perpendicular to the main longitudinal atomic propagation axis (z -axis), reducing the transverse velocity spread $v_{\perp}$ via photon recoil momentum ($\hbar k_{\perp}$). The perpendicular velocity is reduced to about 20 m/s.

1.1.2.1 Operating principle and radiation pressure

The collimation relies on Doppler cooling. An atom moving with a transverse velocity $\vec{v}_{\perp}$ experiences a Doppler shift $\Delta_D = -\vec{k}_{\perp} \cdot \vec{v}_{\perp}$. As illustrated in Fig. 1.6, atomic velocity classes observe shifted transition resonances centered at $\omega_0 \pm \vec{k}_{\perp} \cdot \vec{v}_{\perp}$. When the laser frequency is red-detuned ($\Delta < 0$) with respect to the atomic transition frequency (ω_0 , the center of the purple gaussian), the atom preferentially absorbs photons from the counter-propagating beam, generating a net damping force.

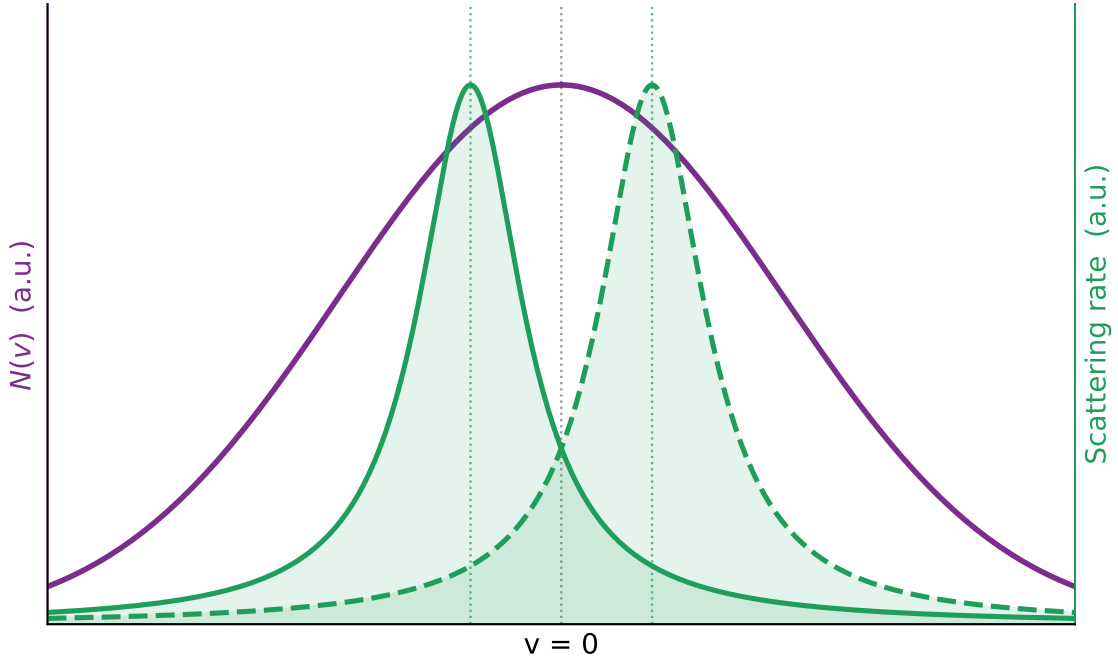


Figure 1.6: Illustration of the transverse cooling effect on the atoms arriving from the oven spread in different velocity classes.

In the low-intensity limit ($s \ll 1$), standard Doppler cooling theory shows that the equilibrium temperature $T(\Delta)$, set by the competition between the velocity-dependent damping and the momentum diffusion from photon scattering, is minimized at a detuning of $\Delta_{\text{eff}} = -\Gamma/2$, yielding the Doppler temperature limit $T_D = \hbar\Gamma/(2k_B)$.

1.1.2.2 Experimental setup and laser configuration

For this stage, we drive the broad-linewidth $^1S_0 \rightarrow ^1P_1$ transition ($\Gamma/2\pi = 30.5$ MHz, $I_{\text{sat}} \approx 40$ mW/cm²). The laser frequency f_{TC} is generated by shifting the main blue

master frequency using an acousto-optic modulator (AOM) driven at $f_{\text{TC}}^{\text{RF}}$:

$$f_{\text{TC}} = f_{\text{Blue,Master}} + f_{\text{TC}}^{\text{RF}} \quad (1.1)$$

The horizontal arm is operated at roughly three times the intensity of the vertical arm, with $I/I_{\text{sat}} = 0.25$ in the horizontal arm: there is no physical reason why it is this way but we observe the best cooling for this ratio. Both arms nonetheless remain below saturation ($I < I_{\text{sat}}$) over an interaction area of approximately 10 cm^2 . The TC laser remains active throughout the subsequent blue-cooling stages to ensure continuous atomic-flux collimation.

1.1.2.3 Detuning and power optimization

While ideal two-level theory predicts an optimal detuning of $\Delta = -\Gamma/2$, experimental constraints alter this optimum:

- **Power limitations over large beam areas:** Operating near saturation ($I \sim I_{\text{sat}}$) would maximize the velocity capture range through power broadening and larger radiation pressure forces. However, illuminating the 10 cm^2 interaction area at saturation would require over 400 mW of optical power. Consequently, the system operates in the low-intensity regime ($I \ll I_{\text{sat}}$).
- **Optimal detuning shift:** Experimental characterization reported in F. Faisant's thesis [25] demonstrated that maximum collimation efficiency (measured by downstream atom yield) is achieved for low intensity at $\Delta = -\Gamma/4$ rather than $-\Gamma/2$, as shown in Fig. 1.7. Indeed considering the residual random motion of the atoms modifies the expression of the total average force: minimum temperature is reached for higher intensities. This plot also illustrates how the maximum atom number gain improves with total beam power.

Future upgrades in progress include replacing the current TC slave diode with a higher-power diode and re-optimizing Δ_{eff} to further enhance atom capture. Following transverse collimation, the atomic beam enters the Zeeman slower stage along the main longitudinal axis.

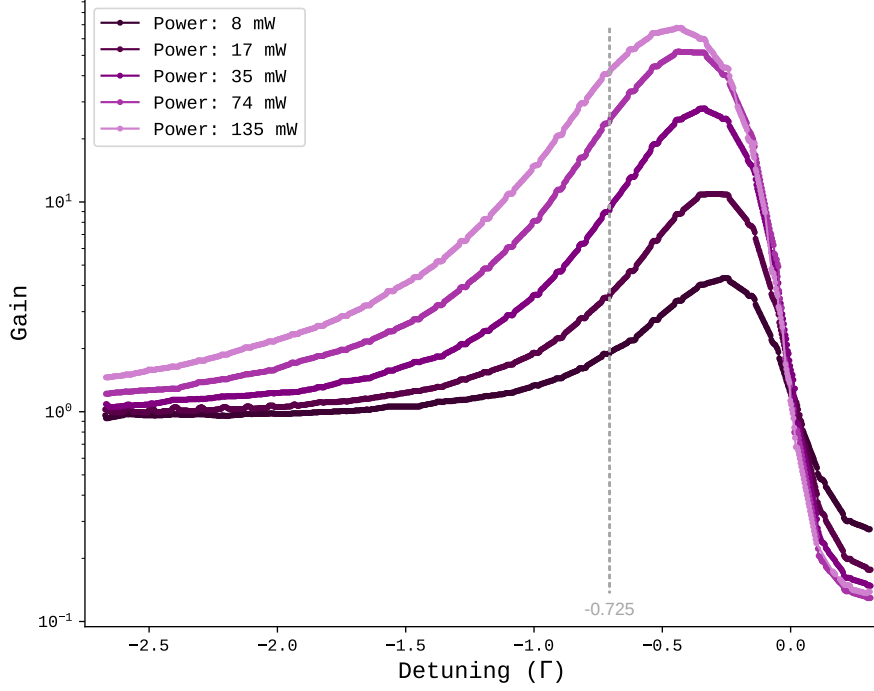


Figure 1.7: Transverse cooling performance as a function of laser detuning and power (adapted from [25]). Curves of atom number gain for different optical powers on TC are showing optimum that are not centered on the same detuning.

1.1.3 Zeeman cooling

Following transverse collimation, atoms are decelerated longitudinally using a Zeeman slower [26]. The mechanism relies on a counter-propagating laser beam maintained on resonance with the decelerating atoms as they travel through a spatially varying magnetic field gradient $B(z)$ generated by a tapered solenoid.

To compensate for the decreasing Doppler shift $kv(z)$ as atoms slow down, the magnetic field $B(z)$ shifts the Zeeman sub-levels to satisfy the resonance condition along the entire deceleration path:

$$\Delta_{\text{eff}}(z) = \omega_{\text{laser}} - \omega_0 + kv(z) - \frac{\mu_B g_J B(z)}{\hbar} = 0 \quad (1.2)$$

Here, $\Delta_{\text{eff}}(z)$ is the local effective detuning experienced by the atom at position z ; ω_{laser} and ω_0 are the angular frequencies of the slowing laser and of the atomic transition, respectively; $k = 2\pi/\lambda$ is the laser wavevector; $v(z)$ is the atom's longitudinal velocity at position z ; μ_B is the Bohr magneton; g_J is the Landé factor of the excited electronic level; and $B(z)$ is the local magnetic field produced by the Zeeman slower.

Only atoms entering the slower with a velocity below the capture velocity $v_0 = 430$ m/s can satisfy Eq. 1.2 and thus be decelerated; faster atoms never come into resonance with the slowing beam and pass through unaffected.

Solving Eq. 1.2 for the required field profile gives

$$B(z) = \frac{\hbar}{\mu_B g_J} \left[kv(z) + (\omega_{\text{laser}} - \omega_0) \right], \quad (1.3)$$

with $\omega_{\text{laser}} - \omega_0 < 0$ (red-detuned laser) and $v(z)$ decreasing progressively along the slower. Whether $B(z)$ eventually changes sign depends on the chosen detuning $|\omega_{\text{laser}} - \omega_0|$ relative to kv_f , where v_f is the target exit velocity: for $|\omega_{\text{laser}} - \omega_0| > kv_f$, the field must cross zero and become negative before the atoms reach v_f .

Because the atom's direction of motion does not change, this sign change reflects only the resonance condition of Eq. 1.2, not a reversal of the atomic trajectory: atoms continue to decelerate in the same direction throughout. Supplementary coils of increasing radii (see Fig.1.2), placed past this zero crossing, produce the required negative field to keep the already-slowed atoms in resonance down to the final capture velocity suitable for MOT trapping.

1.1.4 Blue MOT

The Magneto-Optical Trap (MOT) provides simultaneous spatial confinement and cooling.

The blue MOT combines a magnetic quadrupole field (generated by anti-Helmholtz coils centered on the vacuum chamber, driven at 170 A to produce a gradient of $\Delta B = 51$ G/cm) with three orthogonal pairs of counter-propagating 461 nm laser beams carrying opposite circular polarizations (σ^+/σ^-) [27].

The position-dependent Zeeman shift lifts the degeneracy of the m_F sub-levels:

$$\Delta E(z) = m_F g_F \mu_B B(z) \quad (1.4)$$

Combined with the $\sigma^\pm$ polarization of the counter-propagating, red-detuned beams, this position-dependent shift brings only one beam into resonance on either side of the trap center ($z = 0$), creating a restoring force toward $z = 0$ alongside the velocity-dependent radiation pressure force. The vertical beam waist is twice smaller than the two horizontal one and the mean intensity of each axis is $I/I_{\text{sat}} = 0.3$ (13 mW/cm²).

During this stage, the cloud reaches the Doppler temperature limit [28]:

$$k_B T_D = \frac{\hbar \Gamma_{^1S_0 \rightarrow ^1P_1}}{2} \quad (1.5)$$

Due to the broad linewidth of the 461 nm transition ($\Gamma/2\pi = 30.5$ MHz), this Doppler temperature ($T_D \approx 770$ μ K) remains significantly higher than the single-photon recoil limit T_r .

The atoms are transferred in 1P_1 state that is determined by the equation evolution of the atom number:

$$\frac{dN(t)}{dt} = \alpha \Phi - \gamma_{\text{loss}} N(t) \quad (1.6)$$

where Φ is the incident atomic flux on the MOT capture region, α is the dimensionless capture efficiency (the fraction of that flux actually trapped, depending on experimental parameters), $\gamma_{\text{loss}} = \gamma_{\text{vacuum}} + \gamma_{\text{leak}}$ respectively the loss rate due to the imperfection of the vacuum and the rate leakage to the metastable state. Then the number of atom in time evolves as:

$$N(t) = N_{\text{sat}}(1 - e^{-t/\tau}) \quad (1.7)$$

$N_{sat} = \alpha\Phi\tau$ when the transition reaches its maximum value. Since the blue MOT lifetime is dominated by the leak into the metastable state and amounts to a few tens of milliseconds [29], saturating the blue MOT atom number only requires waiting for a few such lifetimes.

The experimental accumulation time, however, is set by a different consideration: reaching the maximum number of atoms trapped in the metastable state, which takes 8.2 s. Although the blue-MOT population itself saturates within a few tens of milliseconds, only a fraction of the atoms leaking out of the 1P_1 cycle end up trapped in the long-lived metastable state on each pass – the rest decay back to the ground state and re-enter the blue-MOT cycle, where they must leak again before being retained (see branching ratios below). Accumulating a large fraction of the atoms into the metastable reservoir therefore requires many such cycles, hence the much longer 8.2 s duration. Throughout this accumulation, the magnetic field gradient is kept at its maximum, acting as a magnetic trap that confines the atoms in the metastable state, whose magnetic dipole moment is non-zero.

1.1.5 Repumper

A repumping laser system recovers these accumulated atoms into the cooling cycle: of the roughly 1:20000 photon-scattering events on 1P_1 that decay via 1D_2 rather than back to the ground state, about 1:4 reach the long-lived 3P_2 state addressed here (lifetime $\tau \approx 500$ s), with the rest returning through 3P_1 . Without repumping, atoms trapped in 3P_2 remain dark to the 461 nm light and are permanently lost from the cooling cycle.

1.1.5.1 Hyperfine structure of Strontium-87

Because fermionic ^{87}Sr possesses a nuclear spin $I = 9/2$, its electronic states exhibit a rich hyperfine structure that must be addressed for efficient repumping.

Repumping is provided by a 403 nm external-cavity diode laser (TOPTICA). To address multiple F hyperfine sub-levels (see Fig. 1.8), the laser frequency is rapidly swept over a 3 GHz bandwidth for 3.2 ms.

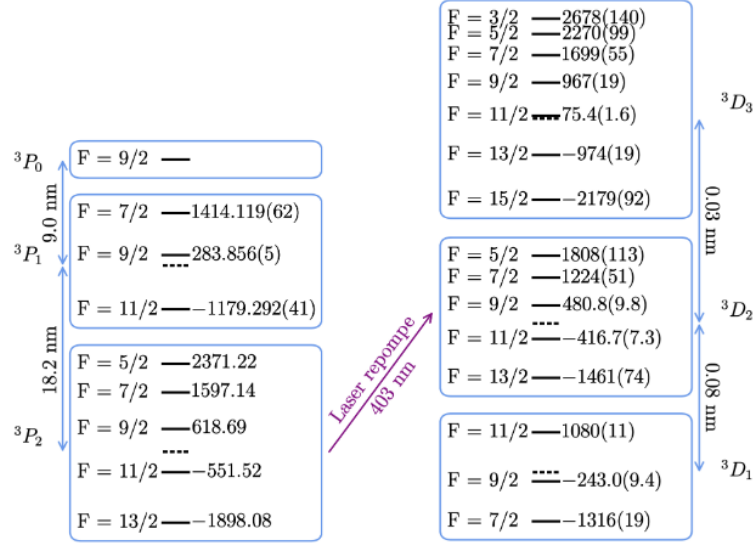


Figure 1.8: Energy level diagram of ^{87}Sr of the transition 3P_2 to the hyperfine states of the 3D_J at 403 nm cooling.

1.1.5.2 Optimization of the repumper sweep

The 3P_2 hyperfine levels are unequally populated during blue MOT decay due to their differing g_F Landé factors, affecting their magnetic trapping efficiency [29].

To recover this population into the cooling cycle, the sequence proceeds as follows: the 461 nm beam is first extinguished at the end of the blue MOT phase, preventing recovered atoms from being re-excited and recycling back onto the blue transition; the 689 nm red MOT lasers are then switched on, so that atoms are captured as soon as they return to the ground state; finally, the 403 nm repumper is pulsed for 3.2 ms at $I \approx 15 \text{ mW/cm}^2$, starting 9.3 ms after the red MOT lasers are switched on, and the red MOT is left running afterwards. The repumper frequency is scanned over a range optimized around 3 GHz to selectively cover the $F = 11/2$ and $F = 13/2$ states; scanning a wider range would reduce total atom recovery by decreasing the dwell time on these critical transitions.

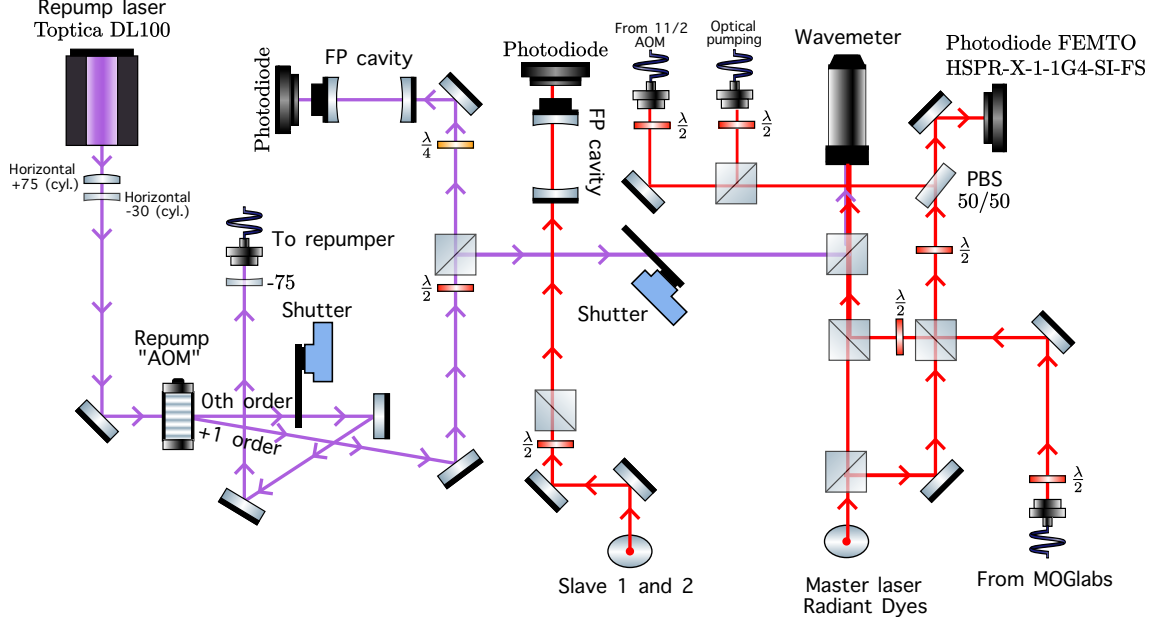


Figure 1.9: Optical setup for the repump and the beat-lock setup of the MOGLabs laser onto the Radiant Dyes.

1.1.6 Laser cooling on the intercombination line : the “red MOT”

1.1.6.1 Optical setup

Because the blue MOT temperature is bounded by $T_D \approx 770 \mu\text{K}$, further cooling requires transferring atoms to the narrow intercombination transition $^1S_0 \rightarrow ^3P_1$ at 689 nm ($\Gamma = 2\pi \times 7.5 \text{ kHz}$).

For this transition, the formal Doppler limit ($T_D = \hbar\Gamma/mk_B = 177 \text{ nK}$) falls below the single-photon recoil limit ($T_r = \hbar^2k^2/2m = 462 \text{ nK}$). Consequently, the ultimate red MOT temperature is bounded by the photon recoil limit.

The 689 nm light is generated by a commercial laser locked to an ultra-stable optical cavity referred as “Red lock” in Fig.1.11 that contains all the frequencies of the red chain. The red master laser, from which the MOT, STIR, and optical pumping beams are all derived, has an optical frequency -220 MHz detuned from the $^{88}\text{Sr}, ^3P_1$. The ‘Red lock’ is the summation of the ‘AOM Inj Slave 1’ that enables to access GHz away transitions from the bosonic state, with the RF frequency of the ultrastable cavity AOM.

Double-pass AOMs then fine-tune the frequency of MOT and Stir beams individually before splitting into three spatial counter-propagating beam pairs, bringing the actual atom-light detuning down to the much smaller values used for trapping (see below).

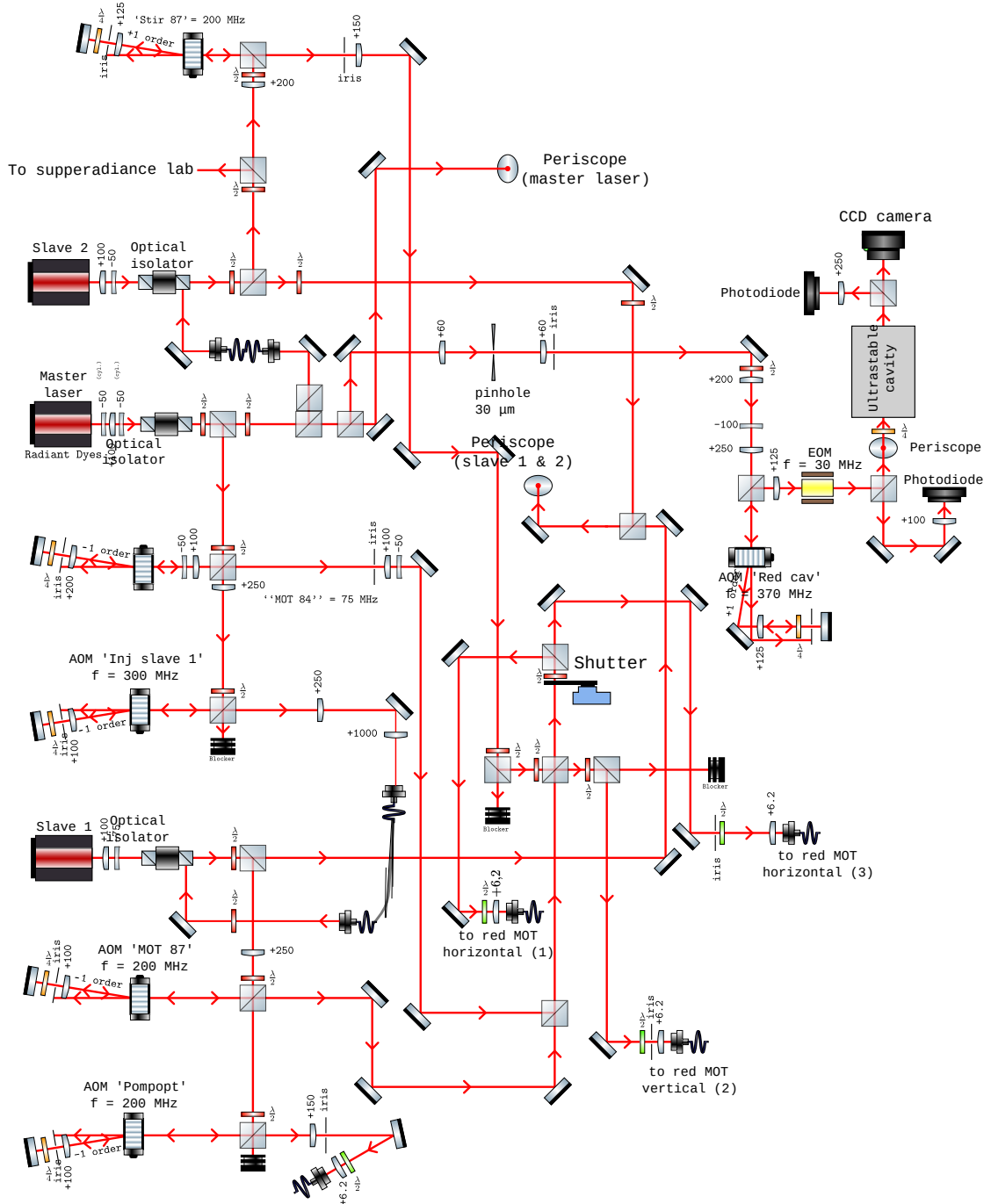


Figure 1.10: Optical setup of the 689 nm laser cooling system.

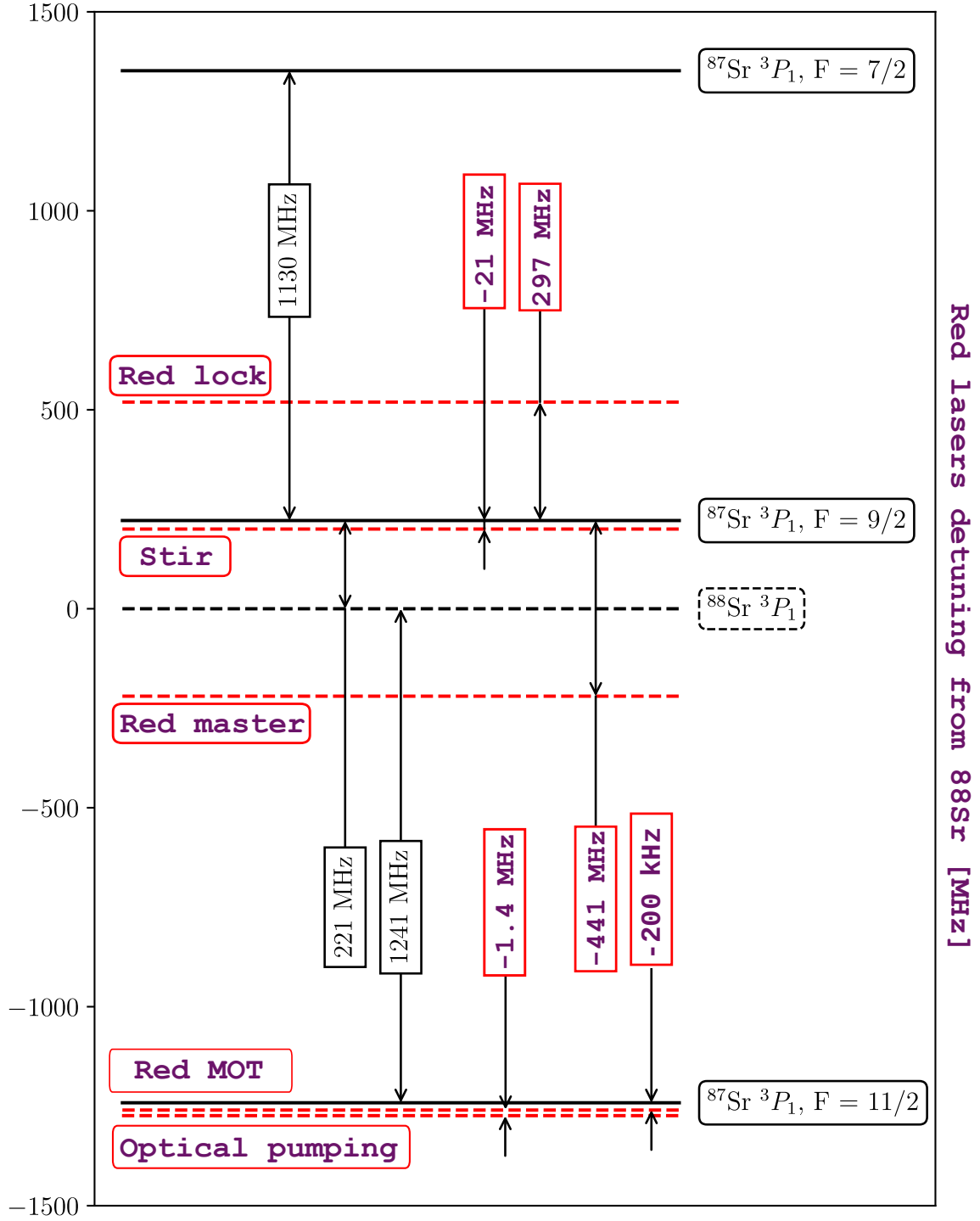


Figure 1.11: Frequencies of the red lasers used for the red MOT cooling stages and optical pumping. The red dashed lines correspond to the tunable frequencies, the dark plain lines for the ^{87}Sr hyperfine states of transition 3P_J and the ^{88}Sr

1.1.6.2 Swap cooling, broadband MOT to narrowband

At the end of the blue MOT, approximately 30 million atoms have been captured; their spatial spread ($\approx 300 \mu\text{m}$, reached at the end of the SWAP stage) and thermal energy exceed the capture velocity of a single-frequency 689 nm MOT. To capture these thermal atoms, a broadband MOT is implemented on the $^1S_0 (F = 9/2) \rightarrow ^3P_1 (F = 11/2)$ transition under a weak magnetic field gradient ($\Delta B \simeq 1 \text{ G/cm}$, bias

coil voltage $V_A = 0.50$), operated in two successive stages. In parallel a redistribution of the spin is assured by a stirring laser, its function explained in the following subsection.

In the first stage, lasting 182 ms, SWAP (Sawtooth Wave Adiabatic Passage) cooling is applied [30]. As illustrated in 1.12 two counter-propagating beams have their frequency swept through resonance, then reset almost instantaneously back to the initial frequency of the ramp. The ramp drives two sequential, stimulated transfers: first absorption from the beam the atom is moving towards. The laser gets on resonance with the Doppler shifted atom at $\omega_L - kv$. The frequency of the laser still increasing then becomes resonant with the opposite polarized laser which affects the atom with a new velocity v' after the absorption: by emission into the opposing beam $2\hbar k$ are removed of momentum per sweep without relying on spontaneous emission. The fast reset, in contrast, is deliberately non-adiabatic so that it drives no transfer, and hence no momentum kick, back the other way; a symmetric ramp would instead undo the cooling just achieved.

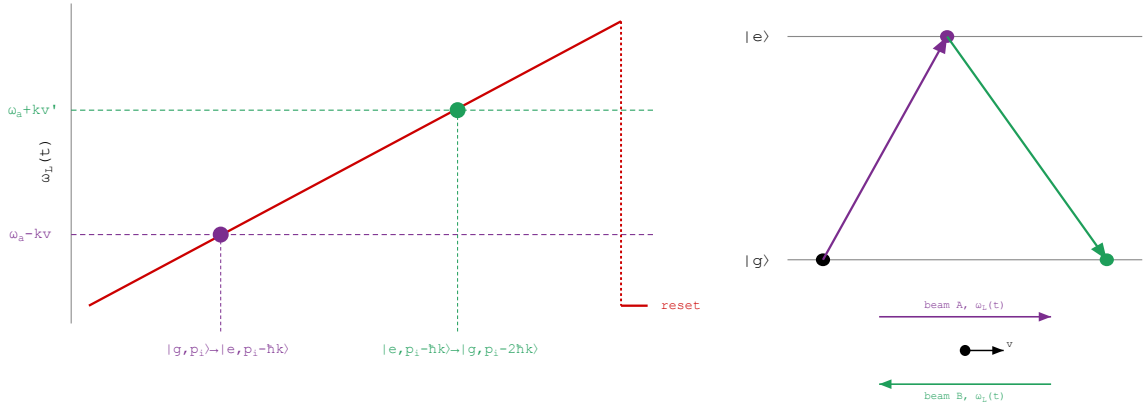


Figure 1.12: Mechanism of SWAP cooling, adapted from [30]. Left: over one slow rise of the sawtooth $\omega_L(t)$, the laser successively crosses the two Doppler-shifted resonance conditions $\omega_a - kv$ and $\omega_a + kv'$, driving first an absorption then a stimulated emission into the opposing beam; the fast reset transfers no population. Top right: the corresponding two-level picture, showing the atom transiently promoted to $|e\rangle$ before returning to $|g\rangle$ with a net momentum change of $-2\hbar k$. Bottom right: geometry, with the atom (velocity v) between two counter-propagating beams sharing the same swept frequency $\omega_L(t)$.

The trapping beam (MOT) is swept between RF frequencies 111.9 and 110.4 MHz – corresponding to a real detuning from the 3P_1 ($F' = 11/2$) resonance (calibrated at 110.34 MHz on this AOM) of -6.24 MHz to -240 kHz – while the stirring beam (STIR, see below) is swept between 105.25 and 103.7 MHz **[TODO: convert to real detuning from $F' = 9/2$ once its resonance is measured on this AOM]**, both starting at an intensity of $I_{A100}(MOT87) = 5 \text{ mW/cm}^2$ and $I_{A100}(STIR) = 2 \text{ mW/cm}^2$.

In principle, since SWAP relies on an adiabatic passage through resonance to generate a deterministic momentum kick on every sweep, the detuning should start blue of resonance, cross it, and end red-detuned by an amount that itself shrinks stage after stage as the cloud cools further. In practice, however, the implemented sweep

range given above (-6.24 to -240 kHz) remained entirely red-detuned relative to the $F' = 11/2$ resonance and never actually crossed it – indicating that this parameter was not fully optimized (already represented in the sequence scheme 1.3).

After 9.3 ms, the repumper pulse described above interrupts the SWAP of the MOT and STIR sweeps for 3.2 ms; the MOT and STIR sweeps then resume for the remainder of the 182 ms stage. As the cloud cools and compresses during this stage, the modulation amplitude Δf is progressively ramped down together with the optical power, which reaches $I_{A85} = 4.3 \text{ mW/cm}^2$ for the MOT beam and still $I_{A85} = 2 \text{ mW/cm}^2$ for the STIR beam by the end of the stage. Toward the end of this first stage, the sweep transitions into a broadband-capture phase, using a symmetric triangular ramp (no fast reset) to sweep continuously across a broader range of Zeeman sub-levels, as illustrated in Fig. 1.13.

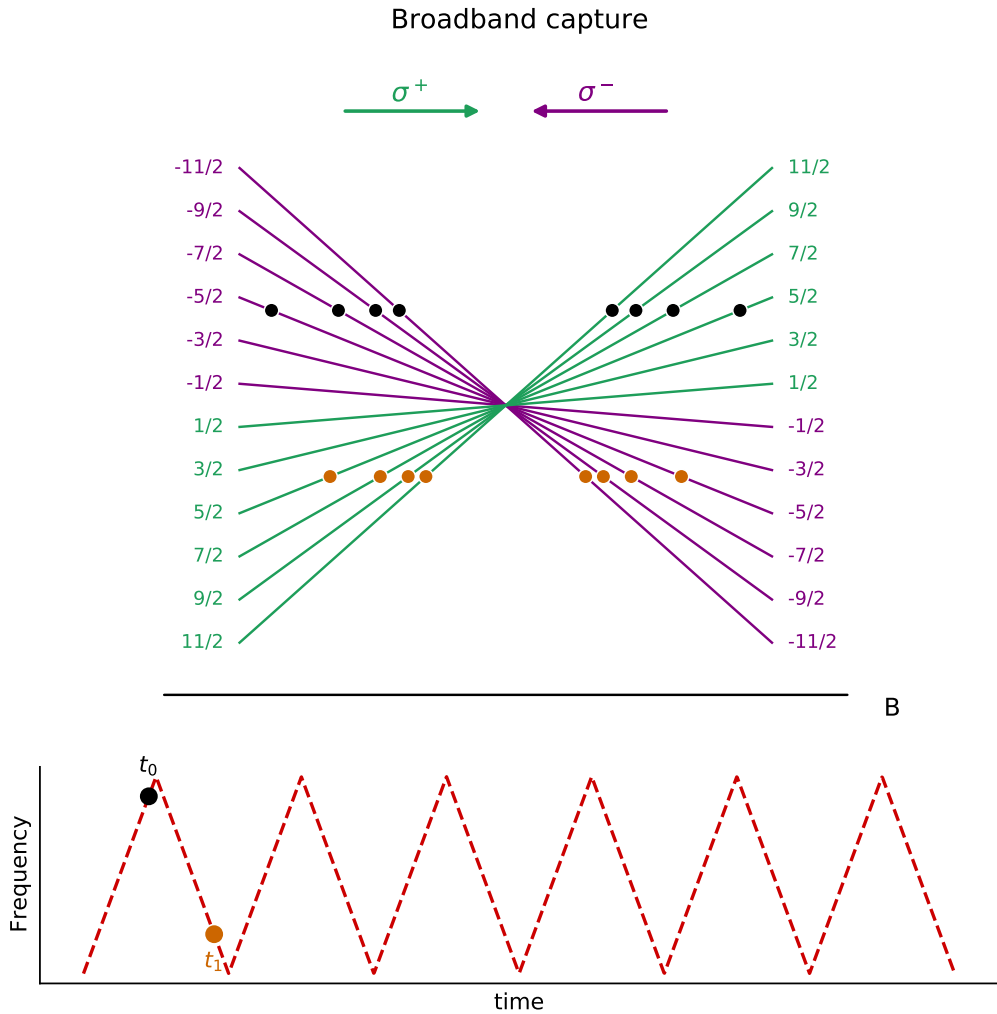


Figure 1.13: Broadband capture stage of the 689 nm red MOT (adapted from [31]). Top: Zeeman sublevels depending on the B field and position since $B \propto x$. Dark dots show the Zeeman states addressed at t_0 by the laser ramp plotted on the bottom figure. The orange dots are the Zeeman states addressed at t_1 by the same laser ramp.

In the second stage, lasting 250 ms and visible on the right of Fig. 1.13, the cool-

ing light is modulated with a conventional sawtooth ramp detuned from resonance – no longer swept symmetrically through it – operating as a standard frequency-modulated Doppler-cooling MOT that further compresses and cools the now colder, denser cloud. The MOT and STIR frequencies are ramped to their narrowband values of 110.395 MHz (a real detuning of -220 kHz from the $F' = 11/2$ resonance, measured after the slight magnetic-field adjustment described below) and 105.309 MHz respectively **[TODO: STIR real detuning from $F' = 9/2$, and the missing lower bound of the stage-2 STIR ramp, to be added]**, while the bias coil voltage is raised to $V_C = 0.51$. Over this stage, the optical power is reduced further, in two additional steps for the MOT beam – from $I_{A85} = 2 \text{ mW/cm}^2$, through $I_{A32} = 1.6 \text{ mW/cm}^2$, down to a final value of $I_{A19} = 1 \text{ mW/cm}^2$ – while the STIR beam is reduced from $I_{A85} = 2 \text{ mW/cm}^2$ to a final value of $I_{A55} = 1.3 \text{ mW/cm}^2$. By the end of this second stage, the cloud has cooled to an average temperature of $T \simeq 30 \text{ } \mu\text{K}$ with a size of $\simeq 300 \text{ } \mu\text{m}$.

1.1.6.3 Stirring laser

The σ^+/σ^- configuration of the red MOT introduces an intrinsic loss channel arising from the large differential Landé g -factor between the ground $^1S_0, F = 9/2$ and excited $^3P_1, F' = 11/2$ hyperfine levels. As illustrated in Fig. 1.14, the position-dependent Zeeman shift ($B \sim x$) brings the two counter-propagating beams into resonance with different m'_F sub-levels at the same position: the σ^+ beam (green) addresses the closed transition $m_F = 9/2 \rightarrow m'_F = 11/2$, while, within the same laser linewidth (grey band), the σ^- beam (purple) simultaneously becomes resonant with the neighbouring, open transition $m_F = 9/2 \rightarrow m'_F = 7/2$. This dual resonance is not by itself a loss mechanism: the relative probability of the two processes, set by their Clebsch-Gordan coefficients, favors the restoring transition by a ratio as high as 55:1 in the most extreme case, so that the force averaged over many scattering events remains restoring [31].

The condition for stable operation of a $F \rightarrow F + 1$ MOT transition requires the double inequality $F/(F + 1) < g_e/g_g < F/(F - 1)$ to hold [31]. For ^{87}Sr , where $g_e/g_g \sim 10^3$ due to the near-vanishing nuclear magnetic moment of the 1S_0 ground state, this condition is violated by several orders of magnitude. As a consequence, the resonance condition $(m'_F g_e - m_F g_g) \mu_B B(x) = h\Delta$ has a widely m_F -dependent slope: low- $|m_F|$ states cross the laser resonance window (grey band) over a broad range of positions, while the most Zeeman-shifted states cross it almost vertically, remaining within resonance over such a narrow range of x that they are, in practice, essentially unaddressed by a fixed-frequency beam (Fig. 1.15, left panel, akin to Fig. 2.8(d) of [31]).

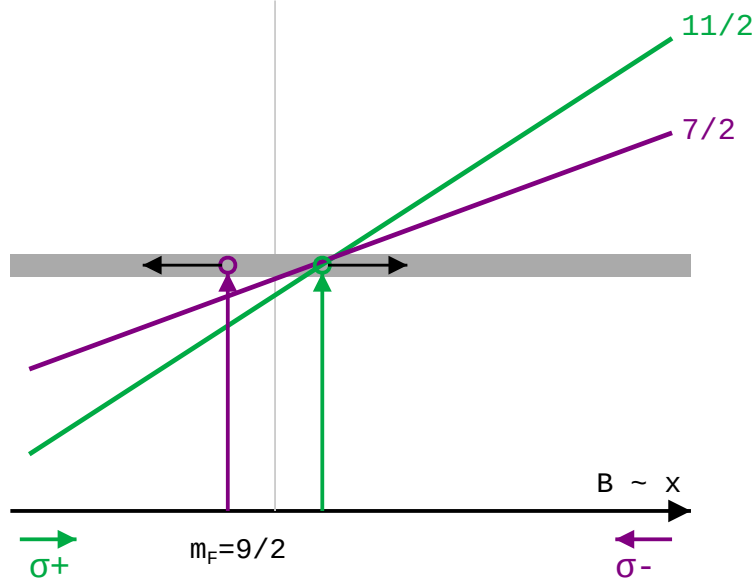


Figure 1.14: (Adapted from [31]) Illustration of the MOT effect on the m_F states of the hyperfine 3P_1 , $F = 11/2$ level. The position-dependent Zeeman shift brings the two counter-propagating beams into resonance with different m'_F sub-levels (σ^+ with the closed transition to $m'_F = 11/2$, σ^- with the open transition to $m'_F = 7/2$) at the same position. The imbalance between the two processes, set by their Clebsch-Gordan coefficients, still favors restoring trapping locally (see text); the actual loss mechanism is discussed below.

The actual loss mechanism therefore appears when an atom finds itself on the *wrong* side of the trap for its m_F state. There, neither beam can come into resonance with it at all: the restoring force for an atom in $m_F = 9/2$ is maximal on one side of the trap only, and on the other side this atom is lost from the trap entirely [31]. This is especially detrimental for the stretched states $m_F = \pm 9/2$, whose cycling transition is *closed* and which therefore have no spontaneous decay path to another m_F state that could rescue them once on the wrong side.

A dedicated stirring laser mitigates this geometric loss by driving the open $^1S_0(F = 9/2) \rightarrow ^3P_1(F = 9/2)$ transition, superimposed on the trapping light and red-detuned such that its resonance region overlaps with that of the trapping transition (Fig. 1.15, left panel, akin to Fig. 2.8(d) of [31]). Because the differential Zeeman shift of $F' = 9/2$ is about five times smaller than that of $F' = 11/2$ [31], driving this transition changes both where and how broadly each m_F state can come into resonance: the resonance position shifts, potentially restoring resonance for atoms otherwise stuck on the wrong side of the trap, and the resonance region in x broadens by the same factor, making it far more likely that an atom actually scatters enough photons near resonance to be redistributed before drifting away. Since this $F' = 9/2$ transition is open, each excitation-decay cycle can additionally redistribute the atom into a different m_F sub-level of the ground state.

In our implementation, we additionally sweep the stirring laser's frequency in a sawtooth pattern over time (Fig. 1.15, right panel): at any instant, the fixed instantaneous frequency is simultaneously resonant with several (m_F, x) pairs across the trap (left panel), and as the frequency sweeps, the ground-state m_F population is continuously and randomly redistributed. This continuous optical pumping brings atoms that would otherwise remain trapped in an unfavorable m_F state on the wrong side of the trap back within reach of the cycling transition, restoring efficiency to the cooling cycle.

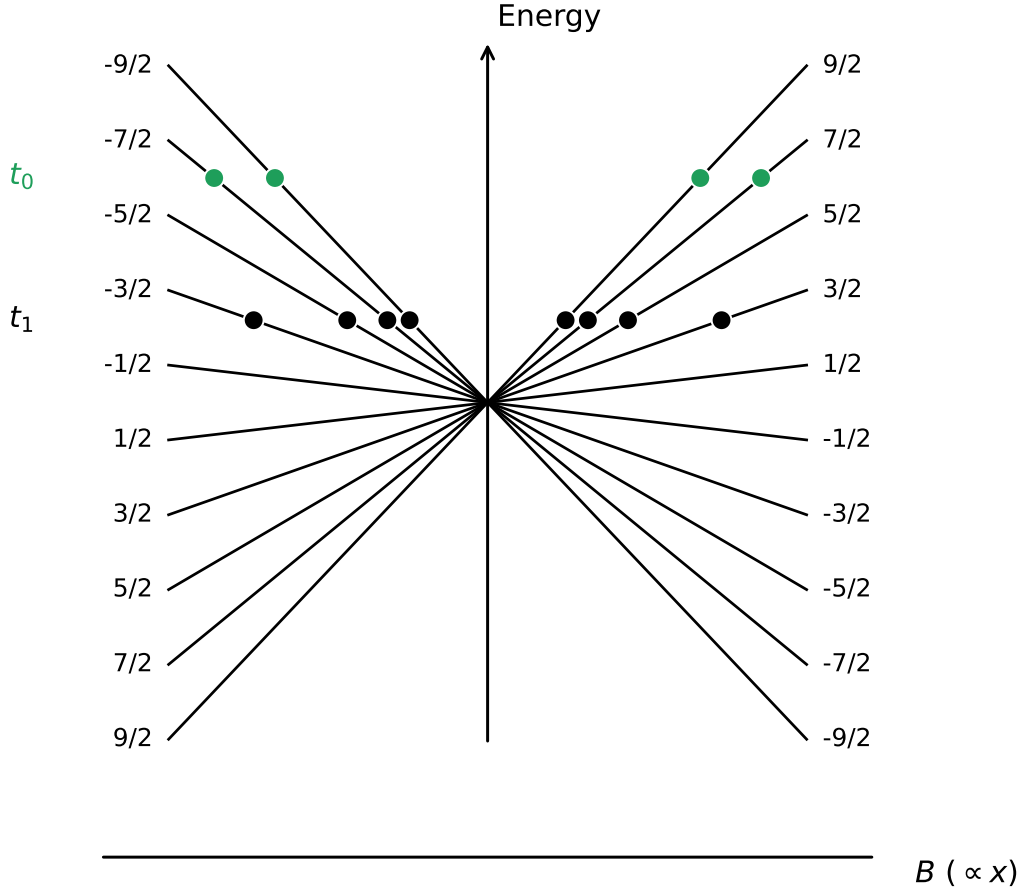


Figure 1.15: Spatial dependence of the resonance condition on each m_F sub-level, as a function of position x ($B \propto x$) – adapted from the principle illustrated in Fig. 2.8(d) of [31].

While the broadened resonance region alone increases the fraction of the trap addressed by the stirring light at any given detuning, a single, fixed stirring frequency still only brings each m_F state into resonance over a finite, localized range of positions around its own resonance point $x_{\text{res}}(m_F)$. An atom in a given m_F state located outside this window at a given time is simply not addressed, and nothing guarantees that its thermal motion within the trap will ever bring it there. In our implementation, we therefore additionally sweep the stirring laser's frequency in a sawtooth pattern over time (Fig. 1.15, right panel): as the frequency ramps, the resonance position $x_{\text{res}}(m_F, t)$ for each m_F sweeps continuously across the trap, so that every m_F state is brought into resonance at every position over the course of one ramp.

This guarantees full coverage of the (m_F, x) phase space, rather than relying on the atom's own motion to bring it within reach of a fixed, broadened resonance window.

1.1.6.4 Narrowband MOT (Single-frequency MOT)

Once the cloud is sufficiently cold, frequency modulation is switched off to operate a single-frequency (monofrequency) MOT during a magnetic field ramp to $V_D = 0.70$ ($\Delta B \simeq 2.5$ G/cm). This stage reduces the temperature to 1–2 μK (near the recoil limit) with a peak density of $n \simeq 10^8$ at/cm³ and a size of 30 μm .

1.1.6.5 Optimization of narrowband MOT Parameters

Cooling efficiency during the single-frequency stage depends on precise frequency, intensity, and timing parameters:

- **Broadband modulation amplitude:** The initial sweep covers $\Delta f = 6$ MHz on the atoms to capture the Doppler distribution from the blue MOT.
- **Intensity ramp and power broadening:** High intensities $I/I_{\text{sat}} = 400$ to 1600 provide a deep trap for initial capture, but causes power broadening ($\Gamma_{\text{eff}} = \Gamma\sqrt{1 + I/I_s}$), raising the achievable Doppler limit. For single-frequency cooling, intensity is adiabatically ramped down over 45 ms to reduce power broadening while maintaining sufficient force to support the cloud against gravity.

1.1.7 Optical Dipole Trap (ODT) and evaporative cooling

1.1.7.1 Loading the crossed ODT

Atoms are adiabatically loaded into the ODT while decreasing the intensity of the single-frequency 689 nm red MOT during a ramp lasting approximately 45 ms. From second-order perturbation theory, the light shift experienced by the energy levels yields the optical dipole potential:

$$U_{\text{dip}} = -\frac{\hbar|\Omega|^2}{4\bar{\Delta}} \quad (1.8)$$

[3], where $1/\bar{\Delta} = 1/(\omega_{\text{laser}} - \omega_{\text{atomic}}) - 1/(\omega_{\text{laser}} + \omega_{\text{atomic}})$ is the effective detuning factor, and $\Omega^2 = \frac{\Gamma^2}{2} \frac{I}{I_{\text{sat}}}$ is the Rabi frequency.

As seen from these expressions, the AC Stark shift induced by the dipole laser is state-dependent. As illustrated in Fig. 1.16, the potential estimated in [3] is deeper for the ground state 1S_0 ($U_{\text{dip}}(^1S_0) \simeq -1.5$ MHz) than for the excited state 3P_1 ($U_{\text{dip}}(^3P_1) \simeq -1.2$ MHz) at the center of the trap, resulting in a differential blue-shift of the $^1S_0 \rightarrow ^3P_1$ transition (black line) compared to the frequency the atoms feel from the red MOT (red curve).

At the center of the trap, the differential light shift increases the transition frequency, so that the MOT laser is red-detuned and keeps cooling and confining the atoms there. Away from the center, the MOT laser becomes blue-detuned with respect to the local transition and heats the atoms located at the edges of the ground-state trap. This could help to remove the most energetic atoms, but it also heats the reservoir

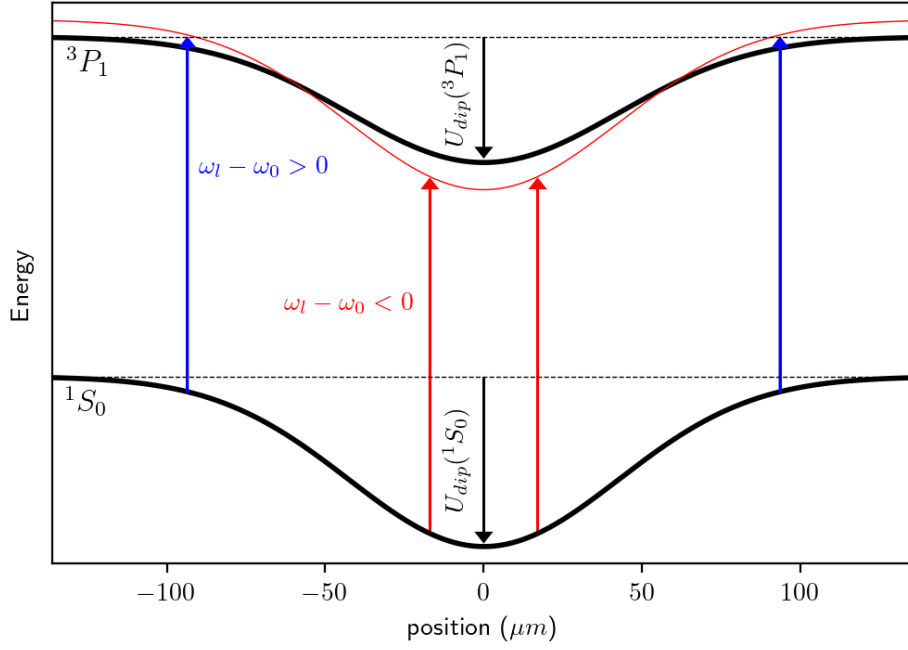


Figure 1.16: Simulation extracted from [3]. Principle of localized 689 nm MOT loading inside the crossed optical dipole trap due to the differential AC Stark shift between 1S_0 and 3P_1 .

of atoms from which the trap is loaded; which of these two effects dominates was not established.

Following loading, the 689 nm red MOT is switched off, and forced evaporative cooling is performed purely by progressively lowering the ODT depth: the most energetic atoms overcome the (now shallower) potential barrier and escape the trap, while the remaining ones re-thermalize to lower temperatures via elastic collisions (Fig. 1.17). Incidentally, as the trap depth decreases, the differential AC Stark shift also shrinks, so the local transition frequency at the trap center relaxes back toward its unperturbed value – though since the MOT laser is off at this stage, this drift plays no role in the evaporation dynamics itself.

The results presented in the following two chapters were obtained with an atomic cloud located at the intersection of a horizontal beam, called the “Reservoir” (waist $w_0^H = 150 \mu\text{m}$), and a vertical beam, called the “Dimple” (waist $w_0^V = 80 \mu\text{m}$). The vertical beam axis is tilted at an angle of $\simeq 30^\circ$ relative to the vertical to facilitate the escape of the hottest atoms from the trap.

In Chapter 2, the horizontal arm is tilted by $\simeq 1^\circ$. For the photoassociation results in Chapter 3, this tilt was canceled: it enables to have bigger volumes to trap the atoms from the crossing of the two arms (see 1.1.7.2).

Both 1S_0 and 3P_1 are shifted downward by the dipole trap potential, but 1S_0 experiences a larger shift due to its higher polarizability, so the energy difference between the two states is much larger at the center of the trap than at the edges. As discussed above, the atoms at the center of the trap are therefore cooled by the red-detuned MOT laser, while the atoms at the edges of the harmonic potential of the ODT are

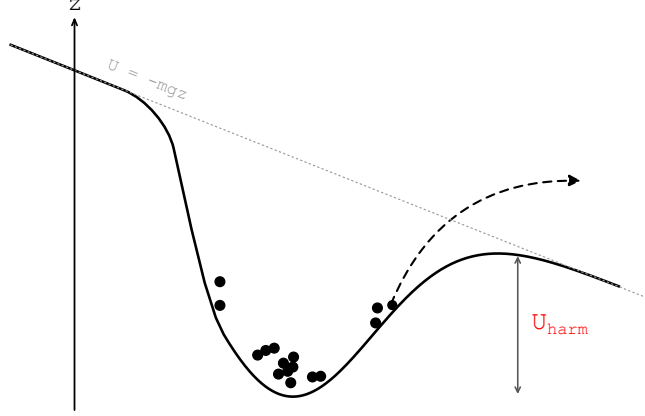


Figure 1.17: Evaporative cooling mechanism in a crossed optical dipole trap, illustrating the escape of high-energy atoms and subsequent re-thermalization.

heated by the slightly blue-detuned laser.

1.1.7.2 Optimization of the evaporation ramps

For a degenerate Fermi gas, the key parameter to optimize is the phase-space density (PSD), which scales as $\text{PSD} \propto N\bar{\omega}^3/T^3$, where N is the atom number, $\bar{\omega}$ the geometrically averaged trap frequency, and T the temperature. While both arms of the dipole trap influence these parameters, their main physical contributions differ:

- **Tight vertical beam (“Dimple”):** Writing the atom number as $N = nV$ (where n is the spatial density and V the trap volume), the elastic collision rate scales as $\Gamma_{\text{el}} = n\sigma_{\text{el}}v_{\text{th}}$ with σ_{el} the elastic collision cross section and v_{th} the thermal speed of the atoms, enhancing collision rates for fast re-thermalization. This beam impacts mostly the temperature of the gas since it acts on the collision rate. The tightly focused waist of the dimple beam mainly dictates the local horizontal confinement, and contributes negligibly to the vertical confinement due to its near-vertical propagation axis [32].
- **Broad horizontal beam (“Reservoir”):** Owing to its narrow vertical waist, the reservoir beam sets the effective trap depth U against gravity-assisted atom loss, thereby controlling the evaporative truncation of the energy distribution and, in turn, the atom number [32]. Increasing the reservoir depth raises N (less truncation, more atoms retained), but also T (less evaporative cooling) and the vertical trap frequency: its net effect on the PSD results from a competition between these terms, and was optimized empirically. In our geometry, the reservoir is also the only beam that compensates gravity, since the dimple propagates nearly vertically.
- **Ramp duration and adiabaticity:** The duration and adiabaticity of the evaporative ramps depend on the targeted state. Short evaporations are preferred when aiming solely for a large atom number at moderate temperatures when it is necessary to detect atom number fluctuations, such as for our experiments on photoassociation (see Chapt.3). Conversely, reaching deep quantum

degeneracy requires a multi-stage ramp lasting approximately 7 seconds which was necessary for the results of Chapt.2.

For further details on loading atoms into the optical crossed dipole trap, the reader is referred to the thesis of A. Litvinov [3]. An additional optical lattice, formed by a 1064 nm ALS laser, was initially implemented to enhance the phase-space density (PSD) of the atomic cloud prior to forced evaporative cooling. While this configuration was employed for the experiments presented in Chapter 2, it was subsequently discarded due to intensity fluctuations in the laser that induced significant heating.

1.1.7.3 Optical setup

The 1075 nm light from the IPG laser is split by a half-wave plate and polarizing beamsplitter into the two arms described above (Fig. 1.18). The horizontal arm forms the “Reservoir” beam: a pair of cylindrical lenses ($f = -50$ mm and $f = -75$ mm) shapes the elliptical beam into an anisotropic ellipse, with nominal waists of $w^h = 147$ μm (horizontal) and $w^v = 34$ μm (vertical) at the lens focal plane. However, the atoms sit $\simeq 3.3$ mm beyond this focus rather than at the waist itself, so that the vertical waist they actually experience is somewhat larger, $w^v \simeq 47$ μm , while the horizontal waist remains close to its nominal value at this distance. The second arm forms the “Dimple” beam, focused by a spherical lens ($f = 300$ mm) and crossing the reservoir arm at the position of the atoms, at $\simeq 30^\circ$ from vertical; astigmatism in this arm means its two transverse waists are not equidistant from the atoms either (measured at $\simeq 4.5$ mm and $\simeq 7$ mm, respectively).

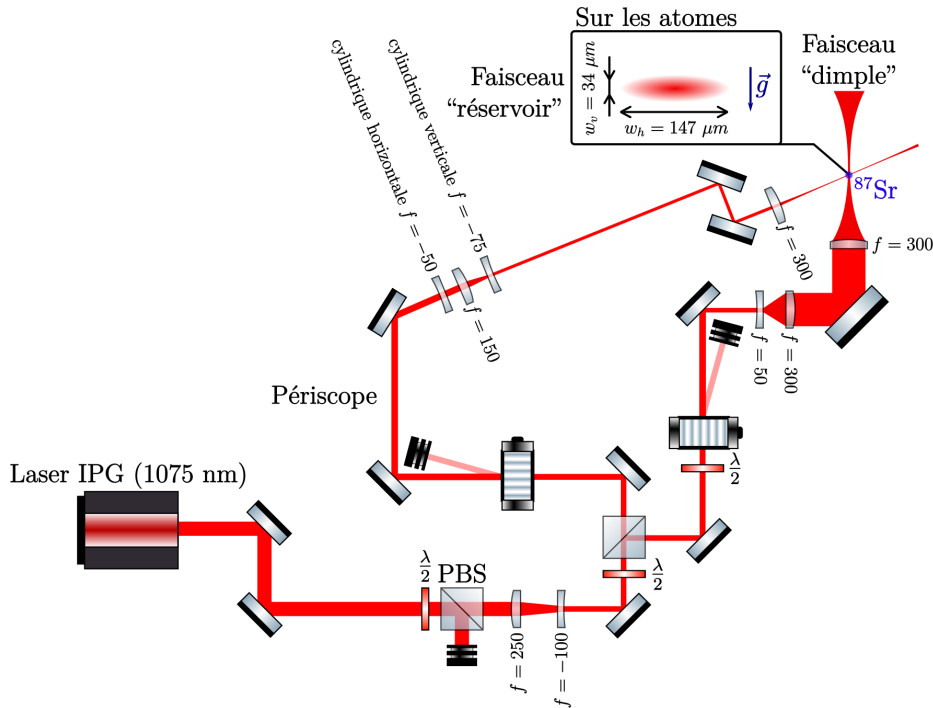


Figure 1.18: Optical setup of the 1075 nm crossed dipole trap laser system.

For the photoassociation measurements presented in Chapter 3, quantum degeneracy is not required, and a high atomic density is favored instead; these experiments are realized with 400 000 atoms of spread $\sigma \simeq 100$ μm at a few μK , as introduced above.

1.2 Spin Selection

1.2.1 Optical Pumping

Optical pumping is implemented prior to the final evaporation stages to prepare the atomic gas in a tailored mixture of nuclear spin states (m_F). Due to the $SU(N)$ symmetry of ^{87}Sr , elastic s -wave collisions between identical fermions in the same m_F state are forbidden by the Pauli exclusion principle. Therefore, at least two distinct spin components must be present to enable elastic collisions and ensure efficient re-thermalization during evaporative cooling. Furthermore, $SU(N)$ symmetry guarantees that elastic collisions conserve the nuclear spin state, preventing spin-exchange collisions during evaporation.

To perform state preparation, a bias magnetic field of 6.05 ± 0.05 G is applied to lift the degeneracy of the $^3P_1, F = 11/2$ hyperfine sub-levels via the linear Zeeman effect. The cloud is illuminated by a σ^- -polarized optical pumping beam tuned to the $^1S_0(F = 9/2) \rightarrow ^3P_1(F = 11/2)$ transition. Its frequency is defined as:

$$f_{11/2} = f_{\text{Radiant Dyes}} \pm f_{\text{Slave 1}}^{\text{RF}} + f_{11/2}^{\text{RF}} \quad (1.9)$$

The sign goes depends on the wanted spin manipulation experiment (more details in chapter 3). The laser frequency is swept across 200 kHz around the targeted $^3P_1, F = 11/2, m'_F$ sub-level – to compensate for day-to-day energy shifts arising from experimental fluctuations – depleting the corresponding $^1S_0, F = 9/2, m_F$ ground-state population over about 5 ms at an intensity of $\simeq 70 I_{\text{sat}}$. Spontaneous emission, governed by the electric-dipole selection rule $\Delta m_F = 0, \pm 1$, then redistributes the excited atoms into adjacent, lower-lying ground-state sub-levels, leading to a progressive accumulation of population in the target spin states.

To prepare a specific mixture (e.g., a 2-spin mixture), unwanted spin populations are sequentially pumped out. One spin component, typically $m_F = -9/2$, is intentionally retained alongside the target mixture to maintain two-body elastic collisions for thermalization during evaporation. Once evaporative cooling is complete, this auxiliary component can be selectively removed using a short **'blast'** pulse. Under σ^- excitation on the $F = 9/2 \rightarrow F' = 11/2$ transition, the $m_F = -9/2$ state couples exclusively to $m'_F = -11/2$. Because no $m_F = -11/2$ ground state exists, spontaneous emission from $m'_F = -11/2$ can only decay back to $m_F = -9/2$, forming a closed cycling transition. The resulting continuous photon scattering exerts a strong radiation pressure that expels the $m_F = -9/2$ population from the trap without disturbing the target single state or mixture. In practice, the blast pulse frequency is swept over a narrow range through resonance (rather than held fixed) to make the removal insensitive to small drifts in laser frequency or magnetic field calibration [24].

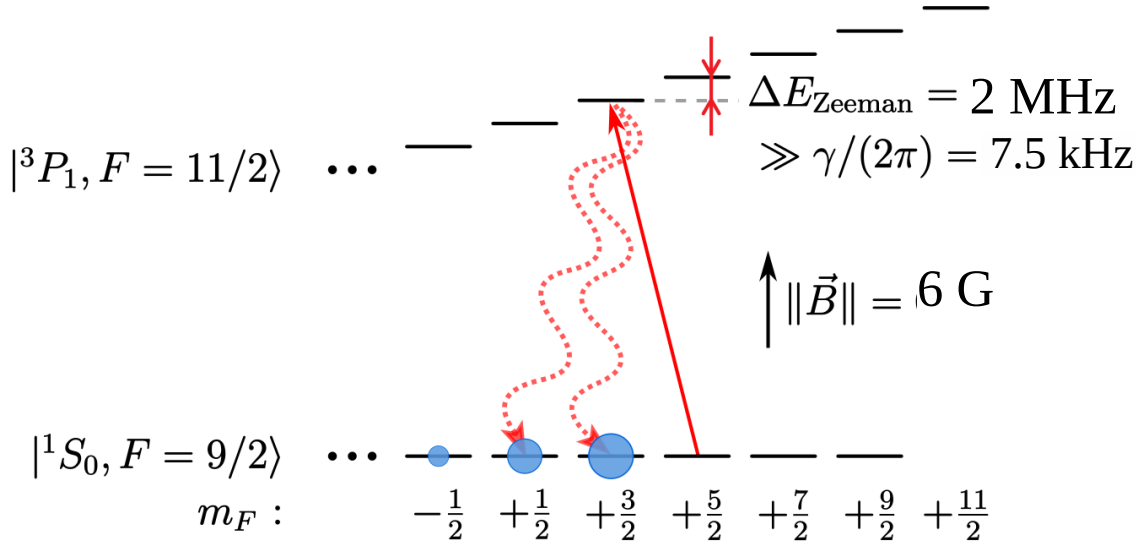


Figure 1.19: Optical pumping scheme on the $^1S_0(F = 9/2) \rightarrow ^3P_1(F = 11/2)$ transition used to prepare a specific nuclear spin mixture. The red laser is ramped through the target energy level of 3P_1 : the atoms decay to neighbours states $m_F \pm 1$ of 1S_0

1.3 Lasers for spin Manipulation

We developed a method demonstrated in [24] for coherent control within the ground state manifold of zeeman sublevels. it relies on tensor shifts enacted by a laser operating near 689 nm, and will be described in detail in chapter 2. At a later stage, we performed photoassociation experiments in the vicinity of the intercombination manifold, see chapter 3. Both experiments actually rely on the same laser system, that I will present here.

1.3.1 External Cavity Diode Laser (ECDL)

1.3.1.1 Laser architecture

The spin-selective method presented in the article [24] reveals the importance of inducing a Tensorial Light Shift (TLS) to be added to the Zeeman degeneracy lift.

For this purpose, an External Cavity Diode Laser (ECDL) from MOGlabs was chosen, whose architecture is shown in Fig. 1.20. A diode operating near 689 nm is first collimated by a collimation lens, before its frequency is roughly selected by an interference filter. The beam is then focused by a cat's-eye lens onto a partially reflective mirror placed at its focus – represented by the dashed rectangular shape on the scheme. Together, this lens and mirror form a **cat's-eye retroreflector**: it compensates for any tilt of the wavefront introduced by the mirror, which would otherwise cause a loss of mode-matching with the light re-injected into the diode, and it closes the external cavity used to narrow and fine-tune the laser frequency. Most of the light reflected on the mirror sustains this resonant cavity, while the small transmitted fraction is collimated by an output lens and leaves the laser head through an optical isolator, with a linewidth below 100 kHz.

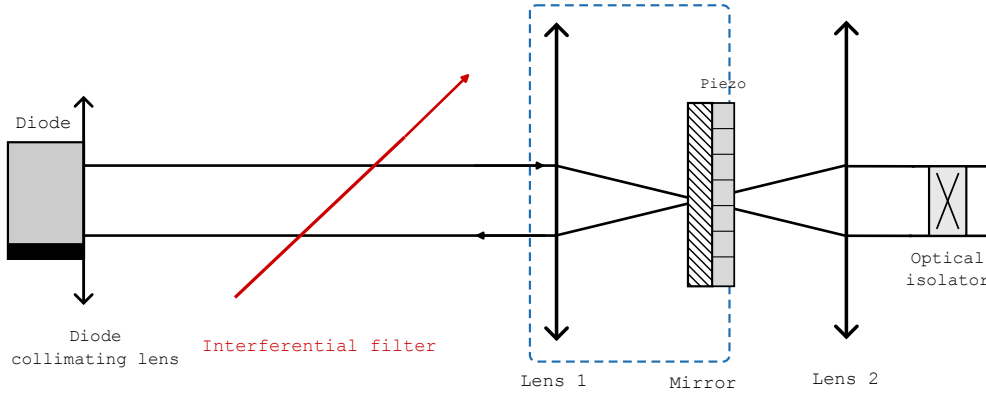


Figure 1.20: MOGlabs External Cavity Diode Laser (ECDL) architecture used for spin manipulation. A diode lasing at 689 nm is first collimated by a lens, then its frequency is roughly selected by an interference filter marked in red. The beam is focused by a second lens onto a partially reflective mirror (hatched rectangle): together they form a cat’s-eye retroreflector that closes the external cavity, feeding most of the light back for resonant frequency filtering while transmitting a small fraction onward. This output is collimated by a last lens and leaves the laser head through an optical isolator.

An internal PID controller robustly compensates for thermal and mechanical drifts of the cavity length as well as diode current fluctuations. The laser is subsequently beat-locked to a master optical reference adapted to the specific experimental requirements.

1.3.1.2 Problems encountered on the MOGlabs ECDL

For years, this ECDL exhibited a recurring instability that directly affected our experiments. We operate this laser far above threshold, at a fixed, high diode current chosen to maximize its output intensity on the external-cavity mode we lock to on a daily basis. As intracavity losses accumulated with time, however, the laser’s entire mode spectrum – the current-dependent map linking diode current to wavelength and output intensity – drifted: at this same fixed operating current, the mode we needed was no longer the one giving maximum intensity, so the laser no longer delivered its usual output power on the intended wavelength. Rather than changing the operating current, we instead had to open the laser head and reduce the intracavity losses; lowering the laser’s threshold in this way shifted the whole mode spectrum back, restoring the correspondence between our fixed operating current and maximum output intensity on the intended mode. This drift was eventually traced to two independent hardware issues, both responsible for a progressive loss of intracavity intensity:

- the interference filter had developed damage at its center, degrading the transmission of the light. It was replaced with a new filter supplied by the manufacturer.
- one of the cavity’s three piezoelectric actuators had failed. The full cat’s-eye assembly was therefore replaced with an MGEF adapter featuring a wider-

range frequency piezo.

Even after these repairs, a slow residual drift of the threshold persisted, now on a timescale of several days rather than hours. Gluing the cat's-eye assembly to the mount holding the diode and interference filter helped reduce this residual drift further, and keeping the laser switched off when not in use also limits it. If the stable mode eventually becomes inaccessible within the available diode current range, the threshold can still be lowered by gently adjusting the knobs of the mount holding the diode barrel. Because this realignment procedure slightly displaces the output beam, we added two optical fibers at the laser output: fiber coupling preserves the alignment to the rest of the spin-manipulation optical setup regardless of small beam-pointing changes.

1.3.2 Optical bench for coherent Raman manipulations

1.3.2.1 Physical Concept

The second chapter of this thesis focuses on spin manipulation within the $SU(10)$ manifold of ^{87}Sr . This operation is realized via a two-photon Raman process driven by two laser beams red-detuned from the $^1S_0 \rightarrow ^3P_1$ intercombination line.

Controlling the dimension of the effective Hilbert space from 1 to 10 requires independent addressing of individual spin states. The two-photon Rabi frequency Ω coupling two ground-state sub-levels $|g, m_F\rangle$ and $|g, m'_F\rangle$ through intermediate excited states $|e\rangle$ takes the form:

$$\hbar\Omega = \sum_e \frac{\langle g, m'_F | \vec{d} \cdot \vec{E}_2^* | e \rangle \langle e | \vec{d} \cdot \vec{E}_1 | g, m_F \rangle}{\Delta_e} \quad (1.10)$$

where $\vec{d}$ is the dipole operator, $\vec{E}_1, \vec{E}_2$ are the electric field amplitudes, and Δ_e is the single-photon detuning relative to the intermediate state $|e\rangle$.

By applying the TLS, the energy difference between two consecutive spin projections m_F and $m_F + 1$ becomes non-linear:

$$\Delta E_{m_F}^{m_F+1} = q(2m_F + 1) + \beta \quad (1.11)$$

where q parameterizes the quadratic (tensor) light shift and β represents the linear Zeeman contribution. This explicit dependence on m_F lifts the energy degeneracy between adjacent transitions, allowing spectral isolation and individual control of any given $m_F \leftrightarrow m_F + 1$ pair.

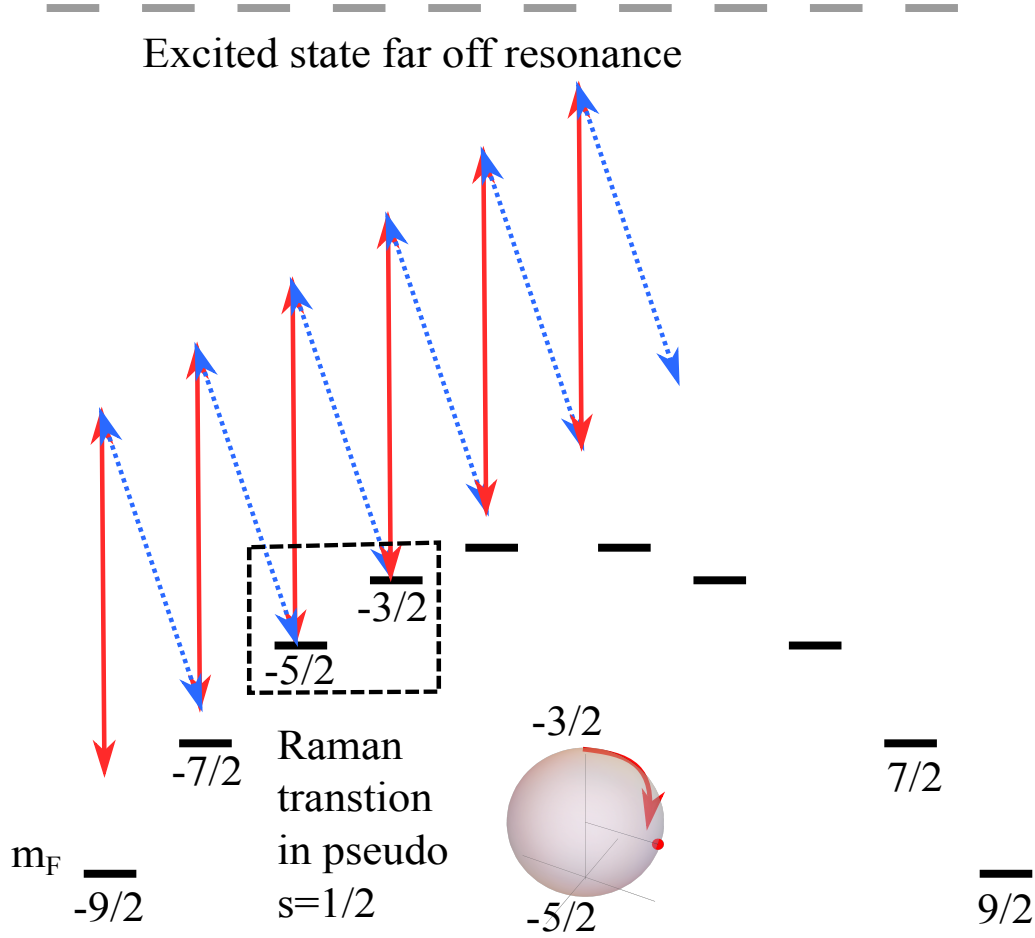


Figure 1.21: Spin manipulation scheme of [2]. The isolation of spin states is allowed by the tensorial shift of the ground states that makes neighbours m_F states isolated from the other resonances. The Bloch sphere represents the reduced Hilbert space of the selected qubit.

For more details this process is discussed in the second chapter of this thesis and also in the article of the group [24].

1.3.2.2 Experimental setup of the Raman and PA process

The two-photon process is driven by two distinct beams: a first beam derived from the “AOM TLS” laser (shown in red in Fig. 1.22) generates the TLS and supplies the first photon. A second beam, frequency-shifted by the “AOM Raman” (shown in blue), completes the Raman transition.

The TLS beam polarization is set parallel to the magnetic field (π transitions), producing a purely scalar/tensor light shift with no vector component. The Raman beam polarization is linear and orthogonal to the magnetic field, so that its decomposition onto σ^+ and σ^- components drives the required $\Delta m_F = \pm 1$ legs of the two-photon transition – one photon absorbed from the π -polarized TLS beam, one photon emitted into the $\sigma^\mp$ component of the Raman beam. Both beams are copropagating in the same polarization maintaining fiber with perpendicular polarizations preserving any interferences that could affect the process. This configuration selectively

couples a single pair of states—for instance, $|^1S_0, F = 9/2, m_F = -7/2\rangle \leftrightarrow |^1S_0, F = 9/2, m_F = -5/2\rangle$ —while remaining far off-resonance for all other m_F states. The two-photon Raman coupling strength scales as $\Omega_R \propto \sqrt{I_D I_R}$, while the spacing between adjacent Raman resonances scales as $2q/h \propto I_D$. Spectral isolation of a single $m_F \leftrightarrow m'_F$ pair therefore requires $I_R \ll I_D$, so that neighbouring resonances remain spectrally resolved; in practice we use $I_R/I_D \lesssim 10^{-4}$ [2].

The complete optical path for this two-photon process, together with the filtering cavity described in Sec. 1.3.3, is shown in Fig. 1.22.

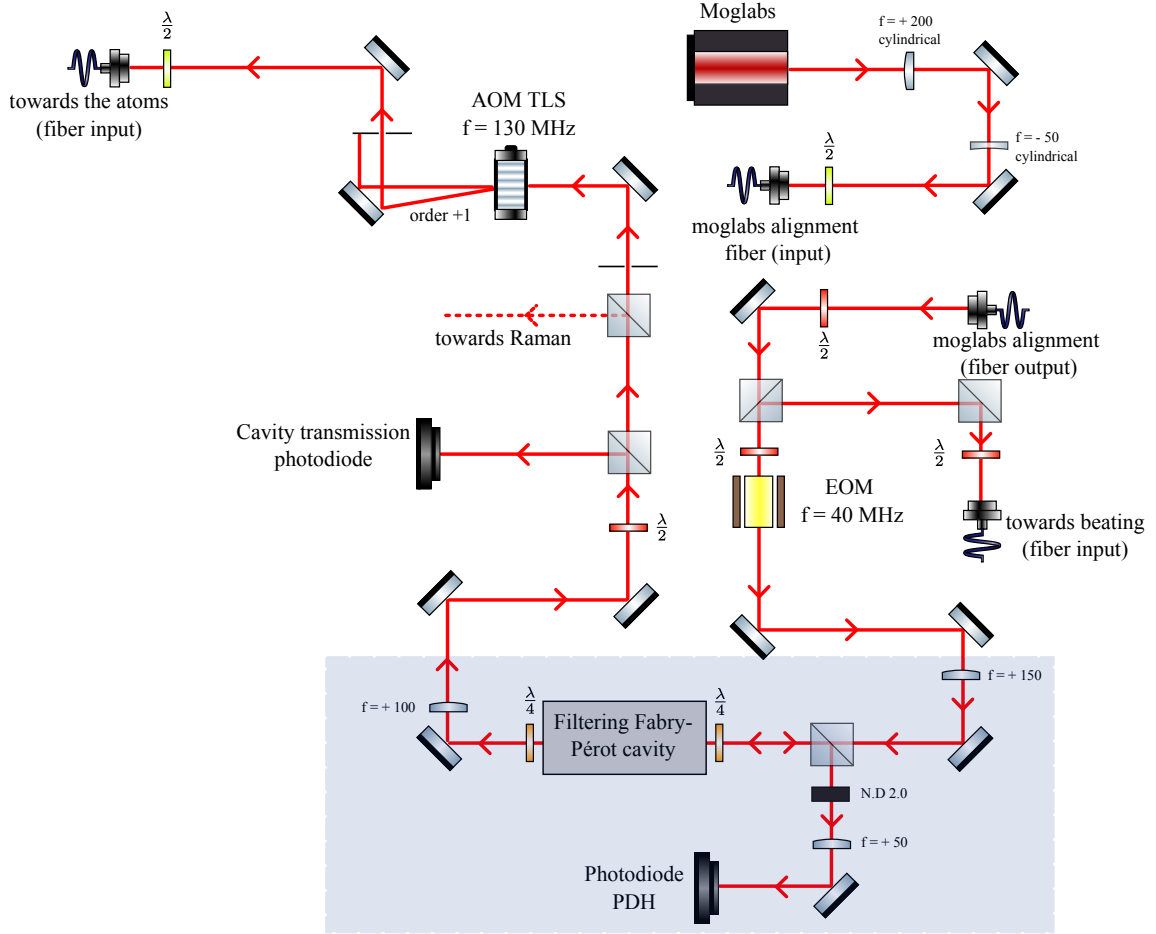


Figure 1.22: Complete optical setup including the Fabry-Pérot cavity and PDH stabilization loop.

To reduce amplified spontaneous emission (ASE) from the laser source – and thereby minimize stray light and photon-recoil heating during spin manipulation – the optical chain incorporates a custom-built, PDH-stabilized confocal Fabry-Pérot cavity, detailed in Sec. 1.3.3. The effect of the ASE has been observed by the heating of the gas while the laser was hundred's of MHz off resonance of transitions where we expect really low spontaneous emission. This cavity was originally built with these Raman TLS experiments in mind; however, by the time its construction was completed the corresponding measurements had already been carried out, and the experimental effort had moved on to the photoassociation spectroscopy presented in Chapter 3 – for which the cavity turned out to be essential, given the higher

sensitivity of these measurements to stray light. It is intended to be used again for future extensions of the Chapter 2 spin-manipulation experiments.

1.3.3 Fabry-Pérot cavity and PDH Locking system

1.3.3.1 Cavity design and assembly

A primary objective of the cavity injection scheme in our optical setup is to maximize overall transmission for spin manipulation. To achieve this, we constructed a confocal cavity configuration ($d_0 = R_C$, where d_0 is the mirror separation and R_C is their radius of curvature), ensuring that all odd transverse modes are degenerate and resonate at the same frequency. If these modes were non-degenerate, off-resonance spatial components would be reflected at the input mirror, thereby reducing the total cavity transmission efficiency.

In addition to the confocal geometry, longitudinal mode resonance requires that the cavity length d_0 satisfies the condition:

$$d_0 = n \frac{\lambda}{2} \quad (1.12)$$

where n is an integer and λ is the wavelength of the injection laser.

To fulfill these requirements, we selected curved mirrors with a diameter of $D = 4.5 \pm 0.15$ mm and a radius of curvature matching d_0 . The manufacturer specified a nominal mirror transmission of $T_{\text{mirror}} = 1.28\%$, though the effective system performance can be influenced by mechanical mounting constraints.

Proper cavity operation requires the mirrors to be aligned precisely parallel to each other and perpendicular to the optical propagation axis. In our initial implementation (Fig. 1.23 top), the mirrors were permanently glued into place. However, this approach introduced major technical issues over time: excess glue migrated onto the optical surfaces, degrading overall transmission, while alignment errors during gluing introduced an uncorrected angular tilt relative to the beam axis.

Because the original design provided no mechanical degrees of freedom to correct these misalignments, we redesigned the mounting mechanism. The updated cavity assembly is presented in Fig. 1.23 bottom. In this configuration, three adjustment knobs on a circular outer mount apply controlled pressure against the central cylindrical spacer housing the mirrors, with intermediate washers providing mechanical compliance. This allows for precise tilt control and fine adjustment of the mirror separation.

An additional parameter that has been adjusted during the construction is the mass of the mirrors: a first generation of the cavity has been constructed with heavier mirrors than actual ones, which presented mechanical resonance in the acoustic frequency range. The lock of the cavity was not robust. Then this new generation is made of smaller and light mirrors which pushes these resonances to higher frequency range we do not bother.

Once locked to the injection laser, the optimized cavity achieves a total power transmission efficiency of $T = 0.63$. This limited efficiency likely results from residual mechanical imperfections inherited from the original glued assembly: a slight angular mis-gluing of a mirror could tilt the cavity axis and cause part of the beam to

clip inside the cavity, while leftover glue or dirt on the mirror surfaces could add further scattering losses. Despite this imperfect transmission, the cavity's spectral properties remain well characterized. The Free Spectral Range (FSR) of the cavity is given by:

$$\text{FSR} = \frac{c}{2d_0} = 3 \text{ GHz} \quad (1.13)$$

with a measured finesse of:

$$\mathcal{F} = \frac{\text{FSR}}{\delta\nu} = 310 \pm 15 \quad (1.14)$$

where $\delta\nu$ represents the full width at half maximum (FWHM) of the transmitted resonance peak measured on a photodiode for the odd transverse modes.

1.3.3.2 Pound-Drever-Hall locking system and PID controller analysis

The cavity is frequency-locked to the laser using the Pound-Drever-Hall (PDH) technique, as detailed by Black [33] and schematized in Fig. 1.24. The scheme utilizes an Electro-Optic Modulator (EOM) to generate sidebands at frequencies $f_0 \pm f_M$ around the optical carrier f_0 , with a modulation frequency of $f_M = 40.48 \text{ MHz}$. The modulated electric field incident on the cavity is expressed as:

$$E_{\text{inc}}^{\text{mod}} \approx E_0 [J_0(\beta)e^{i2\pi f_0 t} + J_1(\beta)e^{i2\pi(f_0+f_M)t} - J_1(\beta)e^{i2\pi(f_0-f_M)t}] \quad (1.15)$$

where $J_0(\beta)$ and $J_1(\beta)$ are the zeroth- and first-order Bessel functions of the first kind, and β is the modulation index.

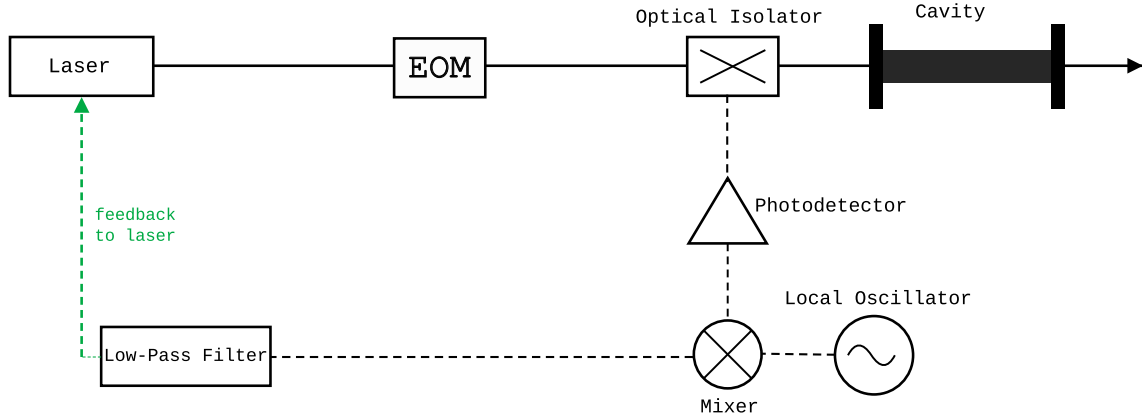


Figure 1.24: Schematic of the Pound-Drever-Hall locking system, extracted and adapted from [33]. The laser is firstly passing through an EOM that is modulating the electric field at frequency f_M , creating two sidebands at frequency $f_{\text{laser}} \pm f_M$. The carrier at f_{laser} and the sidebands are passing through the filtering Fabry-Pérot cavity which is reflecting the light going back to the entrance of the cavity. A beamsplitter serving as an optical isolator is reflecting the light to a photodiode. Then this signal is mixed with a local oscillator to extract an error signal to which a low-pass filter is combined to keep only the signal for the locking.

The carrier enters the cavity and acquires a steep, frequency-dependent phase shift near resonance, whereas the sidebands are promptly reflected from the input mirror.

The interference between the reflected carrier and the sidebands is detected on a fast photodiode, demodulated at f_M , and low-pass filtered to extract the dispersive error signal (Fig. 1.25). The sharp linear slope around resonance provides a robust error signal highly sensitive to the sign of the frequency excursion. From this graph, one measures a signal-to-noise ratio of $\text{SNR} = 187$. Interestingly, the sideband resonance features were not distinctly visible on the scanning error signal, even when driving the EOM with a high modulation depth to maximize the J_1/J_0 ratio.

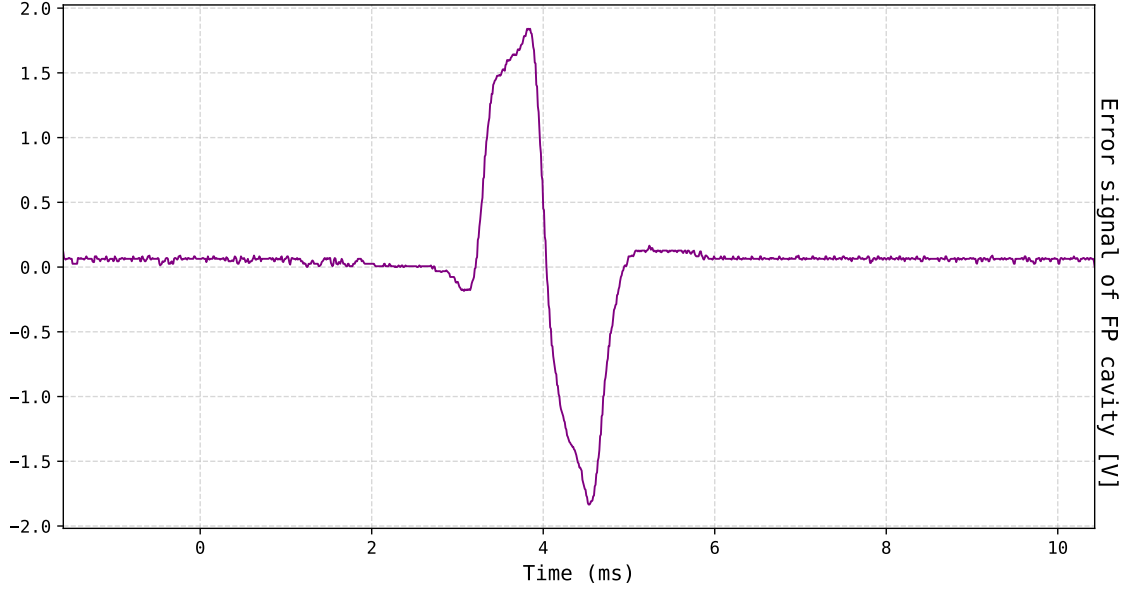


Figure 1.25: Pound-Drever-Hall (PDH) error signal obtained by scanning the cavity length across resonance, enabling the lock cavity for the filtering of MOGlabs laser.

Digital feedback control and noise analysis were implemented primarily using a Red Pitaya STEMLab board, driving a home-made PID controller with separate proportional and integral stages (Fig. 1.26), which we found sufficient for our setup. The PID parameters were optimized by injecting a broadband electronic noise signal into the cavity piezo driver and measuring the closed-loop transfer function. As seen in the frequency response plot (Fig. 1.27), the closed-loop noise spectrum shows a servo bump at 3 kHz, followed by peaks at its multiples (6, 9 and 12 kHz). It is a feature of the feedback loop rather than a mechanical resonance: its frequency increases with the gain (not shown). No mechanical resonance of the cavity is visible in this frequency range. Consequently, the final feedback loop acting on the cavity incorporates a low-pass filter with a cut-off frequency of $f_c = 40$ kHz, a proportional gain strictly bounded between $1 < K_p < 10$, and an integral gain configured between $0 < K_i < 8 \text{ s}^{-1}$.

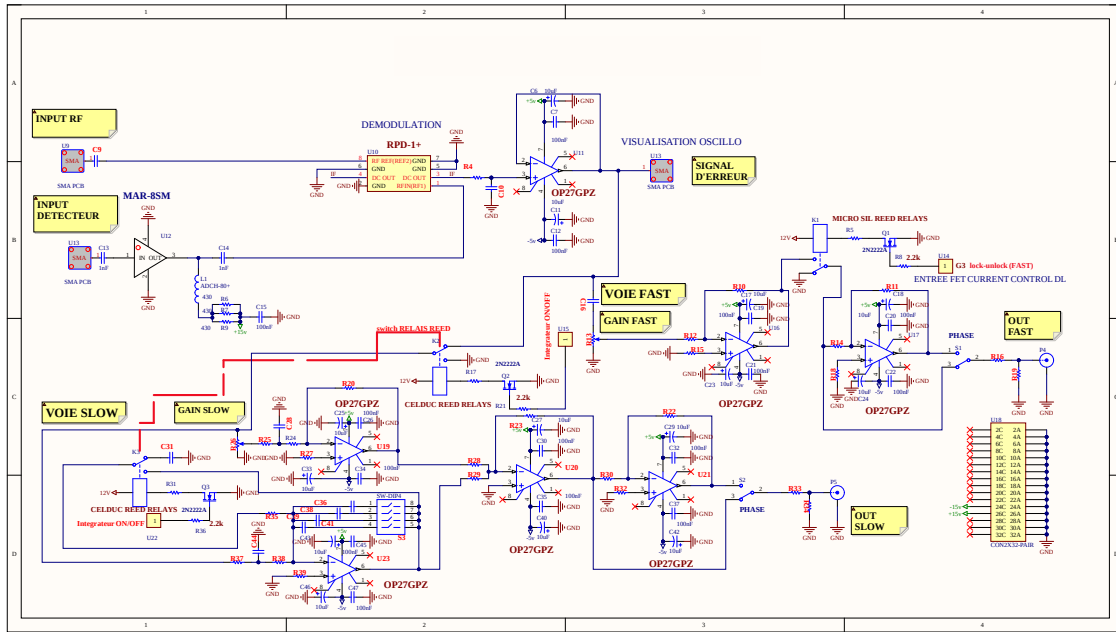


Figure 1.26: Homemade PID scheme constructed by the Fabrice Wiotte from electronic workshop of LPL. It is composed of separate proportional and integral stages, which we considered sufficient for our setup.

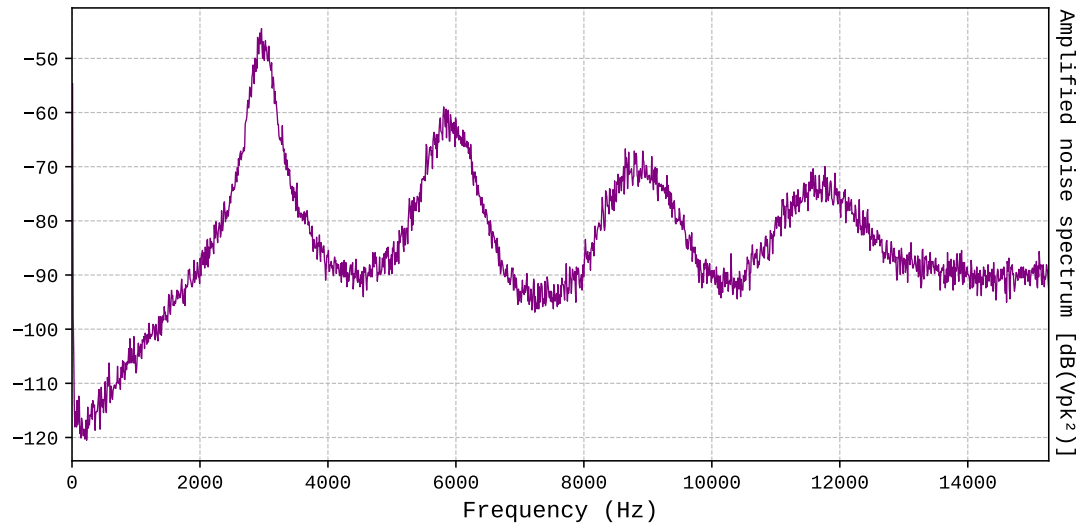


Figure 1.27: Closed-loop noise spectrum measured with the Red Pitaya, showing the servo bump at 3 kHz and its multiples.

1.4 Spin Measurement Scheme

Total atom numbers are determined via standard absorption imaging on the broad $^1S_0 \rightarrow ^1P_1$ transition at 461 nm using a Vortex laser. State-resolved spin detection follows the momentum-transfer (“spin-kick”) technique detailed in [2].

The measurement sequence proceeds as follows:

1. Atoms are illuminated by a vertical, retro-reflected laser beam with opposite circular polarizations (σ^+/σ^-), as illustrated in Fig. 1.28(a), addressing the $|^1S_0, F = 9/2\rangle \rightarrow |^3P_1, F = 11/2\rangle$ transition at an intensity of 5 mW/cm², giving Rabi couplings $\Omega \simeq 2\pi \times 220$ kHz (scaled by the relevant Clebsch-Gordan coefficients). A magnetic field of 16 G is applied along the beam propagation direction, splitting adjacent m'_F sub-levels of $^3P_1, F = 11/2$ by 6.0 MHz – far larger than the 7.5 kHz transition linewidth. The atoms are held against gravity in a weakly confining (125 Hz) horizontal dipole trapping beam during the momentum transfer.
2. A state-selective pulse tuned to a specific $^1S_0 \rightarrow ^3P_1, F = 11/2, m_F$ transition –in the presence of linear Zeeman shift– adiabatically transfers targeted spin populations (Fig. 1.28(b)). The laser frequency is swept, with decreasing frequency only, over a span of 1.4 MHz within 300 μ s.
3. Each absorption-stimulated emission cycle imparts a coherent $2\hbar k$ momentum kick to the selected m_F state.
4. Following a time-of-flight (TOF) expansion duration $t = 6.6$ ms (set for the gas to stay in camera plane), spatial separation between spin components is observed in the absorption image (Fig. 1.28(d)).

Unaddressed spin states remain at the central position of the cloud. In contrast, states subject to the momentum transfer form distinct spatial wavepackets shifted to the left or right depending on the direction of the resonant momentum kick (e.g., $m_F = -3/2$ and $m_F = -7/2$ in Fig. 1.28(d)).

To ensure efficient population transfer, the optical pulse sweep rate must satisfy the adiabaticity criterion across the avoided crossing between coupled adiabatic states (Fig. 1.28(c)).

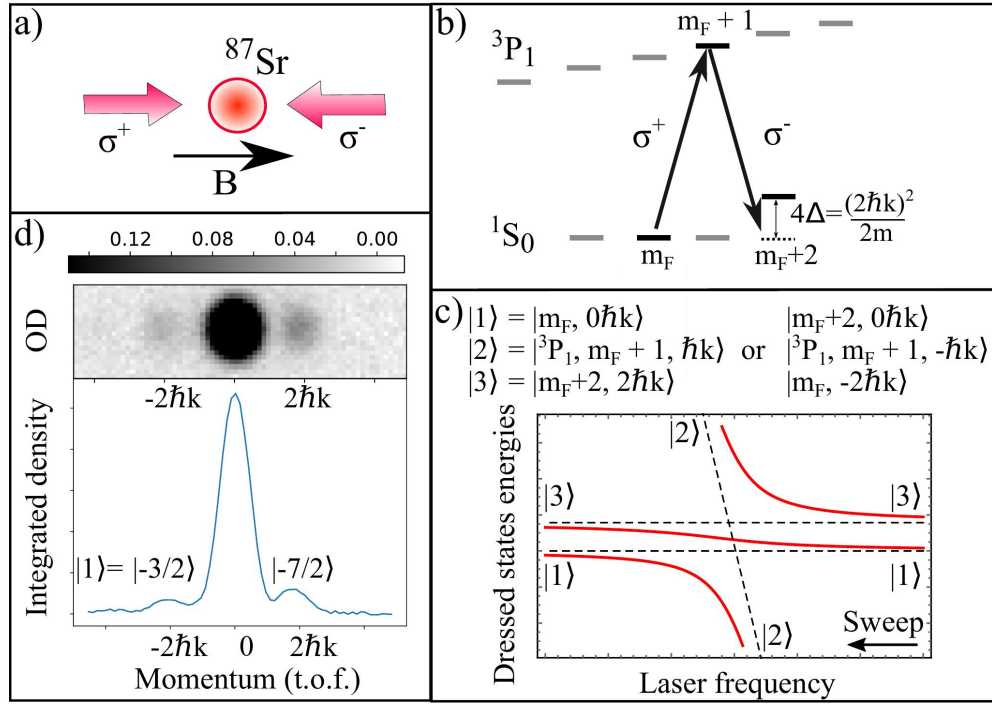


Figure 1.28: Principle of spin-selective detection via coherent momentum transfer. (a) Counter-propagating beam geometry of the lasers used for the TLS and two-photon Raman scheme. (b) Energy levels scheme of the Raman process, with recoil energy shifting the $m_F + 2$ level. (c) Avoided crossing due to the coupling of the states, necessitating adiabatic transfer. (d) Resulting TOF absorption image showing spatially separated spin components and the corresponding momentum distribution.

Chapter 2

Quantum state measurement and characterization in qudit systems

This chapter presents the experimental results published in Ref. [1], centered on the coherent manipulation of the 10-dimensional nuclear spin Hilbert space ($2I+1 = 10$) of ^{87}Sr atoms. By exploiting a tensor light shift to isolate tailored subsystems, we demonstrate control ranging from robust pseudo-spin qubits to high-dimensional qudits ($D = 3$ to 10) displaying multi-second coherence times. Based on this platform, we implement multi-state interferometric schemes over four nuclear levels for multi-parameter sensing. Following section 2.1.1 reproduces article, Section 2.2 is written for this manuscript and provides an extended theoretical derivation of the Positive-Operator-Valued Measures (POVMs) implemented in our non-commuting observables measurement scheme.

2.1 Article ‘Coherent control over the high-dimensional space of the nuclear spin of alkaline-earth atoms’ [1]

2.1.1 Gas preparation and spin control

Large-spin particles open new possibilities for quantum technologies. In the context of quantum many-body physics and quantum simulators, the enlarged spin degrees of freedom correspond to a significantly larger Hilbert space, which leads to qualitatively new physics [34, 35, 36, 37, 38, 39, 40]. Large spins can also enhance quantum information processing [41] and quantum metrology [42, 43, 44, 45]. Conveniently, they emerge from several physical platforms, such as superconducting devices [46, 47], molecular magnets [48], ions [49], multi-mode photons [50], and atoms [51, 52].

Among large-spin atoms, fermionic alkaline-earth and alkaline-earth-like atoms (AEA) are particularly interesting for quantum metrology because they display a weak sensitivity to magnetic fields and possess narrow optical transitions that make them prominent for optical clocks with cold atoms [53]. These atomic species are also remarkable for quantum many-body physics, as the purely nuclear nature of their

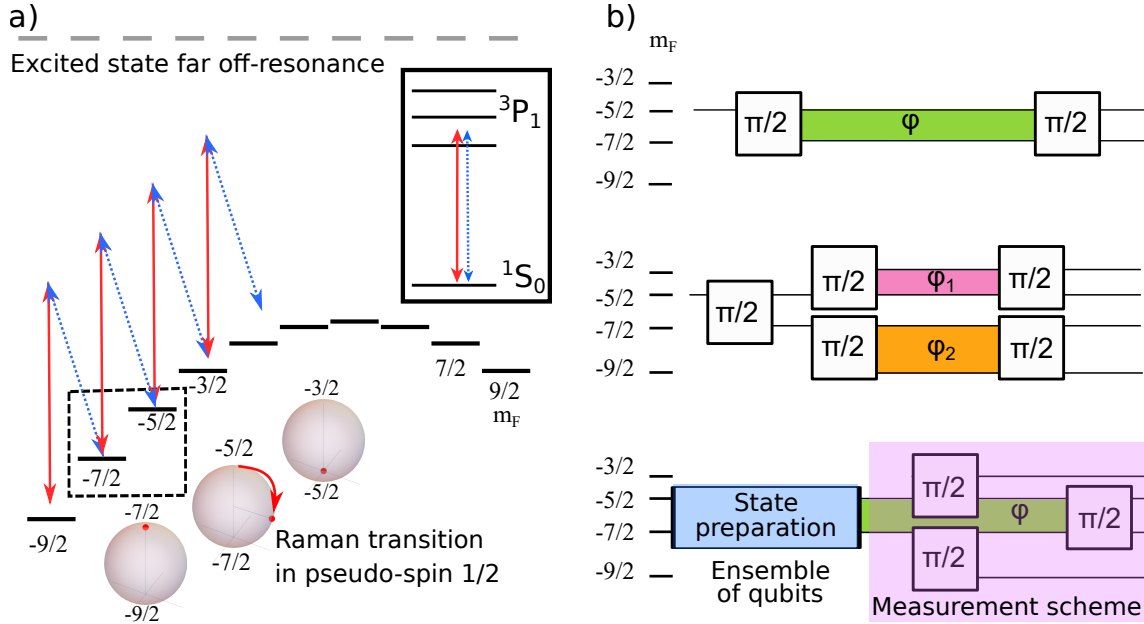


Figure 2.1: **Principle of the experiment.** (a) Atomic levels and laser coupling scheme. A π -polarized light beam tuned within the hyperfine structure of the $^1S_0 \rightarrow ^3P_1$ transition (red arrows) creates an energy shift in the electronic ground state, with significant tensor component $U_{TLS}(m_F) = q m_F^2$. A second laser (here σ^- polarized) enables Raman transitions. Thanks to the tensor light shift, the Raman resonance conditions differ from each other, making it possible to drive Rabi oscillations between two isolated spin states (here $|-5/2\rangle$ and $|-7/2\rangle$), all other processes being off-resonant. The Raman operation can be understood as a rotation in a sub-Bloch sphere. A potential linear energy shift $U_{linear}(m_F) = b m_F$ (in the illustration, $b \simeq -3q$) does not directly affect the capacity to isolate a pair of states. (b) The three experimental schemes in this paper. Top: We operate a Ramsey interferometer between states $|-5/2\rangle$ and $|-7/2\rangle$, decoupled from the other spin states. Middle: We operate two interferometers in parallel, demonstrating multi-parameter simultaneous sensing of the fields acting on the atoms. Bottom: we characterize the collective state of atoms in a superposition of two Zeeman levels ($|-5/2\rangle$ and $|-7/2\rangle$), identified as qubit states. The measurement sequence starts by applying two pulses that map the populations of the qubit states into ancillary states. Combining this with a final rotation in the qubit sub-Bloch-sphere, one can simultaneously measure two orthogonal components of the pseudo-spin of the ensemble of qubits.

spin in the ground state generates an $\mathfrak{su}(N)$ symmetry that introduces frustration and should thus strongly impact quantum magnetism [54]. This possibility has attracted considerable attention from both theoretical [55, 56, 57] and experimental [36, 58, 59, 60] standpoints. To exploit the full potential of large-spin fermionic alkaline-earth atoms and to characterize the $\mathfrak{su}(N)$ phases at low temperatures, it is crucial to develop tools that coherently control the spin states of atoms beyond simple spin precessions (see [61, 24, 62]).

In this paper, we use a tensor light shift [63] to control the resonance conditions of Raman transitions within the manifold of $2F + 1 = 10$ Zeeman states of the spin- $F = 9/2$ ground state of ^{87}Sr . The quadratic energy shift $\propto m_F^2$ enables

us to perform coherent manipulations between isolated pairs of m_F states with a fidelity better than 99 %. These operations demonstrate control over unitary transformations deriving from $\mathfrak{su}(N = 2F + 1)$ generators, beyond the usual precession operations deriving from the spin- F representation of the $\mathfrak{su}(2)$ group. Taking advantage of this new possibility and the atoms' insensitivity to magnetic fields, we operate a Ramsey interferometer between two isolated hyperfine states and demonstrate Ramsey fringes with no discernible loss of contrast over 3s. We further exploit the large spin of the atoms with two new types of interferometers. In the first implementation, we operate two Ramsey interferometers simultaneously and in parallel, using two independent pairs of spin states within the hyperfine structure. This configuration allows simultaneous measurement of two independent parameters (here corresponding to the effective quadratic and linear Zeeman shifts). In the second implementation, we mirror two physical spin states that make a qubit into two ancillary spin states. Selective unitary transformation ("rotation") of the two qubit states and the measurement of all four states then permit the simultaneous measurement of two non-commuting observables of an ensemble of qubits.

2.1.2 Implementation of $\mathfrak{su}(N)$ generators.

One key idea of this paper is to implement coherent rotations between two and only two arbitrary spin states chosen among the N spin states of an N -component Fermi mixture. Rabi oscillation between such a selected pair of states corresponds to rotations driven by infinitesimal generators of the $\mathfrak{su}(N)$ spin algebra, described by an operator σ_{m_F, m'_F}^x . The restriction of σ_{m_F, m'_F}^x to the Hilbert space $\{|m_F\rangle, |m'_F\rangle\}$ is the Pauli matrix σ^x , while all its other elements are zero. The evolution operator after a sequence reads $\sim \exp(-i\theta\sigma_{m_F, m'_F}^x/2)$, where $\theta = \int \Omega_R(t)dt$ is called pulse area and Ω_R is the Rabi frequency. The evolution operator describes a rotation in the restricted Hilbert space associated with the two selected spin states, and is the identity in the orthogonal space consisting of all other levels. Thus, it is distinct from the more usual manipulation of a spin $\vec{F}$ by precession around a magnetic field, $\exp(-i\vec{\alpha} \cdot \vec{F})$, which involves a combination of the three generators $\vec{F} = (F_x, F_y, F_z)$ of the spin- F representation of the $\mathfrak{su}(2)$ algebra. This new tool enables complete control over the spin degrees of freedom, which is necessary to characterize phases of $\mathfrak{su}(N)$ -symmetric quantum matter at low energy (see e.g. [64]) and for computing applications leveraging the vast Hilbert space provided by large spins (see e.g. [41]).

It is challenging to intuitively represent the state of large-spin atoms and the $\mathfrak{su}(N)$ generators of rotations we implement since a spin $F > 1/2$ possesses a rich inner structure that the usual Bloch sphere fails to represent fully. This structure can be captured by, for example, the Majorana stellar representation [65] (see also [66, 67, 68]). However, the motion of the Majorana stars driven by coherent manipulations is generally quite complicated and does not always allow a clear physical interpretation. Qualitatively, $4F$ angles ($2F$ points on a sphere) are sufficient to describe the state. A simple representation, that can be well-suited to visualize some of our interferometric experiments, is to consider each two-level sub-space of adjacent Zeeman levels $(m_F, m_F + 1)$ in a Bloch sphere representation and consider the set of $2F = N - 1$ Bloch spheres as a representation of the state. For each Bloch sphere, the latitude represents the relative populations while the longitude represents the relative phase of the spin wavefunction in the two relevant states.

Experimentally, we perform rotations on a sub-Bloch sphere between selected pairs of nuclear-spin states, either $m_F \longleftrightarrow m'_F = m_F + 1$ or $m_F \longleftrightarrow m'_F = m_F + 2$ - see Fig. 2.1. To this end, we introduce two-photon Raman couplings. We rely on a quadratic dependence of the light shift as a function of m_F (tensor light shift, TLS), to resonantly drive only one of the two-photon resonances, whereas the other couplings are off-resonant and have negligible impact on the atomic state evolution. Provided the Raman coupling is much smaller than the TLS, we can spectrally resolve any pair of m_F states (with $\Delta m_F = 1, 2$), beyond what was achieved by the approach in Ref. [62], which only allowed coherent oscillations between $m_F = F$ and $m_F = F - 1$ (or $m_F = -F$ and $m_F = -F + 1$). Our work is thus analogous to the proposal in Ref. [69] but with Raman couplings instead of radio-frequency couplings between adjacent Zeeman sublevels. It is also very similar to the experiment on the solid-state platform in Ref. [48], where a hyperfine interaction, instead of an external field, lifts the degeneracy.

Pulsing the Raman coupling drives precession on one selected sub-Bloch sphere around the x axis (by phase convention). Furthermore, at any time, all of the sub-Bloch spheres experience σ_z precession at different rates determined by the detuning of the Raman beat note (controlled by an RF local oscillator) to the corresponding two-level transition. In this article, we explore three independent interferometric schemes restricted to a given Hilbert subspace (see the right panels of Fig. 2.1). They illustrate the new possibilities for quantum sensing or simulation that emerge when using simple combinations of coherent rotations between pairs of spin states, complementing previously demonstrated schemes using large-spin atoms [42, 45, 68]).

The intercombination line $^1S_0 \rightarrow ^3P_1$ of the ^{87}Sr isotope is particularly favorable to engineer large tensor shifts with only weak spontaneous emission, thanks to the high ratio of about 10^5 between the hyperfine splitting in 3P_1 and the inverse lifetime of these hyperfine states [70]. Using the much narrower $^1S_0 \rightarrow ^3P_2$ transition might lead to even better performances. Our scheme should readily adapt to other alkaline-earth-like atoms and to other species with similarly rich electronic structures (Yb, Cd, Hg, Er, Dy, ...). Finally, compared to the recent results in Ref. [71, 72], where the spins of erbium and strontium atoms, respectively, are manipulated through coherent oscillations on a narrow optical transition, our scheme also benefits from *not* relying on a clock-like optical transition. This, we achieve because the Raman transitions negligibly populate lossy excited states.

2.1.3 Experimental setting

All the experiments described in the following sections start with a spin-polarized ultracold Fermi gas of ^{87}Sr . To produce this gas, we first laser cool and hold within a dipole trap a 5 μK -cold sample of atoms in the $5s^2^1S_0$ ground state, fully depolarized across the ten spin states of the $F = 9/2$ manifold. We then use optical pumping on the $^1S_0 \rightarrow ^3P_1$, $F = 11/2$ intercombination line to empty all spin states but two (typically, the $m_F = -9/2$ and $m_F = -5/2$ states). We thus prepare a stable spin mixture since the $\mathfrak{su}(N)$ symmetry prevents spin-changing collisions. Ref. [24] gives more details on this optical pumping. We next use forced evaporative cooling to obtain an ultracold gas at a temperature of about

100 nK. Finally, we apply a state-selective radiation-pressure pulse to remove atoms in the $m_F = -9/2$ state, with no noticeable heating of the remaining $m_F = -5/2$ sample. We achieve such state selection by a 4.5 ms light pulse tuned to the $|^1S_0, m_F = -9/2\rangle \rightarrow |^3P_1, F = 11/2, m_F = -11/2\rangle$ transition in a magnetic field of 4.5 G. The polarized gas, of about 10^4 atoms, is then kept in a uniform magnetic field $B\vec{e}_z$ with $B = 5.2$ G that defines the quantization axis and produces a linear Zeeman splitting of 960(5) Hz. With less than 1% of the atoms in each Zeeman state other than the one chosen, the residual spin entropy is below 0.5 (while a fully random spin state has entropy $\log(10) = 2.3$).

After this preparation, we coherently manipulate the atoms' nuclear spin using an external-cavity laser diode whose frequency is tuned within the hyperfine structure of the 3P_1 state, 600 MHz to the red of the $|^1S_0, F = 9/2\rangle \rightarrow |^3P_1, F = 9/2\rangle$ line. We produce two phase-coherent laser beams with independently controlled frequencies from a single laser split into two paths with independent single-pass acousto-optic modulators. The first beam's primary role is to engineer a quadratic energy shift $U_{TLS}(m_F) = q m_F^2$ of the Zeeman states labeled by their spin projection m_F . We will refer to this beam as the "Tensor Light Shift (TLS)" beam. The second beam, labeled the "Raman" beam, acts together with the first (in most of the paper) to drive Raman transitions between two Zeeman states when their frequency difference matches the energy difference between states. We drive the AOMs using two AD9852 direct digital synthesizers synchronized on a shared 20 MHz clock. The beam frequency difference thus has a long term relative instability directly set by the shared clock (RIGOL DG1022), at the 5 ppm level. They are recombined on the same path with orthogonal linear polarizations through a single-mode optical fiber before being sent onto the atoms with a beam waist of 320 μm . This configuration also renders recoil effects negligible.

We set the TLS beam polarization almost co-linear to the magnetic field, corresponding to π transitions in this quantization axis. We ramp its power up to typically 5 mW in 3.5 ms, slowly enough that the atoms' spin state adiabatically follows the slight change of quantization axis. We stabilize the beam intensity to a sub-percent level at the millisecond timescale, using a photodiode positioned after the beam has crossed the atoms. The Raman beam polarization is linear and orthogonal to the magnetic field, allowing σ^+ and σ^- transitions. Throughout this article, we mainly apply $\delta m_F = 1$ spin-changing Raman transitions from absorption and stimulated emission of one π photon from the TLS beam and one σ^- photon from the Raman beam. By setting the two-photon detuning resonant with such transitions, the σ^+ Raman beam photons only drive widely off-resonant processes thanks to the TLS beam's quadratic shift $U_{TLS}(m_F)$ and ultimately remain spectators. Moreover, the σ^- transitions addressed by the Raman laser have higher Clebsch-Gordan coefficients and coupling strengths when manipulating $m_F < 0$ states. In an alternative scheme, the Raman laser is bichromatic, and drives alone $\delta m_F = 2$ two-photon spin-changing transitions using its two circular polarization components. The two optical frequencies are generated on the same AOM by a bichromatic RF drive, produced by frequency mixing.

At the end of an experiment cycle, we adiabatically ramp down the TLS beam and measure the spin-state distribution by spin-selective momentum transfer [24], simultaneously extracting and probing two spin states of choice ($m_F - 1$ and $m_F + 1$)

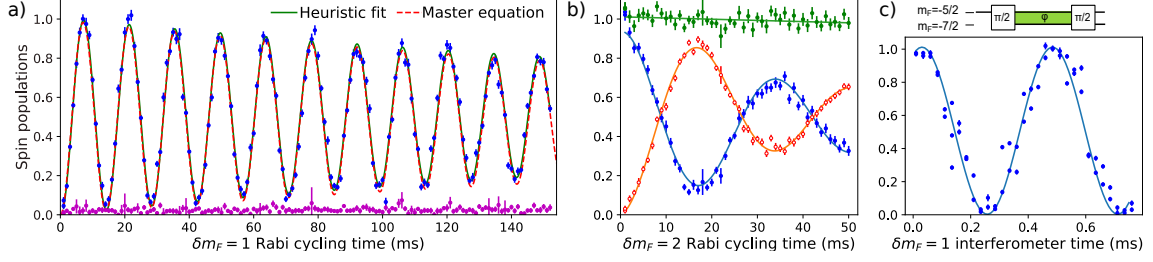


Figure 2.2: **Encoding a qubit on two nuclear spin states.** a) Rabi oscillations between $m_F = -3/2$ (blue) and $m_F = -5/2$ (not shown), driven by a $\delta m_F = 1$ Raman coupling. Here $\hbar\Omega_R/2q$ is small enough that the neighboring states $m_F = -7/2$ (magenta) and $m_F = -1/2$ (not shown) are unaffected. The green line is a heuristic fit to the data with a damped sine, yielding a $1/e$ damping time of 298 (17) ms. The dashed red line is the result of a master equation simulation with two adjustable parameters: the Raman coupling strength and the coherences decay rate. It is almost indistinguishable from the damped sine. (b) Rabi oscillations between $m_F = -3/2$ (blue, filled circles) and $m_F = -7/2$ (red, empty circles) driven by a $\delta m_F = 2$ Raman coupling. Green dots show the total population in these two states. Both populations are measured simultaneously. Solid lines are fits with damped sinusoidal oscillations, yielding a $1/e$ damping time of 40(5) ms. (c) Ramsey interferogram realized between states $m_F = -7/2$ (blue dots) and $m_F = -5/2$, driven by a $\delta m_F = 1$ Raman coupling. The line is an optimal distance regression fit with a sinusoidal interferogram. In all of this figure, each data results from a single shot, and error bars reflect the population measurement statistical uncertainty. Furthermore, in plots (a) and (b), the detection efficiencies $\eta(-3/2)$ are slightly recalibrated to bring maximal $m_F = -3/2$ population estimates down to 1.

for each measurement. Spin populations presented throughout the paper are relative to the total atom number. For this, we calibrate the detection efficiencies $\eta(m_F)$ for each spin state on clouds prepared in that single spin state. For $m_F = -7/2$, $-5/2$, and $-3/2$, we find $\eta(m_F) = 0.65$, 0.70 , and 0.51 , respectively^a. However, $\eta(-3/2)$ fluctuated throughout the experiments shown here, as evidenced by some population estimates larger than one. In such cases, we recalibrate $\eta(-3/2)$ by up to 6 % to bring extremal population estimates down to 1.

2.1.4 Coherent manipulation of a nuclear-spin qubit

Rabi oscillations between two isolated hyperfine states. We now turn to the experimental demonstration of coherent manipulations between selected pairs of states within the ^{87}Sr ground state manifold, which realizes evolutions driven by $\mathfrak{su}(N)$ generators σ_{m_F, m'_F}^x and σ_{m_F, m'_F}^z . We first show Rabi oscillations between two Zeeman sublevels, then characterize the coherence of quantum superpositions by Ramsey interferometry.

We first demonstrate Rabi oscillations with $\delta m_F = 1$. To this end, we prepare the gas in the $m_F = -5/2$ state and turn the quadratic light shift on $U_{TLS}(m_F) = qm_F^2$. For $\delta m_F = 1$ Raman transitions, this splits the resonance conditions of the various

^aThe detection efficiencies are probably limited by a lack of optical power and by using rather warm clouds with momentum spread comparable to the optical recoil.

Raman processes by multiples of $2q/h$ (h is the Planck constant). Consequently, as illustrated in Fig. 2.1, tuning the Raman beam frequency to one of the resonance conditions (e.g., the process $-5/2 \rightarrow -3/2$) produces a state evolution that can be approximated as generated by a Pauli operator coupling two and only two states. To reach this regime, the separation between adjacent resonances $2q/h$ (proportional to the TLS beam intensity I_{TLS}) must be larger than the two photon Raman coupling strength ($\Omega_R \propto \sqrt{I_{TLS}I_R}$, where I_R is the Raman beam intensity), which implies $I_R \ll I_{TLS}$. In practice we use $I_R/I_{TLS} \leq 10^{-4}$.

In the presence of the TLS beam, we pulse the Raman beam at a fixed frequency and monitor the spin state evolution. In Fig. 2.2a, we demonstrate the Rabi oscillation dynamics driven between the states $m_F = -5/2$ and $m_F = -3/2$. Here $q/h = -320$ Hz and $\Omega_R/2\pi = 71$ Hz. We observe well-pronounced coherent oscillations with a short-time contrast consistent with 1, while the population in the neighboring, not targeted state $-7/2$ grows by less than 1% over almost 11 Rabi oscillations. The amplitude of the oscillations exhibits a slow exponential decrease with a characteristic time $\tau_{decay} = 298(17)$ ms $\simeq 21 \frac{2\pi}{\Omega_R}$. As we do not observe significant population diffusion to other states, this damping rate limits the fidelity of a $\pi/2$ pulse to 0.994. The shot-to-shot fluctuations of the Rabi pulse area away from $\pi/2$ (calibrated from complementary experiments) are even less limiting to the fidelity (> 0.998). This demonstration of an evolution well approximated by $\exp(-i\theta\sigma_{m_F, m_F+1}^x/2)$ constitutes a key result of this article, enabling the various experiments described in the following sections.

We also demonstrate Raman transitions with $\delta m_F = 2$ using a slightly different scheme: we detune the Raman beam 1 MHz below the TLS beam and modulate its frequency in two sideband components (suppressing the carrier frequency). In this manner, its two opposed $\sigma^\pm$ polarization components drive $\delta m_F = 2$ transitions, with a detuning controlled by the modulation frequency. We performed this experiment at a much lower $q/h = -95$ Hz (because of the aging of a laser component). In Fig. 2.2b, we observe the Rabi oscillation induced between the states $-7/2$ and $-3/2$. The two spin populations, monitored in parallel, exhibit complementary oscillations at frequency $\Omega_R/2\pi = 29(3)$ Hz with a damping time of 40(5) ms. The total population in the two states decays by 3% within the damping time, so the damping mainly reflects a loss of coherence, not diffusion to other spin states. The limit imposed by this damping on the fidelity of a $\pi/2$ pulse can thus be estimated at 0.90, much lower than what we achieve on the $\delta m_F = 1$ Raman processes. The cause of this more significant damping compared to the $\delta m_F = 1$ coupling experiment is unknown - possibly the laser degrading, which also resulted in the lower value of q . Still, this experiment demonstrates a complementary set of two-level rotations. They are not required to have complete control over the atomic state, but, in principle, an extended set of generators of evolution can be advantageous for some manipulations (assuming similar fidelities). Furthermore, combining $\delta m_F = 1$ and $\delta m_F = 2$ transitions enables conceptually new kinds of experiments, such as engineering dynamics in a spin space with periodic boundary conditions, with phenomena related to Berry phases and non-Abelian gauge fields [73, 61]. In the remainder of the paper, we will use exclusively $\delta m_F = 1$ transitions as the basic building block for interferometers.

Coherence of the Rabi oscillations. Let us characterize the coherence properties of

the nuclear qubit - first in the presence, then in the absence, of the Raman driving field. In the Rabi oscillations presented in Fig. 2.2a, each point is the outcome of an individual realization. The observed damping is seen without averaging different experimental realizations, which implies that it is associated with loss of coherence rather than fluctuations. This decoherence may result from a spatial inhomogeneity or from spontaneous emission. To establish which one dominates, we measure the heating rate of atoms exposed to TLS light in the dipole trap. We find it to be three times higher than the expected heating rate associated with spontaneous emission assuming the laser is monochromatic. We attribute this to a frequency noise pedestal in our laser spectrum due to amplified spontaneous emission (ASE) that causes *resonant* photon scattering. We then model the Rabi dynamics by a master equation describing the evolution of an atom in the laser and magnetic fields. We calculate the separate spontaneous emission terms from and to each spin state, such that the model captures decoherence and spin-state diffusion. To heuristically capture the effect of ASE, we scale these rates of coherence decay and spin diffusion by a factor of three. Typical coherence decay rates are $\sim 1 \text{ s}^{-1}$, and population transfer rates $\sim 0.5 \text{ s}^{-1}$. However, the simulation with these parameters does not produce the evolution observed in Fig. 2.2a, nor does artificially scaling the impact of spontaneous emission by a different factor. Instead, we note that there is TLS/Raman beam inhomogeneity, and empirically add a decay of coherences between spin states (m_1, m_2) with rate $\Gamma_q \times |m_1^2 - m_2^2|$. With $\Gamma_q = 1.2 \text{ s}^{-1}$, we recover excellent agreement with the data, as shown in Fig. 2.2a. We estimate independently, by simulations with a random distribution of (q, Ω_R) , that the observed dynamics is consistent with an inhomogeneity of q and Ω_R of about 1.2 % in standard deviation. Across a $5 \mu\text{m}$ -radius cloud, this can be induced by a beam off-centered by $w/4$, where $w = 320 \mu\text{m}$ is the waist.

Coherence of the hyperfine qubit Ramsey interferometer. We then test the coherence of the qubit in the absence of Raman drive. For this, we implement an interferometer. Keeping the TLS beam on, we apply a set of two resonant $\pi/2$ pulses, separated by a temporal interval T - thus realizing a standard Ramsey sequence restricted to the Hilbert space of a qubit, see the top sequence schematic of Fig. 2.1b. During this time interval T , we alter the local oscillator frequency that controls the phase difference between TLS and Raman beams. The Ramsey interferometer is sensitive to the phase ϕ defined by $\frac{d\phi}{dt} = \Delta E_{m_F, m'_F}(t)/\hbar - \delta_{RF}(t)$, where $\Delta E_{m_F, m'_F}$ is the energy difference between the two qubit states and $\delta_{RF}(t) = \omega_R - \omega_{TLS}$. The quantities $\omega_{TLS}/2\pi$ and $\omega_R/2\pi$ are the (potentially time-dependent) TLS and Raman beam laser frequencies.

Figure 2.2c shows the fringe pattern as a function of T for qubits encoded in the two nuclear-spin states $m_F = -5/2$ and $m_F = -7/2$. Here, $q/\hbar = 190 \text{ Hz}$ and $\Omega_R/2\pi = 93 \text{ Hz}$. The well-resolved oscillations come from the accumulated differential phase in the qubit states, resulting in a rotation (controlled by T) along the sub-Bloch sphere equator. We observe fully contrasted oscillations for the best data, which demonstrates that the coherence of the qubit is well preserved.

We now systematically study the contrast and noise of Ramsey interferometers as a function of T to characterize the sources of decoherence. We will highlight that the main detriment to the qubit's coherence is the TLS beam and that we can adiabatically remove it to recover coherence over seconds. We present this study in

Fig. 2.3. We focus on two effects: decoherence, which reduces the interferometric contrast, and phase fluctuations. In principle, the sources of fluctuations in the data are (i) atomic quantum projection noise; (ii) measurement uncertainties associated with fitting the spin-separated optical density pictures; (iii) reproducibility of the $\pi/2$ pulses; iv) fluctuations of the phase integrated by the interferometers. For our data, the quantum projection noise (i) is smaller than the measurement noise (ii) by typically a factor of 3, so we neglect it. Fluctuations of the area of “ $\pi/2$ ” pulses (iii) are estimated from repeating a sequence without closing the interferometer. We find a standard deviation of the pulse area of 0.063(5) rad (which sets a fidelity upper bound for a $\pi/2$ pulse at 0.999). However, their effect in a *closed* Ramsey sequence remains smaller than the measurement uncertainty. On the interferogram shown in Fig. 2.3b, we notice that the residuals to a sinusoidal model are much larger than the fluctuations expected from combining the effect of (i), (ii), and (iii). We attribute this to uncontrolled fluctuations of the interferometer phase. To estimate their RMS amplitude, we numerically generate trial noisy models of the experiment, including all sources of fluctuations, and compare them with the data^b. In this optimization, the interferometric contrast and the variance of phase fluctuations are the adjustable parameters.

Fig. 2.3c and d show the outcome of this analysis. As shown in Fig. 2.3c, the contrast (defined as peak-to-peak signal amplitude) decays as a function of T with a $1/e$ decay constant of 54(5) ms. Like the damping of Rabi oscillation, this decay most likely results from the TLS inhomogeneity rather than photon scattering. The observed decay is significantly faster than that of the Rabi oscillations seen in Fig. 2.2a. This illustrates the sensitivity of Ramsey interferometers, which are directly sensitive to detunings Δ , in contrast to generalized Rabi frequencies, generically $\sqrt{\Omega_R^2 + \Delta^2}$.

A second observation is that a random phase is integrated for each realization of the interferometer. At the smaller dark time $T = 5$ ms, its standard deviation is about 0.35 rad and gradually increases with T , as shown in Fig. 2.3d. It is about 1.4 rad after 100 ms. We attribute this to polarization fluctuations in the TLS beam, as we will demonstrate in Sec. 2.1.5.1. One could reduce these fluctuations by polarization filtering, although our setup, where we co-propagate the TLS and Raman beams in the same optical fiber, is inadequate for this.

Thus, simply removing the tensor light shift for most of the duration T of the interferometer should significantly mitigate decoherence and phase noise. After the initial Rabi pulse, we adiabatically turn off the TLS in 2 ms, and bring it back on in the last 2 ms before the second $\pi/2$ pulse. The atom’s environment during the interferometer is now primarily the magnetic field and a dipole trap with negligible spin-dependent light shift. The observed decoherence is considerably suppressed, with no visible contrast decay after 3 seconds. The phase noise also grows with T at a much slower rate. Phase fluctuations of $\simeq 0.3$ rad are acquired during the 4 ms with TLS, but the phase-noise contribution of the remaining time without light only reaches 1 radian after about 2 seconds. Although the RMS phase fluctuations are not exactly linear with T , they can qualitatively be interpreted as an insta-

^bWe look for a trial model for which a least-square sinusoidal fit estimates the same amplitude and with the same residuals as the data. Note that the peak-to-peak amplitude of the sinusoidal fit is generally smaller than the contrast of the interferogram: an interferogram with complete loss of control over the phase will be fitted with zero amplitude and large residuals.

bility of the Raman detuning of ~ 0.08 Hz when integrated (averaged) over two seconds. The frequency instability of the DDS-controlled $\delta_{RF} = \omega_R - \omega_{TLS}$ is, in principle, one order of magnitude smaller. Thus, the phase noise probably reflects magnetic field fluctuations of about 0.4 mG at the second timescale, consistent with the specifications of the current supplies for the coils.

In this section, we have demonstrated coherent control of a qubit defined over two selected Zeeman sublevels using σ_{m_F, m_F+1}^x , σ_{m_F, m_F+2}^x , and $\sigma_{m_F, m_F'}^z$ rotations. Our demonstrations have been restricted to the manifold $m_F \in \{-9/2, -7/2, -5/2, -3/2\}$. Note that, when driving atoms in other states, we observed uncontrolled population transfers. We attribute this to the value of our magnetic field $b \simeq -3q$, which causes quasi-degeneracy and sign reversal of the Raman energy differences, see Fig. 2.1a. One could lift this issue and acquire complete control over the entire spin manifold, provided $b > |q|(2F - 1)$. Within the manifold of states used in this paper, we have shown the high sensitivity of the qubits' coherence and phases to the TLS beam fluctuations. Minimizing the exposure time to the TLS beam, we have demonstrated long-lived coherent superpositions over several seconds, without any magnetic shielding. Such coherence time is a feature of the minute Landé factor of the nuclear spins, basically immune to stray magnetic field gradients ($g_I \mu_0 \nabla B < 10^{-4}$ Hz/ μ m), and of the extremely small vector and tensor polarisabilities in the closed-shell ground state, at the optical trapping wavelength (with spin-dependent shifts $\simeq 10^{-4}$ Hz). It can be a great asset, e.g., for high-precision sensing in long-interrogation-time interferometers or quantum information storage.

2.1.5 Beyond pseudo-spin 1/2 schemes: qudit interferometry

The possibility to coherently couple two spin states among the $N = 10$ available ones in ground-state ^{87}Sr atoms enables coherent navigation over the whole ground-state manifold - for example, by simply concatenating a series of resonant pulses between isolated pairs of Zeeman states. Consequently, our experimental toolkit creates new possibilities that use more than two spin states in order to store or retrieve information. In this section, we experimentally investigate two of those new possibilities that use the large nuclear spin as a genuine qudit.

First, we will use well-defined linear-superposition states involving four spin states, in order to simultaneously measure more than one external field at each experimental realization. Such multi-parameter estimates are useful in the context of metrology [74]. Our work thus illustrates the potential of large-spin particles as original quantum sensors.

Second, we present a scheme that simultaneously measures collective observables that do not commute, in a single experimental realization. Building on Ref. [75], our scheme can characterize a many-body state where the coherences are stored in a subset of the fully available ground-state hyperfine manifold. The main idea is to extend the Hilbert space of the atoms for measurement purposes only; the measurement sequence involves mapping the state of the atoms into initially empty “ancillary” states and applying controlled rotations, which provides additional observables, and facilitates tomographic characterization of a quantum many-body state.

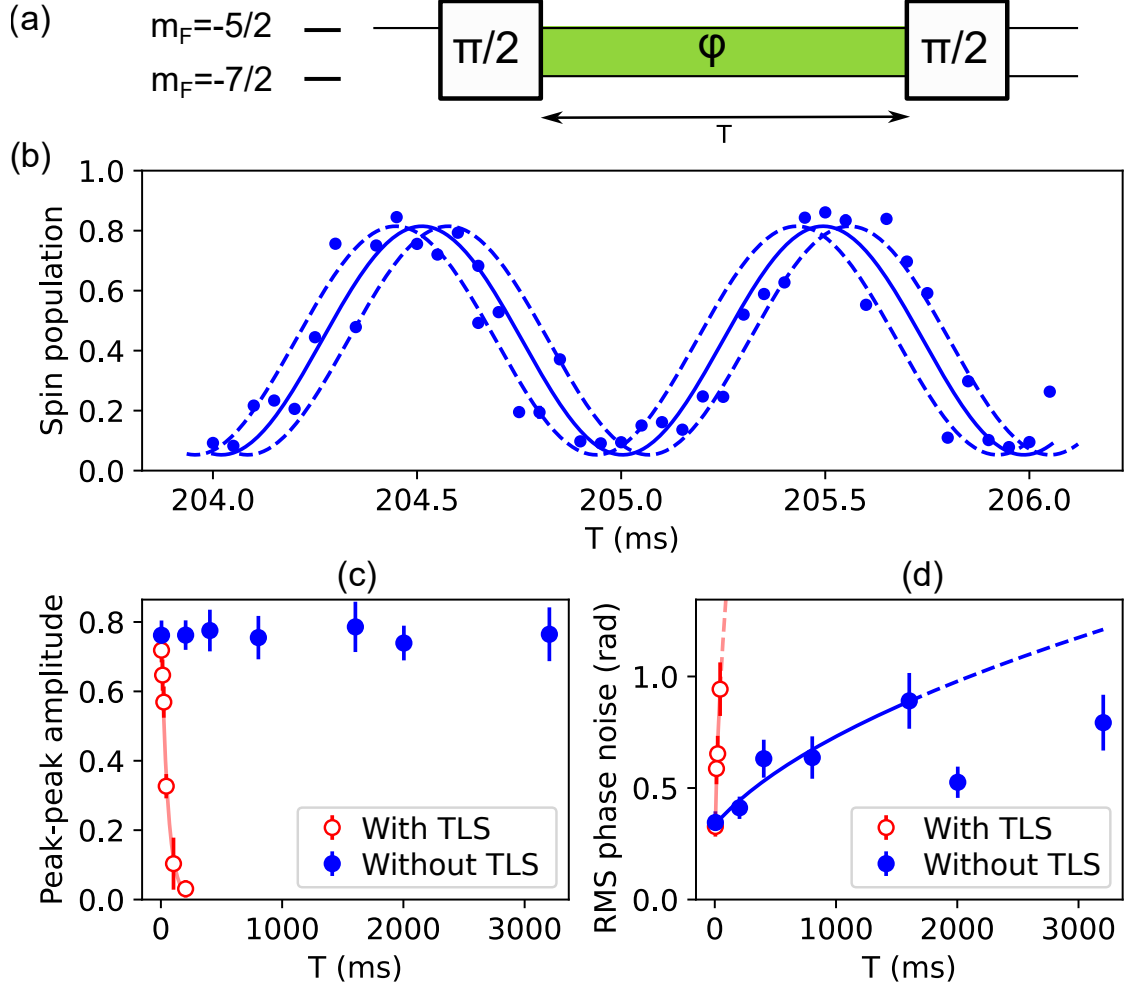


Figure 2.3: **Qubit coherence.** (a) Schematics of the interferometer. (b) Fringe pattern of a Ramsey interferometer operated between $m_F = -5/2$ (measured) and $m_F = -7/2$, with, between the two pulses, 4 ms of exposition to TLS light and $T-4$ ms of absence of any laser light. Error bars, combining population measurement uncertainty and expected impact of pulse area fluctuations, are comparable to the dots' size. The solid line is the underlying interferogram, barring phase fluctuations. The dashed lines interferograms are shifted by $\pm$ one standard deviation of the phase fluctuations. (c) Contrast as a function of T . The red line is a fit to an exponential decay for the data with TLS beam on (red, empty circles), with $1/e$ decay time of 54(5) ms. When the TLS beam is adiabatically turned off during the dark time (blue, filled circles), there is no observable contrast decay for at least 3 s. The shortest measured T (5 ms) corresponds to ramping down and almost immediately back up the TLS. (d) Phase noise of the interferometer as a function of T . Continuous lines are fits assuming $Var(\phi) = \delta\phi_0^2 + Dt$. We have $\sqrt{D} = 0.14(1)$ rad/ms $^{1/2}$ with TLS, and $\sqrt{D} = 0.021(2)$ rad/ms $^{1/2}$ without TLS.

2.1.5.1 Parallel Ramsey interferometers for simultaneous multiple-field sensing.

Performing simultaneous measurements using the same local quantum sensor is a new possibility with large-spin particles that is particularly interesting, for example, because the measurements can then benefit from common noise rejection (see

for example [76, 77, 78, 79]) or because it becomes possible to perform correlations analyses and thus learn from sources of noise (we will show an example in the following). Here, we present simultaneous measurements of two fields in each experimental realization. We drive two $s = 1/2$ Ramsey interferometers in parallel, using two independent sets of pairs of Zeeman states within strontium's ground-state hyperfine manifold. Our scheme using sets of qubits *internally distributed* within the ground-state spin manifold has similarities with the architecture of *spatially distributed* interferometers studied in Ref. [80].

The protocol is shown in Fig. 2.4a. From an initially pure spin state in $m_F = -5/2$ we apply a first $\pi/2$ pulse that creates a coherent superposition equally distributed over $\{-7/2, -5/2\}$ [81]. Next, each state enters the respective input port of two independent Ramsey interferometers, one engineered between $\{-5/2, -3/2\}$ extracting a phase ϕ_1 , the other between $\{-7/2, -9/2\}$ extracting a phase ϕ_2 . The experiment relies on a single RF local oscillator that controls the Raman detuning δ_{RF} . The two interferometers are thus opened sequentially at times t_1 and t_2 by switching the Raman detuning from one resonance to the other, after which they accumulate phase for a time together, then are finally closed sequentially again. Both interferometers are open for the same duration T . The first interferometer's populations are thus sensitive to the phase

$$\begin{aligned}\phi_1 &= \frac{1}{\hbar} \int_{t_1}^{t_1+T} (E_{-5/2} - E_{-3/2}) dt - \int_{t_1}^{t_1+T} \delta_{RF}(t) dt \\ &= \frac{T}{\hbar} (4q - b) - T \langle \delta_{RF} \rangle_1,\end{aligned}\tag{2.1}$$

and the second interferometer's populations, to

$$\begin{aligned}\phi_2 &= \frac{1}{\hbar} \int_{t_2}^{t_2+T} (E_{-9/2} - E_{-7/2}) dt - \int_{t_2}^{t_2+T} \delta_{RF}(t) dt \\ &= \frac{T}{\hbar} (8q - b) - T \langle \delta_{RF} \rangle_2.\end{aligned}\tag{2.2}$$

Note that $\langle \delta_{RF} \rangle_1 \neq \langle \delta_{RF} \rangle_2$. The parallel operation of the Ramsey interferometers provides a measurement of both terms contributing to the single-atom ground-state energies, that are, the linear energy splitting b (due to both the magnetic Zeeman effect and the vector light shift) and the quadratic energy splitting q (due to the tensor light shift). As $\langle \delta_{RF} \rangle_1$ and $\langle \delta_{RF} \rangle_2$ are controlled (here with an accuracy below 0.02 Hz), the difference between the two interferometers' phases $\phi_1 - \phi_2$ provides a measurement of q , which is insensitive to shot-to-shot fluctuations in b , while $2\phi_1 - \phi_2$ provides a measurement of b , which is insensitive to shot-to-shot fluctuations in q ^c.

In Fig. 2.4b, we present the interferograms obtained by measuring the variation of the populations in $m_F = -7/2$ and $m_F = -3/2$, relative to the total atom number, as a function of the interferometer time T . Here, $q/\hbar \simeq -320$ Hz (independently deduced by spectroscopy) and $\Omega_R/2\pi = 77$ Hz on the $(-5/2 \rightarrow -7/2)$ line. Both populations are measured simultaneously at each experimental realization. During

^cTechnically, q and b can also be time-varying during the interferometer sequence, and the interferometers are sensitive to their time-averages. Fluctuations of (q, b) at the timescale of T will be properly retrieved by the combined interferometers provided $T \gg 1/\Omega_R$, or if an additional laser frequency is used to engineer the Raman couplings simultaneously.

the time when both interferometers are simultaneously opened ($\simeq T - 3.5$ ms), we set δ_{RF} to $2\pi \times 1$ Hz, off-resonant with both transitions. We observe that the periods of the two fringe patterns strongly differ, as $E_{-5/2} - E_{-3/2} \neq E_{-9/2} - E_{-7/2}$. From the two periods, we can deduce the mean quadratic and linear Zeeman splittings: $\langle q \rangle/h = -303(8)$ Hz and $\langle b \rangle/h = 1000(45)$ Hz - in agreement with independent estimates from Raman spectra. These are mean values over all realizations, as q, b can fluctuate from shot to shot. Upon knowing the fringe contrast and the two phases ϕ_1, ϕ_2 with better than π accuracy, single realizations of the experiment provide parallel individual simultaneous measurements of ϕ_1 and ϕ_2 , from which we can deduce single-shot measurements of q and b . Given the interrogation time $T \simeq 4$ ms and noise level of these data (affected by Rabi pulse area fluctuations, mostly of the first pulse), the respective precisions in q, b from individual points when at maximal slopes are $h \times 1.8$ Hz and $h \times 11$ Hz.

To illustrate this, we operate the interferometer at a fixed T chosen to be close to the maximum phase sensitivity of the interferometer (*i.e.*, close to mid-fringe for both signals). Fig. 2.4c shows a set of 93 measurements of the two phases (ϕ_1, ϕ_2) taken consecutively. For each measurement, we directly reconstruct the deviations of b and q relative to the mean, as shown in Fig. 2.4d. These data demonstrate that the simultaneous nature of the measurement makes possible correlation analyses that would otherwise be impossible if the experiment were alternating between one interferometer and the other.

As an example of correlation analysis, we have used this data as follows. We observe that the parallel measurements of (q, b) do not seem 2D-Gaussian distributed. Instead, two classes of shots are identifiable. The two groups' separation in $\langle q \rangle$ is $-1.4(5)$ Hz, *i.e.*, a small fraction (0.4%) of $\langle q \rangle$ and below the spread within a single group. In contrast, the separation in $\langle b \rangle$ of $23(3)$ Hz is significant and higher than the spread in a single group. We attribute this to an evolution of the TLS beam polarization that shifts between two configurations; and thus induces a fluctuating vector light shift with values clustering in two groups, while there should be no vector light shift for the ideal π polarization. By protecting the TLS optical fiber from airflow, only one group remains, and the fluctuations become consistent with measurement uncertainties (also impacted by the first pulse's area fluctuation). We acquire most of the data in this paper (including Fig. 2.4b) with fiber protection.

We now discuss the fundamental limit to the sensitivity of the measurements. In the experiments presented here, the dominant uncertainties on b and q arise from Rabi-pulse-area fluctuations and the atom number measurement precision. Nevertheless, the atomic quantum projection noise sets the fundamental limit to the achievable precision of measurements. A key question is what the tradeoff for parallelizing two measurements is. Each one uses, on average, half of the total atom number. In addition, we present in this paper a suboptimal measurement, as we measure only one output port of each interferometer at each realization. Not knowing the exact atom number fed into each interferometer in each realization increases the quantum-projection-noise limit for phase estimation by $\sqrt{3}/2$. However, in principle, we could measure all four states in each realization [82, 83, 24]. The fluctuations in the measurement of the two phases associated with the atomic quantum-projection noise are simply $\sqrt{2}$ times stronger than when using all atoms on the same interferometer. This mild disadvantage vanishes if one considers that at least two runs of the

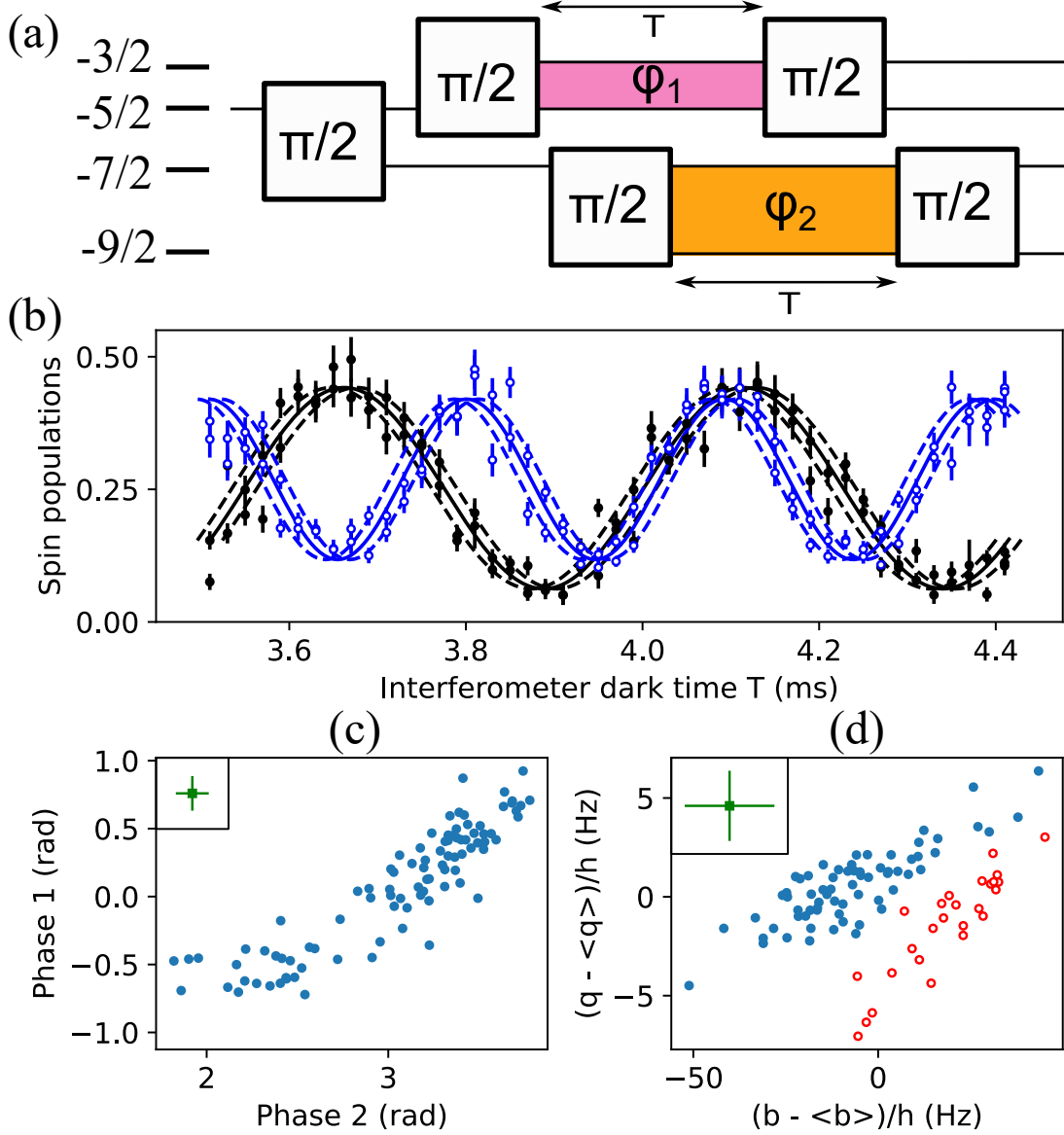


Figure 2.4: **Simultaneous sensing of two external fields.** a) Double interferometer pulse scheme. b) Measured interferogram. We vary the phases ϕ_1 and ϕ_2 together by tuning the interferometers' phase accumulation times T . We measure both states $|-3/2\rangle$ (black, full circles) and $|-7/2\rangle$ (blue, empty circles) in each realization. The differing periods highlight that two different energies are measured. The error bars combine population measurement uncertainties and the expected impact of Rabi pulse area fluctuations. The solid line is the underlying sinusoidal interferogram, barring phase fluctuations. The dashed-line interferograms represent shifts of $\pm$ one standard deviation of the phase fluctuations. c) Correlation analysis of the two phases ϕ_1 and ϕ_2 , measured modulo 2π by repeating experiments at a chosen phase with the highest slope for both interferometric signals. We show the median of the error bars (including measurement uncertainty and pulse area fluctuations) in the top-left corner. d) Reconstruction of the two fields (b, q) . The data split into two groups, identified by color and predominantly separated in b (the split corresponds to $\phi_2 \leq 2.8$ rad). We cancel this experimental fluctuation for most data in the paper, including Fig. 2.4b, by protecting the TLS optical fiber from airflow.

experiment would be required to measure sequentially the two interferometers.

We finally discuss the three main systematic effects that could affect the phase measurements. First, the pulses of the two interferometers are interleaved temporally. Consequently, between the splitting and recombining pulses of the interferometer (1), we realize one Raman pulse addressing interferometer (2), and vice-versa. This interleaving creates AC Stark shifts $\simeq \Omega_R^2/8q$ on the two levels probed by interferometer (1) of the same sign but not exactly compensating due to different Clebsch-Gordan coefficients in the Raman couplings. These shifts can be calculated, or minimized by ensuring the highest possible energy scale separation $2q/\Omega_R$. Second, when the Rabi frequencies are only slightly smaller than the energy difference between states, one should account for effects beyond the rotating-wave approximation, such as the Bloch-Siegert shift, and expect the interferometer to be sensitive to the initial temporal phase of the beat between the two Raman beams [84]. Third, in a scheme spanning many pulses and Zeeman states, off-resonant transfers between spin states can interfere with the resonantly driven population transfers. We will see in the next section evidence for this. Effectively, this affects the fringe patterns and phase estimation.

We have shown in this section an example of a sensor that parallelizes two measurements using an extended Hilbert space basis. Thanks to the simultaneity of the measurements, we have demonstrated that correlation analyses are possible. We illustrated it by tracking in parallel variations of the vector and tensor light shifts acting on the atoms. One can also interpret these experiments as a common-mode noise-rejection scheme, i.e., extracting precise values of q despite the fact that fluctuations of b affect all interferometers. Similar schemes for the simultaneous sensing of multiple fields could be devised, e.g., for measuring multiple spatial components of a vector (magnetic) field.

2.1.5.2 Simultaneous measurements of multiple collective atomic observables.

We will now explore a second possibility offered by the large spin-state manifold: increasing the number of observables accessible in one realization and characterizing the collective spin state of a quantum many-body ensemble.

Let us start by remarking that tracking two spatial components of a magnetic field by monitoring spin precession requires measuring the projection of the collective spin along at least two directions simultaneously. However, this is typically not possible due to the non-commutativity of the two spin-projection observables. More fundamentally, fluctuations of a collective spin in the two directions transverse to its mean orientation are critical to the precision of metrological applications [85, 86] but remain assessed along one given quantization axis at a time. Furthermore, measuring the collective spin properties enables tests of a large class of entanglement witnesses [87, 88, 89, 90]. We also point out that large-spin atoms are compelling due to numerous new scenarios involving entanglement witnesses [91, 92]. For large-spin atoms, only measuring the total population in each spin state can already reveal beyond mean-field dynamics [34, 93], and measuring collective spin fluctuations in a single axis can already be used to characterize correlations [40, 94]. Here, we will show that ensembles of large-spin atoms offer an opportunity for physics

with ensembles of effectively smaller pseudo-spins: measuring two orthogonal spin-projection operators - or generically, two non-commuting observables - in one given experimental realization.

The general idea is to use a larger space of states for each atom in the measurement sequence than during the experiment that came before it, so that a higher number of observables is available. These observables (e.g., populations) in the large Hilbert space commute with each other and can be simultaneously measured; but their expectation values and even higher order statistics reflect those of non-commuting observables of the state produced by the experiment.

One such scheme emerges if one can coherently map the internal states involved in the experiment into other, initially empty, ancillary states used for measurement purposes only. Reference [75] demonstrated this idea, using the complete hyperfine structure of alkali atoms to probe spinor properties in the $F = 1$ manifold. We generalize this approach to the case of ground-state alkaline-earth atoms. In practice, we consider ensembles of effective qubits, i.e., pseudo-spins $1/2$, initially restricted to two Zeeman sublevels. To simultaneously measure two pseudo-spin projections, we first map each state of this qubit into ancillary spin states accessible in the $F = 9/2$ manifold, using a partial coherent transfer. Then, using again the capacity to perform manipulations targeted on specific pairs of states (i.e., rotations on a sub-Bloch-sphere), we perform different basis transformations on the qubit states than on the ancillary states. The final measurement of populations thus contains information formerly inaccessible in the initial populations of the qubit states. We show a typical sequence in Fig. 2.5a.

For our experiments, we define the two qubit states, $|\uparrow\rangle = |-5/2\rangle$ and $|\downarrow\rangle = |-7/2\rangle$. The initial state, which we produce to test the measurement protocol, is a coherent state of qubits $|\psi\rangle^{(N_{at})} = ((|\uparrow\rangle - i|\downarrow\rangle)/\sqrt{2})^{\otimes N_{at}}$, where N_{at} is the atom number. We produce it from a sample polarized in $|\uparrow\rangle$, by one $\pi/2$ Raman pulse between the qubit states. With our phase convention, this coherent state is polarized against the pseudo-spin quantization axis “y”, i.e., the total pseudo-spin of the ensemble has projections $\langle \vec{s}_{\uparrow,\downarrow} \rangle = (0, -N_{at}/2, 0)$.

The measurement sequence then begins. First, two successive $\pi/2$ pulses duplicate the $|\uparrow\rangle$ amplitude to the ancillary state $|a\rangle = |-3/2\rangle$ and the $|\downarrow\rangle$ amplitude to the ancillary state $|b\rangle = |-9/2\rangle$, respectively. Second, we let the phases of all four states evolve in the presence of the TLS and magnetic field. In particular, the qubit sub-Bloch sphere effectively evolves by $\exp(-i\phi\sigma_{\uparrow,\downarrow}^z/2)$, where the angle ϕ is controlled by the duration of the rotation and by the detuning between the RF clock and the qubit transition. Third, we apply a final $\pi/2$ pulse between the two qubit states, corresponding to applying $\exp(-i(\pi/2)\sigma_{\uparrow,\downarrow}^x/2)$. At the end of the sequence, one would ideally measure the populations N_i of all four states. The population difference $N_a - N_b$ provides an estimator for the many-body operator $\hat{s}_{\uparrow,\downarrow}^z$. Similarly, $N_{\uparrow} - N_{\downarrow}$ provides an estimator for $\hat{s}_{\uparrow,\downarrow}^\phi = \cos(\phi)\hat{s}_{\uparrow,\downarrow}^y + \sin(\phi)\hat{s}_{\uparrow,\downarrow}^x$, the total pseudo-spin projection in one selected direction of the plane (x, y) . Both quantities are thus accessible in a single experimental run.

Let us describe more formally the measurement, which will enable us to discuss its statistics. We label U the one-body evolution operator corresponding to the series of three $\pi/2$ pulses and phase evolution within the measurement sequence, outlined

in Fig. 2.5, with $\phi = 0$ for simplicity. We apply the same evolution to all spins in parallel. Thus, the many-body evolution operator is factorizable onto all the modes of the atomic external degrees of freedom λ : $U^{(N)} = \Pi_\lambda U_\lambda$. We define two projective measurement operators,

$$\hat{O}^z = U^{(N)\dagger}(N_a - N_b)U^{(N)}, \text{ and} \quad (2.3)$$

$$\hat{O}^y = U^{(N)\dagger}(N_\uparrow - N_\downarrow)U^{(N)}. \quad (2.4)$$

They correspond to measuring all four atomic populations ($N_a, N_b, N_\uparrow, N_\downarrow$) after the collective evolution described by $U^{(N)}$ and computing the appropriate number differences. Assuming that before the measurement sequence, the only populated states are those of the qubit ($\uparrow, \downarrow$), one can easily show:

$$\langle \hat{O}^z \rangle = \langle \hat{s}_{\uparrow, \downarrow}^z \rangle, \langle \hat{O}^y \rangle = \langle \hat{s}_{\uparrow, \downarrow}^y \rangle, \quad (2.5)$$

$$\hat{O}^z \hat{O}^y = \hat{O}^y \hat{O}^z. \quad (2.6)$$

Note that $\hat{O}^z$ and $\hat{O}^y$ do commute with each other, which highlights the simultaneous nature of the two measurements. Thus, measuring the four populations after the outlined sequence allows us to estimate the expectation values $\langle \hat{s}_{\uparrow, \downarrow}^y \rangle$ and $\langle \hat{s}_{\uparrow, \downarrow}^z \rangle$ in a single physical realization. Furthermore, we have shown using second quantization formalism that

$$\text{Var}(\hat{O}^{\{y,z\}}) = \text{Var}(\hat{s}_{\uparrow, \downarrow}^{\{y,z\}}) + N_{at}/4, \quad (2.7)$$

provided that only the qubit levels ($\uparrow, \downarrow$) are populated before the measurement operations described by $U^{(N)}$. It is remarkable that the scheme only introduces an additive fluctuation term $N_{at}/4$, which could then be subtracted in quadrature to estimate $\text{Var}(\hat{s}_{\uparrow, \downarrow}^{\{y,z\}})$ from the measurement of $\text{Var}(\hat{O}^{\{y,z\}})$. Note that $N_{at}/4$ corresponds to the transverse spin variance of a coherent state.

Figure 2.5b shows the measured interferograms. Here, $q/h = -330$ Hz and $\Omega_R/2\pi = 76$ Hz on the $(-5/2 \rightarrow -7/2)$ line. We control the phase ϕ by varying the Raman detuning during a fixed span of 0.51 ms just before the last Rabi pulse. Ideally, one would simultaneously measure the populations in all four states [82, 83]. In our setup, only two spin states, $m_F = -7/2$ ($\downarrow$) and $m_F = -3/2$ (a), can be measured in the same experimental run, as it is the easiest for our spin-population measurement protocol [24]. We measure the spin state $m_F = -5/2$ ($\uparrow$) separately, whereas, in the current implementation, the Clebsch-Gordan coefficient is too weak to efficiently measure the state $m_F = -9/2$. Since we operate in the regime where the Rabi pulses are well resolved, we can consider that $N_{-9/2} + N_{-3/2} = N_{at}/2$ and $N_{-7/2} + N_{-5/2} = N_{at}/2$ and thus relate each population to a spin observable,

$$\langle \hat{s}_{\uparrow, \downarrow}^z \rangle = \langle U^{(N)\dagger} (2N_{-3/2} - N_{at}/2) U^{(N)} \rangle, \quad (2.8)$$

$$\langle \hat{s}_{\uparrow, \downarrow}^\phi \rangle = -\langle U^{(N)\dagger} (2N_{-7/2} - N_{at}/2) U^{(N)} \rangle. \quad (2.9)$$

However, the fluctuation of these observables will be higher than those of $\hat{O}^z$ and $\hat{O}^\phi$, which are more optimal if all four states can be measured simultaneously.

The striking feature of Fig. 2.5b is that the populations in states $m_F = -7/2$ ($\downarrow$) and $-5/2$ ($\uparrow$) show pronounced oscillations, while the population in state $m_F =$

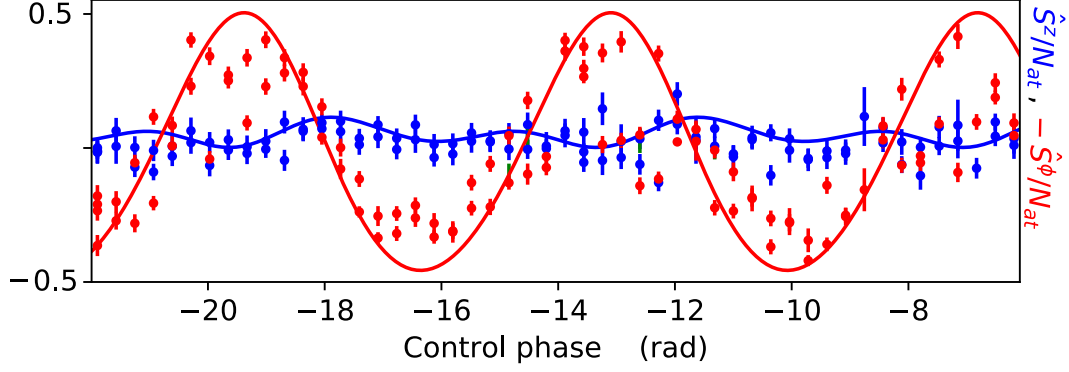


Figure 2.5: **Measuring two non-commuting observables in a single realization** a) Pulse scheme, highlighting one stage for preparing a well-defined state, and one measurement stage that includes unitary operations before the detection of populations. In frame (b), we monitor, as a function of the control phase, the fractional populations $N_{(-7/2, -5/2, -3/2)}/N_{at}$, in red, green, and blue, respectively. Error bars reflect only the population measurement uncertainty. Solid lines are the result of simulations of the master equation. We simultaneously acquire $N_{-7/2}/N_{at}$ and $N_{-3/2}/N_{at}$ during the same runs. From $N_{-3/2}$, we extract an estimate of the projection $\langle \hat{s}_{\uparrow, \downarrow}^z \rangle$ of qubit pseudo-spin, while from $N_{-7/2}$, we extract an estimate of $\langle \hat{s}_{\uparrow, \downarrow}^\phi \rangle$, all during the same runs of the experiment. We acquire the population $N_{-5/2}/N_{at}$ by independent runs. The solid lines result from a master equation simulation with a single adjustable parameter: an 18 Hz correction on the linear Zeeman splitting. (c) and (d) present the dependence of $\langle \hat{O}_z \rangle$ on the energy scale separation $2|q|/\hbar\Omega_R$ for Rabi pulses with a simple square temporal envelope, obtained from numerical simulations. We consider a coherent state with $\langle \hat{s}_{\uparrow, \downarrow}^z \rangle = 0$. We take the maximal, minimal, and mean values of $\hat{O}_z$ over a 2π variation of the control phase. We operate our experiment at $2|q|/\hbar\Omega_R \simeq 9$, providing the best balance between technically-limited decoherence and scheme fidelity.

$-3/2$ (a) barely evolves. The oscillations reflect that we effectively change the spin measurement basis with the interferometric phase. The approximately flat behavior of $N_{-3/2}$ demonstrates the capacity to rotate the spin in the sub-Bloch sphere set by the states $|-7/2\rangle$ and $|-5/2\rangle$ without significantly modifying the population $N_{-3/2}$. We still observe a weak dependence of $N_{-3/2}$ on the control phase ϕ . This signal is a signature of population transfers driven off-resonantly by the Raman couplings, which will be finite as long as $|\hbar\Omega_R/2q| > 0$. Interestingly, we did not see a signature of off-resonant transfers in the experiments restricted to a single pair of states. Indeed, in more complex pulse schemes, “leaked” atomic amplitudes have the opportunity to interfere with macroscopic amplitudes that are resonantly driven, which exacerbates their impact on populations.

The simple approach to tackle these parasitic population transfers is to increase the energy scale separation, $|2q/\hbar\Omega_R|$. We show in Fig. 2.5c,d how the stray signals fall with reduced Rabi couplings, using numerical simulations of the master equation. Here, the fundamental source of dissipation (*i.e.*, spontaneous emission due to a perfectly monochromatic laser) is included. The studied quantity is $\langle\hat{O}_z\rangle/N_{at}$ for a coherent state with $\langle\hat{s}_{\uparrow,\downarrow}^z\rangle = 0$. Provided the phase control is achieved in a time shorter than the Raman pulses, this quantity depends on the ratio $2|q|/\hbar\Omega_R$, irrespective of the values of q and Ω_R . We see that the ϕ -dependence and mean value of $\langle\hat{O}_z\rangle/N_{at}$ both slowly approach the quantum-projection-noise limit ($\sqrt{1/2N_{at}} = 0.007$ for 10^4 atoms) when increasing $2|q|/\hbar\Omega_R$. However, at values $2|q|/\hbar\Omega_R > 10^2$, spontaneous emission introduces spin diffusion and degrades the scheme. Technical imperfections will bring the optimum to faster pulses, *i.e.*, lower values of $2|q|/\hbar\Omega_R$.

To push this limitation, a most promising avenue is the temporal shaping of the Rabi pulses. By deviating from the square amplitude envelopes used here, even with simple shapes [95], one can drastically reduce off-resonant population transfers for a given Raman coupling strength, and thus at constant overall scheme duration. More dramatically, Ref. [69] demonstrates numerically that optimal control methods would enable performing arbitrary unitary $\mathfrak{su}(10)$ operations in a time $\sim 30\hbar/q$. This is about the same duration as the experiments that we present here, and relies on high inter-state couplings: $2q/\hbar\Omega_R \lesssim 1$. One could also use the ultra-narrow transition $^1S_0 \rightarrow ^3P_2$ [96] to engineer the tensor light shift with dramatically reduced spontaneous emission.

Thus, expanding the internal Hilbert space in the measurement sequence enables the parallel estimates of several atomic observables. This follows the idea exposed in [97] for implementing positive-operator-valued measures (POVMs), and the literature on POVMs, *e.g.* to estimate measurement fluctuations [98], applies. Here, we have studied the case where two orthogonal pseudo-spin projections of an ensemble of qubits are measured by expanding the space to four states. This expansion comes at the cost of an increased impact from the quantum-projection noise, owing to smaller populations in each measured state and the stochastic nature of the “mapping” pulses. Given that there are ten possible spin states in strontium’s ground state, one could, in principle, simultaneously measure all three spin projections of an ensemble of pseudo-spins $s = 1/2$ or even $s = 1$. Alternatively, one could simultaneously measure two projections of $s = 1/2$, $s = 1$, $s = 3/2$, or $s = 2$ pseudo-spin ensembles.

2.1.6 Conclusion

In this work, we have demonstrated the implementation of operations effectively driven by generators of the $\mathfrak{su}(10)$ group. We focused on selectively driving Rabi oscillations between adjacent ($m_F \longleftrightarrow m_F + 1$) and next-to-adjacent ($m_F \longleftrightarrow m_F + 2$) levels. We demonstrated these on four out of ten levels, with high fidelity > 0.99 . We expect the generalization to the entire spin manifold to require moderate technical adjustments, such as a higher magnetic field bias. Being able to drive all $\Delta m_F = (1, 2)$ transitions effectively amounts to driving “rotations” around $2N - 3 = 17$ generators of the $\mathfrak{su}(N)$ group. This number exceeds the $N - 1$ number of generators that can be simultaneously diagonalized (which correspond to the $N - 1$ quantization axes). Therefore concatenating $\Delta m_F = (1, 2)$ transitions is in principle sufficient to implement all the $N^2 - 1$ generators.

In practice, this set of rotations is sufficient to measure the relative amplitudes and coherences between all pairs of Zeeman levels, i.e. perform tomography and thus reconstruct the complete state. Combining these schemes with a spin-selective detection capable of giving access to all ten spin components, as in Refs. [83, 82], one would gain comprehensive control of the spin state, with applications to quantum simulation, computation, or sensing schemes requiring operations within the $\mathfrak{su}(N)$ group [41, 69, 48, 49, 50]. Thanks to these tools, the spins of strontium 87 atoms are a fully exploitable qudit resource with $\mathfrak{su}(10)$ character, which one can even complement by other coherently controllable degrees of freedom in electronically-excited states, e.g., as in Ref. [99, 100].

These assertions also apply to species with similar atomic structures - alkaline-earth-like atoms and group IIB transition metals fermionic isotopes - to which the scheme could be implemented. Indeed, they have in common: a large ratio between the hyperfine splitting and the linewidth of the state 3P_1 , favorable for tensor-shift engineering with low spontaneous-emission levels; a vast spin Hilbert space in the 1S_0 ground state; $\mathfrak{su}(N)$ symmetric interactions; extensive coherence times afforded by the Landé factor of nuclear spins.

We have seen that large-spin atoms open up the possibility of parallelized interferometers, which can be used for measuring multiple fields or for common-mode noise-rejection schemes. Similar sequences could readily explore other avenues, such as multiple-path interferometers and quantum random walks. More generally, being able to tailor the spin state by tensor light shifts enables the production of “non-classical” spin states, i.e., single-atom spin states that cannot be modeled by an assembly of uncorrelated spin-1/2 particles [101]. These “non-classical” spin states present specific advantages for quantum sensing [45].

Finally, following the original idea of Ref. [75], we have seen that the vast Hilbert space is an opportunity for the parallel estimation of multiple non-commuting observables of an atomic ensemble. We studied this in the case of two observables of an ensemble of pseudo-spins 1/2. However, it would be most interesting to study the parallel estimation, e.g., of three non-commuting observables or pseudo-spins $> 1/2$. It would come at the cost of more Hilbert space resources and increased quantum-projection-noise effects. The full potential of such measurements deserve investigation. The scheme is analogous to randomly partitioning the system into several sub-ensembles and measuring each along a different quantization axis. One question

is whether this could be amenable to extracting signatures of specific many-body states (chiral states, cluster states [102, 103]), or to characterize general properties of the many-body states (entanglement, purity) in the spirit of, e.g., Refs. [104, 105].

2.2 Outlook to positive-operator-valued measures (POVMs)

In traditional quantum metrology, standard interferometers are bounded by Projective Valued Measurements (PVM), which restrict the device to sensing a single physical quantity per experimental run. Tracking multiple environmental parameters or reconstructing the full state usually requires sequential measurements across different configurations, heavily increasing the statistical overhead and cycle times. The high-dimensional ground state manifold of ^{87}Sr allows us to bypass these rigid constraints through parallelized architectures.

To formally understand how a quantum system can extract such complementary pieces of information simultaneously, one look at the generalization of the measurement postulate via Positive-Operator-Valued Measures (POVM) [98, 106] of the reduced space of a two level object. Indeed, standard PVMs face a fundamental impossibility when trying to obtain complete information on a Hilbert space from a single outcome. Let us consider a system that produces either of these two non-orthogonal states of a two-level object, and try to extract from measurements which state is produced. For instance:

$$\left\{ |\psi_1\rangle = |0\rangle, \quad |\psi_2\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle \right\}$$

because $\langle\psi_1|\psi_2\rangle = 1/2 \neq 0$, a projection-based measurement in the $\{|\psi_1\rangle, |\psi_1^\perp\rangle\}$ basis—with $|\psi_1^\perp\rangle = |1\rangle$ —yields a conditional probability of $P(\psi_1|\psi_2) = |\langle\psi_1|\psi_2\rangle|^2 = 0.25$. Unambiguous state discrimination (USD) addresses this limitation by guaranteeing error-free identification of non-orthogonal states at the expense of introducing a non-zero probability of obtaining an inconclusive result. Because of non-orthogonality, a single PVM outcome cannot deduce with certainty which state was prepared, and crucial information is completely lost.

A POVM overcomes this by defining an ensemble of positive hermitian operators $\{\hat{E}_1, \hat{E}_2, \hat{E}_3\}$ ($\hat{E}_i \geq 0$, of positive eigenvalues) summing to the identity $\sum_i \hat{E}_i = \mathbb{I}$. This enforces orthogonality conditions in the same basis $\{|\psi_1\rangle, |\psi_2\rangle, |\psi_1^\perp\rangle, |\psi_2^\perp\rangle\}$:

$$\langle\psi_2|\hat{E}_1|\psi_2\rangle = 0 \tag{2.10}$$

$$\langle\psi_1|\hat{E}_2|\psi_1\rangle = 0 \tag{2.11}$$

with $|\psi_2^\perp\rangle = \frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle$. Probability conservation requires an inconclusive operator $\hat{E}_3$ such that $\mathbb{I} = \hat{E}_1 + \hat{E}_2 + \hat{E}_3$. Combined with the condition of semi-positivity of the eigenvalues, these constraints force the operators to be proportional to the dual basis projectors:

$$\hat{E}_1 = \lambda |\psi_2^\perp\rangle \langle\psi_2^\perp| \tag{2.12}$$

$$\hat{E}_2 = \lambda |\psi_1^\perp\rangle \langle\psi_1^\perp| \tag{2.13}$$

where the normalization $\lambda = 2/3$ validates the $\hat{E}_3 \geq 0$ condition. This leads to a success probability of $P(\hat{E}_1|\psi_1) = \lambda|\langle\psi_1|\psi_2^\perp\rangle|^2 = 1/2$ (equivalent for $P(\hat{E}_2|\psi_2)$), while the remaining 50% is routed to the inconclusive outcome $\hat{E}_3$ since projecting $\hat{E}_3$ onto ψ_1 or ψ_2 does not lead to a conclusive determination of the initial state.

Our second interferometric scheme, detailed in Section 2.1.5, constitutes a direct physical realization of this POVM framework. The protocol prepares a target qubit state $|\psi\rangle = \alpha|-5/2\rangle + \beta|-7/2\rangle$, and uses targeted coherent pulses to map a controlled fraction of these populations into two auxiliary ancillary states ($|-9/2\rangle$ and $|-3/2\rangle$).

2.2.1 Derivation of the experimental POVM framework

2.2.1.1 Four-level extended space dynamics

To model this sequence, we consider the four-level subspace $\mathcal{H}_4$ spanned by the basis states:

$$\mathcal{H}_4 = \left\{ \left| -\frac{9}{2} \right\rangle, \left| -\frac{7}{2} \right\rangle, \left| -\frac{5}{2} \right\rangle, \left| -\frac{3}{2} \right\rangle \right\}. \quad (2.14)$$

In this basis, the fundamental projective measurements (PVM) associated with state selective fluorescence detection are described by the diagonal projectors:

$$\hat{E}_1 = \left| -\frac{9}{2} \right\rangle \left\langle -\frac{9}{2} \right|, \quad \hat{E}_2 = \left| -\frac{7}{2} \right\rangle \left\langle -\frac{7}{2} \right|, \quad \hat{E}_3 = \left| -\frac{5}{2} \right\rangle \left\langle -\frac{5}{2} \right|, \quad \hat{E}_4 = \left| -\frac{3}{2} \right\rangle \left\langle -\frac{3}{2} \right|. \quad (2.15)$$

Both operators below are single-qubit rotations generated by a Pauli matrix A (with $A^2 = \mathbb{I}$), for which the exponential reduces to

$$\exp(-ixA) = \cos(x)\mathbb{I} - i\sin(x)A, \quad (2.16)$$

obtained by splitting the Taylor series of the exponential into its even and odd powers of A .

The pulse sequence consists of a differential phase accumulation ϕ applied to state $|-5/2\rangle$ (and other states but not our interest here). It is generated by $\exp(-i\phi\sigma_z^{\uparrow,\downarrow}/2)$ (Sec. 2.1.5.2), which by Eq. (2.16) (with $A = \sigma_z$, $x = \phi/2$) reads $\text{diag}(e^{-i\phi/2}, e^{i\phi/2})$ in the $\{|-5/2\rangle, |-7/2\rangle\}$ basis. Only the relative phase between the two levels affects any measurement outcome, so factoring out the (physically irrelevant) global phase $e^{-i\phi/2}$ – and trivially extending to the two unaffected ancillary states – gives the phase operator:

$$\hat{\Phi}(\phi) = \text{diag}(1, 1, e^{-i\phi}, 1), \quad (2.17)$$

followed by resonant $\pi/2$ pulse rotations acting simultaneously on the adjacent pairs $\{|-9/2\rangle, |-7/2\rangle\}$ and $\{|-5/2\rangle, |-3/2\rangle\}$, duplicating the qubit amplitudes onto the ancillary states used for readout. On each addressed pair, this beam-splitter-like operation is generated by a resonant Rabi pulse of area $\pi/2$, $\exp(-i\frac{\pi}{4}\sigma_x)$, which by Eq. (2.16) (with $A = \sigma_x$, $x = \pi/4$) gives $\frac{1}{\sqrt{2}}(\mathbb{I} - i\sigma_x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$. Applied

simultaneously to both pairs, this takes the following block-diagonal form over $\mathcal{H}_4$:

$$\hat{X}_{\pi/2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i & 0 & 0 \\ -i & 1 & 0 & 0 \\ 0 & 0 & 1 & -i \\ 0 & 0 & -i & 1 \end{pmatrix}. \quad (2.18)$$

The measurement sequence is closed by a final resonant $\pi/2$ pulse acting only on the original qubit pair $\{|-7/2\rangle, |-5/2\rangle\}$, leaving the ancillary states $|-9/2\rangle$ and $|-3/2\rangle$ untouched:

$$\hat{X}_{\pi/2}^{qubit} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} & 0 \\ 0 & \frac{-i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.19)$$

The full unitary transformation experienced by the system prior to detection is given by $\hat{U} = \hat{X}_{\pi/2}^{qubit} \hat{\Phi}(\phi) \hat{X}_{\pi/2}$: the duplicating pulse acts first, followed by the phase accumulation, followed by the closing pulse on the qubit pair. In the Heisenberg picture, the bare detection operators transform as $\hat{E}'_i = \hat{U}^\dagger \hat{E}_i \hat{U}$.

Carrying out the matrix multiplication for $\hat{E}'_2 = \hat{X}_{\pi/2}^\dagger \hat{\Phi}^\dagger \hat{X}_{\pi/2}^{qubit\dagger} \hat{E}_2 \hat{X}_{\pi/2}^{qubit} \hat{\Phi} \hat{X}_{\pi/2}$ (and likewise for $\hat{E}'_4$), explicitly, in the basis $\{|-9/2\rangle, |-7/2\rangle, |-5/2\rangle, |-3/2\rangle\}$, yields for the second and fourth detection channels:

$$\hat{E}'_2 = \frac{1}{4} \begin{pmatrix} 1 & i & e^{-i\phi} & -ie^{-i\phi} \\ -i & 1 & -ie^{-i\phi} & -e^{-i\phi} \\ e^{i\phi} & ie^{i\phi} & 1 & -i \\ ie^{i\phi} & -e^{i\phi} & i & 1 \end{pmatrix}, \quad \hat{E}'_4 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{i}{2} \\ 0 & 0 & -\frac{i}{2} & \frac{1}{2} \end{pmatrix}. \quad (2.20)$$

Compared to a sequence with only the duplicating pulse, the closing pulse $\hat{X}_{\pi/2}^{qubit}$ mixes the phase accumulated on $|-5/2\rangle$ back into the $|-7/2\rangle$ channel, so that $\hat{E}'_2$ now also carries ϕ -dependent coherences – a direct signature of the qubit phase, and the key point of this measurement scheme.

$\hat{E}'_2$ and $\hat{E}'_4$ could be measured in the same shot but it would be preferable to extend the space to more m_F states to reduce SQL noise.

2.2.1.2 Reduction to the target qubit subspace

While the set $\{\hat{E}'_i\}$ forms a valid PVM over the full four-level space $\mathcal{H}_4$, the initial physical state $|\psi\rangle = \alpha|-5/2\rangle + \beta|-7/2\rangle$ is strictly prepared inside the two-level target qubit subspace $\mathcal{H}_{qubit} = \{|-\frac{7}{2}\rangle, |-\frac{5}{2}\rangle\}$.

In Bergou [106] framework, measurement probabilities correspond to state overlaps $P(i) = \langle\psi|\hat{E}'_i|\psi\rangle$. Because the prepared state has zero initial population in the auxiliary states $|-9/2\rangle$ and $|-3/2\rangle$, we have:

$$\left\langle -\frac{9}{2} \middle| \psi \right\rangle = 0, \quad \left\langle -\frac{3}{2} \middle| \psi \right\rangle = 0. \quad (2.21)$$

As a consequence, any matrix elements of $\hat{E}'_i$ acting purely on $|-9/2\rangle$ or $|-3/2\rangle$ yield zero contribution when evaluated on $|\psi\rangle$. Restricting the transformed measurement operators to the active qubit manifold by projecting via $\hat{P}_{\text{qubit}} = \left|-\frac{7}{2}\right\rangle\left\langle-\frac{7}{2}\right| + \left|-\frac{5}{2}\right\rangle\left\langle-\frac{5}{2}\right|$ defines the set of effective operators:

$$\hat{M}_i \equiv \hat{P}_{\text{qubit}} \hat{E}'_i \hat{P}_{\text{qubit}}. \quad (2.22)$$

Applying this projection to $\hat{E}'_2$ isolates the populated block, which retains the ϕ -dependent coherence between the two qubit states:

$$\hat{M}_2 = \frac{1}{4} \left[\left|-\frac{7}{2}\right\rangle\left\langle-\frac{7}{2}\right| + \left|-\frac{5}{2}\right\rangle\left\langle-\frac{5}{2}\right| - ie^{-i\phi} \left|-\frac{7}{2}\right\rangle\left\langle-\frac{5}{2}\right| + ie^{i\phi} \left|-\frac{5}{2}\right\rangle\left\langle-\frac{7}{2}\right| \right]. \quad (2.23)$$

Likewise, projecting $\hat{E}'_4$ eliminates the unpopulated auxiliary states $|-9/2\rangle$ and $|-3/2\rangle$, leaving:

$$\hat{M}_4 = \frac{1}{2} \left|-\frac{5}{2}\right\rangle\left\langle-\frac{5}{2}\right|. \quad (2.24)$$

The set of restricted operators $\{\hat{M}_i\}$ on $\mathcal{H}_{\text{qubit}}$ are non-orthogonal ($\hat{M}_i^2 \neq \hat{M}_i$) and positive semi-definite ($\hat{M}_i \geq 0$), yet they fulfill the completeness relation $\sum_i \hat{M}_i = \mathbb{I}_{\text{qubit}}$. Restricting a global projective measurement on an extended Hilbert space to an active target subspace by eliminating unpopulated auxiliary channels provides a direct physical construction of a POVM.

Since $\hat{P}_{\text{qubit}} |\psi\rangle = |\psi\rangle$ for the qubit-restricted state, the click probabilities coincide with those of the full-space operators, $P(i) = \langle\psi| \hat{E}'_i |\psi\rangle = \langle\psi| \hat{M}_i |\psi\rangle$, so that the measured populations obey $\langle N_{-3/2} \rangle = N_{\text{at}} \langle \hat{M}_4 \rangle$ and $\langle N_{-7/2} \rangle = N_{\text{at}} \langle \hat{M}_2 \rangle$. Combined with the definitions of the collective pseudo-spin observables in terms of these same populations, this directly expresses the mean pseudo-spin projections in terms of the POVM outcome probabilities:

$$\langle \hat{s}_{\uparrow,\downarrow}^z \rangle = N_{\text{at}} \left(2\langle \hat{M}_4 \rangle - \frac{1}{2} \right), \quad \langle \hat{s}_{\uparrow,\downarrow}^\phi \rangle = N_{\text{at}} \left(\frac{1}{2} - 2\langle \hat{M}_2 \rangle \right). \quad (2.25)$$

The ϕ -independent term in $\hat{M}_4$ (eq. (2.24)) reflects that $\hat{s}^z$ does not depend on the interferometric phase, while the coherence carried by $\hat{M}_2$ (eq. (2.23)) is precisely what makes $\hat{s}^\phi$ sensitive to ϕ .

Chapter 3

Photoassociation : concept and spectroscopy of ^{88}Sr and ^{87}Sr

While the previous chapter presented the individual evolution of atoms superposed in multiple states, this third chapter focuses on the loss processes through which two atoms are photoassociated to explore the molecular formation of ^{87}Sr . Photoassociation (PA) consists of illuminating colliding atoms at a specific frequency to form a bound molecular state. With the advent of cold and ultracold atoms, PA has become a high-resolution spectroscopic tool, as the extremely low initial kinetic energy significantly reduces thermal broadening and reveals subtle structures [107].

In bosonic strontium, photoassociation near the narrow $^1\text{S}_0 - ^3\text{P}_1$ intercombination line has been studied extensively: the least-bound levels of the 0_u and 1_u excited potentials of ^{88}Sr were measured [5], used to demonstrate optical Feshbach resonances [14, 108], and mass-scaled to ^{84}Sr and ^{86}Sr [15, 16]; such a level was also used as the intermediate state to form ground-state $^{84}\text{Sr}_2$ molecules by STIRAP [29]. The ground-state potential $X^1\Sigma_g^+$ is known precisely, which gives the scattering lengths of all isotopic combinations, including ^{87}Sr [109]. For the fermionic isotope ^{87}Sr , photoassociation has been reported on the broad $^1\text{S}_0 - ^1\text{P}_1$ transition [110], whereas on the intercombination line it has only been briefly mentioned. On this line, the hyperfine structure of the $^3\text{P}_1$ state ($I = 9/2$) splits each molecular level into many components, which makes the spectrum dense and its assignment difficult, despite its promise as a controllable interface for state manipulation.

Beyond fundamental spectroscopy, resolving these narrow molecular transitions is directly motivated by applications in quantum metrology. In this context, a fundamental benchmark for phase estimation with N independent particles prepared in a coherent spin state is the Standard Quantum Limit (SQL), defined by the shot-noise scaling $\Delta\phi_{\text{SQL}} = 1/\sqrt{N}$ [19]. For an ensemble of particles with effective spin s where interferometry is performed on extreme projection states, this limit scales as $\Delta\phi_{\text{SQL}} = 1/\sqrt{2Ns}$.

While standard protocols rely on coherent control, spin squeezing, or quantum gates to generate entangled states, the work of Foss-Feig et al. [23] inspired our team to pursue an alternative route. Rather than viewing dissipation as a decoherence mechanism to be avoided, we aim to harness the two-body reactive losses associated with these narrow molecular potentials as a dissipative engine. As a long-term

prospect, such a process can drive the system toward a stationary Dicke state in a zero-projection spin ($m = J_z = 0$). In this configuration, the collective spin variance scales quadratically ($\propto N^2$) rather than linearly as for uncorrelated states ($\propto N$). Consequently, the resulting phase sensitivity bypasses the standard $1/\sqrt{N}$ shot-noise scaling of the SQL and approaches the fundamental Heisenberg limit ($\Delta\phi \propto 1/N$), offering the ultimate precision allowed by quantum mechanics [19].

To pave the way toward this goal, the work presented in this chapter focuses on an essential prerequisite: the spectroscopic identification of the molecular states involved in this spin-selective dissipative process. The main objective here is to benchmark the photoassociation mechanism itself, laying the necessary spectroscopic foundation for future state-purification protocols.

This chapter opens with the fundamental concepts of scattering theory required to describe photoassociation and atomic density dynamics via one- and two-body loss processes. Next, we present preliminary experimental results obtained with ^{88}Sr , taking advantage of its well-documented molecular potentials to calibrate and benchmark our experimental setup. Building upon these foundations, we then investigate the fermionic isotope ^{87}Sr . We present the experimental spectra and discuss the technical challenges and systematic effects that could account for the weak losses observed on the molecular lines. Finally, we conclude with an analysis of the surprisingly rich line spectrum and line shapes, which suggest yet unexplained physical mechanisms at play.

3.1 Introduction to photoassociation

3.1.1 Scattering theory

Understanding the PA process necessitates an introduction to scattering theory to define key quantities such as the scattering length and the optical length. These parameters emerge naturally from the asymptotic analysis of the radial Schrödinger equation describing the relative motion of two colliding atoms.

In the study of inter-atomic interactions, the Born-Oppenheimer approximation provides a natural simplification. Due to the large mass ratio between the nuclei and the electrons, the nuclei move on a much slower timescale. One can therefore solve the electronic problem for fixed nuclear positions. The resulting electronic energy curve as a function of the inter-nuclear distance r defines the adiabatic potential $\hat{U}(r)$ experienced by the two atoms.

Assuming a spherically symmetric potential $\hat{U}(r)$, the treatment of the two-body collision in the center-of-mass frame allows the total relative wavefunction to be factorized into radial and angular parts:

$$\psi(r, \theta, \phi) = \frac{u_l(r)}{r} Y_l^m(\theta, \phi) \quad (3.1)$$

where the spherical harmonics $Y_l^m(\theta, \phi)$ encode the angular dependence and l denotes the partial wave quantum number. By introducing the reduced mass of the two-atom system $\mu_N = m_{\text{at}}/2$, the Schrödinger equation reduces to a one-dimensional radial equation for each partial wave l :

$$\left(-\frac{\hbar^2}{2\mu_N}\frac{d^2}{dr^2} + \frac{\hbar^2 l(l+1)}{2\mu_N r^2} + \hat{U}(r)\right) u_l(r) = E u_l(r) \quad (3.2)$$

E the eigenvalue associated to the eigenfunction $u(r)$. This one-dimensional structure allows a direct analogy with the 1D quantum framework [111]. To extract the scattering properties of the system, one studies the asymptotic behavior of the solutions at large distances. For $r \gg 1/k$, beyond the range of the inter-atomic potential, the general solution for $u_l(r)$ takes the form [112]:

$$u_l(r) \xrightarrow{r \rightarrow \infty} a_l \sin\left(kr - \frac{l\pi}{2} + \delta_l\right) \quad (3.3)$$

where a_l is a normalization factor, $k = \sqrt{2\mu_N E}/\hbar$ is the collision wavenumber, and δ_l is the phase shift acquired by the l -th partial wave due to the short-range interaction potential.

At low energy, $kb \ll 1$ with b the range of the potential (the distance below which $\hat{U}(r)$ has a significant effect), the radial equation can be solved separately in three regions: for $r \lesssim b$, where $\hat{U}$ dominates over the energy, so that the solution does not depend on k ; for $b \ll r \ll 1/k$, where only the centrifugal term remains and the solution is $\alpha r^{l+1} + \beta r^{-l}$; and for $r \gg 1/k$, where k^2 dominates and one recovers Eq. 3.3. Matching the solutions in the overlap regions, the ratio β/α is set by the inner solution, hence independent of k , while the matching with the outer solution gives $\beta/\alpha \propto \tan \delta_l / k^{2l+1}$ [112]. Therefore

$$\tan \delta_l \propto k^{2l+1} \quad (k \rightarrow 0). \quad (3.4)$$

This result requires a potential decaying fast enough at large distance ($-C_n/r^n$ with $n > 2l + 3$): for van der Waals ($n = 6$), it holds only for $l = 0, 1$ [112].

For s-waves ($l = 0$), this leads to the definition of the scattering length:

$$a = -\lim_{k \rightarrow 0} \tan \delta_0 / k \quad (3.5)$$

The full wavefunction is recovered by summing over all partial waves. At large distances, this superposition can be rewritten as an incoming plane wave and an outgoing scattered spherical wave:

$$\Psi \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k} \cdot \mathbf{z}} + f(\theta) \frac{e^{ikr}}{r} \quad (3.6)$$

where $f(\theta)$ is the scattering amplitude which can model both elastic and inelastic collisions and θ is the angle between the incident and scattered wavevectors. This amplitude expresses for a specific partial wave l as

$$f_l = \frac{1}{2ik} (e^{2i\delta_l} - 1) \quad (3.7)$$

The asymptotical expression of Ψ and e^{ikz} is expressed in a spherical basis as a sum over all the partial waves. Then we can give as in 3.6 a decomposition of Ψ :

$$\Psi \simeq \frac{i}{2kr} \sum_{l=0}^{\infty} (2l+1) ((-1)^l e^{-ikr} - S_l e^{ikr}) \quad (3.8)$$

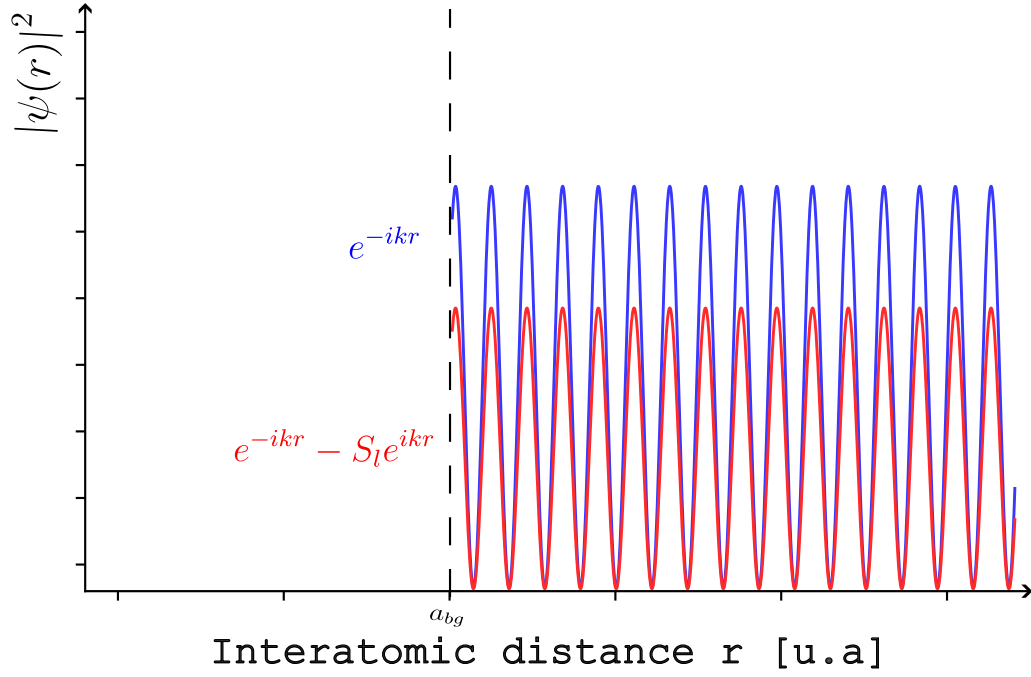


Figure 3.1: The wavefunction Ψ at large inter-atomic distances r behaves as a summation of an incoming plane wave and an outgoing spherical wave with attenuated amplitude $f(\theta)$. Here is a representation of this wavefunction for s-wave collisions ($l = 0$). At large distance, the radial wavefunction behaves as $\sin[k(r - a_{bg})]$, i.e. as the free wavefunction shifted by the background scattering length: a_{bg} is the position of the node of this asymptotic form extrapolated to short distance (for $a_{bg} > 0$, the radius of the equivalent hard sphere), not a distance at which the interaction starts.

where we identify a spherical part and a scattering term with $S_l = e^{2i\delta_l}$ a coefficient characterising the weight of the inelastic collisions.

In the presence of inelastic processes such as photoassociation, the colliding atom pair can be irreversibly lost from the scattering channel-either by forming a bound molecular state or by releasing kinetic energy. This loss of flux from the incoming channel cannot be captured by a purely real phase shift, as $|S_l| = 1$ implies full conservation of the wavefunction amplitude. To account for this absorption of flux, δ_l must be extended to the complex plane, allowing $|S_l| < 1$ and introducing the possibility of inelastic scattering.

The associated elastic cross section representing the surface where the scope of the potential is effective writes as

$$\sigma_{el} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) |1 - S_l|^2 \quad (3.9)$$

But we see in equation 3.8 that $|S_l|$ needs to be < 1 to allow dissipation of the wavefunction. Defining δ_l as a complex number satisfies this condition and the inelastic cross section is

$$\sigma_{inel} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) (1 - |S_l|^2) \quad (3.10)$$

Then from the condition $\delta_0 < 1$ we deduce an approximated scattering amplitude S_0

$$S_0 = e^{2i\delta_0} \simeq 1 + 2i\delta_0 \quad (3.11)$$

Added to the result of 3.5,

$$1 + 2i\delta_0 \simeq 1 - 2ika \quad (3.12)$$

Because δ_0 is complex, a direct result of these equations is that we can write a general scattering length as

$$a = a_{bg} - ia'' \quad (3.13)$$

with $a'' > 0$.

At low temperatures we can re-express equations 3.9 and 3.10 as

$$\sigma_{el} = 4\pi |a|^2 \quad (3.14)$$

$$\sigma_{inel} = \frac{4\pi}{k} |a''| \quad (3.15)$$

A typical example of elastic collision is that of two ground-state atoms, which can only exchange kinetic energy. Photoassociation, in contrast, is an inelastic, dissipative process: a laser couples two ground-state atoms to an excited molecular state, which then decays, leading to the loss of the pair.

3.1.2 PA and complex scattering length

Photoassociation (PA) is a process in which a pair of colliding free atoms absorbs a photon to form a bound molecule in an excited electronic state. Effectively, because the short-lived molecule quickly decays into deeply bound states or dissociates with high kinetic energy, it escapes the optical trap. Consequently, the bound molecule is typically seen as lost, and the process is perceived as a macroscopic two-body loss effect.

In the atomic pair frame, the electric dipole interaction drives the system from the ground electronic state ψ_g of potential $U_g(r)$ to the excited molecular state ψ_e of potential $U_e(r)$. In the absence of a resonant laser, the internal evolution of the system is purely elastic and expresses as:

$$i\hbar \frac{d}{dt} \begin{pmatrix} \psi_g \\ \psi_e \end{pmatrix} = \begin{pmatrix} -\frac{\hbar^2}{2\mu} \nabla^2 + U_g & 0 \\ 0 & -\frac{\hbar^2}{2\mu} \nabla^2 + U_e - i\hbar \frac{\Gamma_{\text{mol}}}{2} \end{pmatrix} \begin{pmatrix} \psi_g \\ \psi_e \end{pmatrix}. \quad (3.16)$$

Both ψ_g and its field-free counterpart $\psi_g^{(0)}$ introduced below are continuum states and cannot be normalized to unity. In what follows, $\psi_g^{(0)}(r, E)$ denotes the s-wave reduced radial wavefunction $u_0(r)$ of Eq. 3.2 for the ground potential U_g alone, at collision energy $E = \hbar^2 k^2 / 2\mu$. According to Eq. 3.3 with $l = 0$, it behaves at large distance as $\psi_g^{(0)}(r, E) \rightarrow A_k \sin(kr + \delta_0^{\text{bg}})$, where δ_0^{bg} is the real s-wave phase shift induced by U_g , related to the background scattering length through Eq. 3.5, $\delta_0^{\text{bg}} \simeq -k a_{\text{bg}}$ at low energy, and A_k is a normalization constant. We adopt the standard energy normalization [113, 107] $\langle \psi_g^{(0)}(E) | \psi_g^{(0)}(E') \rangle = \delta(E - E')$. The normalization integral is dominated by the oscillating asymptotic region, where $\int_0^\infty \sin(kr + \delta_0^{\text{bg}}) \sin(k'r + \delta_0^{\text{bg}}) dr = \frac{\pi}{2} \delta(k - k')$. Since $\delta(k - k') = \frac{dE}{dk} \delta(E - E') = \frac{\hbar^2 k}{\mu} \delta(E - E')$, the condition reads $A_k^2 \frac{\pi}{2} \frac{\hbar^2 k}{\mu} = 1$, which fixes

$$\psi_g^{(0)}(r, E) \xrightarrow{r \rightarrow \infty} \sqrt{\frac{2\mu}{\pi \hbar^2 k}} \sin(kr + \delta_0^{\text{bg}}). \quad (3.17)$$

The amplitude $A_k \propto 1/\sqrt{k}$ is the amplitude of the oscillations at large distance, but the photoassociation overlap is set by the wavefunction at short distance, near the Condon radius. There ($r \ll 1/k$), the sine can be expanded, $\sin(kr + \delta_0^{\text{bg}}) \simeq k(r - a_{\text{bg}})$, so that $\psi_g^{(0)}(r, E) \simeq A_k k(r - a_{\text{bg}})$: the short-range amplitude is $A_k k \propto k/\sqrt{k} = \sqrt{k}$. This $\sqrt{k}$ scaling of the short-range wavefunction is the origin of the Wigner threshold law used below. This normalization also simplifies Fermi's Golden Rule: with energy-normalized continuum states, the density of final states is equal to one, so that the transition rates below can be written as $2\pi |\langle f | H' | i \rangle|^2$, without an explicit density-of-states factor.

In the frame of [114] one treats the case when a laser is tuned near a specific vibrational level of the excited potential $U_e(r)$, coupling the free-atom continuum to the bound molecular state, as illustrated in Fig. 3.2.

This coupling introduces inelastic channels into the system. The effective coupled-channel equation becomes:

$$i\hbar \frac{d}{dt} \begin{pmatrix} \psi_g \\ \psi_e \end{pmatrix} = \begin{pmatrix} -\frac{\hbar^2}{2\mu} \nabla^2 + U_g & \hbar\Omega/2 \\ \hbar\Omega/2 & -\frac{\hbar^2}{2\mu} \nabla^2 + U_e + \hbar\Delta - i\hbar \frac{\Gamma_{\text{mol}}}{2} \end{pmatrix} \begin{pmatrix} \psi_g \\ \psi_e \end{pmatrix} \quad (3.18)$$

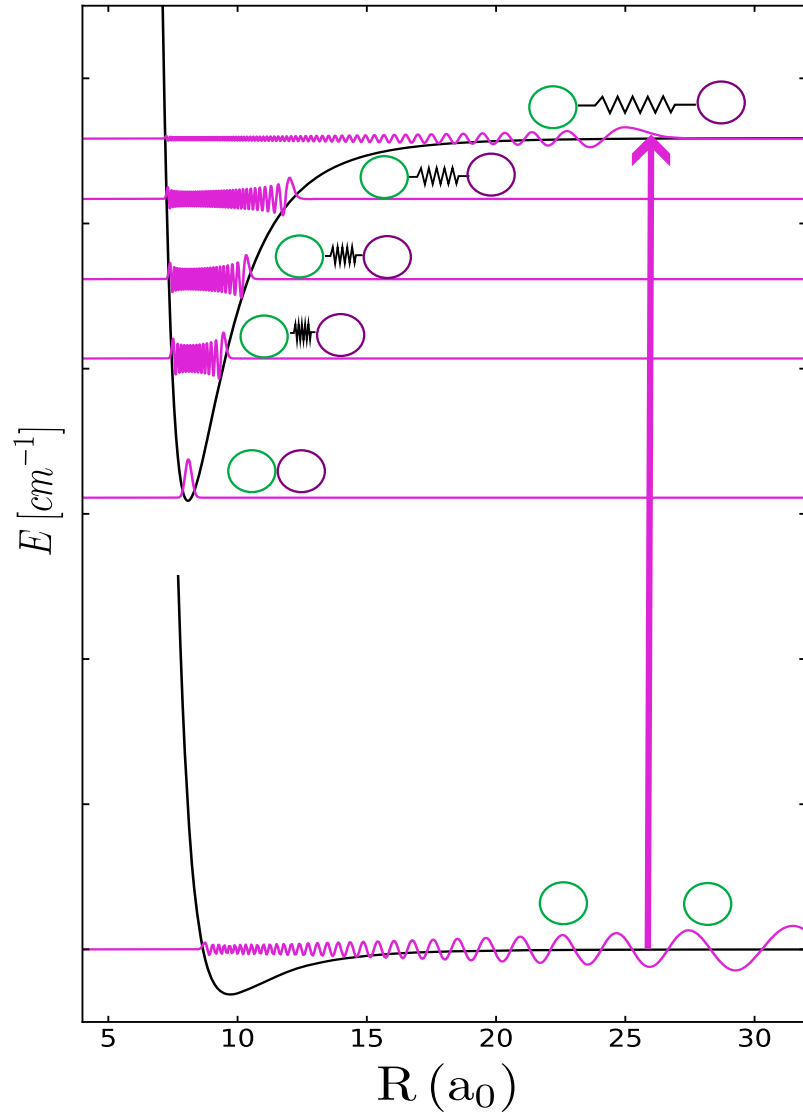


Figure 3.2: Adapted from [115]. General concept of photoassociation: a resonant laser couples a pair of colliding atoms from the free-atom continuum to a bound state of an excited molecular potential. Within this potential, the molecule occupies quantized vibrational states, described by the solutions of the 1D radial Schrödinger equation plotted for each level.

where Γ_{mol} is the total decay linewidth of the excited molecular state (its radiative width Γ_{rad} , defined below, plus possible non-radiative decay channels), $\Delta = \omega_e - \omega_L$ is the laser detuning from the excited bound state, and $\hbar\Omega/2$ represents the coherent optical coupling.

This matrix equation shows that the laser creates a bridge between the two manifolds. During a collision, two atoms approaching each other on the ground potential $U_g(r)$ have a non-zero probability to absorb a photon and transition to the excited molecular potential $U_e(r)$. Once populated, this state decays at the rate Γ_{mol} (spontaneous emission at the rate $\Gamma_{\text{rad}} \leq \Gamma_{\text{mol}}$, plus possible non-radiative channels). To account for this particle loss, the molecular decay appears as an imaginary term $-i\hbar\frac{\Gamma_{\text{mol}}}{2}$ in the Hamiltonian matrix.

The laser is detuned far from the atomic resonance, so that it does not excite single atoms: it is only resonant with the transition from the colliding pair to the bound molecular state. Since this free-bound coupling is weak, the excited molecular state remains weakly populated, and the pair spends most of its time on the ground-state potential. To isolate the response of the excited molecular channel, we expand the closed-channel wavefunction $\psi_e(r)$ onto the bound vibrational eigenstates $|v_e\rangle$ of the excited Hamiltonian $\hat{H}_e = -\frac{\hbar^2\nabla^2}{2\mu} + U_e(r)$, which satisfy $\hat{H}_e|v_e\rangle = E_{v_e}|v_e\rangle$.

Following Bohn and Julienne [113], we keep in the expansion of ψ_e only the vibrational level v_e closest to resonance (isolated-resonance approximation). The other vibrational levels v'_e also contribute to $\langle\hat{V}\rangle$, but each of them is far from resonance and its contribution is reduced by a large energy denominator. For ^{87}Sr , the same argument applies to the hyperfine levels F' of 3P_1 : their splitting ($\gtrsim 1$ GHz [31]) is much larger than Γ_{mol} and than the detuning range explored here, so that each (v_e, F') level can be treated as an isolated resonance.

The coupled wavefunction (ψ_g, ψ_e) always involves the excited molecular state, but with a highly depleted amplitude, since the free-bound coupling is weak. Projecting the second line of Eq. 3.18 onto a specific isolated vibrational state $|v_e\rangle$ yields:

$$E\psi_e = \frac{\hbar\hat{\Omega}}{2}\psi_g + \left(\hat{H}_e + \hbar\Delta - i\hbar\frac{\Gamma_{\text{mol}}}{2}\right)\psi_e \quad (3.19)$$

$$\langle v_e | (E - \hat{H}_e - \hbar\Delta + i\hbar\frac{\Gamma_{\text{mol}}}{2}) |\psi_e\rangle = \langle v_e | \frac{\hbar\hat{\Omega}}{2} |\psi_g\rangle \quad (3.20)$$

We then assume that ψ_e is dominated by this near-resonant level (isolated-resonance approximation, see above), so that

$$\psi_e(r) \approx \frac{\left\langle v_e \left| \frac{\hbar\hat{\Omega}(r)}{2} \right| \psi_g \right\rangle}{E - E_{v_e} - \hbar\Delta + i\hbar\frac{\Gamma_{\text{mol}}}{2}} v_e(r) \quad (3.21)$$

Substituting this spatial projection back into the ground-state channel (first line of

Eq. 3.18) yields:

$$i\hbar \frac{d}{dt} \psi_g = \left(-\frac{\hbar^2 \hat{\nabla}^2}{2\mu} + \hat{U}_g \right) \psi_g + \frac{\hbar \hat{\Omega}}{2} \psi_e \quad (3.22)$$

$$\left(E + \frac{\hbar^2 \hat{\nabla}^2}{2\mu} - \hat{U}_g \right) \psi_g = \frac{\hbar \hat{\Omega}}{2} \psi_e \quad (3.23)$$

$$\left(E + \frac{\hbar^2 \hat{\nabla}^2}{2\mu} - \hat{U}_g \right) \psi_g = \underbrace{\frac{\frac{\hbar \hat{\Omega}}{2} |v_e\rangle \langle v_e| \frac{\hbar \hat{\Omega}}{2} |\psi_g\rangle}{E - E_{v_e} - \hbar \Delta + i\hbar \frac{\Gamma_{\text{mol}}}{2}}}_{\equiv \hat{V} \psi_g} \quad (3.24)$$

$$\left(-\frac{\hbar^2 \hat{\nabla}^2}{2\mu} + \hat{U}_g(r) + \hat{V} \right) \psi_g(r) = E \psi_g(r) \quad (3.25)$$

One wants to isolate the term that concerns the ground state to observe the modifications of it from the coupling to the molecular state. Then by projecting $|\psi_g\rangle$ in 3.25 we define $\langle \hat{V} \rangle$ as:

$$\langle \psi_g | \hat{V} | \psi_g \rangle = \frac{\left| \left\langle v_e \left| \frac{\hbar \hat{\Omega}(r)}{2} \right| \psi_g \right\rangle \right|^2}{E - E_{v_e} - \hbar \Delta + i\hbar \frac{\Gamma_{\text{mol}}}{2}} \quad (3.26)$$

From $\hat{V}$ to a shift of the scattering length.

The general relationship used here between a coupling potential and a resulting shift of the scattering length is the same Green's function argument that underlies the two-channel treatment of Feshbach resonances in [116] (see their Eqs. 13, 17, 19 and 22): there, the coupling matrix element $\langle \phi_{\text{res}} | W | \phi_p^{(+)} \rangle$ plays exactly the role of our $\langle v_e | \frac{\hbar \hat{\Omega}}{2} | \psi_g^{(0)} \rangle$, and their scattering-length shift (their Eq. 22) has the same structure as the one derived below. We reproduce this result directly in position space, working at leading order in the laser coupling.

To relate $\langle \hat{V} \rangle$ to $a(\Delta)$, compare the full solution ψ_g of Eq. 3.25 to the background solution $\psi_g^{(0)}$ of the same equation with $\hat{V} \rightarrow 0$, both evaluated at the same threshold energy $E \rightarrow 0$:

$$-\frac{\hbar^2 \hat{\nabla}^2}{2\mu} \psi_g^{(0)} + \hat{U}_g \psi_g^{(0)} = E \psi_g^{(0)}, \quad (3.27)$$

which is simply Eq. 3.25 with the coupling to the molecular channel switched off, consistent with $\psi_g^{(0)}$ describing purely elastic collisions on U_g alone as in Eq. 3.16. In this derivation only, the two radial wavefunctions are normalized at zero energy, so that they behave as $r - a$ at large distance; the energy normalization of Eq. 3.17 is restored just after. At $E = 0$, the two radial equations read $\psi_g^{(0)''} = \frac{2\mu}{\hbar^2} U_g \psi_g^{(0)}$ and $\psi_g'' = \frac{2\mu}{\hbar^2} (U_g + \hat{V}) \psi_g$. Multiplying the first one by ψ_g , the second one by $\psi_g^{(0)}$, and subtracting gives

$$\frac{d}{dr} \left[\psi_g^{(0)'} \psi_g - \psi_g' \psi_g^{(0)} \right] = \psi_g^{(0)''} \psi_g - \psi_g'' \psi_g^{(0)} = -\frac{2\mu}{\hbar^2} \hat{V} \psi_g \psi_g^{(0)}.$$

Integrating from $r = 0$, where both wavefunctions vanish, to $r \rightarrow \infty$, where $\psi_g^{(0)} \rightarrow r - a_{\text{bg}}$ and $\psi_g \rightarrow r - a$ by definition of the scattering length (so that the bracket tends to $a_{\text{bg}} - a$), yields

$$\delta a \equiv a - a_{\text{bg}} = + \frac{2\mu}{\hbar^2} \int_0^\infty \hat{V}(r) \psi_g(r) \psi_g^{(0)}(r) dr.$$

As expected, a repulsive coupling ($\hat{V} > 0$) increases the scattering length, as for a hard sphere whose scattering length is its radius. This integral still contains the full, unknown ψ_g , so nothing has been solved yet. To make progress, write $\psi_g = \psi_g^{(0)} + \delta\psi$, where $\delta\psi$ is the correction to the wavefunction caused by switching on $\hat{V}$. Since $\hat{V}$ already involves the coupling $\frac{\hbar\Omega}{2}$ twice – once to excite to v_e , once to decay back – it is proportional to Ω^2 ; the correction $\delta\psi$ it induces is therefore also of order Ω^2 . Splitting the integral according to $\psi_g = \psi_g^{(0)} + \delta\psi$ gives one term with $\psi_g^{(0)}$ in place of ψ_g , plus an extra term proportional to $\hat{V} \delta\psi$; since both factors are of order Ω^2 , this extra term is of order Ω^4 – a much smaller correction, which can be dropped consistently with the weak-population assumption already stated above. Keeping only the leading term thus amounts to replacing ψ_g by $\psi_g^{(0)}$ inside the integral:

$$\delta a \approx \frac{2\mu}{\hbar^2} \int_0^\infty \hat{V}(r) [\psi_g^{(0)}(r)]^2 dr,$$

i.e. Eq. 3.26 evaluated on the background threshold state.

We now return to the energy-normalized wavefunction of Eq. 3.17, which is the one entering the Fermi Golden Rule below. The coupling $\hat{V}$ only acts at short distance, where the energy-normalized wavefunction is $A_k k (r - a_{\text{bg}})$ with $A_k = \sqrt{2\mu/(\pi\hbar^2 k)}$: it is equal to the zero-energy-normalized one multiplied by $A_k k$. Replacing the latter by the former therefore multiplies the integral by $1/(A_k k)^2 = \pi\hbar^2/(2\mu k)$, and

$$a(\Delta) = a_{\text{bg}} + \frac{\pi}{k} \langle \hat{V} \rangle = a_{\text{bg}} + \frac{\pi}{k} \frac{\left| \left\langle v_e \left| \frac{\hbar\hat{\Omega}(r)}{2} \right| \psi_g^{(0)} \right\rangle \right|^2}{E - E_{v_e} - \hbar\Delta + i\hbar\frac{\Gamma_{\text{mol}}}{2}}, \quad (3.28)$$

where $\psi_g^{(0)}$ is now energy-normalized.

Explicit form of l_{opt} via the Wigner threshold law.

Following Bohn and Julienne [113], one can define the stimulated rate Γ_{stim} from $|v_e\rangle$ to $|\psi_g^{(0)}\rangle$ by the Fermi Golden Rule as

$$\Gamma_{\text{stim}}(k) = \frac{2\pi}{\hbar} \left(\frac{2\pi I}{c} \right) d^2 |\langle v_e | \psi_g^{(0)} \rangle|^2 \quad (3.29)$$

with I the laser intensity, c the speed of light, and d the transition dipole moment; $\frac{\hbar\hat{\Omega}(r)}{2} = \vec{d} \cdot \vec{E}$ is the dipolar coupling to the laser field $\vec{E}$, so that $\left| \left\langle v_e \left| \frac{\hbar\hat{\Omega}}{2} \right| \psi_g^{(0)} \right\rangle \right|^2 = \left(\frac{2\pi I}{c} \right) d^2 \left| \langle v_e | \psi_g^{(0)} \rangle \right|^2$: the numerator of Eq. 3.28 is identified with $\hbar\Gamma_{\text{stim}}(k)/2\pi$. Γ_{stim} depends on the **Franck-Condon factor** $\langle v_e | \psi_g^{(0)} \rangle$, the overlap between the (energy-normalized) ground-state scattering wavefunction and the excited molecular state.

As shown after Eq. 3.17, the amplitude of $\psi_g^{(0)}$ at short distance, where it overlaps with v_e , scales as $\sqrt{k}$. Writing $\psi_g^{(0)}(r, E \rightarrow 0) \approx \sqrt{k} f_0(r)$ with f_0 a k -independent

function, one obtains $|\langle v_e | \psi_g^{(0)} \rangle|^2 = k |\langle v_e | f_0 \rangle|^2$, so that $\Gamma_{\text{stim}}(k) \propto k$. Writing this proportionality as [117]

$$\Gamma_{\text{stim}}(k) = 2k l_{\text{opt}} \Gamma_{\text{rad}} \quad (3.30)$$

where Γ_{rad} is the radiative linewidth of the molecular state (γ_m in [108]), equal to the total molecular linewidth Γ_{mol} (γ in [108]) when no non-radiative decay (e.g. predissociation) broadens the line, and $\Gamma_{\text{mol}} \geq \Gamma_{\text{rad}}$ otherwise. This relation defines the k -independent **optical length**

$$l_{\text{opt}} \equiv \frac{\pi}{\hbar \Gamma_{\text{rad}}} \left(\frac{2\pi I}{c} \right) d^2 |\langle v_e | f_0 \rangle|^2, \quad (3.31)$$

which is independent of collision energy and scales linearly with the laser intensity I , as required. Throughout this chapter, Γ_{rad} denotes the *radiative* (natural) width of the molecular state, $\Gamma_{\text{rad}} \simeq 2\Gamma_{\text{at}} = 2\pi \times 15$ kHz. The optical length is always defined with respect to this radiative width, even when other mechanisms broaden the observed line (see Sec. 3.2.3.4), so that it keeps its meaning of an intensity-dependent Franck–Condon factor. The normalization by Γ_{rad} rather than Γ_{mol} is natural since $\Gamma_{\text{rad}} \propto d^2$ is set by the same dipole moment as the coupling: d drops out (up to angular factors) and $l_{\text{opt}} \propto \lambda_a^3 I |\langle v_e | f_0 \rangle|^2 / c$, as in [108], whereas additional decay channels do not enter the coupling.

Compact resonant form.

Replacing the numerator of Eq. 3.28 by $\hbar \Gamma_{\text{stim}} / 2\pi = \hbar k l_{\text{opt}} \Gamma_{\text{rad}} / \pi$ (Eq. 3.30), the factors π and k cancel and, after division of the numerator and denominator by $\hbar$,

$$a - a_{\text{bg}} = \frac{l_{\text{opt}} \Gamma_{\text{rad}}}{(E - E_{v_e}) / \hbar - \Delta + i \frac{\Gamma_{\text{mol}}}{2}}.$$

The quantity $(E - E_{v_e}) / \hbar - \Delta$ is the detuning of the laser from the molecular resonance, including the collision energy, counted positively on the blue side. From now on, Δ denotes this detuning, $\Delta = \omega_L - \omega_{\text{res}}$, which is the opposite sign convention from Eq. 3.18 and the one used in the rest of this chapter. The scattering length then takes the compact resonant form

$$a(\Delta) = a_{\text{bg}} + \frac{l_{\text{opt}} \Gamma_{\text{rad}}}{\Delta + i \frac{\Gamma_{\text{mol}}}{2}} \quad (3.32)$$

where a_{bg} is the real background scattering length representing native elastic collisions [117]. This is the expression of [108] for an isolated resonance, $\alpha = a_{\text{bg}} + l_{\text{opt}} \gamma_m / (\Delta + i\gamma/2)$, with $\gamma_m \rightarrow \Gamma_{\text{rad}}$ and $\gamma \rightarrow \Gamma_{\text{mol}}$.

To distinguish how the laser alters the elastic interactions and the loss mechanisms, we multiply the resonant fraction in Eq. 3.32 by the complex conjugate of its denominator. This algebraic separation successfully decouples the scattering length into its real (dispersive) and imaginary (absorptive) components, $a(\Delta) = a_{\text{bg}} + \delta a(\Delta) - i a''(\Delta)$:

$$a(\Delta) = a_{\text{bg}} + \frac{l_{\text{opt}} \Gamma_{\text{rad}} \Delta}{\Delta^2 + \left(\frac{\Gamma_{\text{mol}}}{2}\right)^2} - i \frac{l_{\text{opt}} \Gamma_{\text{rad}} \frac{\Gamma_{\text{mol}}}{2}}{\Delta^2 + \left(\frac{\Gamma_{\text{mol}}}{2}\right)^2} \quad (3.33)$$

The real light-induced shift $\delta a(\Delta)$ governs the modification of the elastic properties of the gas, whereas the imaginary part $a''(\Delta)$ directly quantifies the rate of inelastic two-body losses.

3.1.3 General scaling laws for PA

To connect this microscopic scattering amplitude to observable experimental quantities, one must translate these individual collision events into a population decay that writes as [21]:

$$\frac{dN}{dt} = -\beta(T)N^2 - \Gamma N \quad (3.34)$$

where $\beta(T)$ is related to the two-body loss rate K_2 and Γ is the one-body loss rate that depends experimentally mostly on the vacuum quality.

In quantum scattering theory, the fundamental two-body loss rate coefficient $K_2(k)$ is directly driven by the absorptive (imaginary) part of the scattering length a'' derived from Eq. 3.26:

$$K_2(k) = \frac{16\pi\hbar}{m_{\text{at}}} a''(k, \Delta) \quad (3.35)$$

Combined to Eq. 3.15, $K_2(k) = \frac{4\hbar k}{m_{\text{at}}} \sigma_{\text{inel}}$. Eq. 3.35 holds in the linear regime $\delta_0 \ll 1$ used above. Keeping instead the exact $S_0 = (1 - ika)/(1 + ika)$, whose first-order expansion is the $1 - 2ika$ used above, one has $1 - |S_0|^2 = 4k a''/(1 + 2ka'' + k^2|a|^2)$, with a the complex scattering length of Eq. 3.32, so that Eq. 3.35 is multiplied by $(1 + 2ka'' + k^2|a|^2)^{-1}$. Inserting Eq. 3.32 (neglecting a_{bg}) and $\Gamma_{\text{stim}} = 2k l_{\text{opt}} \Gamma_{\text{rad}}$, this factor equals $[\Delta^2 + \Gamma_{\text{mol}}^2/4] / [\Delta^2 + \frac{1}{4}(\Gamma_{\text{mol}} + \Gamma_{\text{stim}})^2]$. One obtains the complete expression for the loss coefficient as a function of the collision wavevector k , equivalent to Eq. (9) of Blatt et al. [108] (written over the common denominator γ^2) with $\gamma_m \rightarrow \Gamma_{\text{rad}}$, $\gamma \rightarrow \Gamma_{\text{mol}}$ and $\mu = m_{\text{at}}/2$:

$$K_2(k) = \frac{4\pi\hbar}{m_{\text{at}} k} \frac{\Gamma_{\text{mol}} \Gamma_{\text{stim}}(k)}{\Delta^2 + \frac{1}{4}(\Gamma_{\text{mol}} + \Gamma_{\text{stim}}(k))^2} = \frac{8\pi\hbar}{m_{\text{at}}} \frac{l_{\text{opt}} \Gamma_{\text{rad}} \Gamma_{\text{mol}}}{\Delta^2 + \frac{1}{4}(\Gamma_{\text{mol}} + \Gamma_{\text{stim}}(k))^2} \quad (3.36)$$

where m_{at} is the atomic mass, $\Delta = \omega_{\text{L}} - \omega_{\text{mol}}$ is the laser detuning from the molecular resonance (including the collision energy, as in Eq. 3.32), and Γ_{mol} is the total molecular linewidth.

While K_2 characterizes the fundamental two-body collision process, the experimentally observed decay coefficient β depends on the spatial density distribution of the atomic cloud of width $\sigma_x, \sigma_y, \sigma_z$. The relation between K_2 and β is derived by

integrating the local loss over a Gaussian density profile:

$$\frac{dN(t, T)}{dt} = -K_2 \iiint n^2(x, y, z, t, T) d^3r \quad (3.37)$$

$$\text{with } n(x, y, z, t, T) = n_0 \exp \left(-\frac{x^2}{2\sigma_x(T)^2} - \frac{y^2}{2\sigma_y(T)^2} - \frac{z^2}{2\sigma_z(T)^2} \right) \quad (3.38)$$

$$\text{and } n_0 = \frac{N(t, T)}{(2\pi)^{3/2} \sigma_x(T) \sigma_y(T) \sigma_z(T)} \quad (3.39)$$

$$\frac{dN(t, T)}{dt} = -K_2 \frac{N(t, T)^2}{2\sqrt{2}V_e(T)} \equiv -\beta(T)N(t, T)^2 \quad (3.40)$$

$$\text{where } V_e(T) = (2\pi)^{3/2} \sigma_x(T) \sigma_y(T) \sigma_z(T) \quad (3.41)$$

$$\text{Then } \beta(T) = \frac{K_2}{2\sqrt{2}V_e(T)} \quad (3.42)$$

consistent with the definitions found in [4].

For low temperatures, we neglect the contribution of the wavevector since $k \rightarrow 0$. In this threshold regime, the light-induced linewidth becomes negligible compared to the total molecular width ($\Gamma_{\text{stim}}(k) \ll \Gamma_{\text{mol}}$)^a. On resonance ($\Delta = 0$), substituting Eq. 3.30 into Eq. 3.36 allows the wavevector k to cancel out from the prefactor, and the microscopic PA rate coefficient simplifies to:

$$K_2 = \frac{16h}{m_{\text{at}}} l_{\text{opt}} \frac{\Gamma_{\text{rad}}}{\Gamma_{\text{mol}}} \quad (3.43)$$

which is equal to $K_2 = \frac{16h}{m_{\text{at}}} l_{\text{opt}}$ when $\Gamma_{\text{rad}} = \Gamma_{\text{mol}}$. This fundamental result demonstrates that at low temperatures, the maximum photoassociation rate becomes independent of the temperature and scales purely linearly with the laser intensity I through its dependence on l_{opt} . Structurally, K_2 is maximized when the ground- and excited-state wavefunctions exhibit maximum overlap at the Condon radius. In a classical picture, this distance corresponds to the outer turning point of the bound molecular state, where the two atoms of the molecule spend most of their time, maximizing the transition probability.

Eq. 3.43 is the relation through which l_{opt} is defined experimentally. It is however only valid at the *peak* of an ideal line, whose width is the radiative width Γ_{rad} (i.e. $\Gamma_{\text{mol}} = \Gamma_{\text{rad}}$). Any additional broadening (laser noise, thermal distribution, inhomogeneous light shift, extra decay channels of the molecular state) lowers the peak value of K_2 while leaving l_{opt} unchanged; how to account for it is detailed in Sec. 3.2.3.4.

^aThis approximation can fail at high intensity, even in a gas at μK temperatures: since $\Gamma_{\text{stim}}/\Gamma_{\text{rad}} = 2kl_{\text{opt}}$ (Eq. 3.30) and $l_{\text{opt}} \propto I$, the stimulated width reaches the radiative width when $l_{\text{opt}} \sim 1/(2k)$. For Sr at $T = 2 \mu\text{K}$, $k = \sqrt{2\mu k_B T}/\hbar \simeq 10^{-3} a_0^{-1}$, i.e. $l_{\text{opt}} \sim 500 a_0$, which is reached at $I \sim 50 \text{ mW/cm}^2$ for the strongest ^{88}Sr lines ($l_{\text{opt}}/I \sim 10^4 a_0/(\text{W/cm}^2)$); the loss rate then saturates. For the ^{87}Sr lines of this work ($l_{\text{opt}} \lesssim 100 a_0$), $\Gamma_{\text{stim}} \lesssim 0.2 \Gamma_{\text{rad}}$, which is negligible compared with their total width.

Finally, by combining Eq. 3.42 and the low-temperature limit of K_2 , we obtain the total macroscopic evolution equation of the trapped population:

$$\frac{dN}{dt} = -\frac{4\sqrt{2}h}{V_e m_{\text{at}}} l_{\text{opt}} N^2 - \Gamma N \quad (3.44)$$

which highlights that the global loss process is driven by the tunable optical length l_{opt} and enhanced by the peak atomic density $n_0 \propto 1/V_e$. As Eq. 3.43, this expression assumes an unbroadened line probed at its center; in general, the two-body term must be written with the measured loss coefficient K_2 .

3.1.4 Handling multiple molecular potentials

So far, the photoassociation (PA) process has been presented assuming a simplified picture with a single excited-state molecular potential U_e . However, the molecules created via PA exhibit a rich internal structure, making the identification of accessible excited states essential for predicting and optimizing the loss rate coefficient $\beta(T)$.

3.1.4.1 Angular momentum coupling and state enumeration

Spin-orbit coupling plays a dominant role in shaping the excited molecular structure near the $^1S_0 - ^3P_1$ intercombination asymptote. The total angular momentum of the molecule is defined by $T = R + F = R + I + J$, where $F = f_1 + f_2$ is the coupled atomic angular momentum, $J = j_1 + j_2$ is the total electronic angular momentum, and R is the rotational angular momentum.

Along the internuclear axis, the projection of the electronic angular momentum is quantified by:

$$\Omega = \Lambda + \Sigma \quad (3.45)$$

where Λ and Σ represent the internuclear projections of the orbital and spin angular momenta, respectively. Because spin-orbit interaction is strong relative to rotational coupling at short to long ranges, the molecule is appropriately described within Hund's case (c) framework (see Fig. 3.3), where Ω remains a good quantum number.

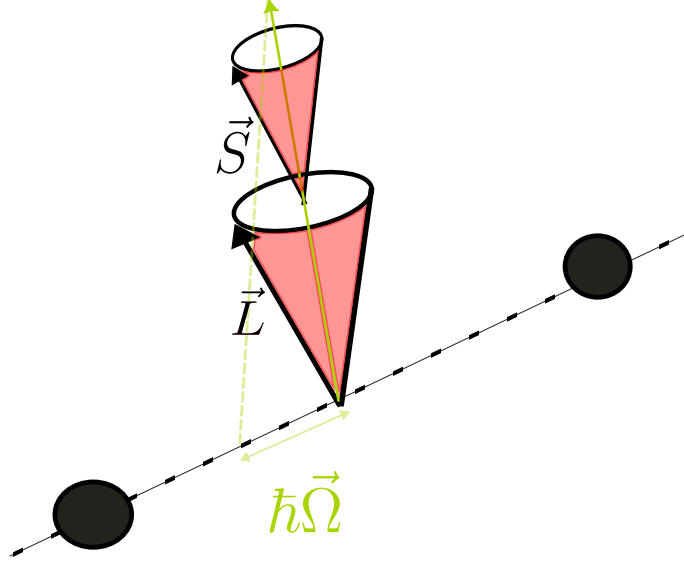


Figure 3.3: Hund’s case (c) coupling scheme, where strong spin-orbit interaction couples the total electronic angular momentum to the internuclear axis with projection $\hbar\Omega$. In most ultracold photoassociation scenarios, the rotational quantum number is low ($R = 0, 1$).

3.1.4.2 Effective hamiltonian and potential modeling

Taking into account the hyperfine-induced mixing between g and u electronic manifolds ($0_u, 1_u, 0_g, 1_g$) at long range—analogue to observations in ^{173}Yb [4]—we estimate a total of 981 accessible (R_f, T^f) molecular configurations. This massive density of states illustrates the extreme complexity of the excited molecular spectrum to be mapped experimentally.

The high density of states expected for ^{87}Sr makes spectral assignment particularly challenging. To move beyond a statistical estimation and accurately predict the energy levels of these 981 configurations, precise modeling of the molecular interaction is required. This involves describing the coupling between the rotation of the nuclear frame, the electronic degrees of freedom, and the hyperfine structure.

Following the molecular picture introduced in Section 3.1.1, the energy of these states is governed by the total hamiltonian $\hat{H}$, which for a Sr-Sr system of reduced mass μ_N can be expressed as:

$$\hat{H} = \frac{p_r^2}{2\mu_N} + \frac{\hbar^2}{2\mu_N r^2} R(R+1) + U_{BO} + \hat{H}_{\text{hf}} \quad (3.46)$$

the first and second terms representing the kinetic energy and radial energy of the molecule. Higher-order electronic interactions have been neglected depending on the relevant Hund’s case. For ^{87}Sr , the Born-Oppenheimer potential $\hat{U}_{BO}$ is approximated using a long-range expansion similar to ^{173}Yb :

$$U_{BO}(r) = -\frac{C_6}{r^6} \left(1 - \frac{\sigma^6}{r^6} \right) - s \frac{C_3^\Omega}{r^3} \quad (3.47)$$

where $s = +1$ and $s = -1$ correspond to g and u potentials, respectively. C_6 and C_3^Ω are Van der Waals and dipole-dipole coefficients [118], Ω still standing for projection

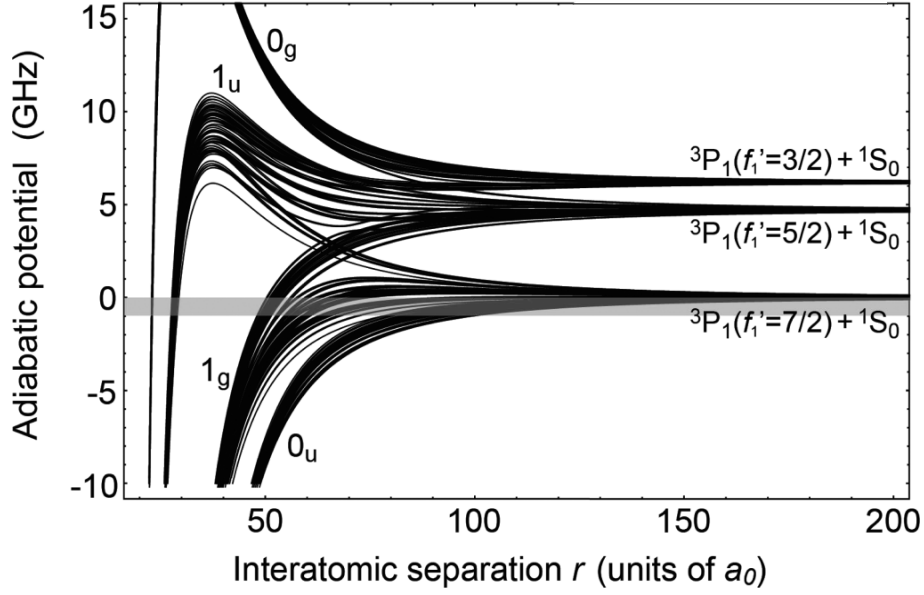


Figure 3.4: Density of molecular potentials for ^{173}Yb with three hyperfine states, adapted from [4], giving rise to numerous triplets of quantum numbers (T, F, R). The spectral lines are identified as $\Omega_{u/g}$, where Ω is the total orbital momentum projection of the molecule and u/g denotes the electronic parity (see Fig. ??).

of angular momentum, and σ is a free parameter to adjust with the experimental results.

The atomic hyperfine Hamiltonian $\hat{H}_{\text{hf}}$ acts solely on the $^3\text{P}_1$ atom (with nuclear spin $i_1 = 9/2$ and electronic angular momentum $j_1 = 1$), as the ground-state $^1\text{S}_0$ atom has $j_2 = 0$ and thus no hyperfine interaction:

$$\hat{H}_{\text{hf}} = A(\mathbf{i}_1 \cdot \mathbf{j}_1) + B \frac{3(\mathbf{i}_1 \cdot \mathbf{j}_1)^2 + \frac{3}{2}(\mathbf{i}_1 \cdot \mathbf{j}_1) - i_1(i_1 + 1)j_1(j_1 + 1)}{2i_1(2i_1 - 1)2j_1(2j_1 - 1)} \quad (3.48)$$

where A and B are the magnetic dipole and electric quadrupole hyperfine constants, respectively.

Among these potential curves, approximately 80 bound internal states were experimentally identified for ^{173}Yb [4]. Atom-loss spectra, recorded as a function of PA laser detuning, reveal the position of bound states relative to the atomic resonance $|^1\text{S}_0, F = 5/2\rangle \rightarrow |^3\text{P}_1, F = 7/2\rangle$. Near this threshold, atoms are illuminated for 30 ms at low intensity ($I = 37 \text{ mW/cm}^2$) to avoid power broadening, resolving bound states closest to the dissociation limit. At larger red detunings, higher laser power and optimized exposure times are used to maximize signal depth.

Theoretically, these experimental loss lines are assigned by computing the eigenvalues of the effective Hamiltonian and adjusting parameters such as C_n coefficients to match observed vibrational levels [118]. Several semi-classical tools aid this assignment: the WKB approximation describes the near-threshold bound states, while LeRoy-Bernstein scaling predicts the spacing of consecutive vibrational lines near the dissociation limit without requiring detailed short-range potential data. Finally,

mass scaling across strontium isotopes [15] serves as a robust benchmark to cross-check these energy assignments.

3.1.5 Vibrational states estimation theories

The definition of the molecular potentials in the previous section is crucial to estimate the bound states of the system. In this subsection one presents the semi-classical methods used to estimate the vibrational energies of a diatomic system, which are particularly useful for predicting the positions of high-lying vibrational states near the dissociation limit.

3.1.5.1 WKB approximation

The Wentzel-Kramers-Brillouin (WKB) approximation is a semi-classical method used to estimate the vibrational energies of a diatomic system. Theoretical potentials alone are not accurate enough to predict ab initio the position of a given level; however, once the position of one bound state has been measured experimentally, WKB-based methods allow the positions of the other levels to be predicted.

In this approach, the wavefunction is assumed to take the form of a modified plane wave of wavevector $k(x)$, $\psi(x) = \frac{C}{\sqrt{k(x)}} e^{i \int_{x_0}^x k(x') dx'}$ [119]. These waves are solutions to the one-dimensional Schrödinger equation for two particles with reduced mass μ and eigenenergy E :

$$-\frac{\hbar^2}{2\mu} \frac{d^2\psi}{dx^2} + U(x)\psi = E\psi \quad (3.49)$$

This leads to the expression of the local wavevector $k(x)$:

$$k(x) = \frac{p(x)}{\hbar} = \frac{\sqrt{2\mu(E - U(x))}}{\hbar} \quad (3.50)$$

The validity of this approximation is governed by the condition $\frac{1}{2\pi} \frac{d\lambda}{dx} \ll 1$, implying that the local de Broglie wavelength λ must vary slowly compared to the scale of the atoms' motion. This can be explicitly rewritten in terms of the potential gradient:

$$\left| \frac{\mu \hbar \frac{dU}{dx}}{[2\mu(E - U(x))]^{3/2}} \right| \ll 1 \quad (3.51)$$

This expression shows where the semi-classical approach fails: near the classical turning points x_1 and x_2 of Fig. 3.5, where $E \approx U(x)$, the kinetic energy vanishes and the local wavelength diverges, so that the condition of Eq. 3.51 is violated. Around these points, connection formulas, typically using Airy functions, are required to match the solutions across the turning points.

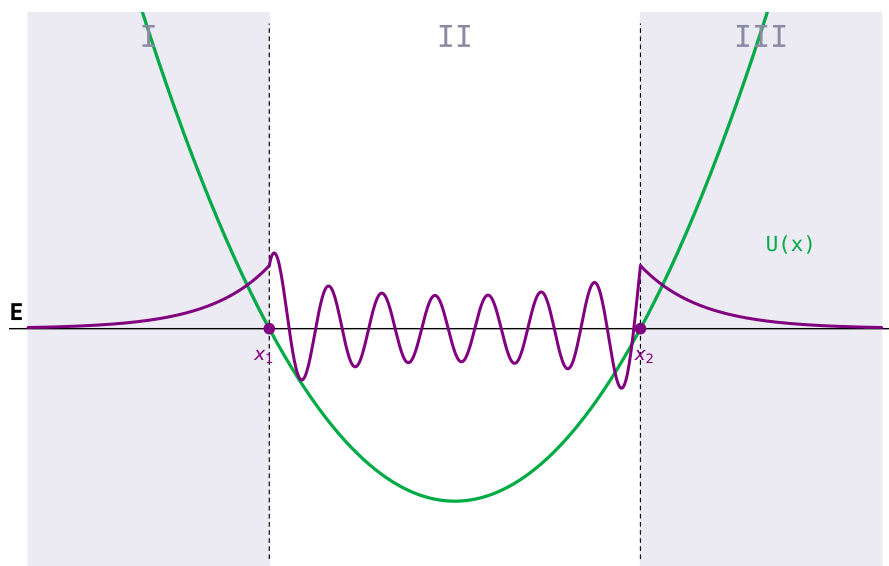


Figure 3.5: Semi-classical (WKB) approximation in a simplified potential well $U(x)$. The energy level E intersects the potential at the turning points x_1 and x_2 . In the classically allowed Region II, the wavefunction oscillates with an amplitude proportional to $1/\sqrt{k(x)}$ where in the forbidden Regions I and III, the wavefunction is evanescent.

At positions $x = x_1$ and x_2 , the momentum reaches zero and becomes imaginary as the energy level E crosses the potential $U(x)$. In the classically forbidden regions I and III, the wavefunction is evanescent and characterized by the decay constant $\kappa(x) = \sqrt{2\mu(U(x) - E)}/\hbar$. Around the turning points, it is standard to linearize the potential to connect the oscillatory and evanescent solutions.

To ensure a physical stationary state, the phase accumulated by the wave as it reflects between the barriers at x_1 and x_2 must satisfy a quantization condition. By ensuring the continuity of the solutions over $[0, +\infty[$, the phase integral must follow:

$$\int_{x_1}^{x_2} k(x)dx = \left(n + \frac{1}{2}\right) \pi \quad (3.52)$$

This formula, known as the Bohr-Sommerfeld quantization condition, enables the identification of the discrete energy eigenvalues E_n that satisfy the integral. these values correspond to the vibrational states of the molecular potential.

3.1.5.2 Leroy-Bernstein scaling

The LeRoy-Bernstein scaling law [120] builds upon the WKB approximation to identify the position of the high-lying vibrational states near the dissociation limit. This model is particularly powerful as it relies solely on the long-range behavior of the potential, bypassing the need for a detailed description of the complex short-range interactions.

Near the dissociation limit D_e , occurring at large internuclear distances r , the potential can be approximated by its leading long-range term:

$$U(r) \approx D_e - \frac{C_n}{r^n} \quad (3.53)$$

where C_n is the long-range coefficient (typically $n = 6$ for Van der Waals or $n = 3$ for resonant dipole-dipole interactions). For a vibrational state ν with energy E_ν , we define the binding energy as $\epsilon_\nu = D_e - E_\nu$. Substituting this into the Bohr-Sommerfeld quantization condition (Eq. 3.52), we seek solutions for:

$$\int_{r_{min}}^{r_{max}} \frac{\sqrt{2\mu}}{\hbar} \sqrt{\frac{C_n}{r^n} - \epsilon_\nu} dr = (\nu_D - \nu)\pi \quad (3.54)$$

where ν_D is a non-integer parameter representing the effective vibrational index at the dissociation limit. After integration, the relationship between the binding energy and the vibrational quantum number is given by:

$$\epsilon_\nu = \left[(\nu_D - \nu) \frac{(n-2) \hbar \sqrt{\pi}}{\sqrt{2\mu} C_n^{1/n}} \frac{\Gamma(1+1/n)}{\Gamma(1/n+1/2)} \right]^{\frac{2n}{n-2}} \quad (3.55)$$

Here, Γ denotes the Euler Gamma function. This expression is often rearranged to show the scaling of the vibrational spacing:

$$E_\nu = D_e - [(\nu_D - \nu) H_n]^{\frac{2n}{n-2}} \quad (3.56)$$

where H_n is defined as:

$$H_n = \frac{(n-2) \hbar \sqrt{\pi}}{\sqrt{2\mu} C_n^{1/n}} \frac{\Gamma(1+1/n)}{\Gamma(1/n+1/2)} \quad (3.57)$$

This notation highlights that the binding energy follows a power law relative to the "distance" $\nu_D - \nu$ from the dissociation limit, where the exponent is purely determined by the nature of the long-range interaction.

3.1.5.3 Mass scaling and isotopic shifts

Mass scaling is a powerful method used to predict the vibrational energy levels of different isotopes from a single known potential. By adjusting the reduced mass μ corresponding to the isotope considered, one can determine the phase constraint on the vibrational states. While this approach neglects the hyperfine structure, it provides a reliable estimation of the spectral range where vibrational resonances are expected.

In particular, ^{88}Sr is a bosonic isotope with a nuclear spin $I = 0$, meaning it possesses no hyperfine structure. Its photoassociation spectrum is significantly simpler than that of fermionic ^{173}Yb (see Fig. 3.4), involving only two accessible potentials in the first excited molecular state: 0_u and 1_u . Based on the results presented in [5], the potential can be modeled as:

$$U(r) \approx -\frac{C_6}{r^6} - \frac{C_3 \Lambda}{r^3} + \frac{\hbar^2 R(R+1)}{2\mu r^2} \quad (3.58)$$

for the two molecular symmetries $\Lambda = 0, 1$. Since the Born-Oppenheimer approximation remains valid across the strontium isotopes (^{84}Sr , ^{86}Sr , ^{88}Sr), the electronic

potential $V_{BO}(r)$ is assumed to be identical for all species. Consequently, only the reduced mass μ in the WKB phase integral varies:

$$\int_{x_1}^{x_2} \frac{\sqrt{2\mu(E - U(x))}}{\hbar} dx = \left(n + \frac{1}{2}\right) \pi \quad (3.59)$$

The experimental positions of the first vibrational states for the 0_u^+ potential in ^{84}Sr , ^{86}Sr , and ^{88}Sr are shown as red crosses in Fig. 3.6. These data points allow for the refinement of the potential shape $U(r)$. The fitted functions (blue lines) then enable the prediction of the frequency ranges where ^{87}Sr resonances should occur. Note that mass scaling applies to the ground-state potential: Martinez de Escobar et al. determined the scattering lengths of all isotopic combinations, including ^{87}Sr - ^{87}Sr [21], but they give none for the excited states. For ^{87}Sr , the hyperfine structure of the excited state prevents a direct mass scaling of the ^{88}Sr levels. Notably, discontinuities observed for the ^{84}Sr isotope correspond to avoided crossings resulting from couplings with the potential correlated to the $^1S_0 + ^1D_2$ asymptote.

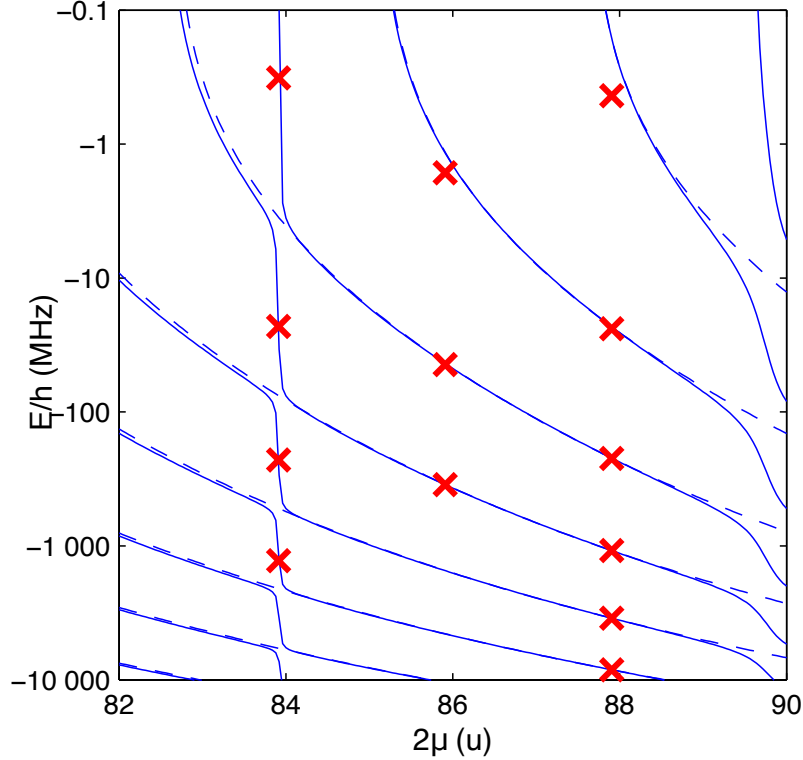


Figure 3.6: This graph - extracted from [15]- confronts the experimental internal states in red crosses to their theoretical predictions for different reduced mass of strontium isotopes. The theoretical model guarantees approximated scales for molecular lines of ^{87}Sr but not precisely because this model neglects the hyperfine interaction of the species.

In conclusion, combining semi-classical WKB theory with LeRoy-Bernstein scaling offers a practical framework to estimate vibrational spacings near the intercombination threshold. This approach serves as a reliable guide for bosonic isotopes like

^{88}Sr , presented in the next section. However, it cannot account for the ^{87}Sr photoassociation spectra, where instead of the dense manifold of lines, only a few narrow resonances are experimentally observed.

In this first part of the chapter, I have presented the theoretical picture of the photoassociation process where the following introduced material will help to understand the experimental results presented in the next section:

- I demonstrated the relation of the two-body collision rate coefficient K_2 with the scattering length, which includes the parameter l_{opt} characterizing the strength of the inelastic loss process.
- I presented the different molecular potentials and their symmetries, which should help for future understanding of the results to identify the molecular lines of ^{87}Sr .
- I presented two ways of estimating the position of the molecular states: the WKB approximation to determine the position of the vibrational states that could be complemented by the LeRoy-Bernstein scaling. The second one is the mass scaling that predicts – without considering the hyperfine structure that changes the position of these lines – an estimation of the energy range where the molecular lines of ^{87}Sr should be found.

In the following section I present the experimental protocol used on the ^{88}Sr species that enables to calibrate the procedure for fermionic molecular lines research.

3.2 Spectroscopy protocol and calibration of the photoassociation laser

3.2.1 Gas preparation and isotope specificities

I now present the experiments carried out to identify photoassociation lines of ^{87}Sr , starting with the molecular lines of ^{88}Sr , which are well documented in the literature. As explained in the introduction, the PA spectrum of ^{87}Sr has not been explored yet, which is surprising since many molecular spectroscopy results have been published for the other strontium isotopes.

The photoassociation (PA) rate strongly depends on the density, as detailed in Sec. 3.1.3. The experimental approach depends on the quantum statistics of the isotope:

- **Fermions (^{87}Sr):** high densities are reached by optimizing the evaporative cooling ramps once the atoms are loaded into the crossed Optical Dipole Trap (ODT).
- **Bosons (^{88}Sr):** the near-zero scattering length of ^{88}Sr prevents efficient evaporative cooling, as it suppresses the elastic collisions needed for thermalization. In this case, the final density depends almost only on the efficiency of the previous cooling and trapping stages (e.g. the MOT).

Once a dense cloud is obtained, the PA process is started in a magnetic field $B = 1.6$ G, chosen to maximize the π optical transitions, by shining on the atoms a dedicated laser beam presented in detail below.

Changing its diffraction order allows switching from ^{88}Sr to ^{87}Sr (see Sec. 3.3);

- $f_{11/2}^{RF}$ is the frequency of the "AOM 11/2", used for optical pumping and for the beat note.

The frequency of the beam sent to the atoms is then given by the following formula:

$$f_{PA} = f_{MOGlabs} + f_{beating}^{RF} - f_{Raman}^{RF} \pm 2 \times f_{slave1}^{RF} - 2 \times f_{11/2}^{RF} \quad (3.60)$$

The MOGlabs lock cannot be used above $f_{beating}^{RF} = 500$ MHz. To address more deeply bound vibrational states, one can change the other frequencies (within their RF limits) and/or invert the diffraction order of the "Raman AOM", which shifts the PA beam by about ± 260 MHz.

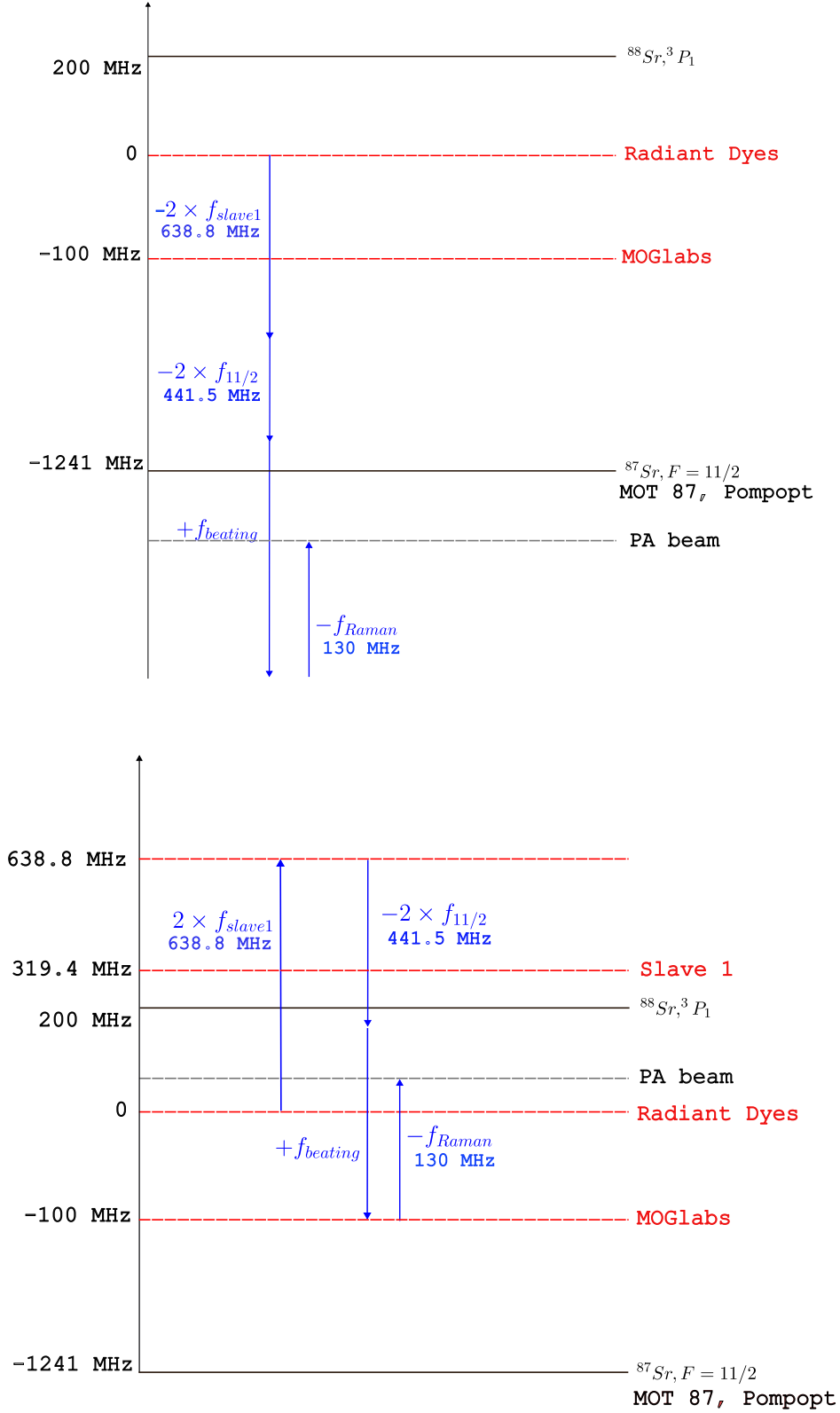


Figure 3.8: Frequency schemes for photoassociation (PA) depending on the targeted isotope: (top) scheme for ^{87}Sr and (bottom) scheme for ^{88}Sr . Switching the diffraction order of 'AOM Slave 1' allows switching between isotopes, with only a small realignment of the beam injection after the AOM.

3.2.2.2 Spectral imperfections from the locking scheme

The beating spectrum of the PA laser locking is presented in Fig. 3.9 at a resolution bandwidth $\text{RBW} = 10 \text{ kHz}$, with a video bandwidth $\text{VBW} = 300 \text{ Hz}$. Since the expected molecular linewidth of ^{87}Sr is $\Gamma_{\text{rad}}/2\pi \simeq 15 \text{ kHz}$, this signal gives the spectrum seen by the molecules. Fitting the beat note with three Lorentzians – a central peak and two symmetric sidebands – gives a central peak of $\text{FWHM} \simeq 13 \text{ kHz}$, close to the RBW of the measurement, and sidebands at $\pm 70 \text{ kHz}$ with a much larger FWHM of $\simeq 80 \text{ kHz}$. Since the vertical axis of the spectrum analyzer is a voltage, the optical power is proportional to its square: integrating the squared signal, about two thirds of the optical power ($\simeq 65\text{--}67\%$) addresses the molecules at the center frequency of the beat note, while about one third is spread over the sidebands. This limits the power at which the atoms can be photoassociated efficiently on resonance. Since the measured width of the central peak is comparable to the RBW itself, this measurement cannot resolve the intrinsic linewidth of the laser – it only gives an upper bound. In any case, these broad sidebands broaden the effective linewidth of the transition addressed by the PA beam, whether atomic or molecular depending on the laser frequency, well beyond Γ_{rad} .

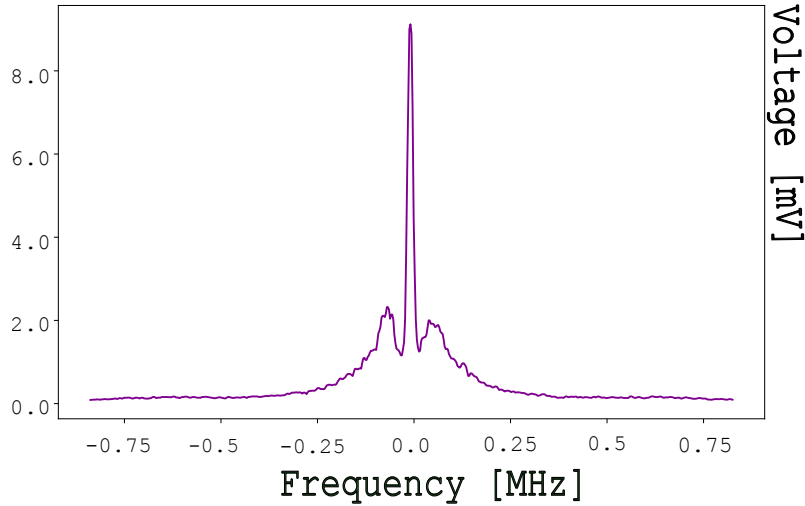


Figure 3.9: Beat note between the MOGlabs laser and the "AOM 11/2" light, measured with the spectrum analyzer at $\text{RBW} = 10 \text{ kHz}$ and $\text{VBW} = 300 \text{ Hz}$. A three-Lorentzian fit (central peak + two sidebands at $\pm 70 \text{ kHz}$) gives sidebands broader than the central peak, which carry about one third of the total optical power (the vertical axis is a voltage, and the power is proportional to its square).

3.2.2.3 Spectral imperfections from the filter cavity system

The MOGlabs laser passes through a Fabry-Perot cavity (see Sec. 1.3.3) stabilized by a Pound-Drever-Hall (PDH) lock. In this scheme, an EOM modulates the phase of the incident electric field, of frequency f_0 , at $f_M = 40.48 \text{ MHz}$ (see [33]). For a modulation depth β , the phase modulation creates two sidebands around the carrier (to first order):

$$E_{\text{inc}}^{\text{mod}} = E_0 \left(J_0(\beta) e^{i2\pi f_0 t} + J_1(\beta) e^{i2\pi(f_0 + f_M)t} - J_1(\beta) e^{i2\pi(f_0 - f_M)t} \right) \quad (3.61)$$

The first term, with the 0th-order Bessel function $J_0(\beta)$, is the carrier at frequency f_0 , with an effective amplitude $E_0 J_0(\beta)$. The second and third terms are two sidebands at $f_0 \pm f_M$, with an amplitude given by the first-order Bessel function $J_1(\beta)$. Figure 3.10 shows how the power is shared between the carrier and the sidebands as a function of β . Experimentally, the modulation depth $\beta = 1$ gives about three times more power in the carrier than in each sideband ($J_0^2(\beta)/J_1^2(\beta) \simeq 3.0$), which is a good compromise for the lock; β is directly tuned by the amplitude of the RF signal.

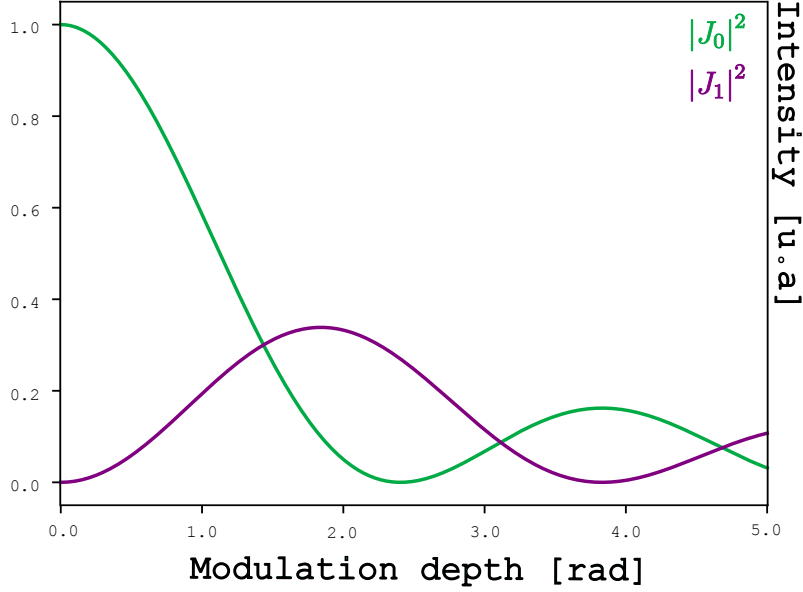


Figure 3.10: Squared modulus of the 0th- and 1st-order Bessel functions J_n , which set the power distribution between the carrier and the PDH sidebands.

The sidebands are essential to the PDH scheme itself: their reflection on the cavity, interfering with the reflected carrier, produces the dispersive error signal used for the lock (see Sec. 1.3.3). Since they are detuned by f_M from the carrier resonance, they are in principle mostly reflected by the cavity rather than transmitted to the atoms. However, this rejection is not perfect, since the cavity linewidth is finite: the sidebands still have a small residual transmission set by the Lorentzian lineshape of the cavity,

$$\frac{T(f_M)}{T(0)} = \frac{1}{1 + \left(\frac{2f_M}{\delta\nu}\right)^2}. \quad (3.62)$$

Using the cavity's measured finesse ($\mathcal{F} = 310 \pm 15$) and FSR (3 GHz), i.e. a linewidth $\delta\nu = \text{FSR}/\mathcal{F} \simeq 10$ MHz, and the PDH modulation frequency $f_M = 40.48$ MHz, this residual transmission amounts to $T(f_M)/T(0) \simeq 1.4\%$. Combined with the sideband-to-carrier power ratio set by the modulation depth ($J_1^2(\beta)/J_0^2(\beta) \simeq 0.33$ at $\beta = 1$), the power reaching the atoms in each sideband is expected to be $\simeq 0.5\%$ of the power reaching them in the carrier. Although small, this residual power is enough to drive a detectable one-body loss signal, displaced from the main photoassociation resonance by $\pm f_M$, as we now demonstrate experimentally.

We measured the effect of these sidebands on a gas of 500 000 atoms, evaporated and

recompressed to a temperature $T = 4.1 \mu\text{K}$, in the two spin components $m_F = -9/2$ and $+9/2$ prepared by optical pumping. The gas was illuminated for 2 s at an intensity $I = 385 \text{ mW/cm}^2$ with the MOGLabs ECDL. As sketched in Fig. 3.11, the sidebands produce "ghost" resonances when the carrier is detuned by $\pm f_M = \pm 40.48 \text{ MHz}$ from the atomic transition ($f'_0 = f_0 \pm f_M$), leading to parasitic one-body losses. In the purple case of Fig. 3.11, the lower sideband at $f'_0 - f_M$ addresses the atomic transition at f_0 , with a smaller amplitude than the carrier (black) when the carrier itself is on resonance.

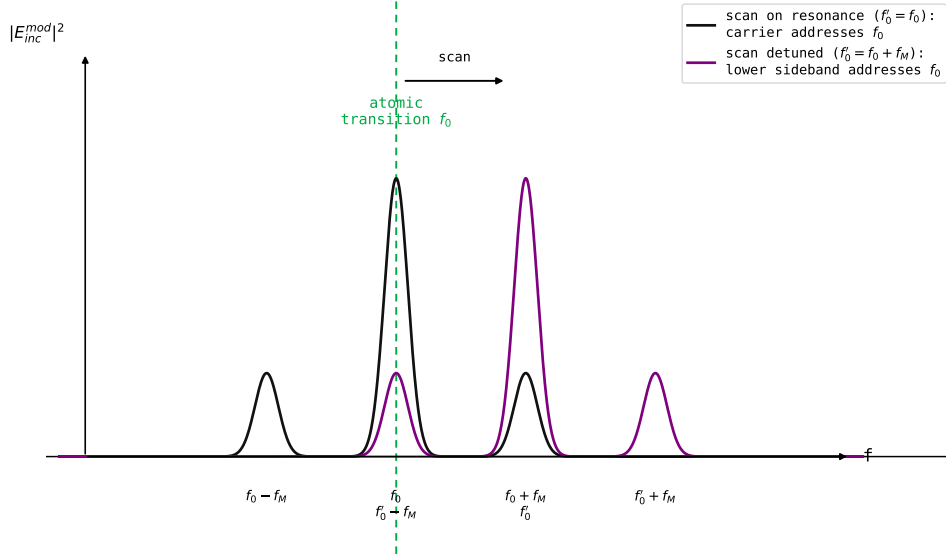


Figure 3.11: Scheme of the PDH sidebands addressing the atomic resonance, shown at two moments of the laser frequency scan. The atomic transition is at the fixed frequency f_0 (green dashed line). In black: the carrier is scanned onto resonance ($f'_0 = f_0$), giving the true signal. In purple: the carrier is detuned to $f'_0 = f_0 + f_M$, so that it is itself off resonance, but its lower sideband coincides with f_0 and drives a parasitic ("ghost") loss resonance. During the scan, this happens again whenever $f'_0 = f_0 \pm f_M$, giving satellite peaks such as the one of Fig. 3.12.

Figure 3.12 shows the resulting ghost resonance, measured on a gas of ^{87}Sr , $F = 9/2$, of density $n = 13 \times 10^{12} \text{ atoms/cm}^3$, addressed here by the resonant lower sideband, as sketched in Fig. 3.11. The carrier and the upper sideband do not appear in this scan, as they remain far from resonance at these frequencies. The green dashed line marks the expected position of the ghost resonance, $|\Delta| = f_M$, close to the observed loss feature (the deepest point is at $|\Delta| \simeq 40.6 \text{ MHz}$).

This resonance is not a single, clean line but shows several overlapping peaks, which we attribute to two possible causes. First, the PA laser polarization may not be perfectly π : a small $\sigma^\pm$ component would drive transitions to neighboring m_F states, each addressed at a slightly different frequency and all overlapping within the scan range. Second, several lines shifted by the Zeeman effect and by the tensor light shift (TLS) may contribute at the same time; this is supported by the same measurement repeated at different magnetic fields, where the splitting between the peaks changed with B . These two explanations can both be true, and separating their contributions would require dedicated measurements.

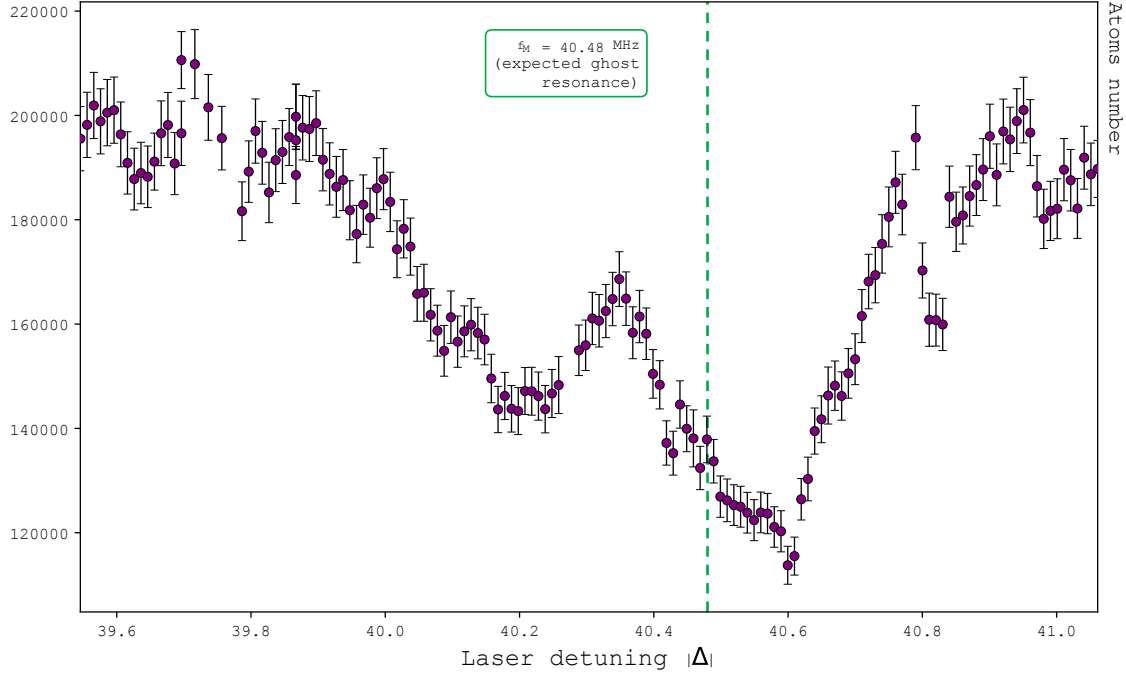


Figure 3.12: Atomic loss due to the FP cavity’s PDH sideband addressing the $^1S_0 \rightarrow ^3P_1$ line at $\Delta = f_0 - f_M$, with f_M the modulation frequency of the PDH EOM. The dashed line marks the expected detuning of the ghost resonance, $|\Delta| = f_M = 40.48$ MHz.

Lowering the modulation depth β reduced the amplitude of this signal, but did not remove it completely: a weak molecular line could still be hidden in this signal. The other laser (injected from ‘AOM 11/2’) could be used to explore this frequency range.

Conclusion on the characterization of the PA spectroscopic protocol

This section described how the PA laser frequency is set and checked. The characterization of the laser system gave two important results:

- **Spectral distribution of the power:** the beat note shows that about one third of the laser power is in unwanted frequency components. This contributes to the observed broadening of both atomic and molecular lines.
- **Parasitic resonances from the modulation:** a modulation of the laser creates sidebands that can drive unexpected atomic or molecular transitions. Here, the sidebands at $\pm f_M = \pm 40.48$ MHz from the carrier can lead to a misinterpretation of the atom loss signals. In our setup, this modulation comes from the electro-optic modulator (EOM) of the Pound-Drever-Hall (PDH) lock, but similar effects can appear with any optical element driven by an RF source, such as acousto-optic modulators (AOMs).

3.2.3 Atomic and molecular lines calibration

3.2.3.1 General experimental framework and baseline parameters

Before scanning for molecular resonances, the PA laser frequency is referenced to the atomic transition, by recording the atom loss as a function of the detuning around the atomic resonance (Fig. 3.13, top). This loss is a one-body process: the atoms scatter photons and are heated out of the trap.

The molecular lines, in contrast, induce two-body losses, $\dot{n} = -K_2 n^2$, so that $1/n(t) - 1/n_0 = K_2 t$, with a characteristic loss time, after which the density is half the initial density,

$$\tau = \frac{1}{K_2 n_0}. \quad (3.63)$$

Once the MOGlabs laser is locked to this reference, f_{beating}^{RF} is scanned to search for the least-bound vibrational states; the two lines studied here are shown in Fig. 3.13.

3.2.3.2 Fitting model for molecular resonances

Each data point of the presented atomic and molecular lines is extracted by fitting the absorption images of the atomic cloud with a 2D Gaussian function of amplitude A and widths σ_x, σ_y . The atom number N is obtained from the integral of the fit, $N \propto A \sigma_x \sigma_y$. The relative statistical uncertainty on N was estimated from the diagonal elements of the fit covariance matrix,

$$\frac{\delta N}{N} \simeq \sqrt{\left(\frac{\delta A}{A}\right)^2 + \left(\frac{\delta \sigma_x}{\sigma_x}\right)^2 + \left(\frac{\delta \sigma_y}{\sigma_y}\right)^2}, \quad (3.64)$$

where δA , $\delta \sigma_x$, and $\delta \sigma_y$ are the standard errors on the amplitude and on the spatial widths along x and y . This expression neglects the off-diagonal covariance terms between the fit parameters (e.g. $\text{Cov}(A, \sigma_x)$), which were not retained from the original fitting routine. In a Gaussian fit, the amplitude and the widths are anti-correlated, since the derivatives of the model with respect to A and to $\sigma_{x,y}$ have the same sign over the whole image: a slightly broader profile with a slightly lower amplitude yields the same integral. These neglected terms are therefore negative, and the expression above only gives a slightly larger, i.e. conservative, uncertainty on N ; a simulation of our fit gives a factor $\simeq 1.6$ between the two estimates. This has no practical consequence, since this fit uncertainty ($\simeq 0.3\text{--}0.5\%$) is negligible compared to the shot-to-shot fluctuation of the atom number included below.

To account for shot-to-shot experimental fluctuations, the relative RMS fluctuation of the atom number, σ_{shot} , measured over ten consecutive realizations, is added in quadrature to the relative fit uncertainty $\delta N/N$ defined above. The total uncertainty on each atom number is thus

$$\sigma_N^{\text{tot}} = N \sqrt{\left(\frac{\delta N}{N}\right)^2 + \sigma_{\text{shot}}^2}. \quad (3.65)$$

The temperature of the cloud is obtained from the same absorption images, by time of flight (TOF), assuming an isotropic thermal velocity distribution. During the

TOF, the cloud expands ballistically, and its Gaussian width σ_y along the vertical axis evolves as

$$\sigma_y^2 = \sigma_{0,y}^2 + v_{\text{th}}^2 \text{TOF}^2, \quad (3.66)$$

where $\sigma_{0,y}$ is the *in-situ* RMS width of the gas in the optical dipole trap (ODT) and v_{th} is the thermal velocity. For a Maxwell-Boltzmann velocity distribution, the equipartition theorem gives $\frac{1}{2}m_{\text{at}}v_{\text{th}}^2 = \frac{1}{2}k_B T$ along each direction, so that the temperature is obtained from the width σ_y measured on the image:

$$T = \frac{m_{\text{at}}(\sigma_y^2 - \sigma_{0,y}^2)}{k_B \text{TOF}^2}. \quad (3.67)$$

Assuming independent uncertainties on the in-situ size $\sigma_{0,y}$ and on the expanded size σ_y , error propagation gives

$$\delta T = \frac{2m_{\text{at}}}{k_B \text{TOF}^2} \sqrt{(\sigma_y \delta \sigma_y)^2 + (\sigma_{0,y} \delta \sigma_{0,y})^2}. \quad (3.68)$$

The atom numbers measured at different detunings form a spectrum, which is then fitted. Each atomic and molecular resonance is fitted with a Lorentzian lineshape,

$$f(x) = y_0 + \frac{A}{1 + \left(\frac{x-x_0}{\Gamma/2}\right)^2}, \quad (3.69)$$

centered at x_0 , with a full width at half maximum (FWHM) Γ , a peak amplitude A (negative for a loss) and an offset y_0 . For the normalized spectra of ^{87}Sr (Sec. 3.3.2), the *depth* of a line is the peak amplitude divided by the offset, $|A|/y_0$.

The resonances exhibit an effective linewidth Γ_{tot} resulting from several physical and technical broadening mechanisms. Some of them can be estimated from first principles, while others – such as the spectral impurities of the laser, an inhomogeneous light shift across the cloud, or additional decay channels of the molecular state like predissociation – cannot be given a simple expression and are only accessible through the measured width. We therefore do not attempt to model Γ_{tot} : it is measured on each spectrum and used as such in the determination of l_{opt} (Sec. 3.2.3.4). The known contributions are nevertheless useful as orders of magnitude.

The first contribution, $\Gamma_0 \equiv \Gamma_{\text{rad}}$, is associated with the radiative lifetime of the excited state. For a photoassociated molecule – formed through a one-photon process – this natural radiative linewidth is, in principle, twice that of an excited atom ($\Gamma_{\text{at}} = 2\pi \times 7.5$ kHz for the $^1S_0 \rightarrow ^3P_1$ atomic line of ^{88}Sr , and $\Gamma_0 \approx 2\pi \times 15$ kHz on average for the molecular states). Due to small Franck-Condon factors, the effective saturation intensity $I_{\text{sat}}^{\text{eff}}$ for molecular transitions is significantly larger than for atomic ones, making power broadening completely negligible for molecular lines [5].

For molecular resonances, the Doppler shift has to be calculated for an atom pair rather than for a single atom. The thermal broadening is thus of order

$$\Gamma_{\text{thermal}} = \frac{1}{\lambda} \sqrt{\frac{k_B T}{2m_{\text{at}}}} \quad (3.70)$$

with $2m_{\text{at}}$ the mass of the pair, λ the laser wavelength, T the gas temperature, and k_B the Boltzmann constant.

Finally, the spectral width of the laser source and any additional technical or physical broadening (light shift inhomogeneity, predissociation, unresolved structure) add to these two contributions. The light shift inhomogeneity comes from the light shift of the molecular level, which is proportional to the local intensity: since the intensity of the Gaussian PA beam, and of the trapping beams when they are on at the same time, varies across the cloud, pairs at different positions – in particular the hotter atoms exploring the edges of the trap – see the line at slightly different frequencies, which broadens the measured line (for the $\nu = -1$ state of ^{88}Sr , the light shift reaches $\simeq 200$ kHz for 0.8 W/cm^2 [14], see Sec. 3.2.4). Since these mechanisms produce different line shapes (Lorentzian widths add linearly, Gaussian widths in quadrature), the difference between Γ_{tot} and the known contributions $\Gamma_0/2\pi$ and Γ_{thermal} only gives an order of magnitude of the excess broadening, and is not decomposed further. It is common for photoassociation lines to appear broader than expected from the radiative and thermal widths alone.

3.2.3.3 Decay dynamics (dN/dt)

The usual test of the inelastic nature of the collisions is the time evolution of the atom number, $N(t)$, discussed in Sec. 3.1.3. This curve allows one to (i) check the presence of two-body losses and (ii) determine the loss coefficient K_2 , hence l_{opt}/I . The fitting function is obtained by integrating the rate equation:

$$\frac{dN(t, T)}{dt} = -\beta(T(t))N(t, T)^2 - \Gamma N(t, T) \quad (3.71)$$

$$\text{Setting } M(t, T) = N(t, T)e^{\Gamma t} \iff N(t, T) = M(t, T)e^{-\Gamma t} \quad (3.72)$$

$$\dot{M}(t, T)e^{-\Gamma t} - \Gamma M(t, T)e^{-\Gamma t} = -\beta(T(t))M(t, T)^2e^{-2\Gamma t} - \Gamma M(t, T)e^{-\Gamma t} \quad (3.73)$$

$$\int_0^t \frac{\dot{M}(t', T)}{M(t', T)^2} dt' = - \int_0^t \beta(T(t'))e^{-\Gamma t'} dt' \quad (3.74)$$

$$-\frac{1}{M(t, T)} + \frac{1}{N_0} = - \int_0^t \beta(T(t'))e^{-\Gamma t'} dt' \quad (3.75)$$

$$M(t, T) = \frac{N_0}{1 + N_0 \int_0^t \beta(T(t'))e^{-\Gamma t'} dt'} \quad (3.76)$$

$$N(t, T) = \frac{N_0 e^{-\Gamma t}}{1 + N_0 \int_0^t \beta(T(t'))e^{-\Gamma t'} dt'} \quad (3.77)$$

Since N depends on the temperature, Eq. 3.77 is rewritten to show explicitly the T -dependence of $\beta(T)$. The atoms are trapped in an ODT with a mean trap frequency $\bar{\omega} = (\omega_x \omega_y \omega_z)^{1/3}$. According to the equipartition theorem, the width σ_i (with $i = \{x, y, z\}$) of the Gaussian distribution is related to the temperature T by $\frac{1}{2}m_{\text{at}}\omega_i^2\sigma_i^2 = \frac{1}{2}k_B T$. This gives the effective volume

$$V_e(T) = (2\pi)^{3/2} \left(\frac{k_B T}{m_{\text{at}}} \right)^{3/2} \frac{1}{\omega_x \omega_y \omega_z} = (2\pi)^{3/2} \left(\frac{k_B T}{m_{\text{at}} \bar{\omega}^2} \right)^{3/2} \quad (3.78)$$

where $\bar{\omega}$ depends on the final power of the ODT beams. Substituting $\beta(T(t')) = \frac{K_2}{2\sqrt{2}V_e(T(t'))}$ into Eq. 3.77, the fitting function becomes

$$N(t, T) = \frac{N_0 e^{-\Gamma t}}{1 + \left[\frac{K_2}{8\pi^{3/2}} (m_{\text{at}})^{3/2} \omega_x \omega_y \omega_z \right] N_0 \int_0^t \left(\frac{1}{k_B T(t')} \right)^{3/2} e^{-\Gamma t'} dt'} \quad (3.79)$$

Since the gas heats up during photoassociation, K_2 is extracted from a measured decay by letting β depend on the measured temperature. The temperature measurements are first linearly interpolated into a continuous function $T(t)$; this interpolated temperature is fixed once and for all, independently of the fit that follows. The non-linear fit then adjusts four free parameters – the one-body loss rate Γ , the two-body loss coefficient K_2 (equivalently l_{opt}), the initial atom number N_0 , and a constant offset added to Eq. 3.79. At each iteration, the current value of K_2 is combined with the interpolated $T(t)$ to compute the instantaneous loss rate $\beta(t) = K_2/(2\sqrt{2}V_e(T(t)))$ at every time step; $\beta(t)$ is then integrated numerically to compute a trial decay curve $N(t)$ with Eq. 3.79. This trial curve is compared to the measured $N(t)$, and the four parameters are updated to reduce the χ^2 , until convergence. The best-fit K_2 is thus a single constant for the whole curve, while $\beta(t)$ changes during the decay only because $T(t)$ changes – not because K_2 itself changes with time. When the temperature changes little during the decay, β is simply taken constant.

From Eq. 3.71, the decay is governed by two dynamics with different characteristic times: the two-body time $1/(\beta N_0)$, which sets the initial decay when the density is high, and the one-body time $1/\Gamma$. As the atom number decreases, the two-body loss rate $\beta N(t)$ decreases, and one-body losses take over when $\beta N(t) \simeq \Gamma$, i.e. when $N \simeq \Gamma/\beta$. There is no sharp break between these two dynamics but a smooth crossover: on a logarithmic scale, the one-body dynamics appears as a straight line, whereas the two-body dynamics appears as a faster, non-exponential decay whose rate continuously decreases.

3.2.3.4 Determining the optical scattering length l_{opt}

The optical length l_{opt} characterizes the strength of a PA line independently of the experimental conditions: it scales linearly with the laser intensity and depends only on the Franck–Condon overlap between the scattering state and the molecular state (Eq. 3.31). It is however never measured directly. Two observables are available for each line: (i) the loss spectrum, i.e. the atom number after a fixed illumination time as a function of the laser detuning, which gives the line center and its total width; (ii) the decay $N(t)$ of the atom number at fixed detuning, which gives the two-body loss coefficient K_2 at that detuning. I first recall how l_{opt} is defined from the loss coefficient of an ideal, unbroadened line, then show how this definition must be adapted to the broadened lines actually observed, and finally describe the two methods used to obtain the required loss coefficient.

Ideal line: definition of l_{opt} .

Following the formulation of [5], written as in [108] to allow for extra molecular decay ($\Gamma_{\text{mol}} \geq \Gamma_{\text{rad}}$), the microscopic two-body loss rate coefficient near a photoassociation resonance is given by:

$$K_2(\Delta) = \frac{4\pi\hbar}{\mu} \frac{\Gamma_{\text{mol}}\Gamma_{\text{rad}} l_{\text{opt}}}{\left(\frac{E}{\hbar} + 2\pi\Delta\right)^2 + \frac{(\Gamma_{\text{stim}} + \Gamma_{\text{mol}})^2}{4}} \quad (3.80)$$

where $\mu = m_{\text{at}}/2$ is the reduced mass of the atom pair, Γ_{mol} is the total molecular linewidth and Γ_{rad} its radiative part (both in rad/s), Γ_{stim} is the light-induced stimulated width (in rad/s), Δ is the laser detuning expressed as a frequency (in Hz), so that $2\pi\Delta$ is the angular detuning used in Eq. 3.36, and E is the relative collision energy.

Experimentally, we measure the relative atomic loss profile $f(\Delta) = \frac{n_0 - n(t)}{n_0}$ as a function of the laser detuning Δ . Considering only the two-body loss term, $n(t) = \frac{n_0}{1 + K_2(\Delta)n_0 t_{\text{PA}}}$ (see Eq. 3.63), so that for a pulse duration t_{PA} and an average atomic density $\langle n \rangle$, in the limit of small loss fraction ($K_2\langle n \rangle t_{\text{PA}} \ll 1$), the relative loss is directly proportional to $K_2(\Delta)$:

$$f(\Delta) \approx \langle n \rangle t_{\text{PA}} K_2(\Delta) \quad (3.81)$$

i.e. the initial loss rate $f(\Delta)/t_{\text{PA}} \simeq \langle n \rangle K_2(\Delta)$ directly gives the two-body loss coefficient. Fitting this measured resonance profile with a Lorentzian function yields:

$$f(\Delta) = \frac{A}{1 + \left(\frac{\Delta + E/\hbar}{\Gamma_{\text{fit}}/2}\right)^2} \quad (3.82)$$

where A is the dimensionless amplitude of the loss peak and Γ_{fit} is the full width at half maximum (FWHM), both Δ and Γ_{fit} being in Hz. Factoring $(2\pi)^2$ out of the denominator of Eq. 3.80 shows that it is indeed a Lorentzian in Δ , centered at $-E/\hbar$ and of FWHM $(\Gamma_{\text{stim}} + \Gamma_{\text{mol}})/2\pi$.

If the molecular line had no other width than its radiative one ($\Gamma_{\text{mol}} = \Gamma_{\text{rad}}$), the fit parameters would map directly onto Eq. 3.80, with $\Gamma_{\text{fit}} = (\Gamma_{\text{stim}} + \Gamma_{\text{mol}})/2\pi \simeq \Gamma_{\text{rad}}/2\pi$ in the threshold regime $\Gamma_{\text{stim}} \ll \Gamma_{\text{mol}}$. The loss coefficient at the line center would then reach

$$K_2(0) = \frac{A}{\langle n \rangle t_{\text{PA}}} = \frac{16h}{m_{\text{at}}} l_{\text{opt}}, \quad (3.83)$$

which is Eq. 3.43 for $\Gamma_{\text{mol}} = \Gamma_{\text{rad}}$. This is how l_{opt} is defined: it is proportional to the peak loss coefficient of the *ideal* line, whose width is the radiative width $\Gamma_{\text{rad}} \simeq 2\pi \times 15$ kHz. In our experiment, however, the measured widths Γ_{fit} exceed $\Gamma_{\text{rad}}/2\pi$ by a factor $\simeq 20$ for ^{88}Sr (Table 3.1) and 10^2 – 10^3 for ^{87}Sr (Table 3.4). Applying Eq. 3.83 to the measured peak loss coefficient would then strongly underestimate l_{opt} , as shown below.

Broadened line: integration over the full linewidth.

The observed lines are much broader than Γ_{rad} , because of the laser frequency noise, the thermal distribution of the atoms, an inhomogeneous light shift or additional decay channels of the molecular state. The decay recorded with the laser set at the line center only collects the losses that occur within a width of order $\Gamma_{\text{rad}}/2\pi \simeq$

15 kHz around the center, i.e. a fraction $(\Gamma_{\text{rad}}/2\pi)/\Gamma_{\text{fit}}$ of the line. The value of K_2 measured at the center, $K_2^{\text{fit}}(0)$, is therefore much lower than the ideal peak value $\frac{16h}{m_{\text{at}}}l_{\text{opt}}$ of Eq. 3.83. To characterize the line, the losses must be considered over its whole fitted width Γ_{fit} .

The broadening does not create or remove losses: it only spreads them over a broader frequency range. The area under the line, i.e. its height times its width (the factor $\pi/2$ of a Lorentzian being the same for both), is thus the same for the ideal line and for the observed one:

$$K_2^{\text{fit}}(0) \times \Gamma_{\text{fit}} = \frac{16h}{m_{\text{at}}} l_{\text{opt}} \times \frac{\Gamma_{\text{rad}}}{2\pi} \iff l_{\text{opt}} = \frac{m_{\text{at}}}{16h} K_2^{\text{fit}}(0) \frac{\Gamma_{\text{fit}}}{\Gamma_{\text{rad}}/2\pi}. \quad (3.84)$$

This relation does not require $\Gamma_{\text{mol}} = \Gamma_{\text{rad}}$: from Eq. 3.36, an extra molecular decay lowers the peak by $\Gamma_{\text{rad}}/\Gamma_{\text{mol}}$ and widens the line by $\Gamma_{\text{mol}}/\Gamma_{\text{rad}}$, so that the area, $\frac{16h}{m_{\text{at}}}l_{\text{opt}}\Gamma_{\text{rad}}/2\pi$, is independent of Γ_{mol} : Γ_{fit} includes this extra decay as well as the other broadening mechanisms. In other words, the optical length obtained from the ideal-line relation must be multiplied by the ratio of the fitted width to the radiative width. This ratio is of order 20 for the ^{88}Sr lines and 10^2 – 10^3 for the ^{87}Sr lines.

This is equivalent to the empirical broadening factor $\eta \geq 1$ used in Ref. [16], where the line remains Lorentzian with a width $\eta\Gamma_{\text{rad}}$ and a peak loss coefficient lowered by η at fixed l_{opt} , here with $\eta = \Gamma_{\text{fit}}/(\Gamma_{\text{rad}}/2\pi)$.

Eq. 3.84 requires two inputs: the width Γ_{fit} and the loss coefficient at the line center $K_2^{\text{fit}}(0)$. The width can only be obtained from a spectrum, i.e. from the Lorentzian fit of the loss spectrum, and is common to both methods below. The two methods differ in the way the peak value $K_2^{\text{fit}}(0)$ is measured: from the depth of the same spectrum (Method 1), or from an independent measurement of the atom number decay with the laser set at the line center (Method 2). Comparing the two methods is therefore a test of the consistency between the spectrum and the decay measurement.

Method 1: Lorentzian fit of the loss spectrum.

The Lorentzian fit of the loss spectrum (Eq. 3.82) gives both the amplitude A and the width Γ_{fit} of the line. In the limit of small losses, the amplitude gives the loss coefficient at the line center,

$$K_2^{\text{fit}}(0) = \frac{A}{\langle n \rangle t_{\text{PA}}}, \quad (3.85)$$

and inserting it into Eq. 3.84 yields

$$l_{\text{opt}} = \frac{m_{\text{at}}}{16h} \frac{A \Gamma_{\text{fit}}}{\langle n \rangle t_{\text{PA}} \Gamma_{\text{rad}}/2\pi}. \quad (3.86)$$

Compared with the ideal-line mapping, in which Γ_{fit} would be the molecular width itself and l_{opt} would scale as $A\Gamma_{\text{fit}}^2$, only one power of Γ_{fit} remains: the width here reflects the broadening of the line, not the lifetime of the molecular state. For the losses of up to 40% observed on ^{87}Sr , the small-loss limit is not accurate. As in Refs. [5, 108], where the loss spectra are fitted with the solution of the two-body rate equation rather than with its linearized form, we use the exact relation given by the derivation of the characteristic time (Eq. 3.63): $1/n(t) - 1/n_0 = K_2 t$, i.e.

$K_2 n_0 t = n_0/n(t) - 1 = A/(1 - A)$ at the line center, where $A = (n_0 - n)/n_0$. In Eq. 3.85, A is therefore replaced by $A/(1 - A)$, the two expressions being equal for small losses. As in these references, the temperature is assumed constant during the pulse, so that the heating of the gas is neglected. Both inputs come from the same spectrum, at the same intensity and density.

Method 2: atom number decay at the line center.

The laser is set at the line center and the decay $N(t)$ is fitted with Eq. 3.77, the temperature being interpolated from the measured $T(t)$ (Sec. 3.2.3.3). This yields $K_2^{\text{fit}}(0)$ while accounting for heating, but relies on a width measured on a separate spectrum. If the two measurements are not performed at the same intensity, a light shift can displace the line so that the decay is no longer recorded at its center, and an intensity-dependent broadening makes the width measured at lower intensity too small: both effects lead to an underestimation of l_{opt} . The fit may also attribute to K_2 a curvature of the decay that is not due to photoassociation (see Fig. 3.19), which would lead to an overestimation.

Since both methods measure the same quantity $K_2^{\text{fit}}(0)$, their comparison under identical conditions is a direct consistency test. In the following, l_{opt} is given for both methods through Eq. 3.84, and their difference is taken as a systematic uncertainty.

3.2.3.5 Estimation of the uncertainties

The quantities used in the following to characterize the PA lines are given with uncertainties estimated as follows; they are combined in quadrature.

Temperature. The temperature is measured by time of flight from the vertical size σ_y of the cloud (Sec. 3.2.3.2). Two uncertainties are used:

- for a single temperature point, the error bar comes from the uncertainty $\delta\sigma_y$ of the Gaussian fit of the image (Eq. 3.68);
- for the temperature of a whole spectrum, we give the mean temperature of all the points of the spectrum, and their standard deviation as uncertainty. This standard deviation, which reflects the shot-to-shot fluctuations, is much larger than the fit uncertainty of a single point.

For the decays, we give the temperatures of the first and last points, with their fit uncertainty. The same relative uncertainty is taken for the ^{88}Sr data. It propagates to the thermal width $\Gamma_{\text{thermal}} \propto \sqrt{T}$ as

$$\frac{\delta\Gamma_{\text{thermal}}}{\Gamma_{\text{thermal}}} = \frac{1}{2} \frac{\delta T}{T}. \quad (3.87)$$

PA intensity. The PA intensity is computed as $I = 2P/(\pi w_x w_y)$, so that

$$\frac{\delta I}{I} = \sqrt{\left(\frac{\delta P}{P}\right)^2 + \left(\frac{\delta w_x}{w_x}\right)^2 + \left(\frac{\delta w_y}{w_y}\right)^2}. \quad (3.88)$$

The uncertainty δw on a waist measured with our camera is estimated from the scatter of repeated measurements around the fitted beam caustic (z-scan of the Reservoir beam), and δP is the resolution of the power meter. For the ^{88}Sr data, only the waist uncertainty is included.

Mean trap frequency and density. The mean trap frequency $\bar{\omega} = (\omega_x \omega_y \omega_z)^{1/3}$ is computed for each measurement from the measured Reservoir and Dimple beam powers – at the end of the evaporation ramps – and the trap geometry, and gives the average density $\langle n \rangle = N/(2\sqrt{2} V_e(T))$ (Eq. 3.41). This estimate has not been cross-checked against a direct trap-frequency measurement (e.g. parametric heating). The uncertainty $\delta\bar{\omega}$ is obtained by varying, in the trap model, the waists of the Reservoir and Dimple beams by their uncertainty and their powers by the resolution of the power meter. Since $V_e \propto \bar{\omega}^{-3}$, it propagates to the density as

$$\frac{\delta\langle n \rangle}{\langle n \rangle} = 3 \frac{\delta\bar{\omega}}{\bar{\omega}}. \quad (3.89)$$

Loss coefficient and optical length. The uncertainty on K_2 combines the statistical uncertainty δK_2^{fit} of the fit (depth of the loss spectrum, or two-body coefficient of the decay) with that of the density,

$$\frac{\delta K_2}{K_2} = \sqrt{\left(\frac{\delta K_2^{\text{fit}}}{K_2}\right)^2 + \left(\frac{\delta\langle n \rangle}{\langle n \rangle}\right)^2}, \quad (3.90)$$

and the uncertainty on l_{opt}/I additionally includes those of the intensity and, when the linewidth is accounted for, of the fitted width:

$$\frac{\delta(l_{\text{opt}}/I)}{l_{\text{opt}}/I} = \sqrt{\left(\frac{\delta K_2}{K_2}\right)^2 + \left(\frac{\delta I}{I}\right)^2 + \left(\frac{\delta\Gamma_{\text{fit}}}{\Gamma_{\text{fit}}}\right)^2}. \quad (3.91)$$

3.2.4 Experimental results on ^{88}Sr

We now present photoassociation data on the isotope ^{88}Sr . We identified two vibrational lines and compared our experimental findings (line center, linewidth) to the literature, before discussing their broadening in more detail. The measured and literature values are summarized in Table 3.1.

	$\nu = -1$	$\nu = -2$
Exposure time t	1000 ms	10 ms
Intensity I	8 ± 1 mW/cm ^{2*}	826 ± 50 mW/cm ²
Measured detuning f	$-23.99(1)$ MHz	$-222.70(1)$ MHz
Literature f [5]	$-23.932(33)$ MHz	$-222.161(35)$ MHz
Γ_{tot} (measured)	344 ± 17 kHz	330 ± 13 kHz
$\Gamma_0/2\pi$ (first principles)	15 kHz	15 kHz
Γ_{thermal} (first principles)	13.4 ± 0.4 kHz	11.8 ± 0.4 kHz
Temperature T	1.8 ± 0.1 μ K	1.4 ± 0.1 μ K
$\bar{\omega}/2\pi$	181 ± 9 Hz	184 ± 9 Hz
Density $\langle n \rangle$	$(1.8 \pm 0.3) \times 10^{13}$ cm ⁻³	$(3.0 \pm 0.4) \times 10^{13}$ cm ⁻³

Table 3.1: Summary of the two vibrational states identified for ⁸⁸Sr, comparing measured line centers and linewidths to the literature values from Zelevinsky et al. [5]. Γ_0 and Γ_{thermal} are estimated from first principles (not extracted from the data). Uncertainties are estimated as described in Sec. 3.2.3.5. *Intensity during the PA pulses: the $\nu = -1$ spectrum was recorded with the ODT/PA chopping sequence (Fig. 3.24, 50% duty cycle), i.e. a time-averaged intensity of 4 ± 1 mW/cm².

Both states were detected using the MOGlabs laser system. A few points lie far from the Lorentzian fits of the molecular lines (Fig. 3.13, middle and bottom). The corresponding absorption images were checked and show a normal cloud, correctly described by the 2D Gaussian fit: these deviations do not come from the image analysis but from uncontrolled shot-to-shot fluctuations of the experiment, and the points are kept in the data.

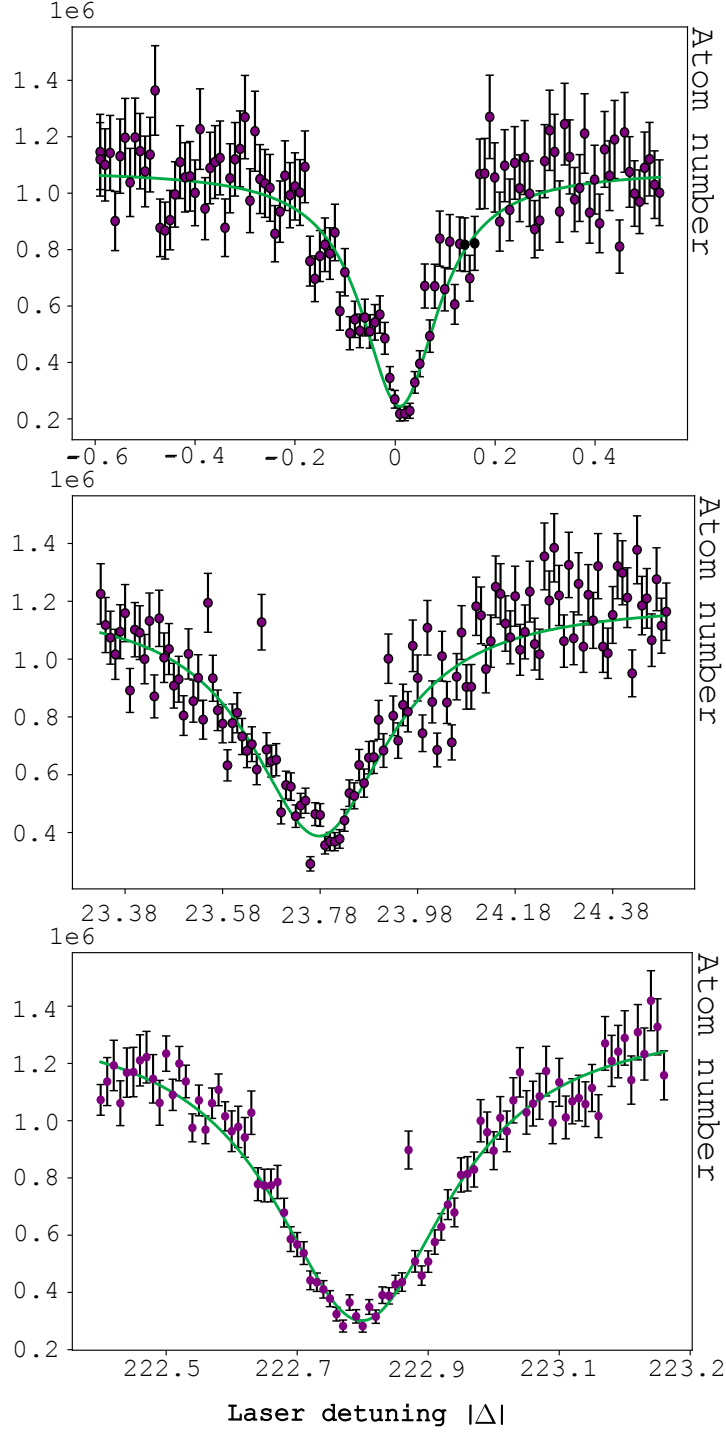


Figure 3.13: ^{88}Sr atom number (in units of 10^6 atoms) across the intercombination line. (Top) Atomic resonance used as the PA frequency reference ($I = 276 \pm 16 \mu\text{W}/\text{cm}^2$, 80 ms, $\bar{\omega}/2\pi = 213 \pm 10$ Hz, $\langle n \rangle = (4.5 \pm 0.6) \times 10^{13} \text{ cm}^{-3}$, $T = 1.40 \pm 0.05 \mu\text{K}$). Data points are extracted from 2D Gaussian fits of absorption images; error bars combine the uncertainty of the image fit and the shot-to-shot RMS fluctuation of the atom number measured without PA light. (Middle) $\nu = -1$ and (bottom) $\nu = -2$ vibrational states, recorded in the conditions of Table 3.1. Uncertainties: see Sec. 3.2.3.5. Solid lines: Lorentzian fits; the error bars combine the diagonal terms of the covariance matrix of the image fit and the shot-to-shot fluctuations.

$\nu = -1$ state For the $\nu = -1$ state (their $n = -2$), we measure $-23.99(1)$ MHz, in agreement with their $-23.98(1)$ MHz (and close to the $-23.932(33)$ MHz of Zelevinsky et al. [5]). The light shift of this state, (-251 ± 67) kHz per W/cm^2 of PA intensity [14], i.e. a red shift of $\simeq 200$ kHz for $0.8 \text{ W}/\text{cm}^2$, amounts to only $\simeq -2$ kHz at the $8 \text{ mW}/\text{cm}^2$ used here.

$\nu = -2$ state For the $\nu = -2$ state (their $n = -3$), we measure $-222.70(1)$ MHz, about 0.5 MHz further from the asymptote than their $-222.18(5)$ MHz and the $-222.161(35)$ MHz of Zelevinsky et al. This difference is not explained by the light shift of this state, (29 ± 2) kHz per W/cm^2 , i.e. only $\simeq +24$ kHz at the $826 \text{ mW}/\text{cm}^2$ used here, and towards the blue. Blatt et al. also report a light shift induced by the trapping light, which broadens the lines on their blue side [108].

The measured linewidths ($\Gamma_{\text{tot}} \simeq 330\text{--}344$ kHz) are much larger than the sum of the natural and thermal widths, $\Gamma_0/2\pi + \Gamma_{\text{thermal}} \simeq 27$ kHz, power broadening being negligible for molecular lines (Sec. 3.2.3.2): the excess broadening is of order 300 kHz. Part of this broadening could come from the spectral impurities of the MOGlabs laser. However, convolving the power spectrum of the beat note (Fig. 3.9) – a narrow central peak (FWHM $\simeq 13$ kHz) with sidebands at ± 70 kHz of FWHM $\simeq 80$ kHz – with the expected clean molecular lineshape ($\Gamma_0/2\pi \oplus \Gamma_{\text{thermal}} \simeq 19$ kHz) only gives an apparent linewidth of $\simeq 30$ kHz. The structure of this beat note therefore explains only a small part of the observed broadening: other technical noise sources, not captured by this beat-note measurement, must contribute. To confirm the technical origin of this broadening, we measured the same molecular line with an alternative, higher-stability laser source.

3.2.4.1 PA experiments with an alternative laser source

To test whether the broadening observed with the MOGlabs ECDL has a technical origin, we compared the MOGlabs results with a more stable laser source: a diode laser injection-locked on the Radiant Dyes master ECDL. The measured parameters for the $\nu = -1$ molecular state are summarized in Table 3.2 and illustrated in Figure 3.14.

Table 3.2: Comparison of the photoassociation line of the bound state $\nu = -1$ measured with two different laser sources. The excess width is obtained by subtracting in quadrature the natural and thermal widths from the measured one, $\sqrt{\Gamma_{\text{tot}}^2 - (\Gamma_0/2\pi)^2 - \Gamma_{\text{thermal}}^2}$; it gives an order of magnitude of the technical broadening.

	Injection-locked laser	MOGlabs ECDL
Detuning f (MHz)	-24.145 ± 0.003	-24.147 ± 0.004
Intensity I (mW/cm^2)	795 ± 45	231 ± 13
Natural width $\Gamma_0/2\pi$ (kHz)	15	15
Γ_{thermal} , calculated (kHz)	11.8 ± 0.2	11.8 ± 0.2
Measured width Γ_{tot} (kHz)	122 ± 13	231 ± 18
Excess width (kHz)	120 ± 13	230 ± 18

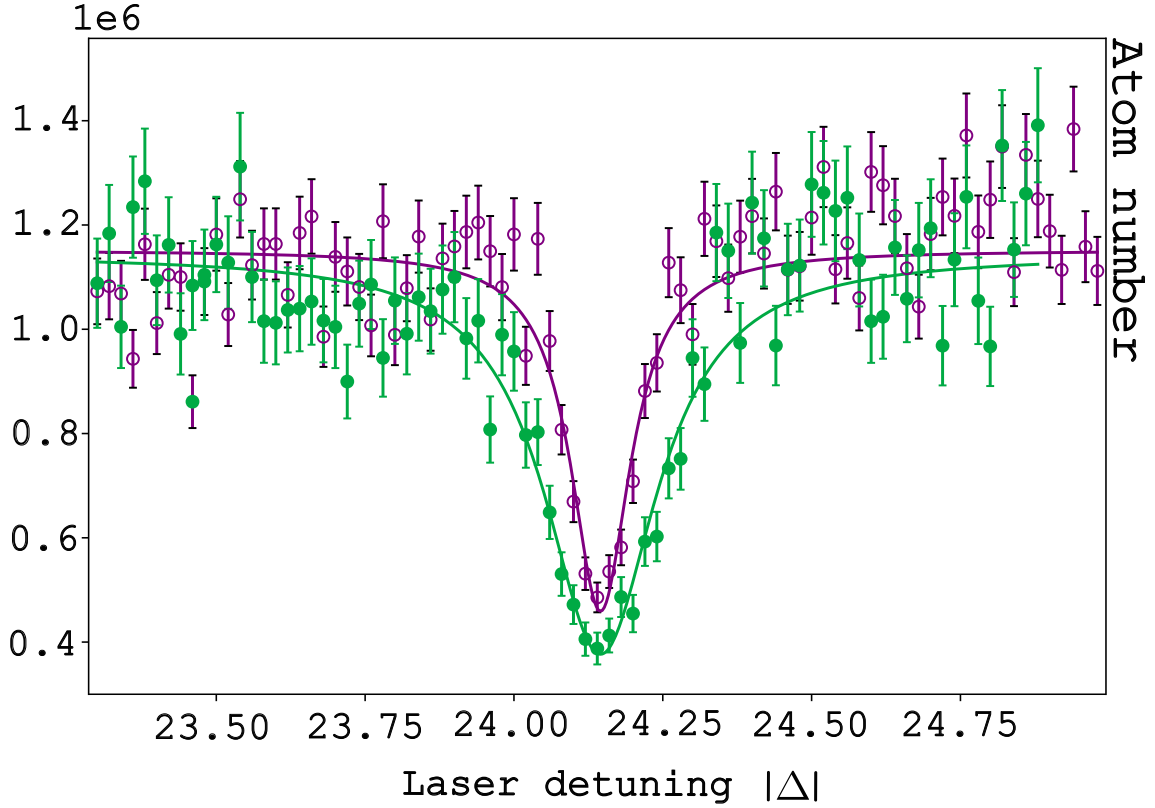


Figure 3.14: Comparison of the $\nu = -1$ molecular line profile with two different sources. The MOGlabs dataset (green) gives a line about twice as broad as the more stable injection-locked laser (purple).

The MOGlabs source gives a total linewidth about twice as large as the injection-locked laser, for similar radiative and thermal contributions, which confirms the technical origin of the excess broadening. Even with the injection-locked laser, the line remains about 5 to 8 times broader than expected (122 kHz instead of $\simeq 15$ –27 kHz), which our measurements do not explain. The MOGlabs laser remains essential for broad spectroscopic searches thanks to its wide frequency tuning range.

3.2.4.2 Noise of the diode

When comparing the two lasers, extra loss peaks appeared with the injection-locked source, at ± 6 MHz and ± 12 MHz from f_0 . Their symmetric position, on both the red and the blue side, points to sidebands, which come from noise on the diode.

At $\delta = -6$ MHz, on the red side of the resonance (Fig. 3.15), no structure is resolved: the atom number only shows noise, with large shot-to-shot fluctuations caused by the heating of the SU(10) gas at $T = 1.6$ μ K (with a fit uncertainty of $\simeq 0.1$ μ K, see Sec. 3.2.3.5).

The same scan, taken on the blue side (Fig. 3.16), clearly resolves the m_F projections: the twelve Zeeman sublevels $-11/2 \leq m_F \leq 11/2$ of the $^3P_1, F = 11/2$ excited state appear around $+6$ MHz and $+12$ MHz. This confirms that the peaks are atomic, not molecular. Each of these resonances removes up to $\simeq 50\%$ of the atoms, although in the SU(10) gas a single Zeeman-resolved transition only addresses one of the ten ground-state spin components, i.e. $\simeq 10\%$ of the atoms. The atoms that

scatter photons are heated by the recoil and share this energy with all the other spin components through elastic collisions; the whole gas then heats up and atoms of all spin states are lost by evaporation from the finite-depth trap. A small resonant fraction can thus deplete a large part of the cloud. None of these features – carrier, red sideband, blue sideband – is a single line: each covers a rather wide range of detunings. This width comes from many single-atom transitions, one for each m_F sublevel, broadened by heating, rather than from the laser noise itself. The whole pattern is simply displaced in frequency, appearing at ± 6 MHz and ± 12 MHz as a consequence of a sideband in the laser spectrum. As a result, searching for molecular lines near ± 6 MHz and ± 12 MHz with the $F = 11/2$ beam is compromised: a weak molecular signal there would be hidden within this atomic structure.

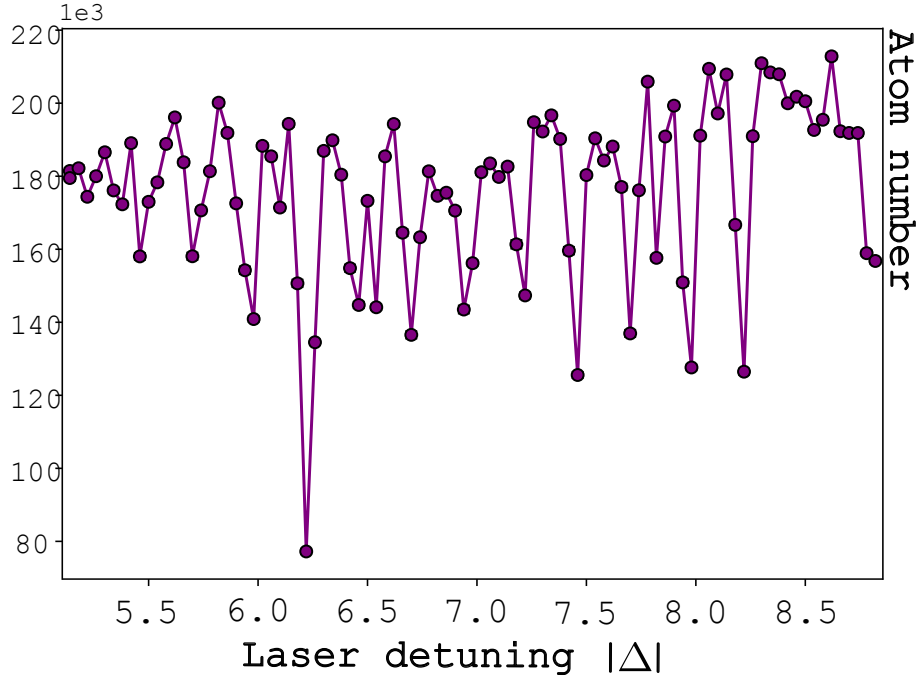


Figure 3.15: Atom number measurement on $^3P_1, F = 11/2$ hyperfine excited state, measured in an SU(10) gas at $\delta = -6$ MHz from the atomic resonance (red side), where a sideband of the diode laser addresses the atomic transition. No structure is resolved: the signal is dominated by noise and by the heating of the gas.

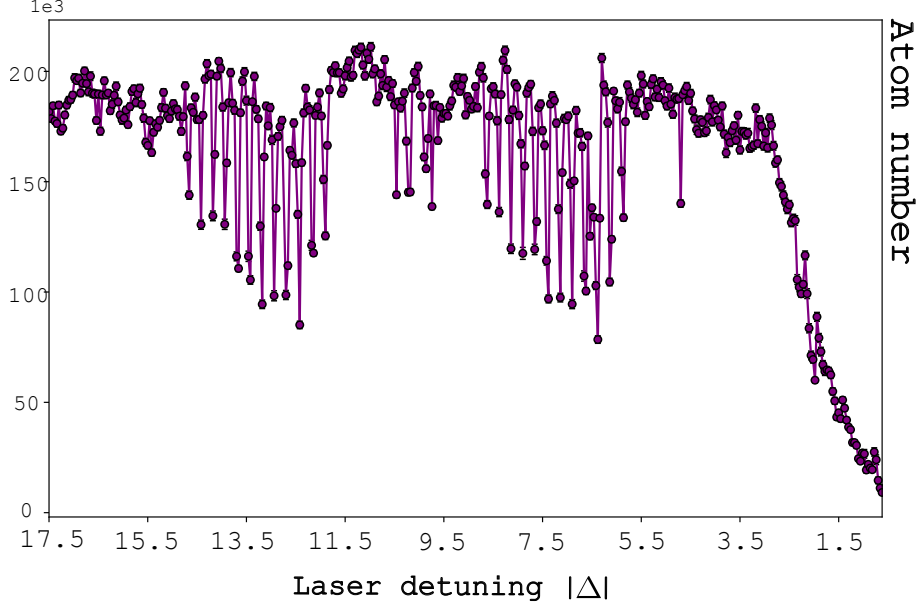


Figure 3.16: Twelve m_F states of the $^3P_1, F = 11/2$ hyperfine excited state, measured in an SU(10) gas at $\delta = +6$ MHz and $\delta = +12$ MHz from the atomic resonance (blue side): the m_F projections are clearly resolved. This heating-broadened pattern of single-atom lines is displaced in frequency by a sideband in the diode laser spectrum.

The two previous subsections characterized the laser used for photoassociation, showing how spectral impurities – frequency components other than the carrier – can address free atoms directly instead of binding them into molecules, and can therefore lead to a misinterpretation of the observed inelastic collisions. I now apply the decay analysis of Sec. 3.2.3.3 to the two ^{88}Sr lines, to extract K_2 and l_{opt} with Method 2 of Sec. 3.2.3.4.

3.2.4.3 Decay measurements on ^{88}Sr

For the ^{88}Sr decays of Fig. 3.17, the two-body times are $\simeq 10$ and $\simeq 20$ ms, and the crossover would occur at $N \simeq 4 \times 10^3$ and 10^4 atoms, well below the measured atom numbers: these decays are dominated by two-body losses over the whole observation window, and β – hence l_{opt} – is extracted from the fit of Eq. 3.77 for each resonance.

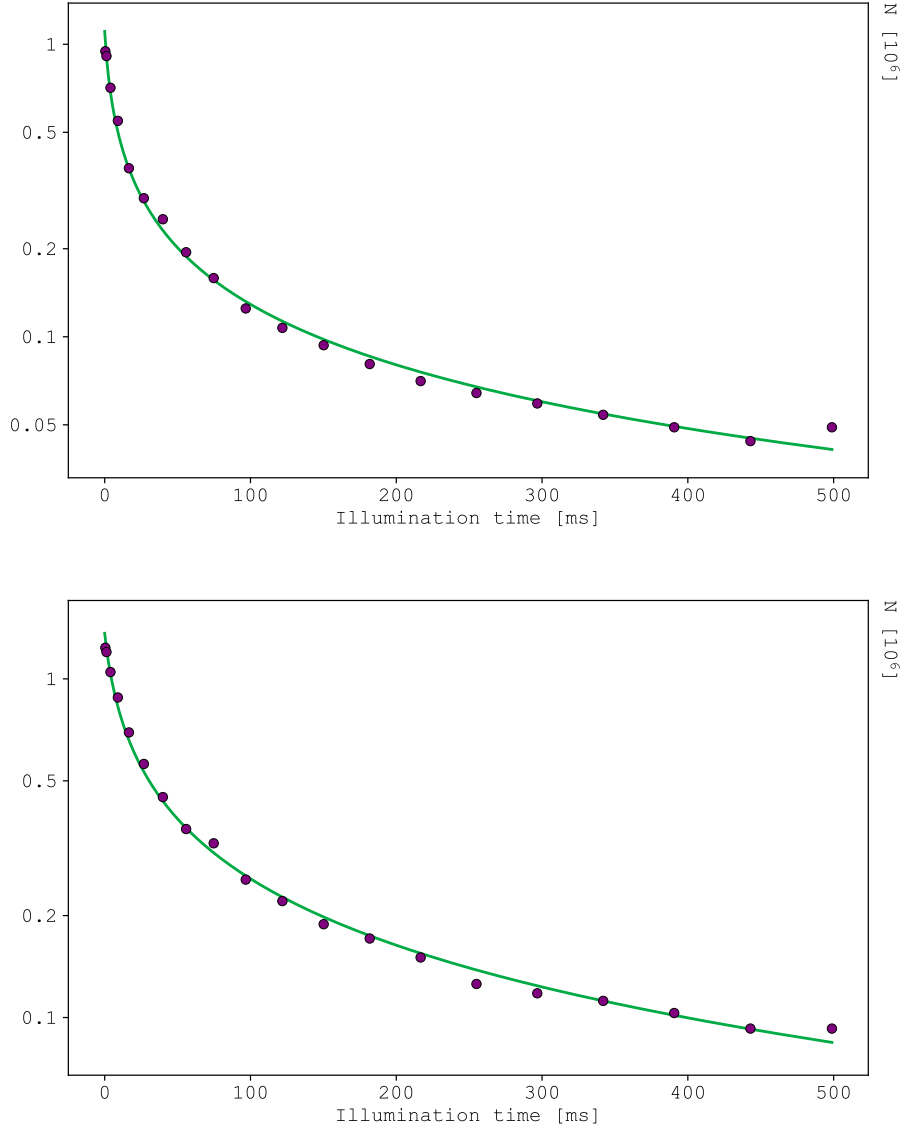


Figure 3.17: Top: atom number decay of ^{88}Sr on the $\nu = -1$ line (Fig. 3.13, middle). Bottom: atom number decay on the $\nu = -2$ line (Fig. 3.13, bottom). Both plots use a logarithmic vertical scale. Solid lines: fit with two-body losses computed locally over the Gaussian density profile, without rethermalization, as expected for the nearly ideal ^{88}Sr gas [108]. The values of K_2 and l_{opt} given in the text were obtained with the global fit of the analysis software (Gaussian profile maintained); the local fit shown here gives a value larger by a factor ~ 2 , i.e. the same order of magnitude. Error bars are not shown, as the original raw data could not be recovered.

For both lines, the two-body coefficient β is fitted on the decay with a constant temperature, and converted into the loss coefficient at the center of the broadened line, $K_2 = 2\sqrt{2} V_e \beta$ (Eq. 3.42). The optical length is then obtained with Method 2 (Eq. 3.84), using the ratio $\Gamma_{\text{tot}}/(\Gamma_0/2\pi)$; the value given by the ideal-line relation (Eq. 3.83) is also given for comparison. Both decays were recorded in the same trap, with $\bar{\omega}/2\pi = 185 \pm 9$ Hz and $\langle n \rangle = (1.4 \pm 0.2) \times 10^{13} \text{ cm}^{-3}$, at $T = 1.8 \pm 0.1 \text{ } \mu\text{K}$ ($\nu = -1$) and $2.0 \pm 0.1 \text{ } \mu\text{K}$ ($\nu = -2$), and at intensities $I = 4 \pm 1 \text{ mW/cm}^2$ ($\nu = -1$)

and $303 \pm 17 \text{ mW/cm}^2$ ($\nu = -2$).

$\nu = -1$ line For this line, the fit of the decay gives $\beta = (1.0 \pm 0.2) \times 10^{-4} \text{ s}^{-1}$ per atom, i.e. $K_2 = 2\sqrt{2} V_e \beta = (6.3 \pm 1.6) \times 10^{-12} \text{ cm}^3/\text{s}$. The ideal-line relation then gives $l_{\text{opt}}^{\text{ideal}} = 1.6 \pm 0.4 a_0$ at $I = 4 \pm 1 \text{ mW/cm}^2$, i.e. $l_{\text{opt}}^{\text{ideal}}/I = (4.1 \pm 1.1) \times 10^2 a_0/(\text{W/cm}^2)$. Accounting for the measured width ($\Gamma_{\text{tot}}/(\Gamma_0/2\pi) = 23.0 \pm 1.2$), the optical length becomes $l_{\text{opt}} = 37 \pm 9 a_0$, i.e. $l_{\text{opt}}/I = (9.2 \pm 2.4) \times 10^3 a_0/(\text{W/cm}^2)$. This lies between the theoretical value $\xi = 6110 a_0/(\text{W/cm}^2)$ reported by Blatt et al. [108] for the corresponding resonance (their $n = -2$ state), which it exceeds by a factor 1.5 ± 0.4 , and their measured value $15.8(8) \times 10^3 a_0/(\text{W/cm}^2)$, a factor 1.7 ± 0.5 above it. The spectrum of this line was recorded with the same MOGlabs laser, but with the ODT/PA chopping sequence active (time-averaged intensity $4 \pm 1 \text{ mW/cm}^2$, Table 3.1), whereas the decay was not; the chopping itself was checked to have a negligible effect on the atoms and on the linewidth. Even at the same time-averaged intensity, the spectrum and the decay are not consistent for this line: the loss coefficient deduced from the depth of the spectrum (Method 1), $(1.2 \pm 0.3) \times 10^{-13} \text{ cm}^3/\text{s}$, i.e. $l_{\text{opt}}/I = (1.8 \pm 0.6) \times 10^2 a_0/(\text{W/cm}^2)$, is a factor 50 ± 15 lower than the one of the decay, whereas the two methods differ only by a factor $\simeq 1.5$ for the $\nu = -2$ line. **At the time of writing, the origin of this factor $\simeq 50$ is not understood.**

$\nu = -2$ line For this line, the fit of the decay gives $\beta = (4.0 \pm 0.3) \times 10^{-5} \text{ s}^{-1}$ per atom, i.e. $K_2 = (2.9 \pm 0.5) \times 10^{-12} \text{ cm}^3/\text{s}$, and the ideal-line relation gives $l_{\text{opt}}^{\text{ideal}} = 0.77 \pm 0.12 a_0$ at $I = 303 \pm 17 \text{ mW/cm}^2$, i.e. $l_{\text{opt}}^{\text{ideal}}/I = 2.5 \pm 0.4 a_0/(\text{W/cm}^2)$. Accounting for the measured width ($\Gamma_{\text{tot}}/(\Gamma_0/2\pi) = 22.0 \pm 0.9$), the optical length becomes $l_{\text{opt}} = 17 \pm 3 a_0$, i.e. $l_{\text{opt}}/I = 56 \pm 10 a_0/(\text{W/cm}^2)$, in agreement with the value $68(3) a_0/(\text{W/cm}^2)$ measured by Blatt et al. [108] for this line (their $n = -3$ state), and a factor 1.7 ± 0.3 above the theoretical value $\xi = 32.8 a_0/(\text{W/cm}^2)$. Method 1, using the depth of the spectrum ($K_2^{\text{fit}}(0) = (1.17 \pm 0.17) \times 10^{-11} \text{ cm}^3/\text{s}$ at 826 mW/cm^2), gives $l_{\text{opt}}/I = 81 \pm 13 a_0/(\text{W/cm}^2)$, a factor $\simeq 1.5$ above Method 2. Our values of l_{opt}/I are therefore close to those of Blatt et al., but a quantitative agreement cannot be confirmed: their theoretical values ξ are given without uncertainty, and the uncertainties quoted on their measured values only come from their fit. They do not include the uncertainty on the atomic density, which scales as $\bar{\omega}^3$ with the mean trap frequency and is therefore large; the uncertainties on their measured values of l_{opt}/I are thus likely underestimated.

The results obtained on the two ^{88}Sr lines are summarized in Table 3.3.

	$\nu = -1$	$\nu = -2$
Measured detuning f (MHz)	$-23.99(1)$	$-222.70(1)$
Literature f (MHz) [5]	$-23.932(33)$	$-222.161(35)$
Measured width Γ_{tot} (kHz)	344 ± 17	330 ± 13
l_{opt}/I , Method 1 (spectrum) ($a_0/(\text{W}/\text{cm}^2)$)	$(1.8 \pm 0.6) \times 10^2^*$	81 ± 13
l_{opt}/I , Method 2 (decay) ($a_0/(\text{W}/\text{cm}^2)$)	$(9.2 \pm 2.4) \times 10^3$	56 ± 10
Literature l_{opt}/I ($a_0/(\text{W}/\text{cm}^2)$) [108]	$15.8(8) \times 10^3$	$68(3)$

Table 3.3: Summary of the results obtained on the two ^{88}Sr lines. Both methods account for the measured linewidth (Eq. 3.84): Method 1 uses the depth of the spectrum, Method 2 the decay of the atom number (Sec. 3.2.3.4). *Computed with the time-averaged intensity of the spectrum, $4 \pm 1 \text{ mW}/\text{cm}^2$; this value disagrees with Method 2 by a factor $\simeq 50$ (see text).

To conclude, the positions of the ^{88}Sr lines are close to the literature values (within 0.5 MHz). Once the linewidth is accounted for, the optical lengths also agree with those of Blatt et al. [108] within a factor $\lesssim 2$. This calibration gives a reliable basis to search for ^{87}Sr molecular lines, provided the following experimental constraints are taken into account:

- **Parasitic sidebands:** the injection-locked laser has noise components at $\pm 6 \text{ MHz}$ and $\pm 12 \text{ MHz}$, which can drive parasitic transitions away from the carrier frequency.
- **Laser-induced broadening:** the MOGlabs laser gives molecular lines about twice as broad as the injection-locked laser (231 against 122 kHz).
- **Loss dynamics:** whether the temperature changes or not during the decay, the time evolution of the atom number is well described by Eq. 3.77, obtained by integrating the combined one- and two-body loss equation $\frac{dN}{dt} = -\beta N^2 - \Gamma N$.
- **Blind zones:** near $\pm 40.48 \text{ MHz}$ from the atomic line, the PDH sidebands of the MOGlabs beam excite the atomic transition, which can hide a molecular line.
- **Extraction of l_{opt} :** the loss coefficient measured at the center of a line is lowered by the broadening. Extracting l_{opt} requires both this loss coefficient and the measured linewidth (Eq. 3.84), ideally measured at the same intensity.

The two-body loss coefficient can only be measured accurately if two-body losses dominate over one-body losses. Having validated our detection, fitting and calibration procedures on the well-known ^{88}Sr lines, we now turn to the fermionic isotope ^{87}Sr , the main subject of this thesis. Its richer hyperfine structure and $SU(N)$ symmetry make the spectroscopy more complex – see for instance the twelve overlapping m_F resonances identified above. The next section uses the same method, taking into account the experimental constraints listed here.

3.3 ^{87}Sr inelastic collisions

The calibration on the well-known ^{88}Sr isotope in the previous section allowed us to choose the experimental parameters needed for photoassociation – exposure times, laser intensities – before turning to the fermionic isotope. However, some signals found while searching for ^{87}Sr molecular lines were first wrongly interpreted as molecular resonances. The checks made on these signals, detailed at the beginning of this section, led us to a set of measurements that test whether a resonance is really molecular.

Molecular line searches on ^{87}Sr are performed only with the MOGlabs source. To address the fermionic species, the diffraction order of the red optical chain was changed from -1 to $+1$, shifting the addressed frequency by 1.441 GHz – the isotopic shift between ^{88}Sr and the $F = 11/2$ hyperfine state of ^{87}Sr (see Fig. 3.18).

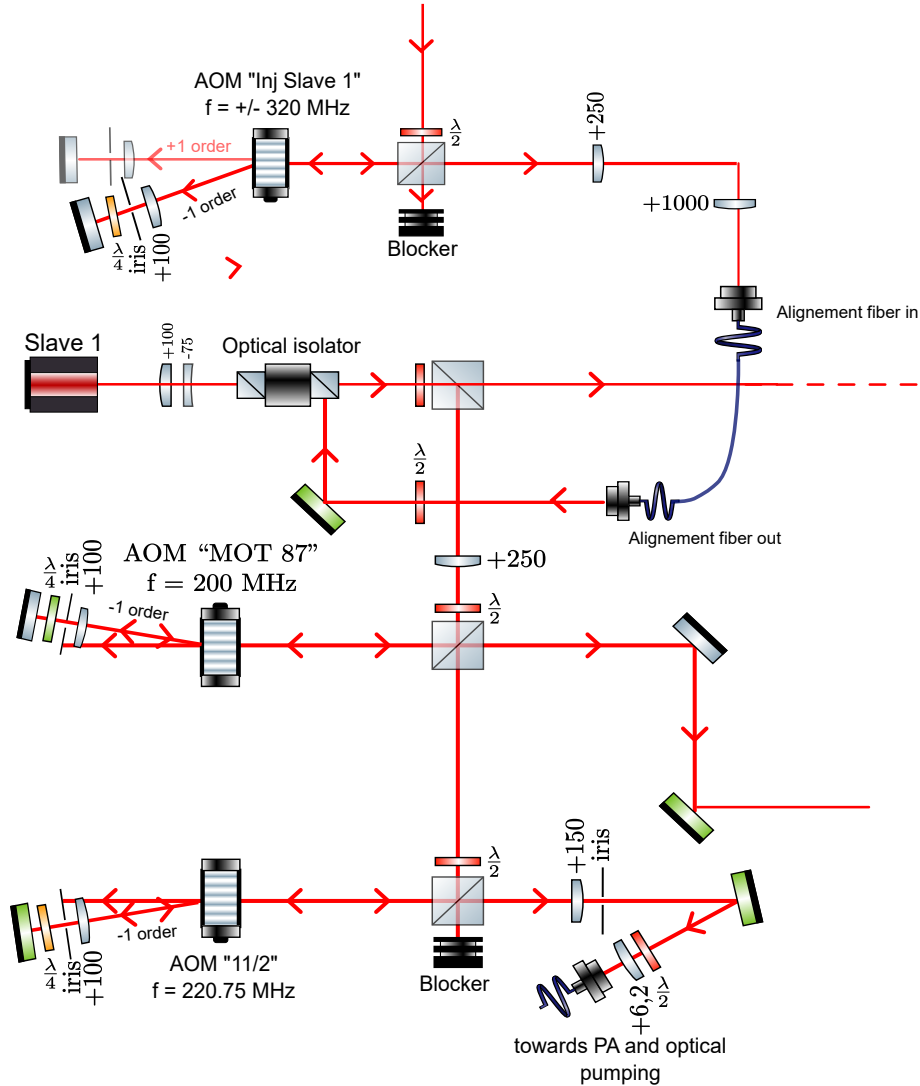


Figure 3.18: Zoom on the part of the red laser chain used for PA. For both isotopes, the frequency is set by 'AOM Inj Slave 1' and adjusted by 'AOM 11/2' before the light is sent to the atoms.

All the resonances presented in this section are recorded with a shot-to-shot nor-

malization, to remove slow drifts of the atom number (due, e.g., to temperature or mechanical drifts of the setup): in each pair of cycles, the first atom number is measured without the PA laser, and the second after exposure to it. This is repeated throughout the scan, and the plotted signal is the ratio of the second to the first atom number.

To increase the two-body loss signal, data were taken for several PA beam waists and atomic densities.

Because of the hyperfine structure of ^{87}Sr , about a thousand molecular channels (981) are expected, as discussed in the theoretical part of this chapter, each line having a natural linewidth of about 15 kHz. The density of these lines can be estimated from the LeRoy-Bernstein law (Eq. 3.1.5.2). Near the $^1S_0 + ^3P_1$ asymptote, the binding energies of the 0_u levels of ^{88}Sr (0.435, 23.932, 222.161 and 1084.093 MHz [5]) follow the law for a C_3/r^3 potential ($n = 3$), with $E_\nu^{1/6}$ linear in ν and $H_3 \simeq 0.78 \text{ MHz}^{1/6}$; mass-scaled to ^{87}Sr ($H_3 \propto \mu^{-1/2}$), this gives about three vibrational levels per molecular channel within the $\simeq 380 \text{ MHz}$ explored below the asymptote, the next one lying around 1 GHz. Distributing the 981 accessible molecular configurations, each with $\simeq 3.5$ levels, over this range, we expect a dense spectrum of about 3×10^3 molecular lines with a mean separation of only $\sim 100 \text{ kHz}$ – a few times larger than the natural linewidth of a single line, but much smaller than the several-MHz linewidths we actually observe (Sec. 3.3.2). This estimate assumes that all channels converge to the same asymptote, and is therefore an upper bound on the line density.

The rest of this section first analyzes the signals that first misled us, and the test protocol they led us to set up. It then presents the molecular lines found for ^{87}Sr , with the associated temperature and atom number evolution. The unexpected shape of these lines leads to a discussion of their possible technical and physical origins. The section ends with the tests needed in the future to check the current hypotheses.

3.3.1 A protocol to identify molecular lines, learned from misleading signals

This section presents the study of the loss signal observed at -6 MHz from the ^{87}Sr atomic resonance (Fig. 3.15), which was first interpreted as a molecular resonance but actually comes from a sideband of the red diode laser addressing the atomic transition, as shown by the resolved m_F structure observed at $+6 \text{ MHz}$ (Fig. 3.16). For these measurements, the atoms were evaporated for 500 ms, recompressed, and illuminated for 2000 ms.

These tests led to a protocol for identifying molecular lines, in which we check the decay model and the temperature evolution.

3.3.1.1 Interpretation of the decay $N(t)$

Without the PA laser, the losses are exponential, as expected for one-body losses due, e.g., to collisions with the background gas. In the presence of a PA laser, we typically observe a non-exponential decay (Fig. 3.19). However, in this specific case, the non-exponential decay was observed both for a positive and a negative 6 MHz detuning with respect to the atomic resonance. An interpretation is that the loss is

then due to a sideband in the laser spectrum: it heats the gas, and the atoms can be lost from the trap at a rate that depends on the atom density. Therefore, this process also leads to a non-exponential loss. As a consequence, each non-exponential loss needs to be carefully checked, which we do by searching for losses at an equal and opposite detuning from the atomic resonance. The same problem is pointed out in Ref. [16]: since the saturation intensity of the atomic transition is very small ($I_{\text{sat}} = 3 \mu\text{W}/\text{cm}^2$), a small amount of sideband noise on the PA laser can cause losses easily confused with a PA resonance, which may explain previously reported lines that the authors could not reproduce.

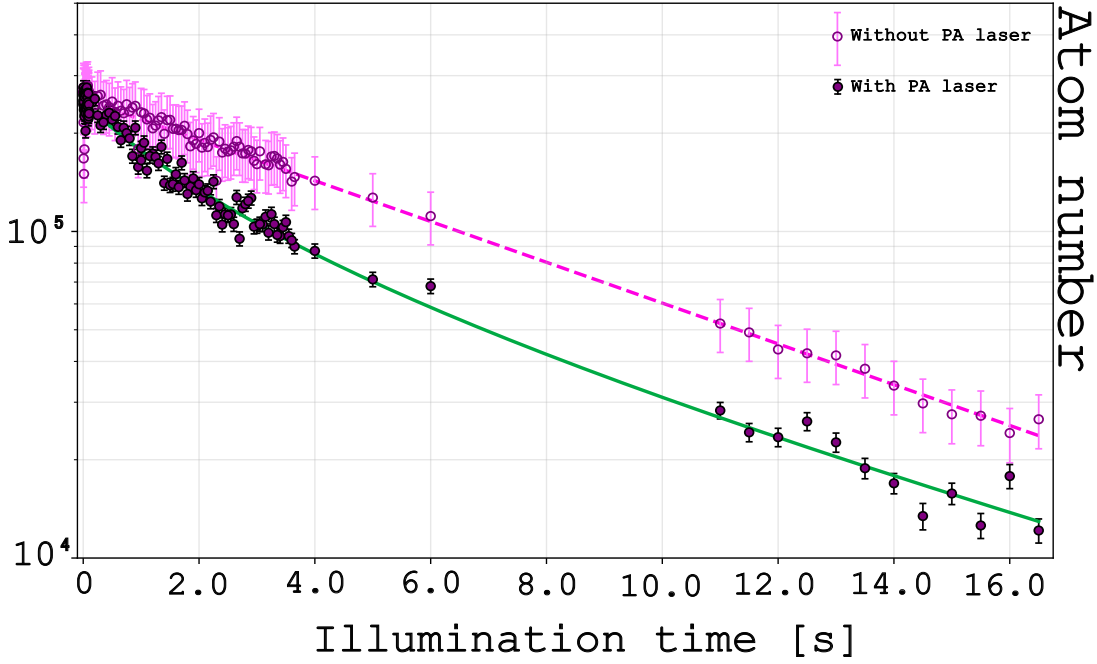


Figure 3.19: Evolution of the atom number (on a logarithmic scale) as a function of the illumination time, measured at a fixed laser detuning of $f = -6$ MHz from the atomic resonance (see Sec. 3.2.4.2). Filled circles: data with the photoassociation (PA) laser, fitted with Eq. 3.77 (combined one- and two-body losses). Open circles: reference measurement at the same frequency without the PA laser. The two decays are different, which shows that a single decay measurement is not enough to identify the loss mechanism.

3.3.1.2 Temperature analysis

Other tests that could indicate inelastic collisions include measuring the temperature of the gas, which we expect to be hotter near the line center. The coldest atoms sit at the bottom of the trap, where the density is highest, and are therefore most likely to photoassociate; once removed, the remaining atoms re-thermalize via elastic collisions, raising the mean temperature, and hence the volume, of the gas.

Measuring the temperature as a function of illumination time provides a further test: as an empirical model, not derived here, we describe the heating of the gas by the saturating exponential

$$T(t) = T_{\infty} - (T_{\infty} - T_0) e^{-\Gamma t} \quad (3.92)$$

with T_0 and T_∞ the temperature without the PA laser and in the limit of infinite exposure time, and Γ the one-body loss rate extracted from the $N(t)$ fit – since this rate sets the timescale over which the loss dynamics transition from the two-body-dominated regime (responsible for the heating) to the one-body-dominated regime. This shape is not a discriminating test: photon scattering on an atomic transition also heats the gas, and the resulting evaporation, which depends non-linearly on $T(t)$ and on the density, also leads to a non-exponential decay and to a curved $T(t)$. In practice, the temperature variations measured here remain small, and no firm conclusion on the loss mechanism can be drawn from them.

Figure 3.20 shows $T(t)$ for the same experiment as Fig. 3.19. The gas heats up during the illumination, but the shape of $T(t)$ alone does not allow one to distinguish atomic scattering from molecular photoassociation: the atomic origin of this signal is established by the symmetric losses observed at equal and opposite detunings (Sec. 3.3.1.1).

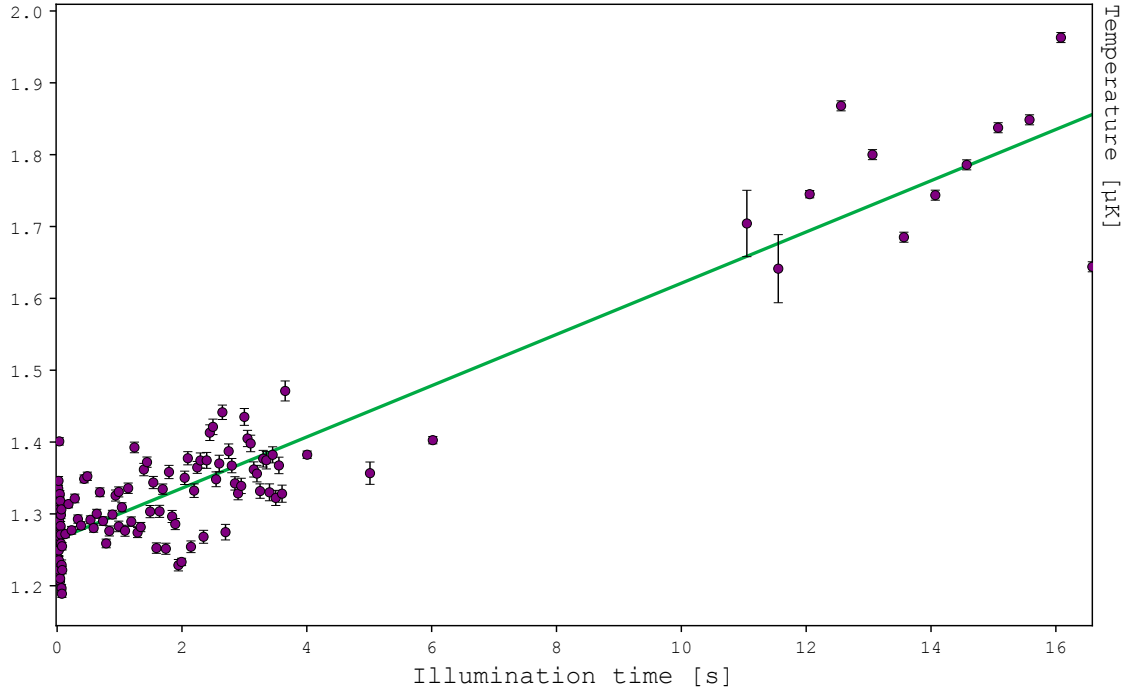


Figure 3.20: Temperature of the gas during the experiment of Fig. 3.19, where the atom number decays as for two-body losses although the signal is atomic, not molecular. The gas heats up during the illumination; the shape of $T(t)$ alone does not distinguish atomic from molecular losses. The green line shows the linear behavior of the temperature.

All the temperatures are measured by time of flight, as described in Sec. 3.2.3.2 (Eq. 3.68 for their uncertainty).

From the analysis of the diode noise at -6 MHz, we conclude that three complementary tests can help to identify the nature of the collisions observed on the ^{87}Sr signals:

- **dN/dt measurement:** the theoretical part of this chapter derives the two dynamics of the atomic decay, with different characteristic times, dominated

by two-body losses at high density and by one-body losses at low density. In practice, comparing this decay in the presence and absence of the PA laser was not on its own sufficient to conclude on the nature of the collisions.

- **$T(t)$ measurement:** the heating of the gas during the illumination is measured, although its shape alone does not discriminate between loss mechanisms.
- **$T(f)$ measurement:** the temperature as a function of the laser detuning is shown for each ^{87}Sr line below; in practice, it only shows negligible variations.

3.3.2 Molecular lines on ^{87}Sr

While searching for molecular resonances on ^{87}Sr , we found only three resonances, instead of the 981 expected molecular channels; they are presented in this section. All the spectra presented in this section have an off-resonant level (baseline) below unity. Close to the atomic resonance, this is expected, since at high intensity the atomic line is power broadened. But even far from the atomic resonance ($\simeq 200$ MHz), the baseline of the normalized data is below unity: we attribute it to atomic losses induced by the PA light, which still scatters photons on the atomic transition. For $|\Delta| \gg \Omega, \Gamma$, the scattering rate per atom is $R_{\text{sc}} \simeq \Gamma \Omega^2 / (4\Delta^2) = \frac{\Gamma}{2} s \left(\frac{\Gamma}{2\Delta}\right)^2$, with $s = I/I_{\text{sat}} = 2\Omega^2/\Gamma^2$ and $I_{\text{sat}} \simeq 3 \mu\text{W}/\text{cm}^2$. For $I \simeq 19 \text{ W}/\text{cm}^2$, i.e. $s \simeq 6 \times 10^6$, and $\Delta/2\pi \simeq -260$ MHz, this gives $R_{\text{sc}} \sim 10$ photons per second and per atom (the Clebsch–Gordan coefficients of the π transitions $F = 9/2 \rightarrow F' = 11/2$, averaged over the ten m_F states of the unpolarized gas, reduce the line strength by a factor $\simeq 0.4$). Each photon heats the atom by about two recoil energies ($E_r/k_B \simeq 0.23 \mu\text{K}$), which heats the gas by $\sim 1 \mu\text{K}/\text{s}$, of the order of the heating measured on $T(t)$, and leads to losses by evaporation: this is responsible for the baseline below unity. Since $R_{\text{sc}} \propto I/\Delta^2$, this effect decreases quickly as the laser is detuned further from the atomic resonance.

Line at -259.01 MHz

We now present the spectra in which PA lines were found.

The molecular state closest to the atomic resonance on the red side is found at $f_1 = -259.01 \pm 0.12$ MHz. It was obtained for an illumination time of $t = 1000$ ms on a recompressed gas of average density $\langle n \rangle = (7.0 \pm 1.0) \times 10^{12}$ atoms/ cm^3 and temperature $T = 2.38 \pm 0.08 \mu\text{K}$ at intensity $I = 19.0 \pm 1.1 \text{ W}/\text{cm}^2$. The PA beam has a waist $w \simeq 60 \mu\text{m}$; all uncertainties are estimated as described in Sec. 3.2.3.5. The Lorentzian fit of the normalized data (each point is the ratio of the atom numbers measured with and without PA light in two consecutive cycles, see the beginning of this section; Fig. 3.21, top left) gives a surprisingly large linewidth $\Gamma_{\text{fit}} = 7.4 \pm 0.9$ MHz, i.e. $\Gamma_{\text{fit}}/(\Gamma_{\text{rad}}/2\pi) \simeq 490$, with a depth of $40 \pm 2\%$ of the off-resonant atom number at resonance. The off-resonant level of the normalized spectrum is itself below unity (0.91 ± 0.02): away from the molecular line, the PA light induces an additional loss of $\simeq 9\%$ in 1 s which is not molecular, as already observed in Fig. 3.19: off-resonant photon scattering heats the gas, which then loses atoms by evaporation. The depth is measured relative to this off-resonant level, which removes this one-body loss from the two-body signal.

Figure 3.21 (bottom left) shows the temperature of the cloud measured in the same experiment, to help confirm the molecular nature of the losses. It only shows negli-

gible variations across the resonance ($2.43 \pm 0.07 \mu\text{K}$ within the half width against $2.34 \pm 0.08 \mu\text{K}$ away from it), comparable to the shot-to-shot scatter, from which no conclusion can be drawn.

Since the gas heats up during photoassociation and the temperature trace shows large fluctuations, the decay (Fig. 3.21, top right) is fitted with the interpolated-temperature procedure of Sec. 3.2.3.3, with four free parameters (Γ , K_2 , N_0 and a constant offset). The same procedure is applied to the -276.83 MHz line.

The decay was recorded in the same trap as the spectrum, the gas temperature rising from 2.65 ± 0.04 to $4.70 \pm 0.13 \mu\text{K}$ during the decay: from the measured trap beam powers, the mean trap frequency is estimated at $\bar{\omega}/2\pi = 224 \pm 10 \text{ Hz}$, giving an average density $\langle n \rangle = (7.0 \pm 1.0) \times 10^{12} \text{ atoms/cm}^3$. The fit yields $K_2 = (5.0 \pm 2.8) \times 10^{-13} \text{ cm}^3/\text{s}$ at $I = 19.0 \pm 1.1 \text{ W/cm}^2$, the same intensity and density as for the spectrum. The ideal-line relation (Eq. 3.83) would give $l_{\text{opt}}^{\text{ideal}}/I = (6.8 \pm 3.8) \times 10^{-3} a_0/(\text{W/cm}^2)$. Accounting for the linewidth through Eq. 3.84, Method 2 gives $l_{\text{opt}}/I = 3.4 \pm 1.9 a_0/(\text{W/cm}^2)$. Method 1, using the depth of the normalized spectrum (Eq. 3.85), gives $K_2^{\text{fit}}(0) = (9.4 \pm 1.5) \times 10^{-14} \text{ cm}^3/\text{s}$, i.e. $l_{\text{opt}}/I = 0.63 \pm 0.13 a_0/(\text{W/cm}^2)$ (including the intensity and density uncertainties). The two methods differ by a factor $\simeq 5$ on l_{opt}/I , but the two loss coefficients, $(9.4 \pm 1.5) \times 10^{-14} \text{ cm}^3/\text{s}$ for the spectrum and $(5.0 \pm 2.8) \times 10^{-13} \text{ cm}^3/\text{s}$ for the decay, remain compatible within $\simeq 1.5\sigma$, owing to the large uncertainty of the decay fit for this line.

The two measurements can also be compared directly, independently of the decay model. Since the spectrum and the decay share the same intensity and density, the $(40 \pm 2)\%$ depth at the line center after 1 s, i.e. $K_2 \langle n \rangle t = A/(1 - A) = 0.66 \pm 0.05$, predicts the same loss after 1 s on the decay. The decay, however, also contains the non-molecular PA-induced loss, which is removed from the spectrum by measuring its depth relative to the off-resonant level. Since the solution of Eq. 3.77 is the product of the one-body factor $e^{-\gamma t}$, with γ the one-body loss rate induced by the off-resonant PA light, and of a two-body factor, the off-resonant level of the spectrum, $b = e^{-\gamma t}$, cancels in the depth measured relative to it, and dividing the decay by b isolates the same two-body factor. The raw $(49 \pm 3)\%$ loss read on the decay at 1 s ($K_2 \langle n \rangle t = 0.96 \pm 0.12$) becomes $(44 \pm 3)\%$ once divided by the off-resonant level of the spectrum recorded at the same intensity, i.e. $K_2 \langle n \rangle t = 0.79 \pm 0.12$, in agreement with the spectrum within $\simeq 1\sigma$. The larger factor between the two methods on K_2 is most likely due to the decay model: the fitted K_2 relies on the interpolated temperature $T(t)$, which rises up to $4.7 \mu\text{K}$ and increases V_e , whereas the direct comparison at 1 s does not. The gas heats up during the decay (Fig. 3.21, bottom right), but, as discussed in Sec. 3.3.1.2, the shape of $T(t)$ does not allow one to identify the heating mechanism.

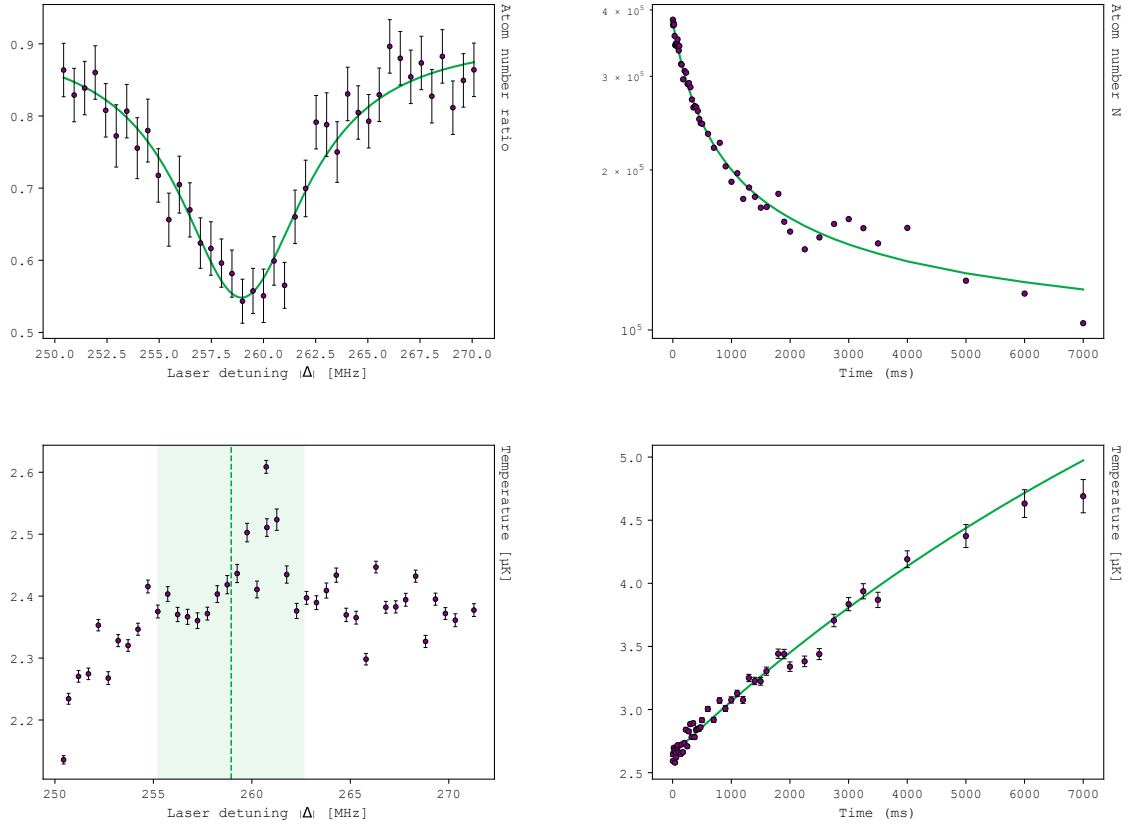


Figure 3.21: Characterization of the molecular line at -259.01 ± 0.12 MHz in ^{87}Sr . Top left: resonance spectrum ($I = 19.0 \pm 1.1$ W/cm 2 , $t = 1000$ ms, $\langle n \rangle = (7.0 \pm 1.0) \times 10^{12}$ cm $^{-3}$), with Lorentzian fit (green). Top right: atom number decay $N(t)$ (same intensity and density), fitted via Eq. 3.77 with interpolated $T(t)$. Bottom left: temperature of the cloud versus detuning; the shaded region and dashed line mark the resonance width and center from the top-left fit. Bottom right: temperature versus illumination time, fitted with the empirical saturating exponential of Eq. 3.92, with the one-body loss rate Γ fixed to the value obtained from the fit of $N(t)$. All detunings are measured from the ^{87}Sr atomic resonance.

Line at -276.83 MHz

The second line is detected at $f_2 = -276.83 \pm 0.03$ MHz from the atomic resonance (Fig. 3.22, top left). The gas, prepared at an average density of $\langle n \rangle = (1.6 \pm 0.2) \times 10^{13}$ atoms/cm 3 and temperature $T = 1.25 \pm 0.03$ μK by evaporation at the crossing of the reservoir and dimple beams, was illuminated for 1000 ms at intensity $I = 5.2 \pm 0.3$ W/cm 2 . The measured linewidth, $\Gamma_{\text{fit}} = 5.0 \pm 0.3$ MHz ($\Gamma_{\text{fit}}/(\Gamma_{\text{rad}}/2\pi) \simeq 330$), is again surprisingly large, and the depth reaches $34.5 \pm 0.7\%$ of the off-resonant atom number, whose normalized level is 0.969 ± 0.010 . Two points of the spectrum, at 276.3 and 281.8 MHz, lie more than 3σ away from the fit (6σ and 4σ respectively, σ being the scatter of the residuals). Their absorption images are normal, but the corresponding shots show an abnormally low temperature (see below): these two points are therefore not considered and are removed from the spectrum and from the fit. The reduced χ^2 of the fit is 1.6, and the uncertainties given here are scaled accordingly.

The temperature (Fig. 3.22, bottom left) only shows negligible variations across the

resonance ($1.27 \pm 0.02 \mu\text{K}$ within the half width against $1.23 \pm 0.01 \mu\text{K}$ away from it), from which no conclusion can be drawn. Three points of this temperature spectrum, at 274.3, 276.3 and 281.8 MHz, were measured at $T \simeq 1.03 \mu\text{K}$, i.e. about 7σ below the dispersion of the other points ($1.25 \pm 0.03 \mu\text{K}$): they are not considered, and are removed from Fig. 3.22. The shots at 276.3 and 281.8 MHz are the two points removed from the atom number spectrum above.

As for the first line, the atom number decay is fitted with the interpolated-temperature procedure of Sec. 3.2.3.3.

For the same trap parameters as for the spectrum (the gas temperature rising from 2.31 ± 0.04 to $4.72 \pm 0.18 \mu\text{K}$ during the decay), the mean trap frequency is estimated at $\bar{\omega}/2\pi = 192 \pm 9 \text{ Hz}$, giving an average density $\langle n \rangle = (7.8 \pm 1.1) \times 10^{12} \text{ atoms/cm}^3$ at the start of the decay. The fit yields $K_2 = (1.1 \pm 0.2) \times 10^{-12} \text{ cm}^3/\text{s}$ at $I = 31.0 \pm 1.8 \text{ W/cm}^2$. The ideal-line relation (Eq. 3.83) would give $l_{\text{opt}}^{\text{ideal}}/I = (9.2 \pm 1.5) \times 10^{-3} a_0/(\text{W/cm}^2)$. Accounting for the linewidth measured at 5.2 W/cm^2 through Eq. 3.84, Method 2 gives $l_{\text{opt}}/I = 3.1 \pm 0.5 a_0/(\text{W/cm}^2)$; since the decay and the spectrum were recorded at different intensities, this value is a lower bound if the width increases with intensity (light shift inhomogeneity) or if the line is light-shifted away from the decay frequency. Method 1, using the depth of the spectrum at 5.2 W/cm^2 , gives $K_2^{\text{fit}}(0) = (3.3 \pm 0.4) \times 10^{-14} \text{ cm}^3/\text{s}$, i.e. $l_{\text{opt}}/I = 0.54 \pm 0.08 a_0/(\text{W/cm}^2)$. As for the first line, the factor $\simeq 6$ between the two methods is most likely due to the decay model, the temperature rising up to $4.7 \mu\text{K}$ during the decay. Compared directly at 1 s, the two measurements are compatible. Assuming $K_2 \propto I$, the $(34.5 \pm 0.7)\%$ depth at the line center after 1 s at 5.2 W/cm^2 , i.e. $K_2 \langle n \rangle t = A/(1-A) = 0.53 \pm 0.02$, corresponds to $K_2 \langle n \rangle t = 1.52 \pm 0.32$ at 31 W/cm^2 and for the density of the decay, that is a loss of $(60 \pm 5)\%$ after 1 s. On the decay, a raw $(74 \pm 2)\%$ loss is read at the same time. As for the first line, this raw loss includes the non-molecular PA-induced loss, measured on the off-resonant level of the spectrum (0.969 ± 0.010 in 1 s at 5.2 W/cm^2). Scaled to 31 W/cm^2 assuming a rate proportional to I , this loss amounts to a factor 0.83 ± 0.05 in 1 s; once it is removed, the decay gives a $(69 \pm 3)\%$ loss, i.e. $K_2 \langle n \rangle t = 2.2 \pm 0.3$ (uncertainties from the fit of the spectrum, from the densities, from the intensities and from the 5% uncertainty on each atom number of the decay). The two measurements thus differ by a factor $\simeq 1.45$ on $K_2 \langle n \rangle t$, i.e. by $\simeq 1.5\sigma$, and remain compatible within 2σ . The residual difference may come from the scaling between the two measurements, recorded at different intensities, densities and temperatures (1.25 and $2.31 \mu\text{K}$). The gas heats up during the decay (Fig. 3.22, bottom right); the saturating-exponential model of Eq. 3.92 does not reproduce the last three points of the temperature trace, and the shape of $T(t)$ does not allow one to identify the heating mechanism (Sec. 3.3.1.2).

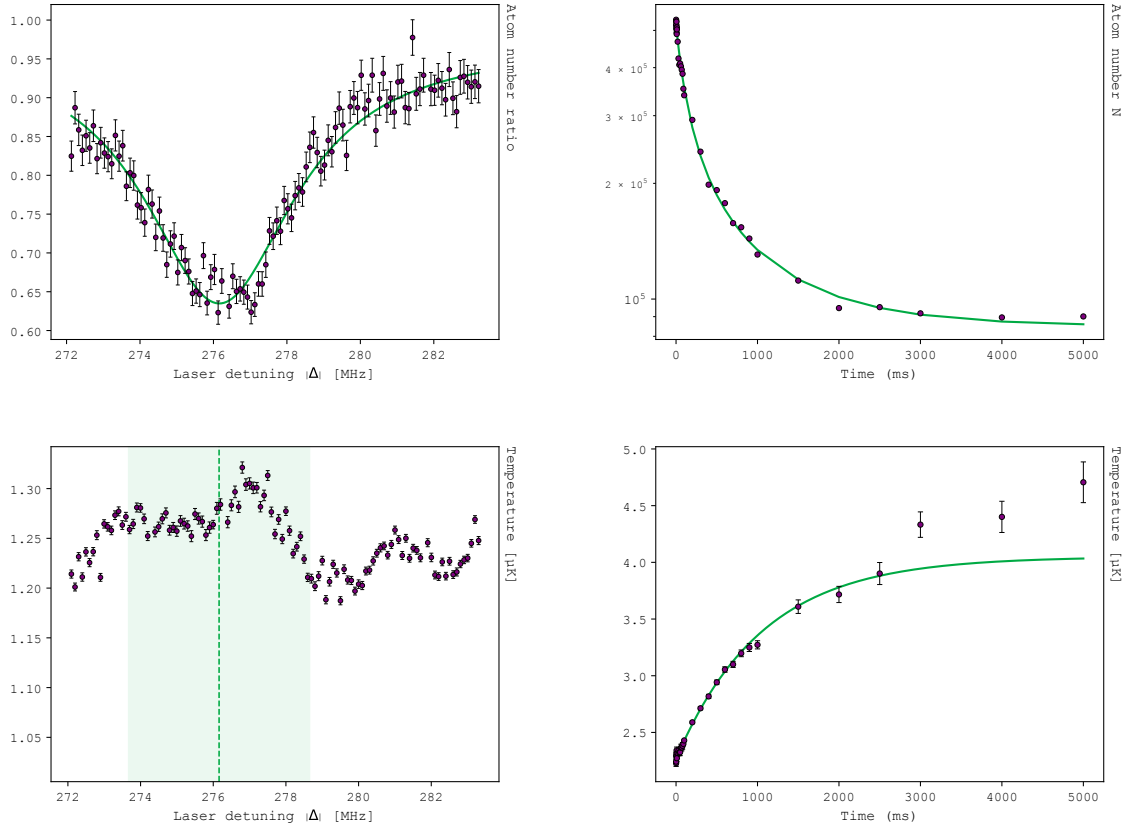


Figure 3.22: Characterization of the molecular line at -276.83 ± 0.03 MHz in ^{87}Sr . Panel layout as in Fig. 3.21, with $I = 5.2 \pm 0.3$ W/cm² for the spectrum and $I = 31.0 \pm 1.8$ W/cm² for the $N(t)/T(t)$ fits, and $\langle n \rangle = (1.6 \pm 0.2) \times 10^{13}$ cm⁻³ (spectrum) and $(7.8 \pm 1.1) \times 10^{12}$ cm⁻³ (decay). Brought to the same intensity and density ($K_2 \propto I$), the loss at the line center after 1 s in the spectrum ($(60 \pm 5)\%$ expected) is compatible within 2σ with the one of the decay once the non-molecular PA-induced loss is removed ($(69 \pm 3)\%$ at 1 s, $(74 \pm 2)\%$ raw). All detunings are measured from the ^{87}Sr atomic resonance.

Line at -377.2 MHz

The last molecular line is measured at $f_3 = -377.2 \pm 0.5$ MHz, for a recompressed gas of average density $\langle n \rangle = (7.2 \pm 1.0) \times 10^{12}$ atoms/cm³ and temperature $T = 2.38 \pm 0.04$ μK illuminated for 1000 ms at intensity $I = 5.6 \pm 0.4$ W/cm² (Fig. 3.23, top left). It has a notably small depth ($8.9 \pm 0.5\%$ loss) and, again, a large linewidth (FWHM) $\Gamma_{\text{fit}} = 20.2 \pm 2.3$ MHz ($\Gamma_{\text{fit}}/(\Gamma_{\text{rad}}/2\pi) \simeq 1.3 \times 10^3$). Each point is the ratio of the atom number with and without PA light for consecutive shots. The Lorentzian fit is performed after an iterative 3σ clipping of the residuals, $\sigma = 0.033$ being estimated from their median absolute deviation: 24 of the 651 points deviate by more than 3σ from the fit; they are removed from the data and are not shown in Fig. 3.23. The error bars combine in quadrature the uncertainty of the 2D Gaussian fits of the images (diagonal of their covariance matrix) and the relative RMS fluctuation of the atom number ratio, 3.4%, evaluated on the first 100 points of the scan, far from the resonance. Because of this shot-to-shot fluctuation, individual ratios above unity are expected even though the mean ratio cannot exceed one; these points are therefore kept, since removing them would bias the off-resonant level downwards

and underestimate the depth of the line (removing them lowers the depth from 8.9% to 7.6%). With these error bars, the fit gives a reduced χ^2 of 1.1. The temperature (Fig. 3.23, bottom left) only shows negligible variations across the resonance ($2.41 \pm 0.03 \mu\text{K}$ within the half width against 2.36 to 2.39 μK on either side), from which no conclusion can be drawn. Since the temperature changes less during the decay of this line than for the two other lines, the decay (Fig. 3.23, top right) is well described by Eq. 3.77 with a constant β .

The decay was recorded in the same trap as for the first line (the gas temperature rising from 2.28 ± 0.03 to $3.23 \pm 0.04 \mu\text{K}$ during the decay): the mean trap frequency is estimated at $\bar{\omega}/2\pi = 224 \pm 10 \text{ Hz}$, giving an average density $\langle n \rangle = (7.0 \pm 1.0) \times 10^{12} \text{ atoms/cm}^3$, close to that of the spectrum. The fit yields $K_2 = (2.7 \pm 1.9) \times 10^{-14} \text{ cm}^3/\text{s}$ at $I = 19.0 \pm 1.1 \text{ W/cm}^2$. The ideal-line relation (Eq. 3.83) would give $l_{\text{opt}}^{\text{ideal}}/I = (3.6 \pm 2.6) \times 10^{-4} a_0/(\text{W/cm}^2)$. Accounting for the linewidth measured at 5.6 W/cm^2 through Eq. 3.84, Method 2 gives $l_{\text{opt}}/I = 0.49 \pm 0.35 a_0/(\text{W/cm}^2)$, with the same caveat on the different intensities as for the previous line. Method 1 gives $K_2^{\text{fit}}(0) = (1.4 \pm 0.2) \times 10^{-14} \text{ cm}^3/\text{s}$ at 5.6 W/cm^2 , i.e. $l_{\text{opt}}/I = 0.84 \pm 0.17 a_0/(\text{W/cm}^2)$, consistent with Method 2 within uncertainties. The two measurements also agree directly once brought to the same intensity and density: assuming $K_2 \propto I$, the $(8.9 \pm 0.5)\%$ loss of the spectrum at 5.6 W/cm^2 ($K_2 \langle n \rangle t = 0.098 \pm 0.006$) corresponds to $K_2 \langle n \rangle t = 0.32 \pm 0.07$ at 19 W/cm^2 and for the density of the decay, i.e. a loss of $(24 \pm 4)\%$ after 1 s, in agreement with the $(27 \pm 5)\%$ raw loss read on the decay (same sources of uncertainty as for the previous line). Removing the non-molecular PA-induced loss as for the two other lines (off-resonant level 0.974 ± 0.004 at 5.6 W/cm^2 , i.e. a factor 0.91 ± 0.01 in 1 s at 19 W/cm^2) gives a $(20 \pm 6)\%$ loss on the decay, still in agreement with the spectrum. The gas heats up moderately during the decay (Fig. 3.23, bottom right), from 2.28 to 3.23 μK ; as for the other lines, the shape of $T(t)$ does not allow one to identify the heating mechanism.

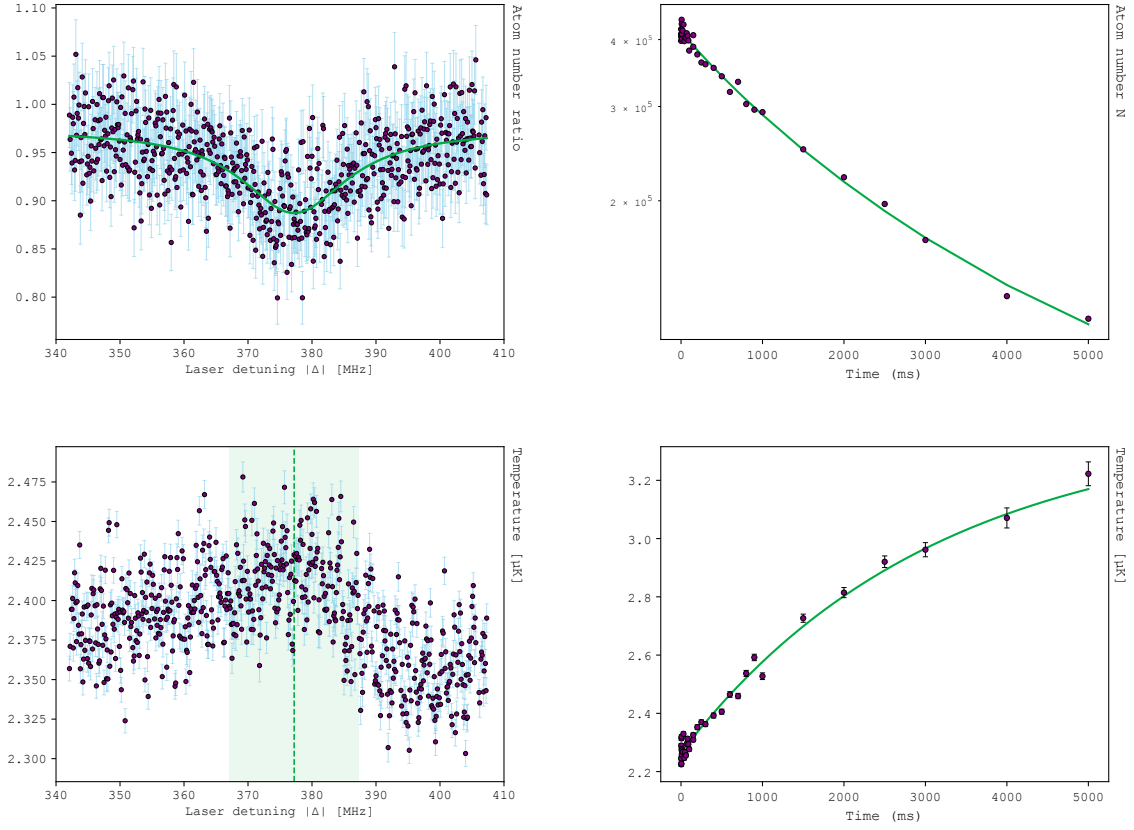


Figure 3.23: Characterization of the molecular line at -377.2 ± 0.5 MHz in ^{87}Sr . Panel layout as in Fig. 3.21, with $I = 5.6 \pm 0.4$ W/cm² and $\langle n \rangle = (7.2 \pm 1.0) \times 10^{12}$ cm⁻³ for the spectrum, and $I = 19.0 \pm 1.1$ W/cm² and $\langle n \rangle = (7.0 \pm 1.0) \times 10^{12}$ cm⁻³ for the $N(t)/T(t)$ fits. Brought to the same intensity ($K_2 \propto I$), the loss at the line center after 1 s in the spectrum ($(24 \pm 4)\%$ expected at 19 W/cm²) agrees with the decay ($(27 \pm 5)\%$ at 1 s). Left panels: ratio of the atom numbers of consecutive shots with and without PA light, and temperature of the PA shots computed from the vertical cloud size after time of flight. The 24 of 651 points lying more than 3σ away from the Lorentzian fit (green) are removed, $\sigma = 0.033$ being the scatter of the fit residuals. Error bars: 2D Gaussian fit uncertainty, combined in quadrature for the atom number ratio with its relative RMS fluctuation (3.4%) over the first 100 off-resonant points of the scan. All detunings are measured from the ^{87}Sr atomic resonance.

Table 3.4 summarizes the experimental conditions and the resulting fit parameters for the three resonances, together with the derived density $\langle n \rangle = N/(2\sqrt{2}V_e)$ and characteristic time τ (defined in Eq. 3.63). The mean trap frequency $\bar{\omega}$ is computed for each measurement from the measured Reservoir and Dimple beam powers, and all uncertainties are estimated as described in Sec. 3.2.3.5. The depth is the peak amplitude of the Lorentzian fitted on the normalized data (ratio of the atom numbers with and without PA light), divided by its off-resonant level (Sec. 3.2.3.2); the off-resonant level, below unity, measures the non-molecular loss induced by the PA light. The densities of the spectrum and of the decay are computed separately: they are identical for the first line and differ for the last two lines, as do the intensities. $l_{\text{opt}}^{\text{ideal}}/I$ is obtained from the ideal-line relation (Eq. 3.83), which neglects the linewidth, while

the last two rows use Eq. 3.84 with $K_2^{\text{fit}}(0)$ from the spectrum depth (Method 1) or from the decay (Method 2).

Table 3.4: Summary of the fit parameters and experimental conditions for the three ^{87}Sr lines.

Line f (MHz)	-259.01 ± 0.12	-276.83 ± 0.03	-377.2 ± 0.5
<i>Spectrum</i>			
Depth	0.40 ± 0.02	0.35 ± 0.01	0.09 ± 0.01
Off-resonant level	0.91 ± 0.02	0.97 ± 0.01	0.98 ± 0.01
Width Γ_{fit} (MHz)	7.4 ± 0.9	5.0 ± 0.3	20.2 ± 2.3
Exposure time t (ms)	1000	1000	1000
Temperature T (μK)	2.38 ± 0.08	1.25 ± 0.03	2.38 ± 0.04
Density $\langle n \rangle$ (10^{12} cm^{-3})	7.0 ± 1.0	16 ± 2	7.2 ± 1.0
Intensity I (W/cm^2)	19.0 ± 1.1	5.2 ± 0.3	5.6 ± 0.4
<i>Decay $N(t)$</i>			
Intensity I (W/cm^2)	19.0 ± 1.1	31.0 ± 1.8	19.0 ± 1.1
$\bar{\omega}/2\pi$ (Hz)	224 ± 10	192 ± 9	224 ± 10
Density $\langle n \rangle$ (10^{12} cm^{-3})	7.0 ± 1.0	7.8 ± 1.1	7.0 ± 1.0
K_2 ($10^{-13} \text{ cm}^3/\text{s}$)	5.0 ± 2.8	11 ± 2	0.27 ± 0.19
Characteristic time τ	286 ms	117 ms	5.3 s
$l_{\text{opt}}^{\text{ideal}}/I$ ($10^{-3} a_0/(\text{W}/\text{cm}^2)$)	6.8 ± 3.8	9.2 ± 1.5	0.36 ± 0.26
<i>Optical length (Eq. 3.84)</i>			
$\Gamma_{\text{fit}}/(\Gamma_{\text{rad}}/2\pi)$	493 ± 60	332 ± 17	1349 ± 155
$K_2^{\text{fit}}(0)$, spectrum ($10^{-14} \text{ cm}^3/\text{s}$)	9.4 ± 1.5	3.3 ± 0.4	1.4 ± 0.2
l_{opt}/I , Method 1 ($a_0/(\text{W}/\text{cm}^2)$)	0.63 ± 0.13	0.54 ± 0.08	0.84 ± 0.17
l_{opt}/I , Method 2 ($a_0/(\text{W}/\text{cm}^2)$)	3.4 ± 1.9	3.1 ± 0.5	0.49 ± 0.35

PA lines found on ^{87}Sr

These results show several unexpected features:

- First, although about 3×10^3 molecular lines, separated by ~ 100 kHz on average, are expected within the $\simeq 380$ MHz explored below the asymptote, we did not detect any two-body loss closer to the asymptote than 259.01 MHz. This is also not consistent with the line positions expected for ^{87}Sr from the mass scaling of Ref. [15].
- The linewidth of each line is large: 7.4 ± 0.9 , 5.0 ± 0.3 and 20.2 ± 2.3 MHz, while we expect $\Gamma_{\text{rad}}/2\pi \simeq 2 \times \Gamma_{\text{at}}/2\pi \simeq 15$ kHz, i.e. two to three orders of magnitude larger.
- As a direct consequence of this broadening, the losses at resonance remain small: the depths of the three lines range from $8.9 \pm 0.5\%$ to $40 \pm 2\%$ after 1 s of illumination (Table 3.4). At fixed l_{opt} , a line 10^2 – 10^3 times broader than Γ_{rad} has a peak loss coefficient lowered by the same factor (Eq. 3.84): the small depths and the large linewidths are therefore not two independent observations: both reflect a peak loss rate lowered by the broadening.

- Once the linewidth is accounted for, the analysis of these data gives optical lengths $l_{\text{opt}}/I \simeq 0.5\text{--}3.5 \text{ } a_0/(\text{W}/\text{cm}^2)$ (Table 3.4), i.e. one to two orders of magnitude below the $\nu = -2$ line of ^{88}Sr measured in the same setup ($56 \pm 10 \text{ } a_0/(\text{W}/\text{cm}^2)$), and about two orders of magnitude below the ^{84}Sr line of similar binding energy ($228(42) \text{ } a_0/(\text{W}/\text{cm}^2)$ at 228 MHz [16], where l_{opt} is also corrected for the broadening through a factor η). They are of the order of the weakest strontium lines, such as the ^{84}Sr line at 1143 MHz ($0.95(12) \text{ } a_0/(\text{W}/\text{cm}^2)$ [16]), which is suppressed by a node of the ground-state wavefunction; such a node is however unlikely for our lines (Sec. 3.3.3.2).

The high intensities needed to observe the ^{87}Sr lines (5 to 31 W/cm^2 , much higher than for ^{88}Sr) come from both a smaller optical length and a peak loss coefficient lowered by the $10^2\text{--}10^3$ broadening. The ideal-line relation, which neglects the width, underestimates l_{opt} by this same factor: once the width is accounted for, the main anomaly of these lines is their width rather than their optical length. For comparison, the broadening factor $\eta = \Gamma_{\text{fit}}/(\Gamma_{\text{rad}}/2\pi)$ is only $\simeq 1.4\text{--}1.6$ for the ^{84}Sr lines of Ref. [16] and $\simeq 22$ for our ^{88}Sr lines, against 330–1350 for ^{87}Sr . The next subsection discusses possible explanations for these unexpected features.

3.3.3 Interpretation of the large linewidth and weak loss rate

3.3.3.1 Possible technical causes

The weak loss rate and the large linewidth could first come from technical limitations of the experiment:

- **Coupling to higher-energy states by the ODT laser:** the ODT laser at $\lambda = 1075 \text{ nm}$ can induce unwanted photoassociation (PA) or transitions to higher-energy molecular states if it is resonant with a transition starting from the intermediate excited state. To suppress these parasitic multi-photon processes, the ODT and the PA laser are switched on and off in opposite phase (chopping sequence, Fig. 3.24).

This chopping meets three constraints:

1. **Suppression of multi-photon processes:** since the ODT (1075 nm) and the PA laser (689 nm) are never on at the same time, photoassociation paths involving both photons are suppressed.
2. **Preservation of the atomic cloud:** the trap is switched off for a time short enough that the cloud does not fall significantly under gravity.
3. **AOM response and spectral purity:** the switching period is long enough for the acousto-optic modulators (AOMs) to follow, so the PA laser spectrum is not significantly broadened or distorted.

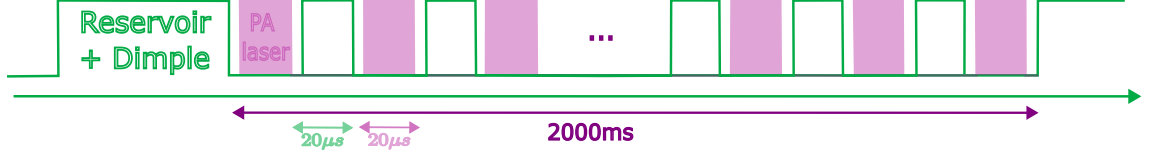


Figure 3.24: Experimental sequence of the ODT and PA laser chopping used to prevent multi-photon transitions to higher energy levels. The RF signals driving the ODT and Raman AOMs are modulated at 25 kHz in opposite phase. This ensures that the two lasers do not overlap while keeping the extinction time short enough to avoid atom loss due to gravity.

The chopping sequence was only used for one ^{88}Sr spectrum (the $\nu = -1$ line, Sec. 3.2.4), where a comparison with and without chopping showed only a negligible effect on the linewidth. It was not used for the ^{87}Sr measurements: this possibility therefore cannot be ruled out for ^{87}Sr .

- **Noise of the MOGLabs laser:** on the ^{88}Sr molecular lines, this laser gave widths about 20 times larger than the expected $\simeq 15$ kHz, which is not enough to explain the MHz linewidths observed on ^{87}Sr .
- **Light shift by the trapping light:** the ODT light also shifts the molecular level, by an amount that depends on the local trap intensity. Atoms at different positions in the trap, in particular the hotter ones exploring its edges, therefore see the line at slightly different frequencies, which broadens it. This broadening does not depend on the PA intensity. On the $\nu = -1$ line of ^{88}Sr , switching the chopping on and off had only a negligible effect on the linewidth, which suggests that this effect is small. However, the ^{87}Sr lines were recorded with the trap on during the PA pulses, so that it cannot be ruled out for them; it would be suppressed by the chopping sequence.
- **Light shift by the PA beam:** the intensities used on ^{87}Sr (5–31 W/cm²) are much higher than on ^{88}Sr . The PA beam itself shifts the molecular level, by an amount proportional to its intensity, as measured on ^{88}Sr : -251 ± 67 kHz per W/cm² for the $\nu = -1$ state, i.e. ~ 200 kHz for 0.8 W/cm² [14]. This has two effects. First, the whole line is shifted: with this coefficient, the shift would reach $\simeq -5$ MHz at 19 W/cm², so that lines measured at different intensities, or a spectrum and a decay recorded at different intensities, are not at exactly the same frequency. Second, since the intensity of the Gaussian PA beam varies across the cloud, atoms at different positions see different shifts, which broadens the line (inhomogeneous light shift of the PA beam). With this coefficient, the waist $w \simeq 60$ μm of the PA beam and a cloud of rms size $\sigma \simeq 11$ μm , the intensity varies by $\simeq 25\%$ over $\pm 2\sigma$, which spreads the resonance over $\simeq 1$ MHz at 19 W/cm² and $\simeq 0.3$ MHz at 5.6 W/cm². Coefficients about 10 to 60 times larger would be needed to explain the observed widths of 5 to 20 MHz, and the broadest line (20 MHz) was measured at the lowest intensity. Both the shift and the broadening scale with the PA intensity, and could be tested by measuring the position and the width of a line at several intensities.
- **Thermal width:** a high gas temperature could broaden the lines, but the thermal width at similar temperatures ($\simeq 12$ –13 kHz, Tables 3.1 and 3.2) is negligible compared with the MHz linewidths observed.

Some of these technical causes, related to the trapping light or to the PA beam, cannot be ruled out yet; we now consider physical explanations.

3.3.3.2 Physical hypotheses for the weak loss rate and large linewidth

As discussed in Sec. 3.2.3.4, the measured peak loss coefficient combines two factors: the optical length of the line, set by the coupling between the colliding pair and the molecular state, and the ratio $\Gamma_{\text{rad}}/\Gamma_{\text{mol}}$ between the radiative and the total width of the line. We first discuss the effects that can lower the optical length without broadening the line, then those that can broaden it.

Effects lowering the optical length.

- **Node of the ground-state wavefunction:** if the Condon radius of a transition coincides with a node of the ground-state scattering wavefunction, the Franck–Condon factor vanishes and the corresponding line cannot be excited. This is unlikely here: the Condon radius of our lines (roughly 60–70 a_0 , scaling as $E_b^{-1/3}$ the Condon radius $R_c \simeq 150 a_0$ of the ^{84}Sr line at 23 MHz [16]) is close to that of the ^{84}Sr line at 228 MHz, whose optical length is not suppressed (228(42) $a_0/(\text{W}/\text{cm}^2)$ [16]). Since the scattering lengths of the two isotopes are close ($a_{87} = 96 a_0$ [110], $a_{84} = 123 a_0$ [16]), their ground-state wavefunctions are similar at this distance, although only a calculation of the Franck–Condon factors could fully rule out a node. Moreover, this mechanism reduces l_{opt} through the Franck–Condon factor but does not broaden the line: it cannot explain the observed linewidths.
- **Quantum statistics:** quantum statistics alone lowers the optical length of ^{87}Sr by a factor $g_f^{(2)}/g_b^{(2)} \simeq 0.9/2 \simeq 0.45$ with respect to the bosonic ^{88}Sr [110], where $g^{(2)}$ is the pair correlation function at short distance. For thermal bosons, $g_b^{(2)} = 2$. In the unpolarized gas, two atoms drawn at random are in the same m_F state with probability $1/N$, with $N = 10$: they are then identical fermions, and the Pauli principle gives $g^{(2)} = 0$. With probability $1 - 1/N$, they are in different m_F states and behave as distinguishable particles, with $g^{(2)} = 1$. Hence $g_f^{(2)} = (1/N) \times 0 + (1 - 1/N) \times 1 = 1 - 1/N = 0.9$, and not simply 1. An expected value of l_{opt}/I cannot however be estimated from this factor, since the Franck-Condon factor of each line differs between the two isotopes.
- **Spin selection:** spin selection further lowers the measured loss coefficient. In the unpolarized gas, each atom is in one of the ten m_F states of $F = 9/2$ with probability $1/10$. Two identical fermions colliding in the s-wave must be in an antisymmetric spin state, $(|m_{F,1}m_{F,2}\rangle - |m_{F,2}m_{F,1}\rangle)/\sqrt{2}$ with $m_{F,1} \neq m_{F,2}$: there are 45 such states, each with a weight $1/10 \times 1/10$, which gives the factor $45/100 \simeq 0.45$ above. For π light, a resolved molecular line has a well-defined total projection $T = m_{F,1} + m_{F,2}$ and can only be reached from the antisymmetric states with this projection. Among the colliding pairs, the loss coefficient is therefore lowered by the factor

$$\frac{1}{45} \sum_{\substack{m_{F,1} < m_{F,2} \\ m_{F,1} + m_{F,2} = T}} |C_{m_{F,1}m_{F,2}}|^2 \leq \frac{1}{45} \left\lfloor \frac{10 - |T|}{2} \right\rfloor, \quad (3.93)$$

where $C_{m_F,1m_F,2}$ is the Clebsch–Gordan coefficient of the coupling between the pair spin state and the molecular state, and $\lfloor \cdot \rfloor$ the integer part. For $T = -8$, only the pair $\{-9/2, -7/2\}$ contributes, i.e. at most $1/45$; near $T = 0$, five pairs contribute, i.e. at most $1/9$. This gives a reduction by one to two orders of magnitude depending on T , which could explain the small values of l_{opt} ; the exact factor depends on the hyperfine structure of each molecular state, which is not known here. This factor comes in addition to the quantum-statistics factor above.

Quantum statistics and spin selection together can explain optical lengths one to two orders of magnitude below those of the bosonic isotopes, but none of these effects explains the observed linewidths.

Effects broadening the lines.

- **Coupling to higher molecular potentials via potential crossings:** molecular potentials correlated to higher atomic asymptotes, such as $^1S_0 + ^1D_2$, can cross the potentials of the intercombination line. Such an avoided crossing between the 0_u^+ potentials of the $^1S_0 + ^3P_1$ and $^1S_0 + ^1D_2$ asymptotes has been shown to strongly perturb the 0_u^+ levels of ^{84}Sr [15, 16]. A molecular level lying close to such a crossing acquires part of the character of the other state, which shifts it and can open additional decay channels. Artlce [15] shows that the 84 isotope is coupled to an excited molecular state which induces avoided crossing, but it does not predicts this effect on the ^{87}Sr . We have no quantitative estimate of this effect for the ^{87}Sr levels observed here.
- **Predissociation:** a small number of broad PA lines with weak losses, as for ^{87}Sr , was also observed in Ref. [121] for $^{87}\text{Rb}^{170}\text{Yb}$ molecules photoassociated near the $^1S_0 \rightarrow ^3P_1$ intercombination line of Yb. In a search over binding energies from 0.1 to 11 GHz, only a single pair of lines was detected, around 3.1 GHz. The loss coefficient of the strongest line is only $K_{\text{PA}} \simeq 5 \times 10^{-14} \text{ cm}^3/\text{s}$ at $I \simeq 40 \text{ W/cm}^2$, of the same order as our K_2 values (Table 3.4). These lines are also broad: their width extrapolated to zero magnetic field is 3 MHz (FWHM), about 17 times the atomic linewidth of Yb ($2\pi \times 181 \text{ kHz}$); it increases with the magnetic field and does not decrease when the PA intensity is lowered. Since K_{PA} is measured at the line center without accounting for this width, the corresponding line strength is underestimated by the same factor (Sec. 3.2.3.4).

The authors give two interpretations. The first is a strong radiative decay to bound states of the electronic ground state, proposed for RbSr; for RbYb, they estimate that the transition dipole moment of the excited state increases by less than 10% of the atomic one, which makes this mechanism unlikely.

The second interpretation is a non-radiative decay within the 3P manifold, for example a predissociation towards the 3P_0 state, illustrated in Fig. 3.25. Two free atoms in potential A are photoassociated to a vibrational state of a molecular potential B. A potential C crosses potential B above its dissociation limit, which couples the two potentials [122]. For ^{87}Sr , potential C would be one of the potentials associated with the hyperfine structure of the 3P_0 states. The bound pair can then decay into the continuum of potential C. This adds a decay channel to the excited molecular state: its lifetime is shortened, and the

molecular line is broadened beyond its radiative width Γ_{rad} (the total width Γ_{mol} exceeds Γ_{rad}). Following Fermi's golden rule, this additional width is proportional to the density of states of the continuum of potential C at the energy of the bound level, and to the square of the coupling between the two potentials. Since the pair is lost in both decay channels, predissociation does not change l_{opt} , which is defined with respect to the radiative width Γ_{rad} , but lowers the peak loss coefficient by the ratio $\Gamma_{\text{rad}}/\Gamma_{\text{mol}}$ (Eq. 3.43), the area under the line remaining unchanged (Eq. 3.84). A broad line with a weak peak loss rate is therefore precisely the signature expected from strong predissociation. Contrary to the inhomogeneous light shift of the PA beam, this broadening does not depend on the PA intensity. In Ref. [121], such a decay could broaden most of the lines, especially in a magnetic field, enough to make them unobservable, which would explain why a single pair of lines was found. Contrary to RbYb, however, the width of our ^{87}Sr lines does not change with the magnetic field (Sec. 3.3.3.3).

In Fig. 3.4, the linewidths do not seem to be affected by predissociation. We expected that, as for Yb, photoassociation below the lowest hyperfine asymptote would avoid this effect, but our ^{87}Sr data suggest the contrary: we do not know why ^{87}Sr would be affected by this process while Yb, which also has a hyperfine structure, is not.

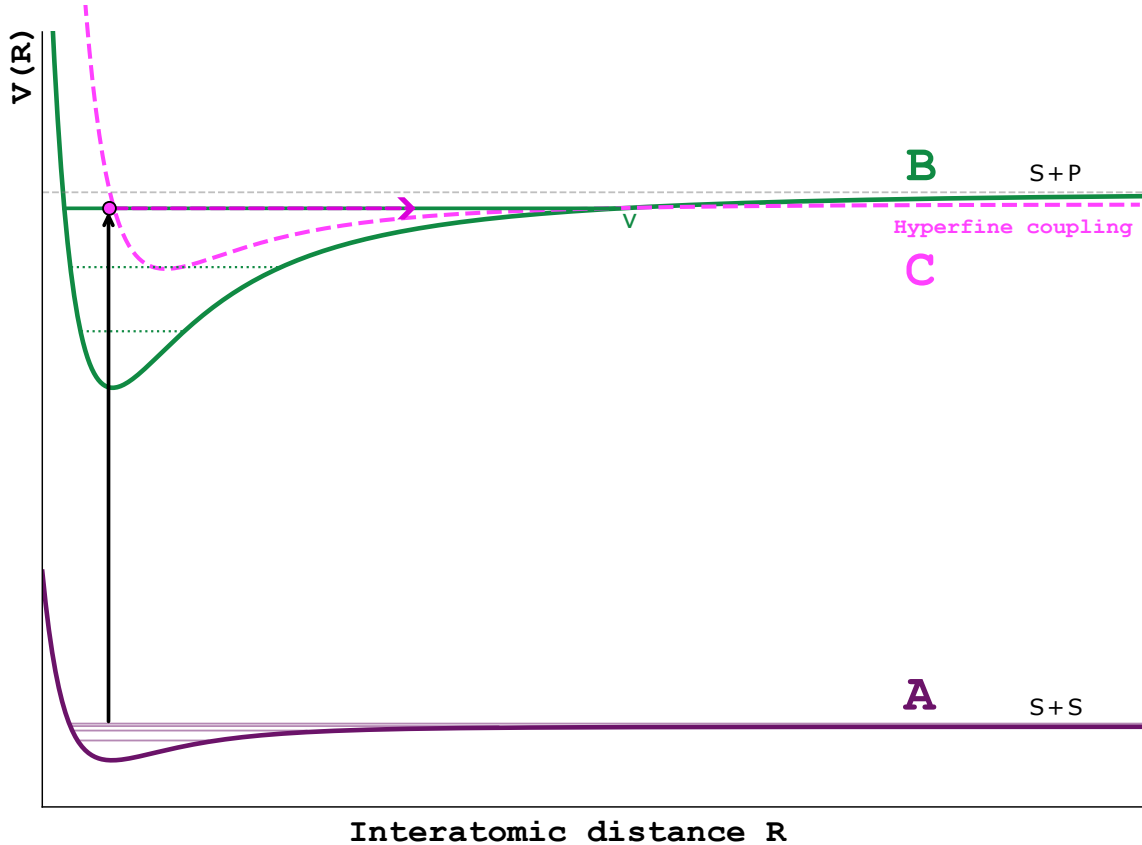


Figure 3.25: Scheme of predissociation. It occurs when the dissociation limit of an excited molecular potential C lies below the bound state ν_B of the excited potential B. The ν_B state is then coupled to the continuum of potential C, which broadens the molecular line beyond Γ_{rad} .

In ^{87}Sr , the hyperfine structure plays both roles. It shares the line strength among many spin components, which lowers the optical length of each resolved line (spin selection above), and it provides lower-lying channels, correlated to other hyperfine asymptotes, into which the bound pair can predissociate, which broadens the lines.

Together with the inhomogeneous light shift of the PA beam and the light shift by the trapping light (Sec. 3.3.3.1), predissociation is the most likely explanation of the broad linewidth of the molecular lines, and hence of their weak loss rate at the line center. Two measurements can distinguish them: the width of a line as a function of the PA intensity, which grows with the intensity only for the light shift of the PA beam, and the width of a line recorded with the chopping sequence, which suppresses the effects of the trapping light. Predissociation is not affected by either of them.

This section ends with the experiments that could clarify this point.

3.3.3.3 Tests to be done in the future

Some tests have already been made on the lines we found, and others remain to be done:

- Changing the polarization of the PA laser could change the coupling strength with the atomic continuum. This test showed no significant change of the line.
- Changing the magnetic field could change the energy difference between the hyperfine states and so decrease the density of states. This was also tested: neither the width nor the position of the lines changed significantly, and no individual molecular lines were resolved.
- Measuring the position and the width of a given line at several PA intensities would give the light shift of the line and discriminate an intensity-dependent broadening (inhomogeneous light shift of the PA beam) from an intensity-independent one (predissociation, light shift by the trapping light).
- Applying the ODT/PA chopping sequence to the ^{87}Sr lines would test the effects of the trapping light (multi-photon transitions, light shift by the trapping light).
- Recording the spectrum and the atom number decay at the same intensity would allow a direct comparison of the two methods of Sec. 3.2.3.4 for every line.
- Optically pumping the atoms to reduce the size of the Hilbert space would reduce the number of atomic potentials and so the probability of predissociation. Experiments are in progress to test this hypothesis. A magnetic field switch has already been implemented in the experiment to populate the extremal m_F states. So far, the measurements of $T(t)$ and $T(f)$ have not given any information on the loss mechanism.

To conclude, this chapter first introduced the theory of the PA process (scattering length, optical length, two-body loss dynamics). It then presented PA measurements on the intercombination line of ^{88}Sr , used as a reference to characterize the experimental procedure. It ends with the measurement of molecular lines in an un-

polarized gas of ^{87}Sr , with linewidths two to three orders of magnitude larger than the radiative width, which account for their weak loss rate at the line center: the loss coefficients measured at the line center are $K_2 \sim 10^{-14}\text{--}10^{-12} \text{ cm}^3/\text{s}$ for PA intensities of 5 to 31 W/cm^2 (Table 3.4), against a few $10^{-12} \text{ cm}^3/\text{s}$ for the ^{88}Sr lines at intensities below 1 W/cm^2 . Once this broadening is taken into account, the optical lengths reach $l_{\text{opt}}/I \sim 0.5\text{--}3.5 \text{ a}_0/(\text{W/cm}^2)$, one to two orders of magnitude below the bosonic lines of similar binding energy, which can be explained by quantum statistics and spin selection. The main anomaly of these lines is therefore their width. Predissociation, the inhomogeneous light shift of the PA beam and the light shift by the trapping light remain possible explanations; measuring the width of a line as a function of the PA intensity, and with the trap switched off during the PA pulses, would distinguish them.

Conclusion

Chapter 1 In the first chapter, I presented the physical concepts underlying the cooling scheme of our experiment, providing the elements necessary for a new PhD student – or a similar cold-atom experiment – to operate the setup and understand the physics behind it. Building on the previous theses from our team, this work also presents a new optical setup for filtering the MOGlabs ECDL laser used for the photoassociation (PA) process. The purpose of this filtering cavity is to suppress spontaneous emission on far-detuned states, which otherwise arises from the broadening of the laser spectrum due to amplified spontaneous emission.

Chapter 2 The second chapter presents the results of our team’s article published in PRX Quantum during my PhD. This work describes the quantum platform we engineered to measure physical properties of our Fermi gas using quantum-enhanced protocols. The article begins with the characterization of the pseudo-spin qubits we engineered: their coherence time reaches more than three seconds, owing to the low sensitivity of ^{87}Sr ’s ground state to external perturbations.

The robustness of these pseudo-spin qubits then allowed us to extend the scheme to a multi-state interferometer – made possible by the large Hilbert space accessible through the nuclear spin $I = 9/2$ of ^{87}Sr – giving access to physical quantities such as the linear and quadratic components of the light shift induced by the two-photon Raman process used for spin manipulation.

Finally, we demonstrate a measurement scheme for the non-commuting observables $\hat{s}^x, \hat{s}^y, \hat{s}^z$, which we show is equivalent to the implementation of a Positive-Operator-Valued Measure (POVM); this equivalence is derived in a dedicated section following the included article.

Chapter 3 The last chapter presents a photoassociation spectroscopy study of ^{87}Sr on the $^1S_0 - ^3P_1$ intercombination line, a process that had only been briefly mentioned in the literature [110]. The chapter opens with the molecular-physics background needed to interpret the measurements, and with the method used to extract the optical length l_{opt} of broadened lines, either from the depth of a loss spectrum or from the decay of the atom number at the line center.

The protocol was first calibrated on two known vibrational lines of ^{88}Sr : their positions agree with the literature, and, once their linewidth is accounted for, their optical lengths agree with the measurements of Blatt et al. within a factor $\lesssim 2$. This calibration, and the first attempts on ^{87}Sr , also revealed several pitfalls that should be kept in mind for this type of spectroscopy: sidebands of the PA laser can address atomic transitions and mimic molecular losses; the PA light induces

non-molecular losses, which are removed from the spectra by measuring the depth relative to the off-resonant level, and must also be removed from the decay curves before comparing them with the spectra; and a non-exponential decay alone is not a proof of photoassociation, so that each candidate line has to be checked by searching for losses at an equal and opposite detuning from the atomic resonance. Once these non-molecular losses are taken into account, the spectra and the decays give consistent loss coefficients.

Three molecular lines were then observed on ^{87}Sr , at -259 , -276 and -377 MHz from the atomic resonance. They are two to three orders of magnitude broader than the expected radiative width (from 5 to 20 MHz instead of 15 kHz). Their loss coefficients at the line center, $K_2 \sim 10^{-14}$ – 10^{-12} cm³/s for PA intensities of 5 to 31 W/cm², are at most comparable to those of the ^{88}Sr lines (a few 10^{-12} cm³/s), although they were measured at intensities 10 to 10^4 times higher, and their optical lengths, $l_{\text{opt}}/I \simeq 0.5$ – 3.5 a_0 /(W/cm²), lie one to two orders of magnitude below that of a ^{88}Sr line at a comparable detuning, which explains the high intensities needed to observe them. The weak loss rates are a direct consequence of this broadening. Three explanations remain open: predissociation of the molecular states towards the continuum of another potential, an inhomogeneous light shift of the PA beam across the cloud, and a light shift by the trapping light, which was on during the ^{87}Sr measurements. Measuring the linewidth as a function of the PA intensity, and with the trap switched off during the PA pulses, would discriminate between them.

Outcome Several improvements to the experiment are already underway. To improve the resolution of the cloud imaging, a microscope has been built by a member of the team and is ready to be implemented; this system would also give access to the atom number before time-of-flight, providing a more accurate estimate at every step of the sequence. In parallel, a full replacement for the current transverse cooling system has been constructed and is ready to be installed on the main experiment table, which will consequently increase the number of atoms delivered to the vacuum cell.

On the scientific side, the interferometry scheme developed in this thesis also suggests a way to probe the SU(N) symmetry of the fermionic species **compléter : lien précis avec la mesure des interactions**, opening a possible new diagnostic tool for interactions in this system.

Understanding the physical origin of the line broadening – expected from the ongoing optical pumping studies – is the next necessary step: it should make a larger fraction of the ^{87}Sr spectroscopy accessible, and in turn allow the observed molecular lines to be unambiguously assigned to their associated molecular potentials. This identification is the last missing piece toward the original goal of this work: driving the system, through spin-selective photoassociative losses, into a highly-entangled Dicke state with well-defined spin components.

Bibliography

- [1] H. Ahmed, A. Litvinov, P. Guesdon, E. Maréchal, J. H. Huckans, B. Pasquiou, B. Laburthe-Tolra, and M. Robert-de Saint-Vincent, “Coherent control over the high-dimensional space of the nuclear spin of alkaline-earth atoms,” *PRX Quantum*, vol. 6, p. 020352, June 2025. arXiv:2501.01731 [quant-ph].
- [2] P. Bataille, “Réalisation de gaz quantiques de grand spin,”
- [3] A. Litvinov, *Manipulation of the nuclear spin states of ^{87}Sr in degenerate $SU(N)$ -symmetric Fermi gases*. Theses, Sorbonne Paris Nord, Jan. 2023.
- [4] J. H. Han, J. H. Kang, M. Lee, and Y.-i. Shin, “Photoassociation spectroscopy of ultracold ^{173}Yb atoms near the intercombination line,” *Physical Review A*, vol. 97, p. 013401, Jan. 2018. arXiv:1710.03072 [physics].
- [5] T. Zelevinsky, M. M. Boyd, A. D. Ludlow, T. Ido, J. Ye, R. Ciurylo, P. Naidon, and P. S. Julienne, “Narrow Line Photoassociation in an Optical Lattice,” *Physical Review Letters*, vol. 96, p. 203201, May 2006. Number: 20 arXiv:physics/0602135.
- [6] C. H. Payne, *Stellar Atmospheres; a Contribution to the Observational Study of High Temperature in the Reversing Layers of Stars*. PhD thesis, Radcliffe College, Cambridge, MA, USA, 1925. Harvard Observatory Monograph No. 1.
- [7] R. E. Franklin and R. G. Gosling, “Molecular configuration in sodium thymonucleate,” *Nature*, vol. 171, no. 4356, pp. 740–741, 1953.
- [8] C. J. Bordé, “La métrologie du XXI^e siècle, la physique quantique et le Système international d’unités,” *Comptes Rendus Physique*, vol. 18, no. 9-10, pp. 580–601, 2017.
- [9] R. P. Feynman, “Simulating physics with computers,” *International Journal of Theoretical Physics*, vol. 21, no. 6-7, pp. 467–488, 1982.
- [10] M. Greiner, O. Mandel, T. Esslinger, T. W. Hänsch, and I. Bloch, “Quantum phase transition from a superfluid to a mott insulator in a gas of ultracold atoms,” *Nature*, vol. 415, no. 6867, pp. 39–44, 2002.
- [11] A. Mazurenko, C. S. Chiu, G. Ji, M. F. Parsons, M. Kanász-Nagy, R. Schmidt, F. Grusdt, E. Demler, D. Greif, and M. Greiner, “A cold-atom fermi–hubbard antiferromagnet,” *Nature*, vol. 545, no. 7655, p. 462, 2017.
- [12] H. Sun, B. Yang, H.-Y. Wang, Z.-Y. Zhou, G.-X. Su, H.-N. Dai, Z.-S. Yuan, and J.-W. Pan, “Realization of a bosonic antiferromagnet,” *Nature Physics*, vol. 17, no. 9, pp. 990–994, 2021.

- [13] W. S. Bakr, J. I. Gillen, A. Peng, S. Fölling, and M. Greiner, “A quantum gas microscope for detecting single atoms in a hubbard-regime optical lattice,” *Nature*, vol. 462, no. 7269, pp. 74–77, 2009.
- [14] S. Blatt, T. L. Nicholson, B. J. Bloom, J. R. Williams, J. W. Thomsen, P. S. Julienne, and J. Ye, “Measurement of Optical Feshbach Resonances in an Ideal Gas,” *Physical Review Letters*, vol. 107, p. 073202, Aug. 2011.
- [15] M. Borkowski, P. Morzyński, R. Ciuryło, P. S. Julienne, M. Yan, B. J. DeSalvo, and T. C. Killian, “Mass scaling and nonadiabatic effects in photoassociation spectroscopy of ultracold strontium atoms,” *Physical Review A*, vol. 90, p. 032713, Sept. 2014.
- [16] B. J. Reschovsky, B. P. Ruzic, H. Miyake, N. C. Pimenti, P. S. Julienne, and G. K. Campbell, “Narrow-line photoassociation spectroscopy and mass-scaling of bosonic strontium,” Oct. 2018. arXiv:1808.06507 [physics].
- [17] A. Derevianko and H. Katori, “Colloquium: Physics of optical lattice clocks,” *Reviews of Modern Physics*, vol. 83, no. 1, pp. 331–347, 2011.
- [18] H. Katori, M. Takamoto, V. G. Pal’chikov, and V. D. Ovsiannikov, “Ultra-stable optical clock with neutral atoms in an engineered light shift trap,” *Physical Review Letters*, vol. 91, no. 17, p. 173005, 2003.
- [19] L. Pezze’ and A. Smerzi, “Quantum theory of phase estimation,” Nov. 2014. arXiv:1411.5164 [quant-ph].
- [20] Y. A. Yang, M. Miklos, Y. M. Tso, S. Kraus, J. Hur, and J. Ye, “Clock precision beyond the standard quantum limit at 10^{-18} level,” *Physical Review Letters*, vol. 135, no. 19, p. 193202, 2025.
- [21] Y. N. Martinez De Escobar, P. G. Mickelson, P. Pellegrini, S. B. Nagel, A. Traverso, M. Yan, R. Côté, and T. C. Killian, “Two-photon photoassociative spectroscopy of ultracold Sr 88,” *Physical Review A*, vol. 78, p. 062708, Dec. 2008.
- [22] S. Stellmer, F. Schreck, and T. C. Killian, “Degenerate quantum gases of strontium,” vol. 2, pp. 1–80, May 2014. arXiv:1307.0601 [cond-mat.quant-gas].
- [23] M. Foss-Feig, A. J. Daley, J. K. Thompson, and A. M. Rey, “Steady-state many-body entanglement of hot reactive fermions,” *Physical Review Letters*, vol. 109, p. 230501, Dec. 2012. arXiv:1207.4741 [cond-mat.quant-gas].
- [24] P. Bataille, A. Litvinov, I. Manai, J. Huckans, F. Wiotte, A. Kaladjian, O. Gorceix, E. Maréchal, B. Laburthe-Tolra, and M. Robert-de Saint-Vincent, “Adiabatic spin-dependent momentum transfer in an SU(N) degenerate Fermi gas,” *Phys. Rev. A*, vol. 102, p. 013317, 2020.
- [25] F. Faisant, *Building a strontium quantum gas microscope*. Theses, Université Paris-Saclay, Dec. 2024. Issue: 2024UPASP147.
- [26] W. D. Phillips and H. Metcalf, “Laser Deceleration of an Atomic Beam,” *Physical Review Letters*, vol. 48, pp. 596–599, Mar. 1982.
- [27] W. D. Phillips, “Laser cooling and trapping of neutral atoms,” *Rev. Mod.*

Phys., vol. 70, no. 3, 1998.

- [28] J. Dalibard, “Atomes ultra-froids.” Notes de cours de Master 2 / DEA, École Normale Supérieure, 2008.
- [29] S. Stellmer, B. Pasquiou, R. Grimm, and F. Schreck, “Creation of ultracold Sr2 molecules in the electronic ground state,” *Physical Review Letters*, vol. 109, p. 115302, Sept. 2012. arXiv:1205.4505 [cond-mat].
- [30] J. P. Bartolotta, M. A. Norcia, J. R. K. Cline, J. K. Thompson, and M. J. Holland, “Laser cooling by sawtooth-wave adiabatic passage,” *Phys. Rev. A*, vol. 98, p. 023404, 2018.
- [31] S. Stellmer, F. Schreck, and T. C. Killian, “DEGENERATE QUANTUM GASES OF STRONTIUM,” in *Annual Review of Cold Atoms and Molecules*, vol. 2, pp. 1–80, WORLD SCIENTIFIC, May 2014.
- [32] S. Stellmer, R. Grimm, and F. Schreck, “Production of quantum degenerate strontium gases: Larger, better, faster, colder,” Dec. 2012.
- [33] E. D. Black, “An introduction to Pound–Drever–Hall laser frequency stabilization,” *American Journal of Physics*, vol. 69, pp. 79–87, Jan. 2001.
- [34] C. Klempt, O. Topic, G. Gebreyesus, M. Scherer, T. Henninger, P. Hyllus, W. Ertmer, L. Santos, and J. J. Arlt, “Parametric amplification of vacuum fluctuations in a spinor condensate,” *Phys. Rev. Lett.*, vol. 104, p. 195303, May 2010.
- [35] J. S. Krauser, J. Heinze, N. Fläschner, S. Götze, O. Jürgensen, D.-S. Lühmann, C. Becker, and K. Sengstock, “Coherent multi-flavour spin dynamics in a fermionic quantum gas,” *Nature Physics*, vol. 8, pp. 813–818, Sept. 2012.
- [36] S. Taie, R. Yamazaki, S. Sugawa, and Y. Takahashi, “An SU(6) Mott insulator of an atomic Fermi gas realized by large-spin Pomeranchuk cooling,” *Nat. Phys.*, vol. 8, no. 11, p. 825, 2012.
- [37] A. de Paz, A. Sharma, A. Chotia, E. Maréchal, J. H. Huckans, P. Pedri, L. Santos, O. Gorceix, L. Vernac, and B. Laburthe-Tolra, “Nonequilibrium quantum magnetism in a dipolar lattice gas,” *Phys. Rev. Lett.*, vol. 111, p. 185305, Oct 2013.
- [38] G. Pagano, M. Mancini, G. Cappellini, P. Lombardi, F. Schäfer, H. Hu, X.-J. Liu, J. Catani, C. Sias, M. Inguscio, and L. Fallani, “A one-dimensional liquid of fermions with tunable spin,” *Nat. Phys.*, vol. 10, no. 3, pp. 198–201, 2014.
- [39] A. Patscheider, B. Zhu, L. Chomaz, D. Petter, S. Baier, A.-M. Rey, F. Ferlaino, and M. J. Mark, “Controlling dipolar exchange interactions in a dense three-dimensional array of large-spin fermions,” *Phys. Rev. Res.*, vol. 2, p. 023050, Apr 2020.
- [40] Y. A. Alaoui, B. Zhu, S. R. Muleady, W. Dubosclard, T. Roscilde, A. M. Rey, B. Laburthe-Tolra, and L. Vernac, “Measuring correlations from the collective spin fluctuations of a large ensemble of lattice-trapped dipolar spin-3 atoms,” *Phys. Rev. Lett.*, vol. 129, p. 023401, Jul 2022.
- [41] E. Kiktenko, A. Fedorov, A. Strakhov, and V. Man’ko, “Single qudit real-

- ization of the Deutsch algorithm using superconducting many-level quantum circuits,” *Phys. Lett. A*, vol. 379, no. 22, pp. 1409–1413, 2015.
- [42] T. Fernholz, H. Krauter, K. Jensen, J. F. Sherson, A. S. Sørensen, and E. S. Polzik, “Spin squeezing of atomic ensembles via nuclear-electronic spin entanglement,” *Phys. Rev. Lett.*, vol. 101, p. 073601, Aug 2008.
 - [43] C. Hamley, C. Gerving, T. Hoang, E. Bookjans, and M. Chapman, “Spin-nematic squeezed vacuum in a quantum gas,” *Nat. Phys.*, vol. 8, pp. 305–308, 2 2012.
 - [44] D. Kajtoch and E. Witkowska, “Spin squeezing in dipolar spinor condensates,” *Phys. Rev. A*, vol. 93, p. 023627, Feb 2016.
 - [45] T. Chalopin, C. Bouazza, A. Evrard, V. Makhalov, D. Dreon, J. Dalibard, L. A. Sidorenkov, and S. Nascimbene, “Quantum-enhanced sensing using non-classical spin states of a highly magnetic atom,” *Nat. Commun.*, vol. 9, no. 1, p. 4955, 2018.
 - [46] M. J. Peterer, S. J. Bader, X. Jin, F. Yan, A. Kamal, T. J. Gudmundsen, P. J. Leek, T. P. Orlando, W. D. Oliver, and S. Gustavsson, “Coherence and decay of higher energy levels of a superconducting transmon qubit,” *Phys. Rev. Lett.*, vol. 114, p. 010501, Jan 2015.
 - [47] E. Svetitsky, H. Suchowski, R. Resh, Y. Shalibo, J. M. Martinis, and N. Katz, “Hidden two-qubit dynamics of a four-level Josephson circuit,” *Nat. Commun.*, vol. 5, no. 1, p. 5617, 2014.
 - [48] C. Godfrin, R. Ballou, E. Bonet, M. Ruben, S. Klyatskaya, W. Wernsdorfer, and F. Balestro, “Generalized Ramsey interferometry explored with a single nuclear spin qudit,” *npj Quantum Inf.*, vol. 4, no. 1, p. 53, 2018.
 - [49] J. Randall, S. Weidt, E. D. Standing, K. Lake, S. C. Webster, D. F. Murgia, T. Navickas, K. Roth, and W. K. Hensinger, “Efficient preparation and detection of microwave dressed-state qubits and qutrits with trapped ions,” *Phys. Rev. A*, vol. 91, p. 012322, Jan 2015.
 - [50] B. P. Lanyon, M. Barbieri, M. P. Almeida, T. Jennewein, T. C. Ralph, K. J. Resch, G. J. Pryde, J. L. O’Brien, A. Gilchrist, and A. G. White, “Simplifying quantum logic using higher-dimensional Hilbert spaces,” *Nat. Phys.*, vol. 5, no. 2, pp. 134–140, 2008.
 - [51] D. M. Stamper-Kurn and M. Ueda, “Spinor Bose gases: Symmetries, magnetism, and quantum dynamics,” *Rev. Mod. Phys.*, vol. 85, pp. 1191–1244, 2013.
 - [52] L. Chomaz, I. Ferrier-Barbut, F. Ferlaino, B. Laburthe-Tolra, B. L. Lev, and T. Pfau, “Dipolar physics: a review of experiments with magnetic quantum gases,” *Rep. Prog. Phys.*, vol. 86, no. 2, p. 026401, 2022.
 - [53] A. D. Ludlow, M. M. Boyd, J. Ye, E. Peik, and P. O. Schmidt, “Optical atomic clocks,” *Rev. Mod. Phys.*, vol. 87, pp. 637–701, Jun 2015.
 - [54] A. V. Gorshkov, M. Hermele, V. Gurarie, C. Xu, P. S. Julienne, J. Ye, P. Zoller, E. Demler, M. D. Lukin, and A. M. Rey, “Two-orbital SU(N) magnetism with ultracold alkaline-earth atoms,” *Nat. Phys.*, vol. 6, no. 4, pp. 289–295, 2010.

- [55] M. Hermele, V. Gurarie, and A. M. Rey, “Mott insulators of ultracold fermionic alkaline earth atoms: Underconstrained magnetism and chiral spin liquid,” *Phys. Rev. Lett.*, vol. 103, p. 135301, 2009.
- [56] P. Corboz, K. Penc, F. Mila, and A. M. Läuchli, “Simplex solids in $SU(N)$ Heisenberg models on the kagome and checkerboard lattices,” *Phys. Rev. B*, vol. 86, p. 041106, 2012.
- [57] M. Cazalilla and A. Rey, “Ultracold Fermi gases with emergent $SU(N)$ symmetry,” *Rep. Prog. Phys.*, vol. 77, p. 124401, 11 2014.
- [58] C. Hofrichter, L. Riegger, F. Scazza, M. Höfer, D. R. Fernandes, I. Bloch, and S. Fölling, “Direct probing of the Mott crossover in the $SU(N)$ Fermi-Hubbard model,” *Phys. Rev. X*, vol. 6, p. 021030, 2016.
- [59] H. Ozawa, S. Taie, Y. Takasu, and Y. Takahashi, “Antiferromagnetic spin correlation of $SU(N)$ Fermi gas in an optical superlattice,” *Phys. Rev. Lett.*, vol. 121, p. 225303, 2018.
- [60] D. Tusi, L. Franchi, L. Livì, K. Baumann, D. Benedicto Orenes, L. Del Re, R. Barfknecht, T.-W. Zhou, M. Inguscio, G. Cappellini, M. Capone, J. Catani, and L. Fallani, “Flavour-selective localization in interacting lattice fermions,” *Nat. Phys.*, vol. 18, pp. 1201–1205, 9 2022.
- [61] F. Leroux, K. Pandey, R. Rehbi, F. Chevy, C. Miniatura, B. Grémaud, and D. Wilkowski, “Non-Abelian adiabatic geometric transformations in a cold strontium gas,” *Nat. Commun.*, vol. 9, p. 3580, 9 2018.
- [62] K. Barnes, P. Battaglino, B. Bloom, K. Cassella, R. Coxe, N. Crisosto, J. King, S. Kondov, K. Kotru, S. Larsen, J. Lauigan, B. Lester, M. McDonald, E. Megidish, S. Narayanaswami, C. Nishiguchi, R. Notermans, L. Peng, A. Ryou, T.-Y. Wu, and M. Yarwood, “Assembly and coherent control of a register of nuclear spin qubits,” *Nat. Commun.*, vol. 13, p. 2779, 5 2022.
- [63] I. Deutsch and P. Jessen, “Quantum control and measurement of atomic spins in polarization spectroscopy,” *Opt. Commun.*, vol. 283, pp. 681–694, 3 2010.
- [64] S. Chen, C. Wu, S.-C. Zhang, and Y. Wang, “Exact spontaneous plaquette ground states for high-spin ladder models,” *Phys. Rev. B*, vol. 72, p. 214428, 2005.
- [65] E. Majorana, “Atomi orientati in campo magnetico variabile,” *Il Nuovo Cimento*, vol. 9, pp. 43–50, 2 1932.
- [66] R. Barnett, A. Turner, and E. Demler, “Classifying novel phases of spinor atoms,” *Phys. Rev. Lett.*, vol. 97, p. 180412, 2006.
- [67] P. Bruno, “Quantum geometric phase in Majorana’s stellar representation: Mapping onto a many-body Aharonov-Bohm phase,” *Phys. Rev. Lett.*, vol. 108, p. 240402, 2012.
- [68] A. Evrard, V. Makhalov, T. Chalopin, L. A. Sidorenkov, J. Dalibard, R. Lopes, and S. Nascimbene, “Enhanced magnetic sensitivity with non-Gaussian quantum fluctuations,” *Phys. Rev. Lett.*, vol. 122, p. 173601, May 2019.
- [69] S. Omanakuttan, A. Mitra, M. J. Martin, and I. H. Deutsch, “Quantum op-

- timal control of ten-level nuclear spin qudits in ^{87}Sr ,” *Phys. Rev. A*, vol. 104, p. L060401, 2021.
- [70] D. Burba, H. Dunikowski, M. Robert-de-Saint-Vincent, E. Witkowska, and G. Juzeliūnas, “Effective light-induced Hamiltonian for atoms with large nuclear spin,” *Phys. Rev. Res.*, vol. 6, p. 033293, 2024.
 - [71] F. Claude, L. Lafforgue, J. J. A. Houwman, M. J. Mark, and F. Ferlino, “Optical manipulation of spin states in ultracold magnetic atoms via an inner-shell hz transition,” *Phys. Rev. Res.*, vol. 6, p. L042016, 2024.
 - [72] X. Zheng, J. Dolde, and S. Kolkowitz, “Reducing the instability of an optical lattice clock using multiple atomic ensembles,” *Phys. Rev. X*, vol. 14, p. 011006, Jan 2024.
 - [73] J. Ruseckas, G. Juzeliūnas, P. Öhberg, and M. Fleischhauer, “Non-Abelian gauge potentials for ultracold atoms with degenerate dark states,” *Phys. Rev. Lett.*, vol. 95, p. 010404, Jun 2005.
 - [74] Y. Baamara, M. Gessner, and A. Sinatra, “Quantum-enhanced multiparameter estimation and compressed sensing of a field,” *SciPost Phys.*, vol. 14, p. 050, 2023.
 - [75] P. Kunkel, M. Prüfer, S. Lannig, R. Rosa-Medina, A. Bonnin, M. Gärttner, H. Strobel, and M. K. Oberthaler, “Simultaneous readout of noncommuting collective spin observables beyond the standard quantum limit,” *Phys. Rev. Lett.*, vol. 123, p. 063603, Aug 2019.
 - [76] S. L. Campbell, R. B. Hutson, G. E. Marti, A. Goban, N. D. Oppong, R. L. McNally, L. Sonderhouse, J. M. Robinson, W. Zhang, B. J. Bloom, and J. Ye, “A Fermi-degenerate three-dimensional optical lattice clock,” *Science*, vol. 358, p. 90, 2017.
 - [77] D. Savoie, M. Altorio, B. Fang, L. A. Sidorenkov, R. Geiger, and A. Landragin, “Interleaved atom interferometry for high-sensitivity inertial measurements,” *Sci. Adv.*, vol. 4, p. eaau7948, 2018.
 - [78] M. Schioppo, R. C. Brown, W. F. McGrew, N. Hinkley, R. J. Fasano, K. Beloy, T. H. Yoon, G. Milani, D. Nicolodi, J. A. Sherman, N. B. Phillips, C. W. Oates, and A. D. Ludlow, “Ultrastable optical clock with two cold-atom ensembles,” *Nat. Photonics*, vol. 11, p. 48, 2016.
 - [79] P. Cheiney, L. Fouché, S. Templier, F. Napolitano, B. Battelier, P. Bouyer, and B. Barrett, “Navigation-compatible hybrid quantum accelerometer using a Kalman filter,” *Phys. Rev. Appl.*, vol. 10, p. 034030, 2018.
 - [80] M. Gessner, A. Smerzi, and L. Pezzè, “Multiparameter squeezing for optimal quantum enhancements in sensor networks,” *Nat. Commun.*, vol. 11, p. 3817, 7 2020.
 - [81] We could equally have used an incoherent mixture of these two states.
 - [82] S. Taie, Y. Takasu, S. Sugawa, R. Yamazaki, T. Tsujimoto, R. Murakami, and Y. Takahashi, “Realization of a $\text{SU}(2) \times \text{SU}(6)$ system of fermions in a cold atomic gas,” *Phys. Rev. Lett.*, vol. 105, p. 190401, Nov 2010.

- [83] S. Stellmer, R. Grimm, and F. Schreck, “Detection and manipulation of nuclear spin states in fermionic strontium,” *Phys. Rev. A*, vol. 84, p. 043611, 2011.
- [84] C. Bidinosti, G. Tassevin, and P.-J. Nacher, “Generating accurate tip angles for NMR outside the rotating-wave approximation,” *J. Magn. Reson.*, vol. 345, p. 107306, 12 2022.
- [85] M. Kitagawa and M. Ueda, “Squeezed spin states,” *Phys. Rev. A*, vol. 47, pp. 5138–5143, Jun 1993.
- [86] B. Yurke, S. L. McCall, and J. R. Klauder, “SU(2) and SU(1,1) interferometers,” *Phys. Rev. A*, vol. 33, pp. 4033–4054, Jun 1986.
- [87] J. Estève, C. Gross, A. Weller, S. Giovanazzi, and M. K. Oberthaler, “Squeezing and entanglement in a Bose–Einstein condensate,” *Nature*, vol. 455, no. 7217, pp. 1216–1219, 2008.
- [88] M. Lewenstein, B. Kraus, J. I. Cirac, and P. Horodecki, “Optimization of entanglement witnesses,” *Phys. Rev. A*, vol. 62, p. 052310, Oct 2000.
- [89] B. Lücke, J. Peise, G. Vitagliano, J. Arlt, L. Santos, G. Tóth, and C. Klempt, “Detecting multiparticle entanglement of Dicke states,” *Phys. Rev. Lett.*, vol. 112, p. 155304, Apr 2014.
- [90] O. Hosten, N. J. Engelsen, R. Krishnakumar, and M. A. Kasevich, “Measurement noise 100 times lower than the quantum-projection limit using entangled atoms,” *Nature*, vol. 529, no. 7587, pp. 505–508, 2016.
- [91] A. S. Sørensen and K. Mølmer, “Entanglement and extreme spin squeezing,” *Phys. Rev. Lett.*, vol. 86, pp. 4431–4434, May 2001.
- [92] G. Vitagliano, P. Hyllus, I. L. Egusquiza, and G. Tóth, “Spin squeezing inequalities for arbitrary spin,” *Phys. Rev. Lett.*, vol. 107, p. 240502, Dec 2011.
- [93] S. Lepoutre, J. Schachenmayer, L. Gabardos, B. Zhu, B. Naylor, E. Maréchal, O. Gorceix, A. M. Rey, L. Vernac, and B. Laburthe-Tolra, “Out-of-equilibrium quantum magnetism and thermalization in a spin-3 many-body dipolar lattice system,” *Nat. Commun.*, vol. 10, p. 1714, apr 2019.
- [94] Y. Aziz Alaoui, S. R. Muleady, E. Chaparro, Y. Trifa, A. M. Rey, T. Roscilde, B. Laburthe-Tolra, and L. Vernac, “Measuring bipartite spin correlations of lattice-trapped dipolar atoms,” *Phys. Rev. Lett.*, vol. 133, p. 203401, 2024.
- [95] C. Leprince, V. Gondret, C. Lamirault, R. Dias, Q. Marolleau, D. Boiron, and C. I. Westbrook, “Coherent coupling of momentum states: selectivity and phase control,” *arXiv*, p. 2411.09284, 2024.
- [96] O. Onishchenko, S. Pyatchenko, A. Urech, C.-C. Chen, S. Bennetts, G. A. Siviloglou, and F. Schreck, “Frequency of the ultranarrow 1S_0 – 3P_2 transition in ^{87}Sr ,” *Phys. Rev. A*, vol. 99, p. 052503, May 2019.
- [97] A. Peres and W. K. Wootters, “Optimal detection of quantum information,” *Phys. Rev. Lett.*, vol. 66, pp. 1119–1122, Mar 1991.
- [98] S. Massar, “Uncertainty relations for positive-operator-valued measures,” *Phys. Rev. A*, vol. 76, p. 042114, Oct 2007.

- [99] S. Pucher, V. Klüsener, F. Spriestersbach, J. Geiger, A. Schindewolf, I. Bloch, and S. Blatt, “Fine-structure qubit encoded in metastable strontium trapped in an optical lattice,” *Phys. Rev. Lett.*, vol. 132, p. 150605, Apr 2024.
- [100] Y. Nakamura, T. Kusano, R. Yokoyama, K. Saito, K. Higashi, N. Ozawa, T. Takano, Y. Takasu, and Y. Takahashi, “Hybrid atom tweezer array of nuclear spin and optical clock qubits,” *Phys. Rev. X*, vol. 14, p. 041062, Dec 2024.
- [101] T. Satoor, A. Fabre, J.-B. Bouhiron, A. Evrard, R. Lopes, and S. Nascimbene, “Partitioning dysprosium’s electronic spin to reveal entanglement in nonclassical states,” *Phys. Rev. Res.*, vol. 3, p. 043001, Oct 2021.
- [102] G. Bornet, G. Emperauger, C. Chen, F. Machado, S. Chern, L. Leclerc, B. Gély, Y. T. Chew, D. Barredo, T. Lahaye, N. Y. Yao, and A. Browaeys, “Enhancing a many-body dipolar Rydberg tweezer array with arbitrary local controls,” *Phys. Rev. Lett.*, vol. 132, p. 263601, 2024.
- [103] M. Mamaev, R. Blatt, J. Ye, and A. M. Rey, “Cluster state generation with spin-orbit coupled fermionic atoms in optical lattices,” *Phys. Rev. Lett.*, vol. 122, p. 160402, Apr 2019.
- [104] T. Brydges, A. Elben, P. Jurcevic, B. Vermersch, C. Maier, B. P. Lanyon, P. Zoller, R. Blatt, and C. F. Roos, “Probing Rényi entanglement entropy via randomized measurements,” *Science*, vol. 364, no. 6437, pp. 260–263, 2019.
- [105] I. Frérot and T. Roscilde, “Optimal entanglement witnesses: A scalable data-driven approach,” *Phys. Rev. Lett.*, vol. 127, p. 040401, Jul 2021.
- [106] J. A. Bergou, “Quantum state discrimination and selected applications,” *J. Phys.: Conf. Ser.*, vol. 84, p. 012001, 2007.
- [107] K. M. Jones, E. Tiesinga, P. D. Lett, and P. S. Julienne, “Ultracold photoassociation spectroscopy: Long-range molecules and atomic scattering,” *Reviews of Modern Physics*, vol. 78, pp. 483–535, May 2006.
- [108] S. Blatt, T. L. Nicholson, B. J. Bloom, J. R. Williams, J. W. Thomsen, P. S. Julienne, and J. Ye, “Systematic study of Optical Feshbach Resonances in an ideal gas,” *Physical Review Letters*, vol. 107, p. 073202, Aug. 2011. arXiv:1104.0210 [physics].
- [109] A. Stein, H. Knöckel, and E. Tiemann, “The $^1S+^1S$ asymptote of Sr_2 studied by Fourier-transform spectroscopy,” *The European Physical Journal D*, vol. 57, pp. 171–177, Apr. 2010. arXiv:1001.2741 [physics].
- [110] J. C. Hill, W. Huie, P. Lunia, J. D. Whalen, S. K. Kanungo, Y. Lu, and T. C. Killian, “Photoassociative spectroscopy of $\text{Sr } 87$,” *Physical Review A*, vol. 103, p. 023111, Feb. 2021.
- [111] L. et E. Lifchitz, *Mécanique Quantique Théorie non relativiste*. 1967.
- [112] J. Dalibard, “Les interactions entre atomes dans les gaz quantiques,”
- [113] J. L. Bohn and P. S. Julienne, “Prospects for influencing scattering lengths with far-off-resonant light,” *Physical Review A*, vol. 56, pp. 1486–1491, Aug 1997.

- [114] T. L. Nicholson, S. Blatt, B. J. Bloom, J. R. Williams, J. W. Thomsen, J. Ye, and P. S. Julienne, “Optical Feshbach resonances: Field-dressed theory and comparison with experiments,” *Physical Review A*, vol. 92, p. 022709, Aug. 2015.
- [115] A. Guttridge, *Photoassociation of Ultracold CsYb Molecules and Determination of Interspecies Scattering Lengths*. Springer Theses, Cham: Springer International Publishing, 2019.
- [116] K. Góral, T. Köhler, S. A. Gardiner, E. Tiesinga, and P. S. Julienne, “Adiabatic association of ultracold molecules via magnetic-field tunable interactions,” *Journal of Physics B: Atomic, Molecular and Optical Physics*, vol. 37, no. 17, pp. 3457–3500, 2004.
- [117] M. Yan, B. J. DeSalvo, B. Ramachandhran, H. Pu, and T. C. Killian, “Controlling Condensate Collapse and Expansion with an Optical Feshbach Resonance,” *Physical Review Letters*, vol. 110, p. 123201, Mar. 2013. arXiv:0906.1837 [physics].
- [118] M. Borkowski, R. Ciurylo, P. S. Julienne, S. Tojo, K. Enomoto, and Y. Takahashi, “Line shapes of optical Feshbach resonances near the intercombination transition of bosonic Ytterbium,” May 2009. arXiv:0905.0958 [physics].
- [119] J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*. Cambridge University Press, 3rd ed., 2020.
- [120] R. J. Le Roy and R. B. Bernstein, “Dissociation energy and long-range potential constants from vibrational energies of multiatomic molecules,” *The Journal of Chemical Physics*, vol. 52, no. 8, pp. 3869–3879, 1970.
- [121] T. Franzen, B. Pollklesener, C. Sillus, and A. Görlitz, “Intercombination line photoassociation spectroscopy of $^{87}\text{Rb}^{170}\text{Yb}$,” *Physical Review A*, vol. 107, p. 023114, Feb. 2023. arXiv:2212.00913 [physics.atom-ph].
- [122] B. Mukherjee, M. D. Frye, and J. M. Hutson, “Feshbach resonances and molecule formation in ultracold mixtures of Rb and Yb ($P\ 3$) atoms,” *Physical Review A*, vol. 105, p. 023306, Feb. 2022.

